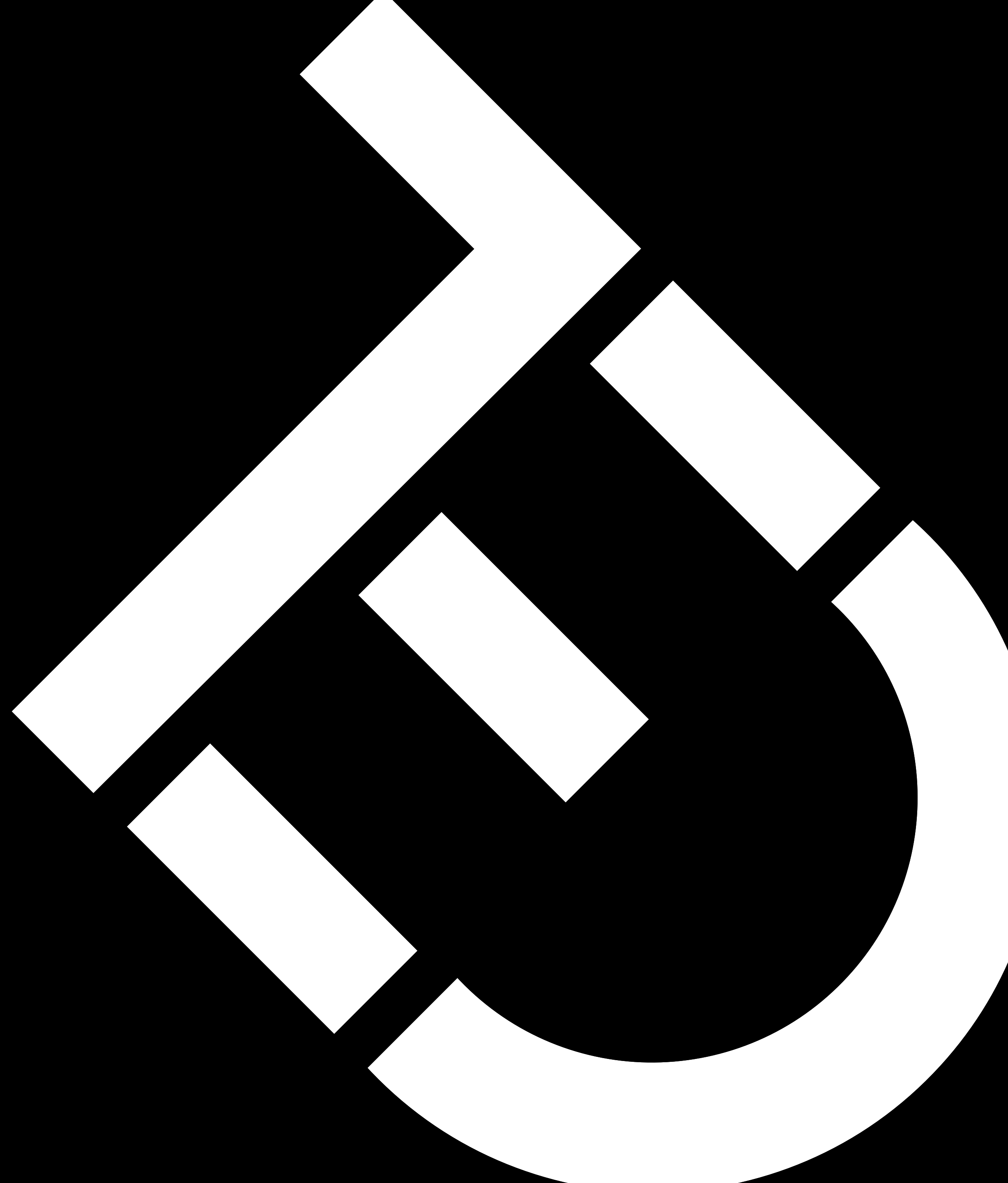
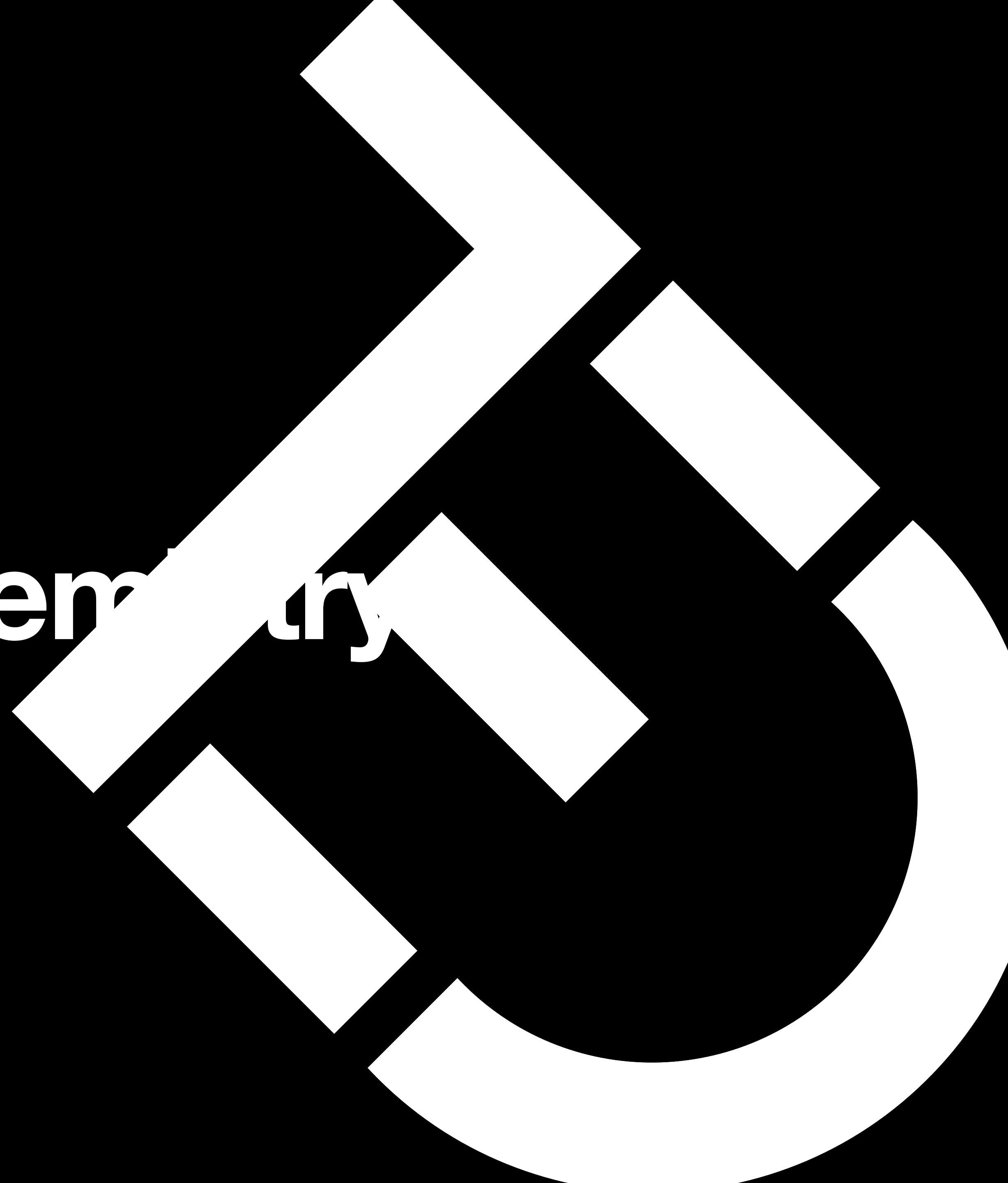


CHem is try

funi



Experimental Chemistry



Measurements

what is this chapter about

Measurements can either be quantitative or qualitative.

Quantitative data is a numerical measurement: i.e volume, mass, temperature,

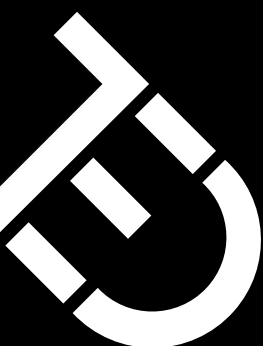
Qualitative data is a measurement of the qualities something has: i.e colour, odour, aka not a numerical measurement

Measurements can also vary in accuracy and precision

Accuracy refers to how close your data is to the true value (how correct your data is)

Precision refers to how close different measurements are to each other (how consistent your data is)

**data can be accurate but not precise, or precise but not accurate



Measurements: Units

what is this chapter about

There are 7 base SI units that are used:

mass: kilograms [kg]

time: seconds [s]

amount of substance: moles [mol]

temperature: Kelvin [K]

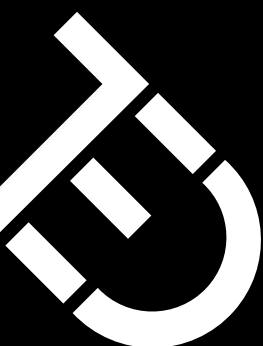
distance: meters [m]

current: Ampere [A]

light intensity: candela [cd]

uncommon in chemistry

All other units like volume, or density, are derived (e.g. volume is m^3 , density is kg/m^3)



Laboratory Apparatus

what is this chapter about

To measure the different quantities, we use the following apparatus

mass: electronic mass balance, recorded to nearest 0.01g

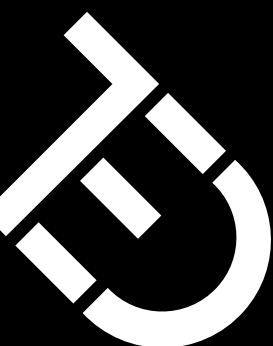
time: stopwatch, recorded to nearest 0.01s

temperature: thermometer, recorded to nearest 0.5°C

volume(containing liquids): measuring cylinder, recorded to nearest 0.5cm³

volume(transferring liquids): burette/pipette, burette is recorded to nearest 0.05cm³, while pipette has fixed amount of liquid that can be transferred (20cm³, 25cm³, etc.)

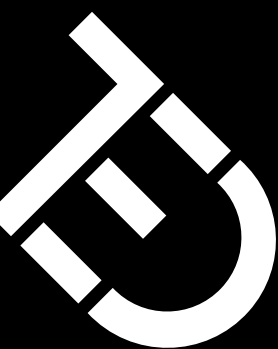
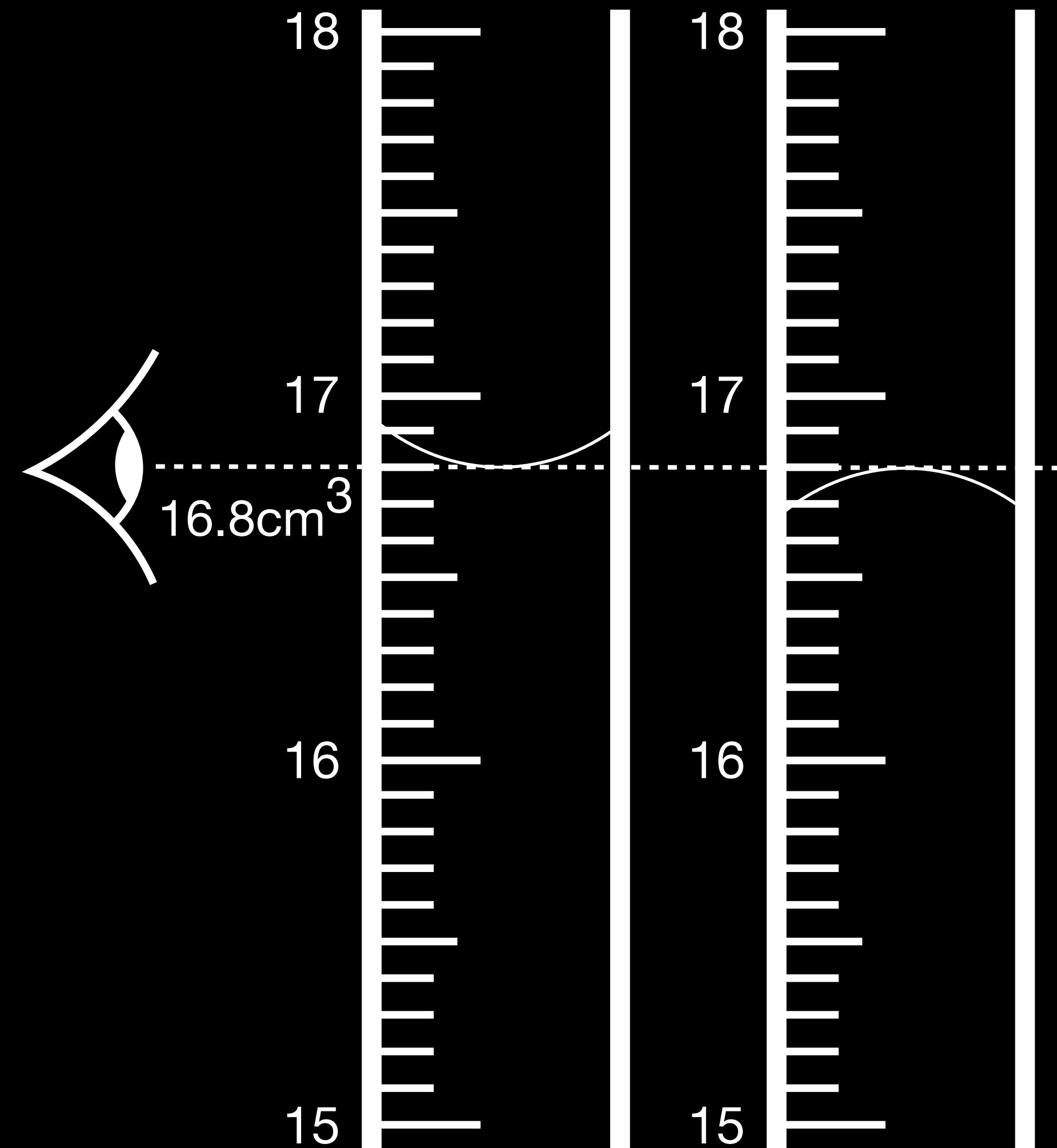
volume(gases): syringe, measured to nearest 0.5cm³



Laboratory Apparatus

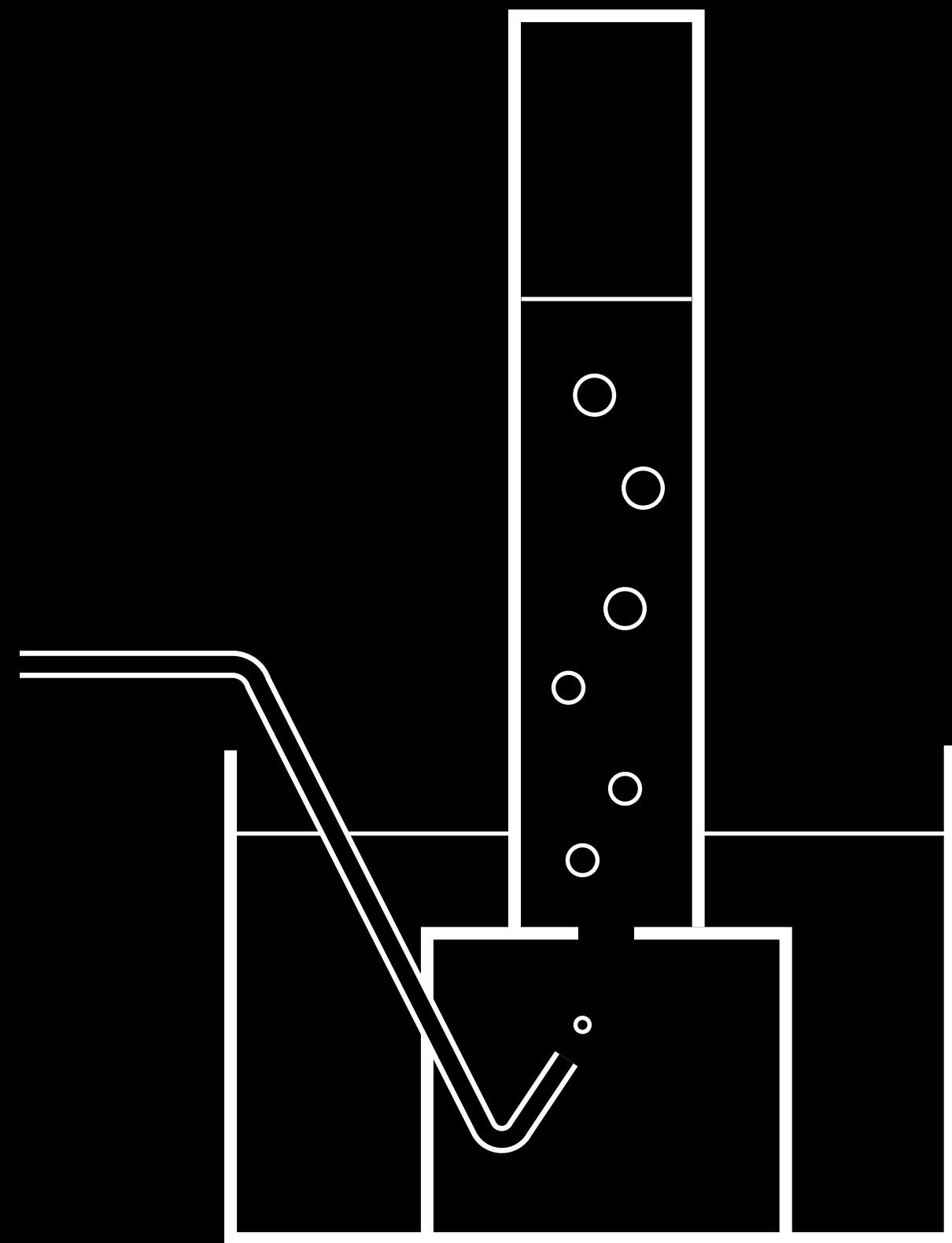
what is this chapter bru

When measuring liquids, make sure to read the meniscus at eye level. Most of the time, the meniscus curves down, so we read at the minimum of the meniscus. However, in cases where the substance has a meniscus that curves up, we read at the maximum of the meniscus.

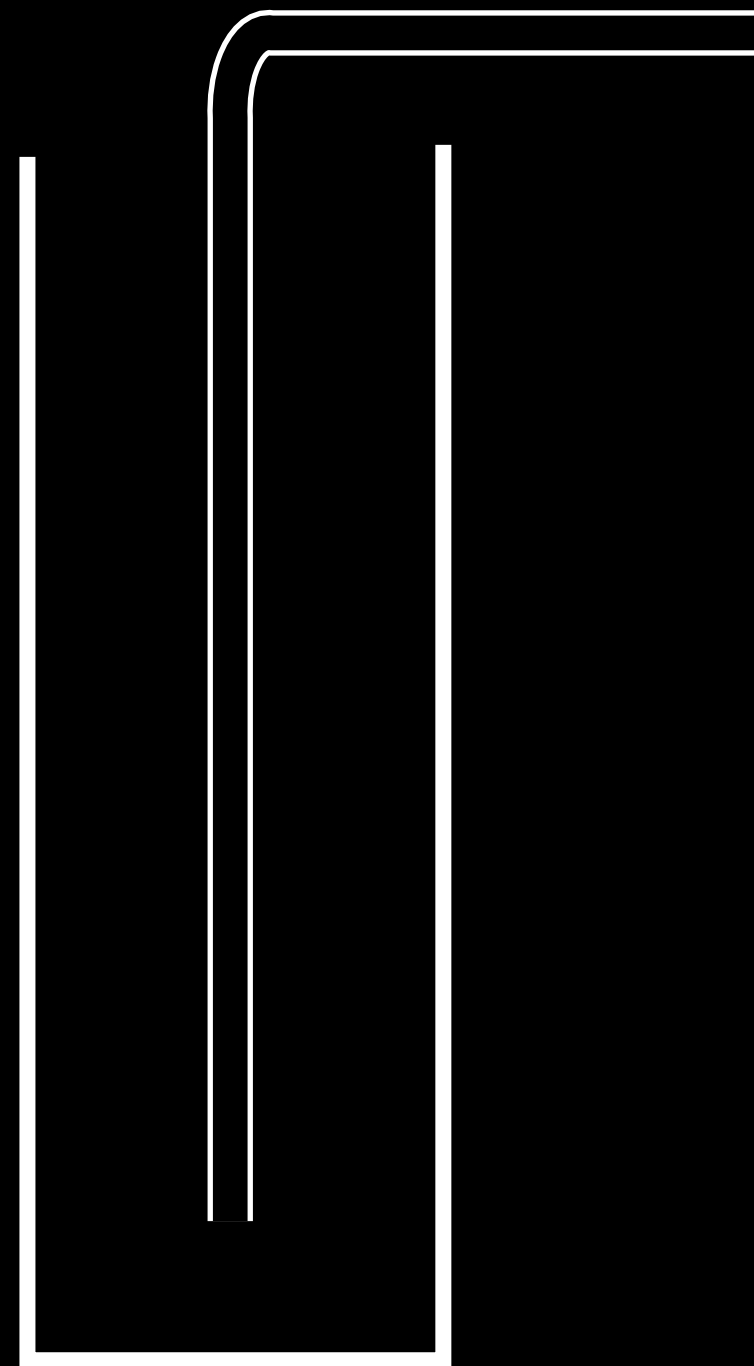


Gas Collection

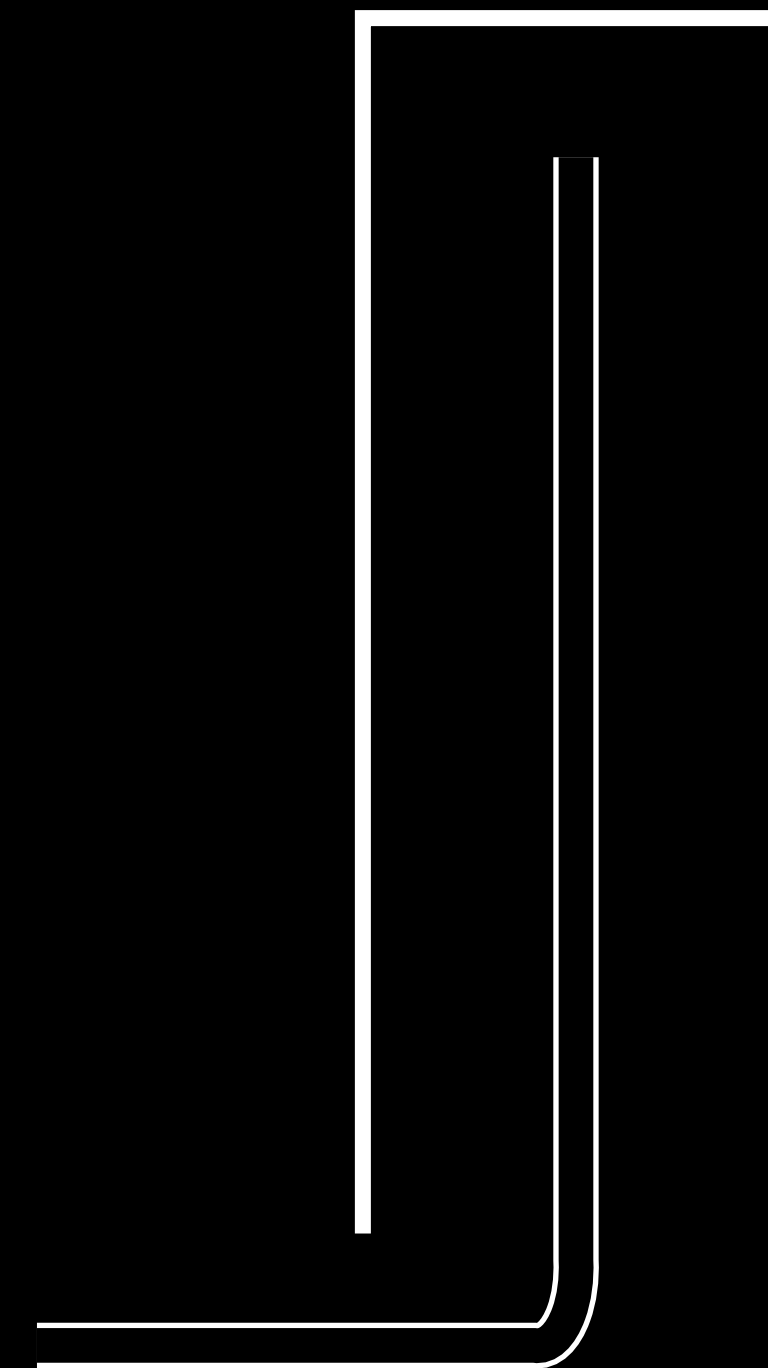
what is this chapter about



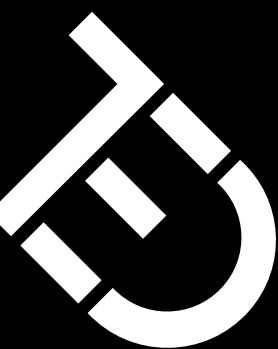
Water Displacement
for collecting gases insoluble in water
e.g. nitrogen



Downward Collection
for collecting gases denser than air
e.g. carbon dioxide



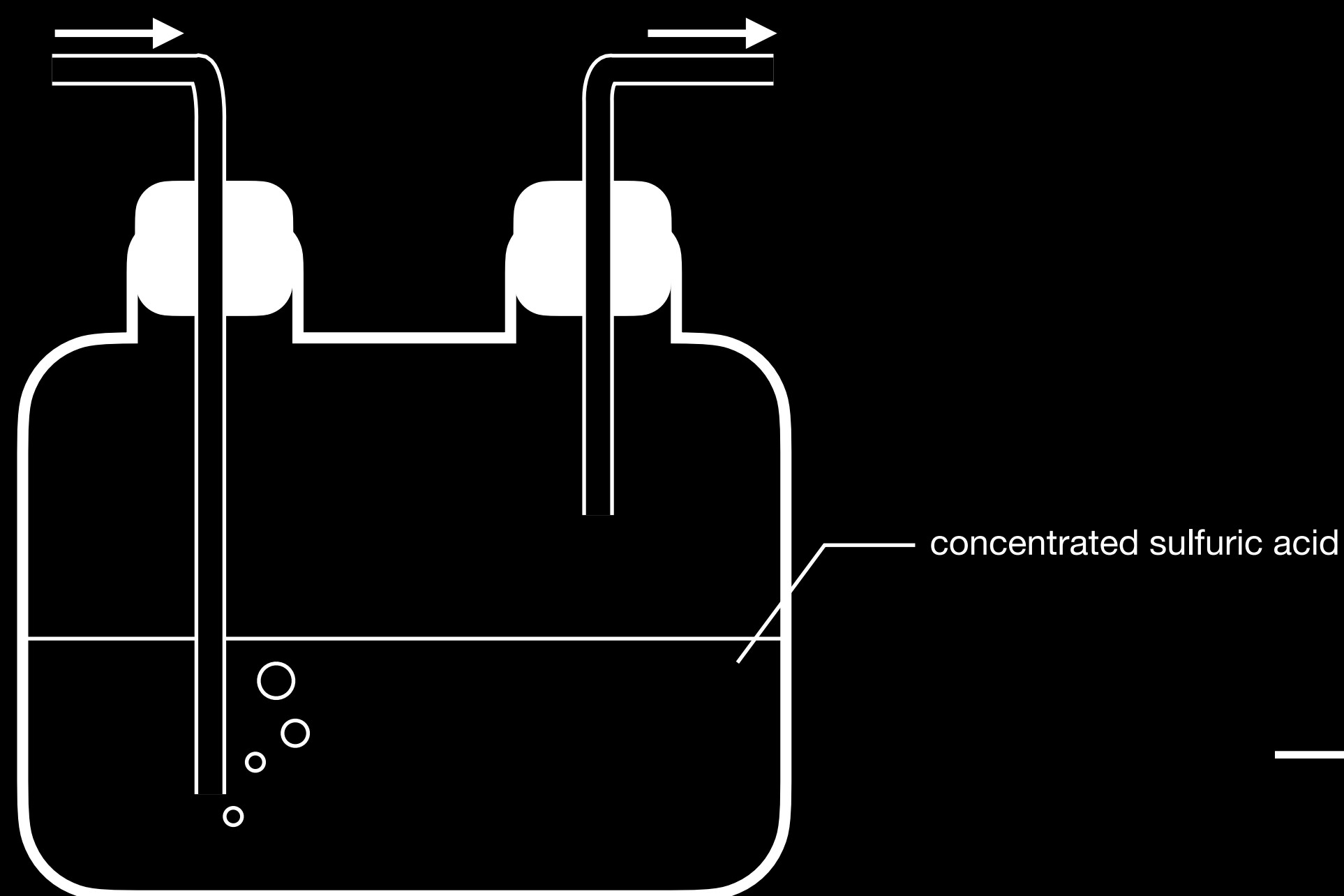
Upward Collection
for collecting gases less dense than air
e.g. hydrogen



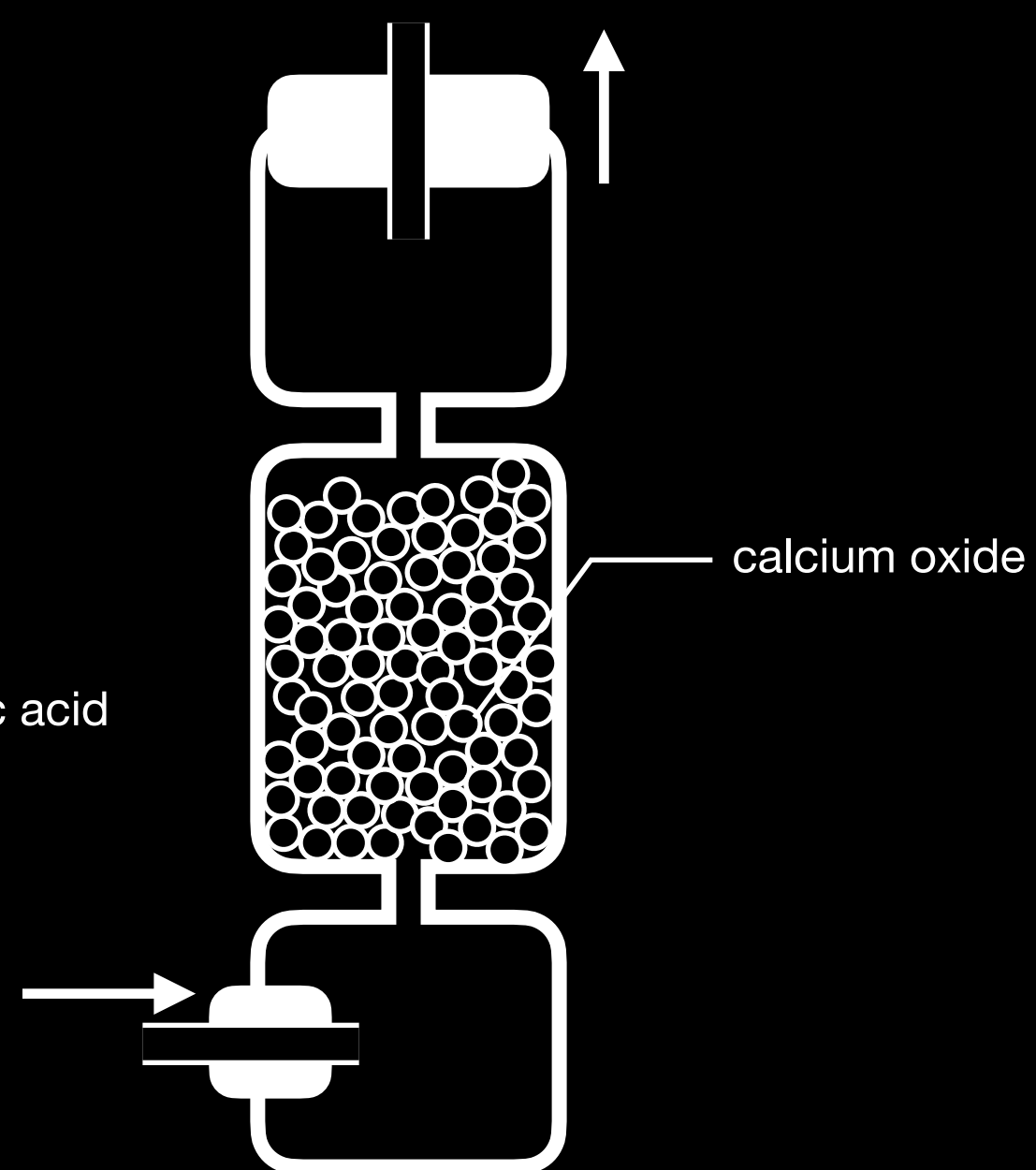
Gas Drying

what is this chapter about

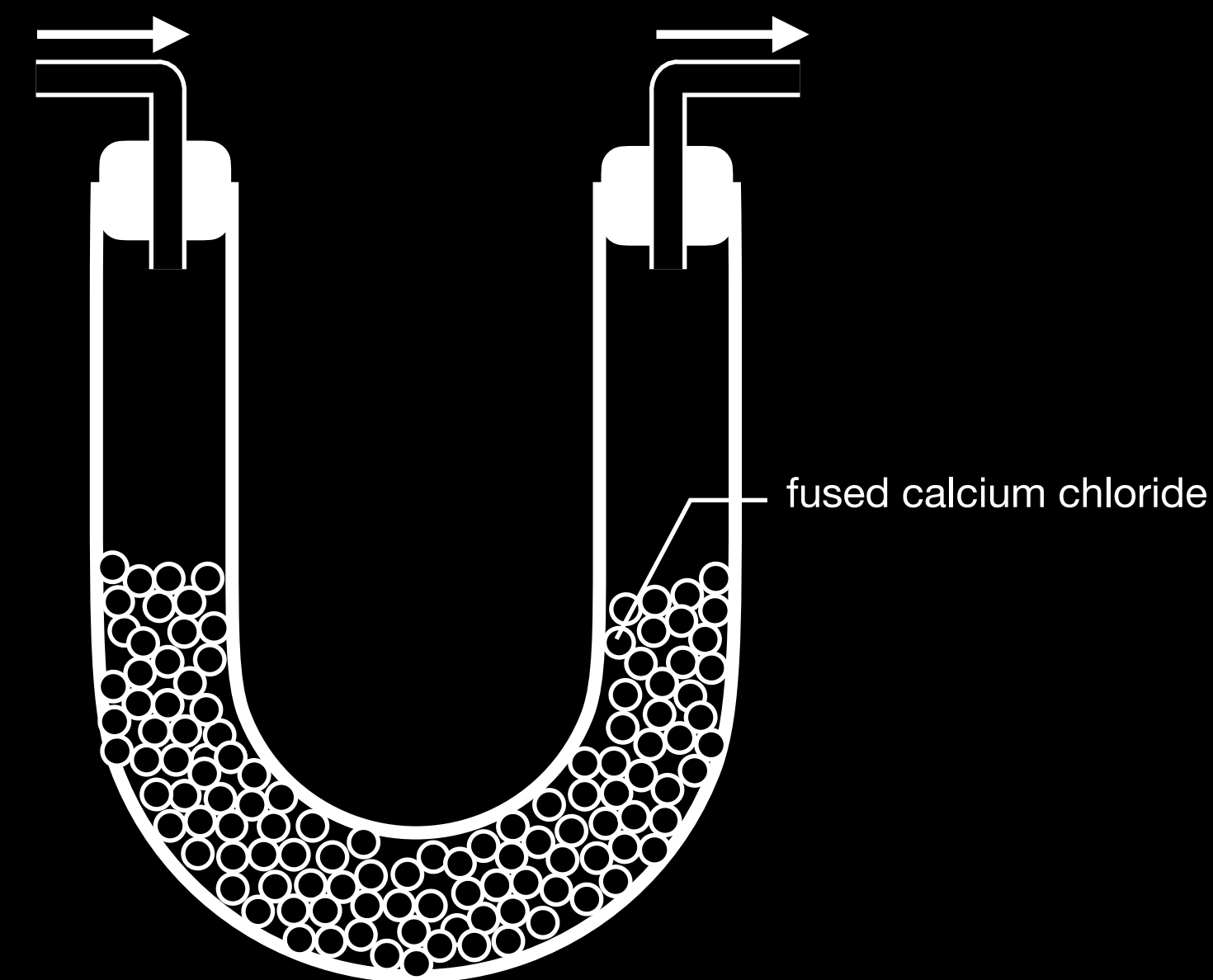
Drying gases refers to removing water vapour from the gas



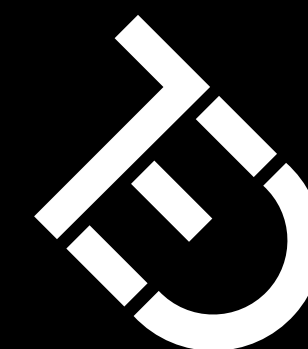
With concentrated sulfuric acid
for drying acidic gases
e.g. sulfur dioxide



With calcium oxide
for drying alkaline gases
e.g. ammonia gas



With fused calcium chloride
for drying neutral(non-acidic/non-alkali) gases
e.g. hydrogen

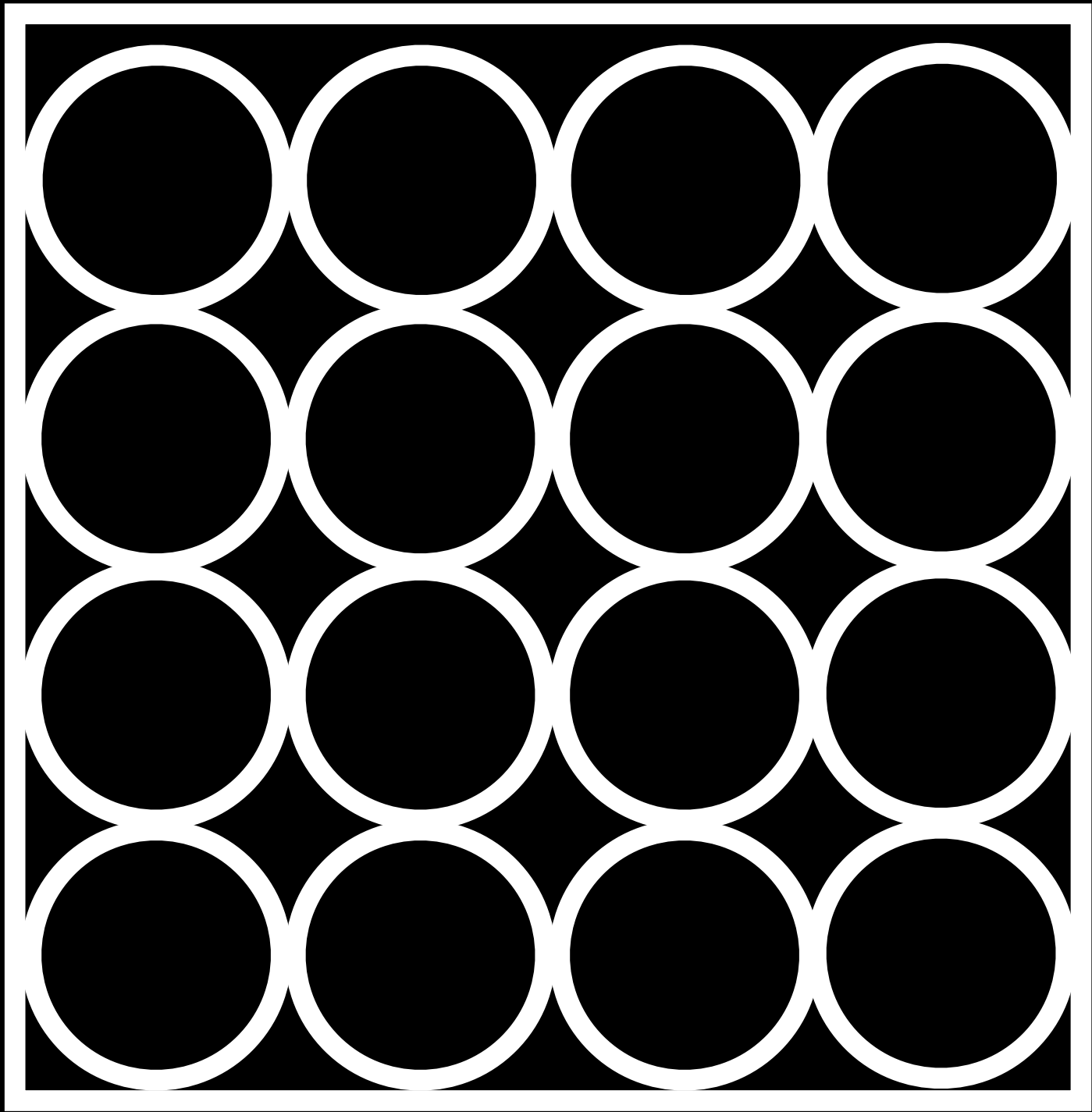


Kinetic Particle Theory



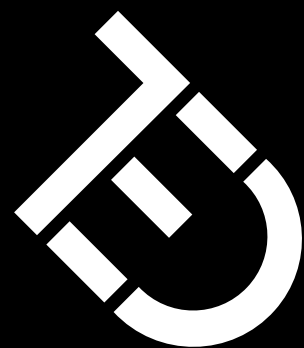
States of Matter - Solid

b



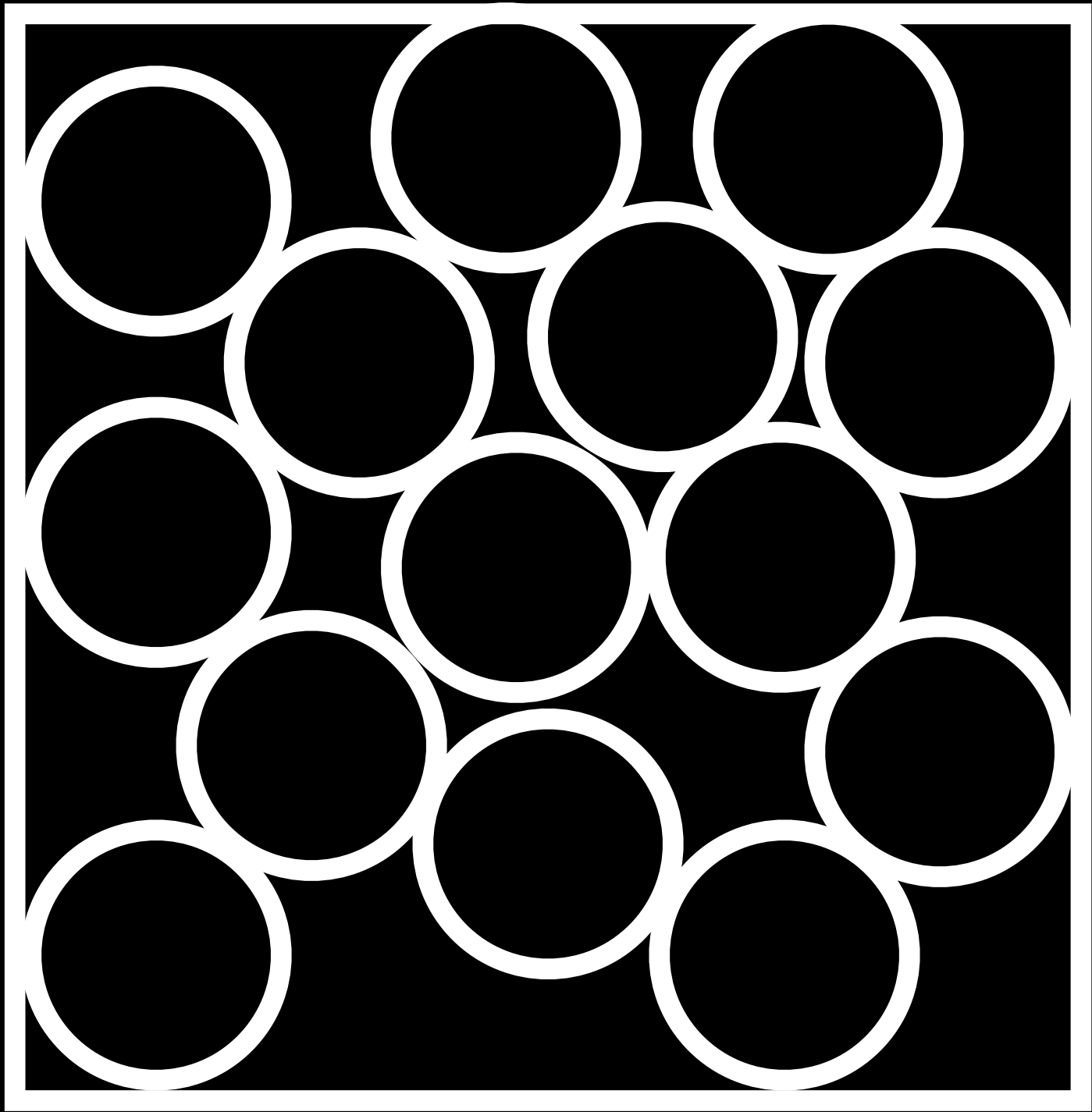
Physical Property	Particle Properties
Fixed Shape	Vibrate about its own position Regular Arrangement
Fixed Volume/Incompressible	Closely packed together Very strong forces of attraction

particle properties causes physical properties



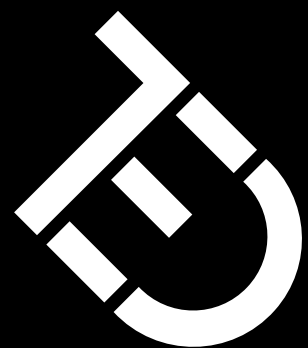
States of Matter - Liquid

b



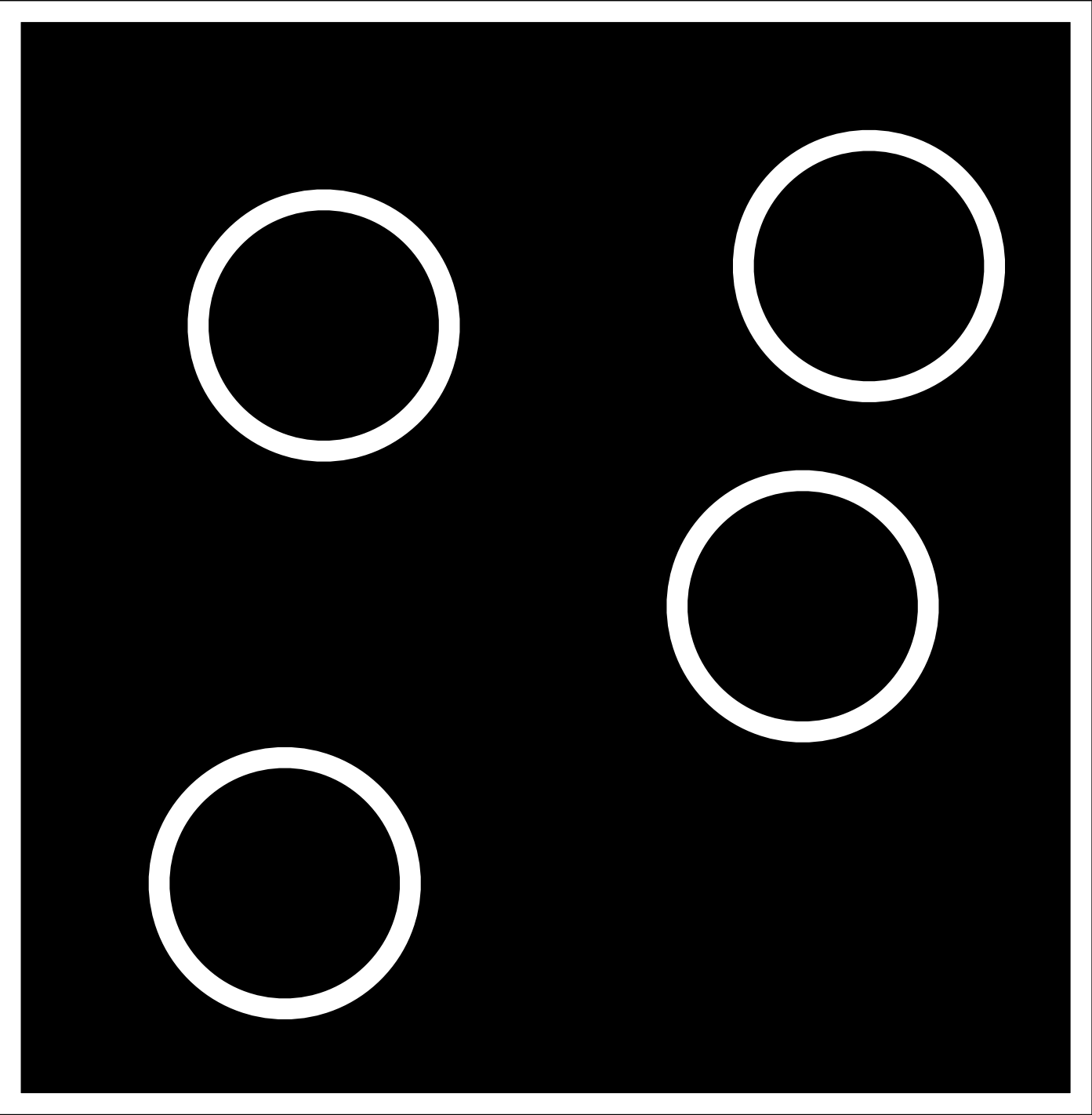
Physical Property	Particle Properties
No fixed Shape	Particles can slide past one another Irregular arrangement
Fixed Volume/Incompressible	Closely packed together strong forces of attraction

particle properties causes physical properties



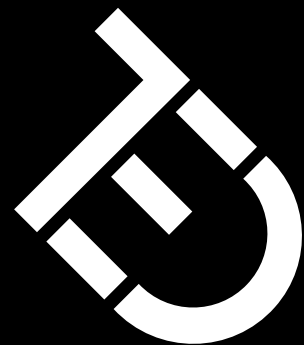
States of Matter - Gas

b



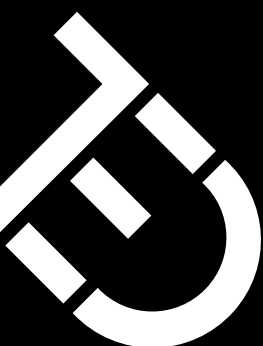
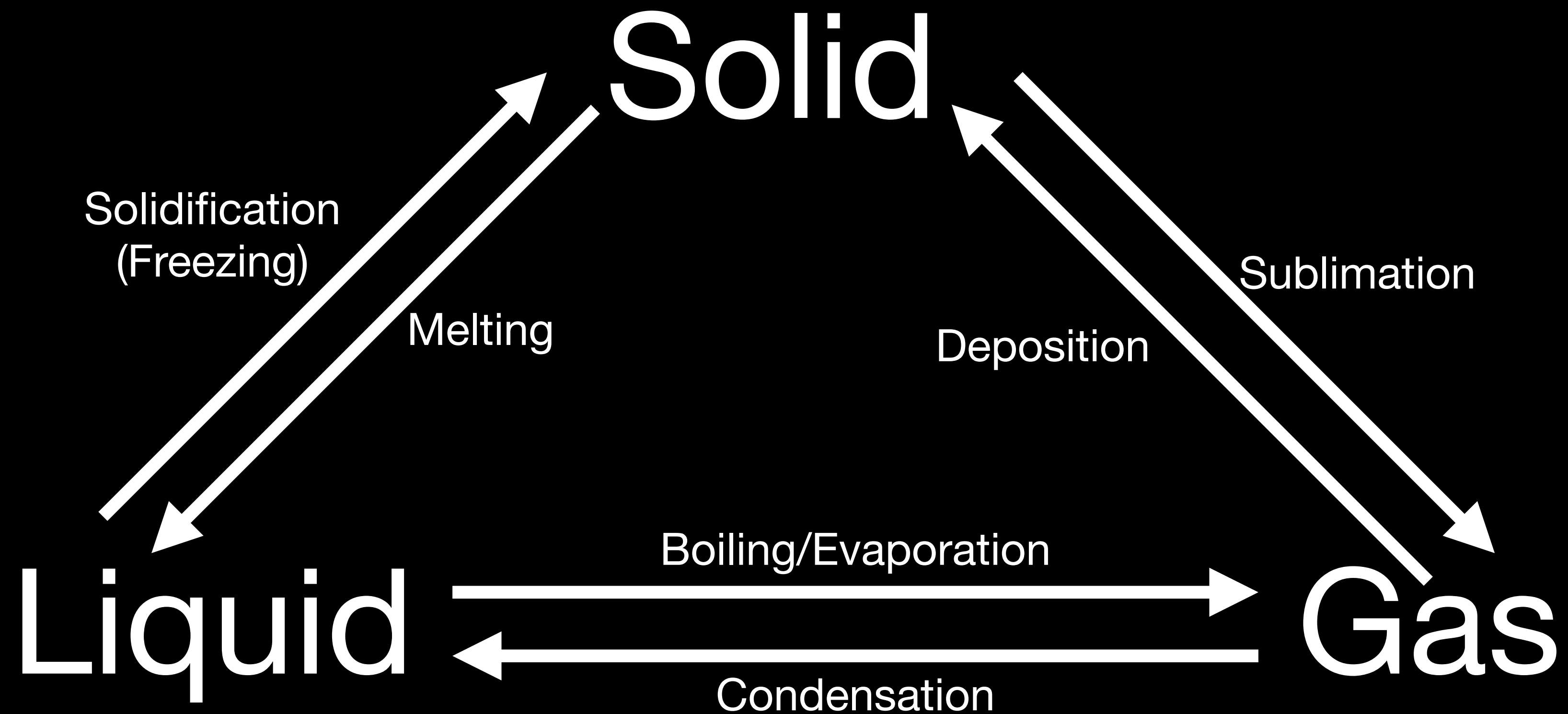
Physical Property	Particle Properties
No fixed Shape	Move in all directions at high speeds Irregular arrangement
No fixed Volume/Compressible	Particles very far spaced from one another Very weak forces of attraction

particle properties causes physical properties



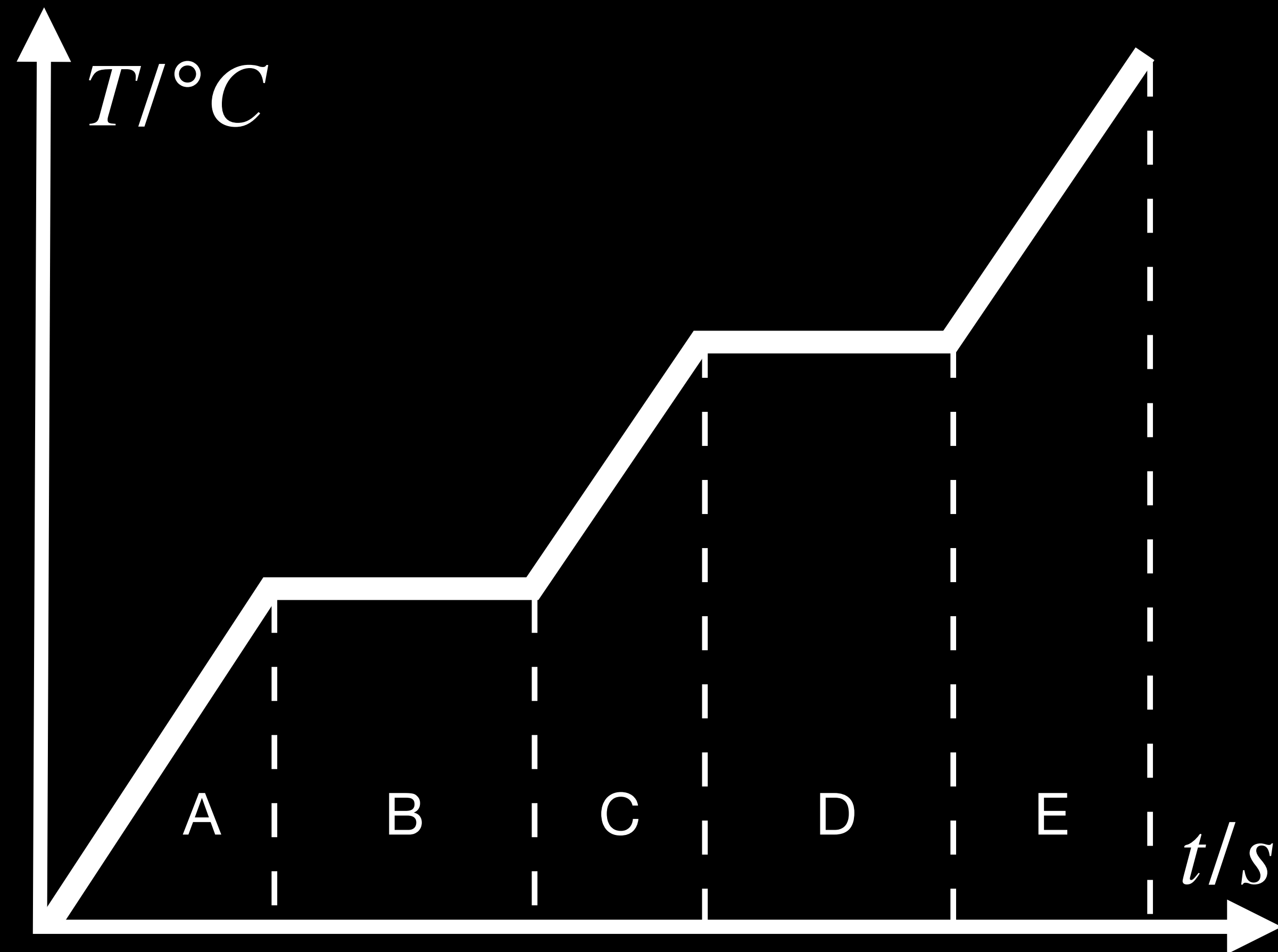
States of Matter - Changes

me



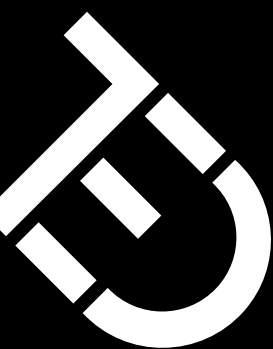
Heating Curve

b



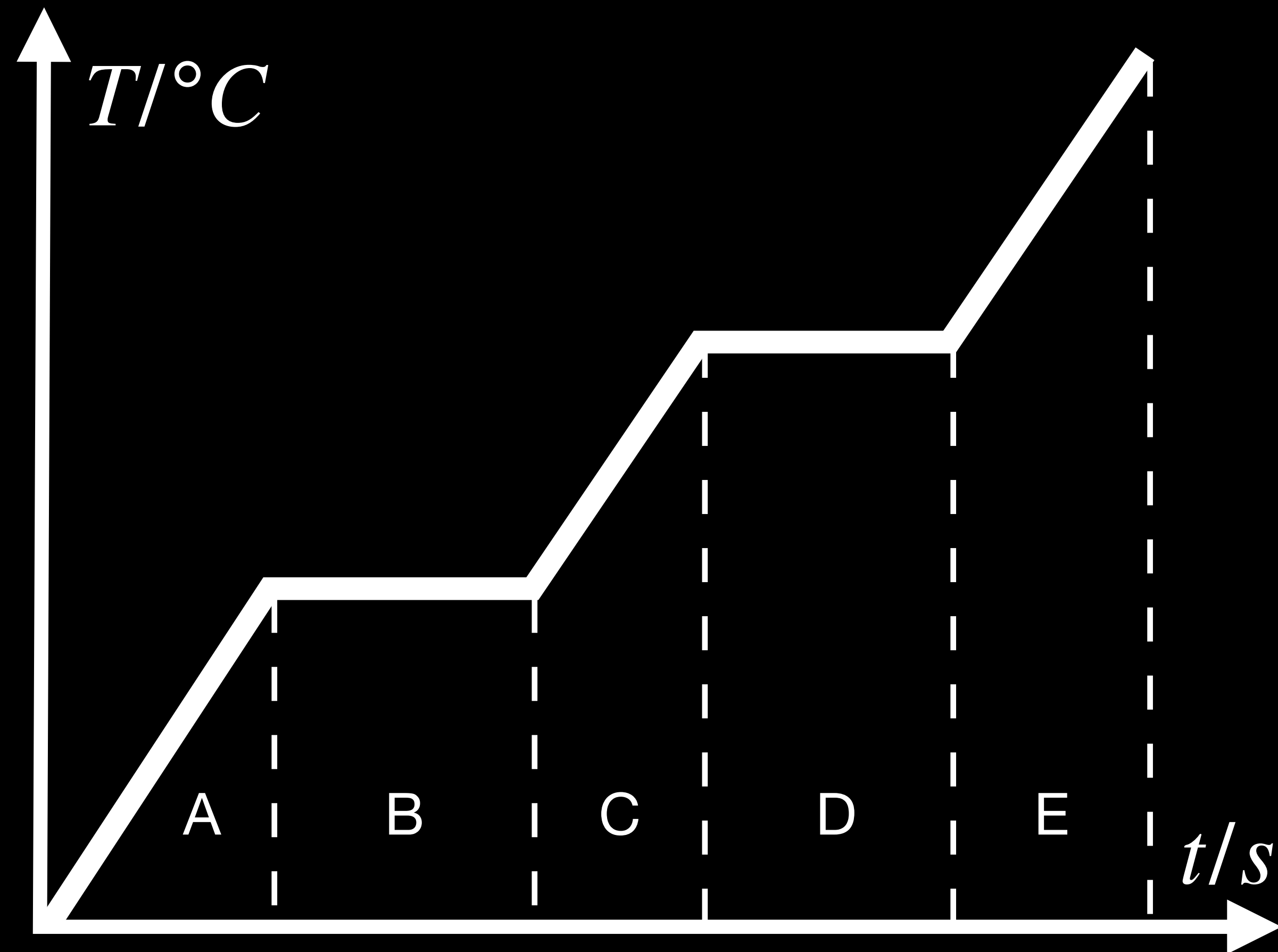
A:(Solid) As temp. increases, the particles in the substance starts to gain K.E and vibrate more about their positions

B: (Melting, Solid-Liquid) As the temperature reaches [melting point], the particles gain enough energy to overcome the forces of attraction on the particles, causing the particles to slide past one another, without increasing the K.E of the particles, keeping the temperature constant



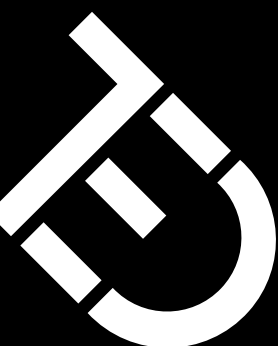
Heating Curve

b



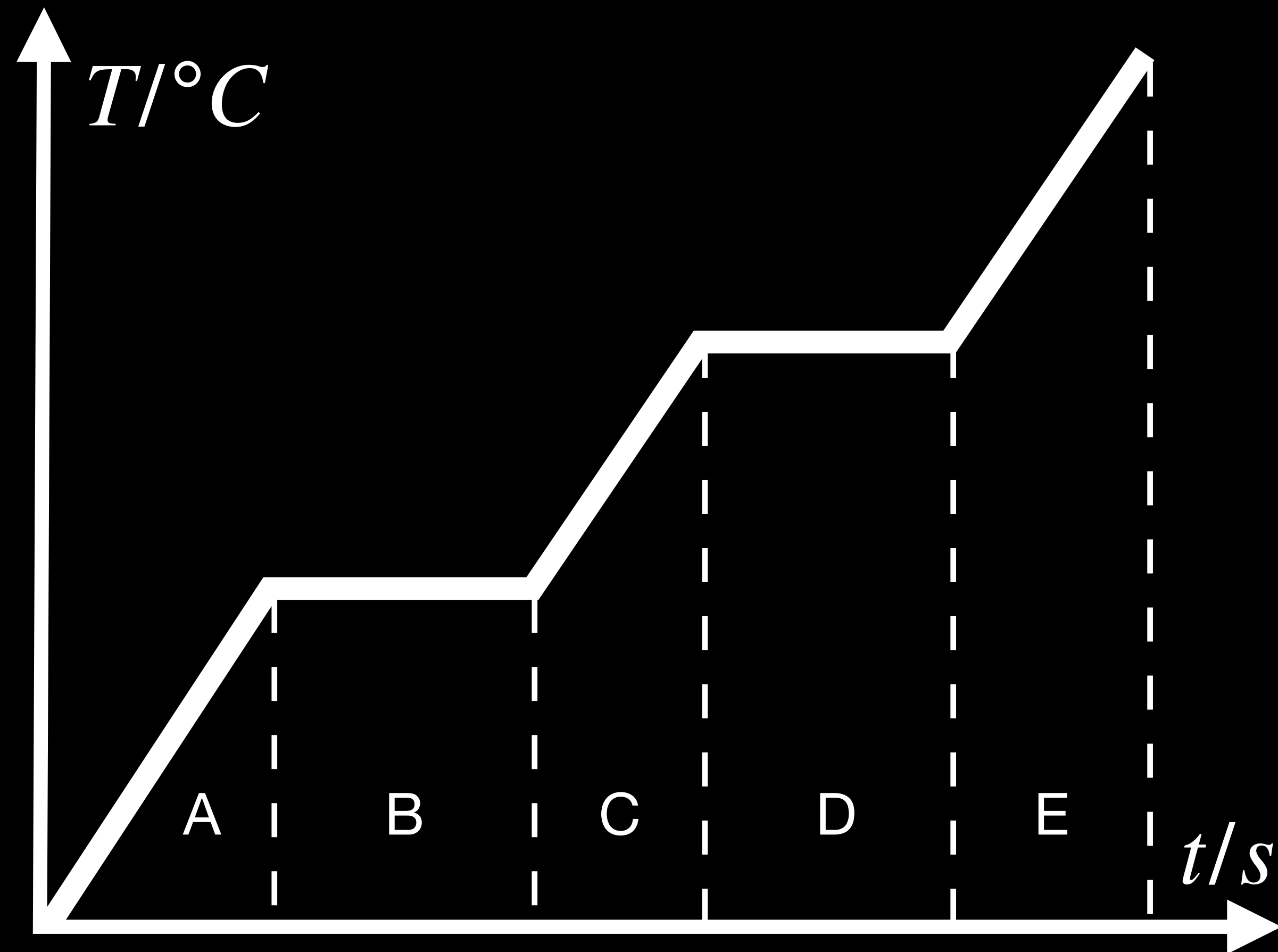
C: (Liquid) As temp increases, the particles in the substance start to gain more K.E and slide faster past each other

D: (Boiling, Liquid-Gas) As the temp reaches [boiling point], the particles gain enough energy to overcome the forces of attraction on the particles, causing the particles to become further apart from one another and move at faster speeds, without increasing the K.E of the particles, keeping the temp. constant



Heating Curve

b

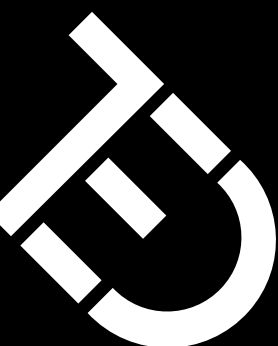


E: (Gas) As temp increases, the particles in the substance start to gain more K.E and move at faster speeds

Rule of thumb:

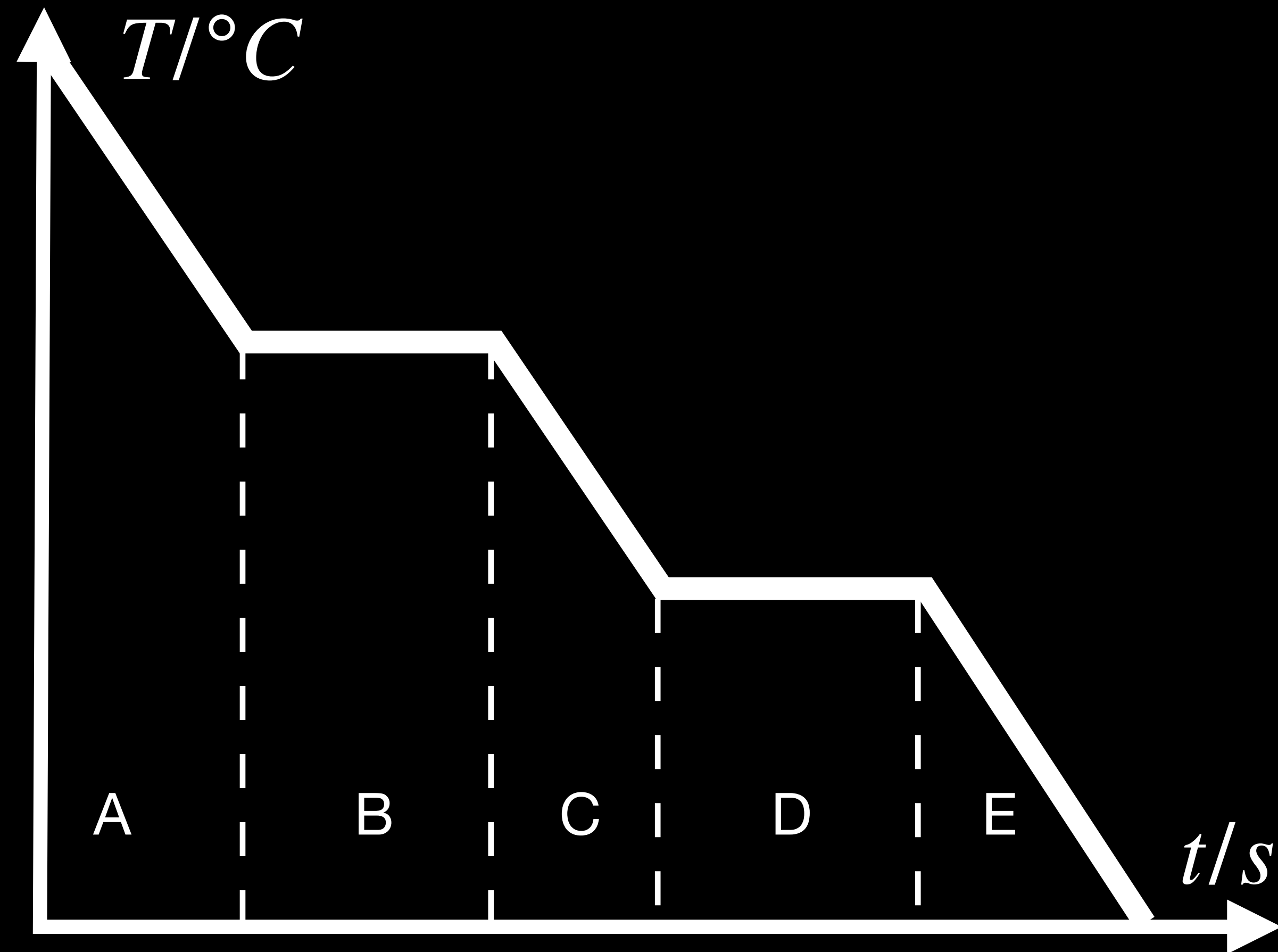
Temp increases $\rightarrow$ K.E increases $\rightarrow$ particles move faster (depending on its state)

Temp reaches m.p/b.p $\rightarrow$ particles gain enough energy $\rightarrow$ overcome forces of attraction $\rightarrow$ become further apart (depending on its state) $\rightarrow$ K.E stays constant, temp stays constant (only write for questions which ask about constant temp)



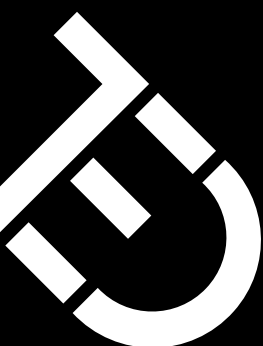
Cooling Curve

Im so col



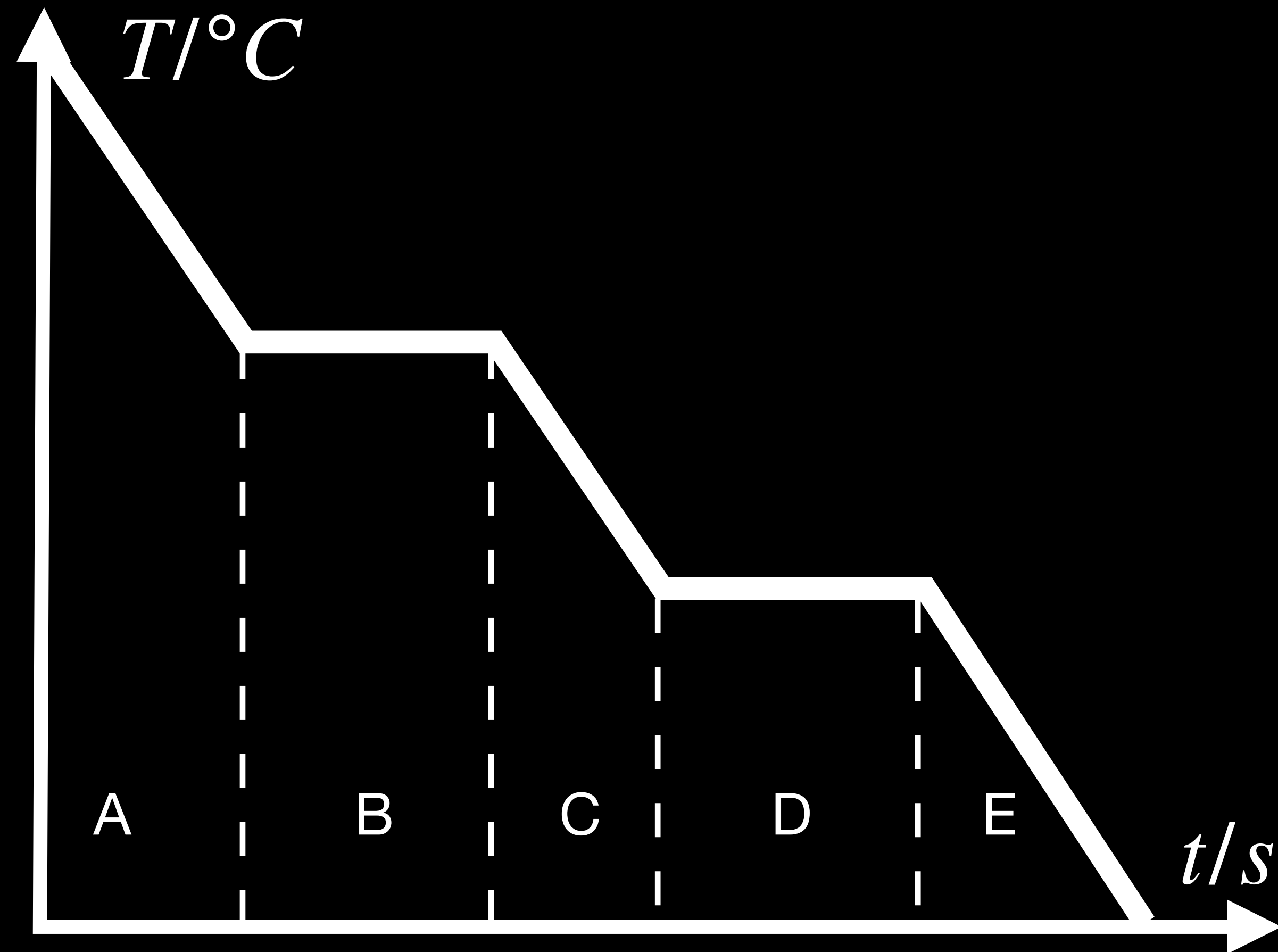
A: (Gas) As the temp. decreases, the K.E decreases and the particles start to move slower

B: (Condensation, Gas-Liquid) As the temp reaches [boiling point], the particles start to lose energy, allowing the forces of attraction to pull the particles together, causing them to be closely packed together in an irregular arrangement without decreasing the K.E. of the particles, keeping the temp. constant



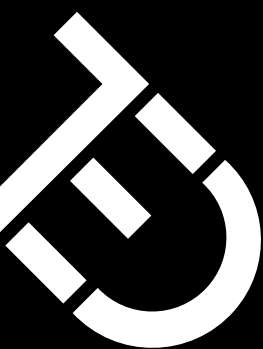
Cooling Curve

Im so col



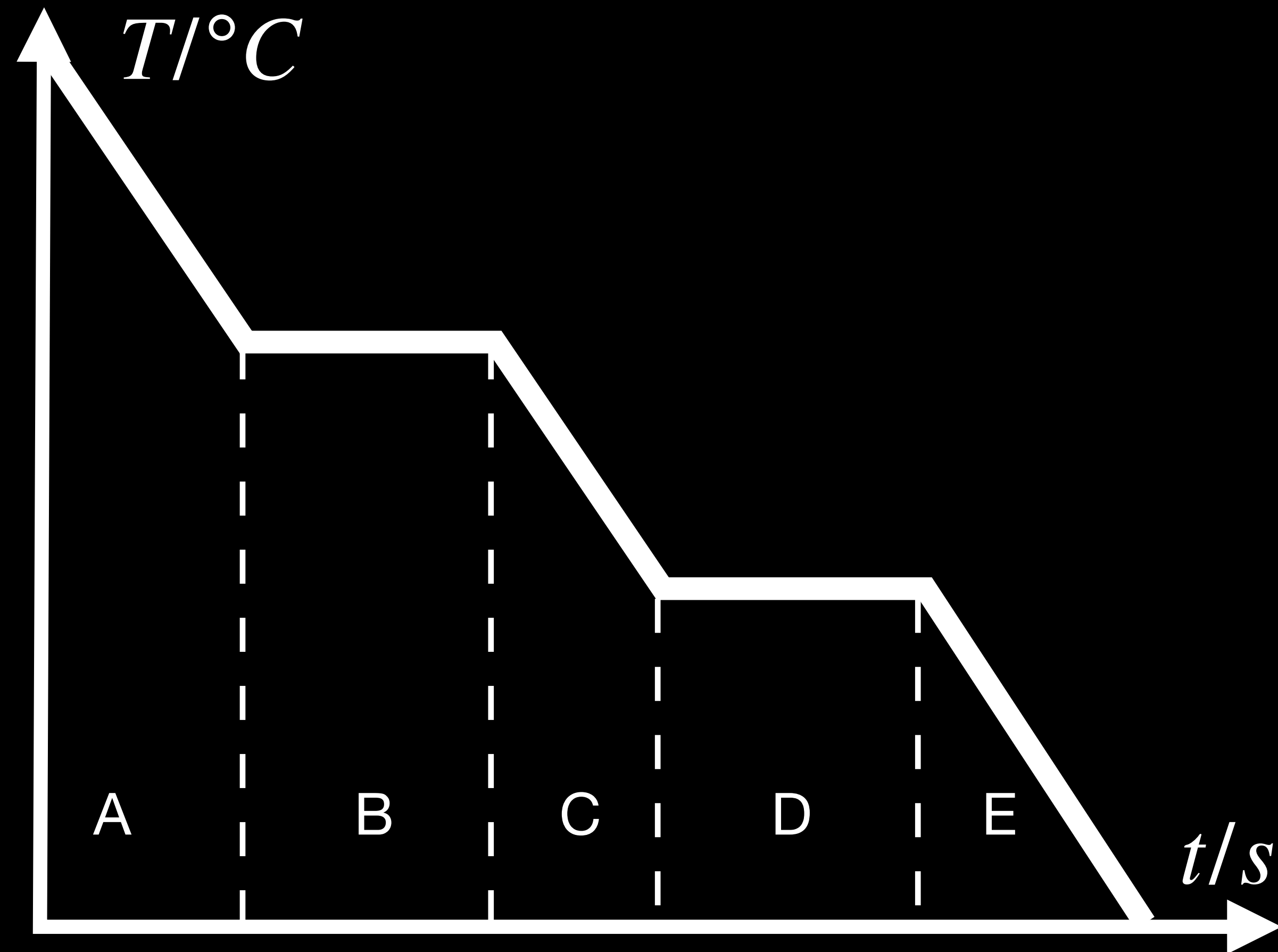
C: (Liquid) As the temp. decreases, the K.E decreases and the particles start to slide past each other slower

D: (Freezing/Solidification , Liquid-Solid) As the temp reaches [melting point], the particles start to lose energy, allowing the forces of attraction to pull the particles together, causing them to be closely packed together in a regular arrangement without decreasing the K.E. of the particles, keeping the temp. constant



Cooling Curve

Im so col

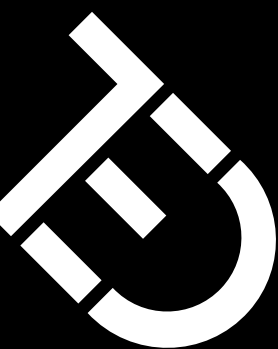


E: (Solid) As the temp. decreases, the K.E decreases and the particles start to vibrate about their positions slower

Rule of thumb:

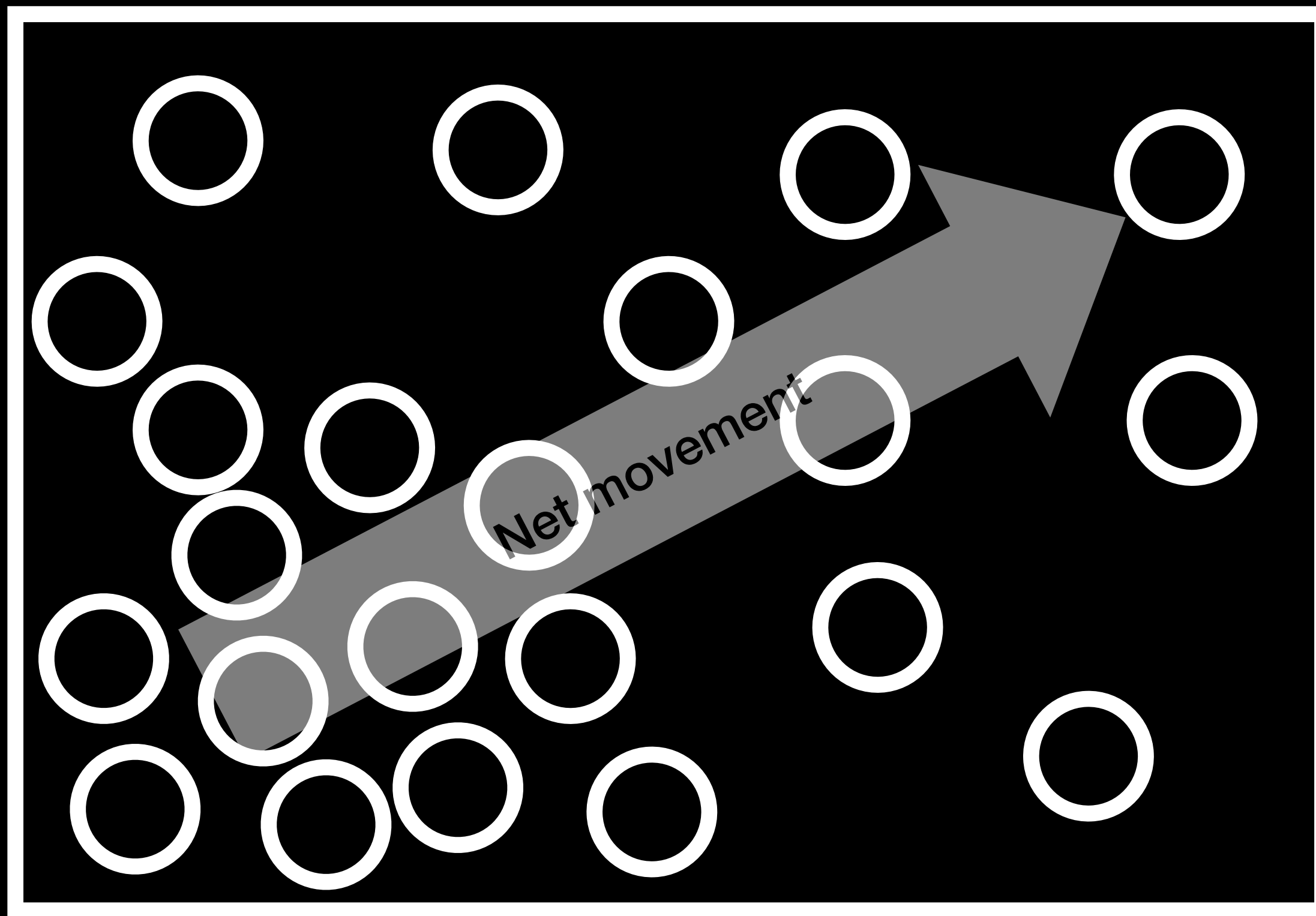
Temp decreases $\rightarrow$ K.E decreases $\rightarrow$ particles move slower (depending on its state)

Temp reaches m.p/b.p $\rightarrow$ particles lose enough energy $\rightarrow$ allow forces of attraction to act on particles $\rightarrow$ become closer together (depending on its state) $\rightarrow$ K.E stays constant, temp stays constant (only write for questions which ask about constant temp)



Diffusion

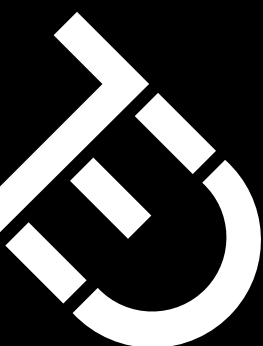
emtpo



High concentration $\rightarrow$ Low concentration

Diffusion Definition:

The net movement of particles from a region of high concentration to a region of low concentration



Factors Affecting Diffusion

d

Temperature:

As temp increases, the rate of diffusion is higher

When the temperature is higher, the particles have more K.E and move faster, allowing the particles to move faster into the low concentration regions.

Temp increase $\rightarrow$ more K.E $\rightarrow$ move faster into low concentration regions $\rightarrow$ faster diffusion rate

Relative Molecular Mass (M_r):

As relative molecular mass increases, rate of diffusion decreases

When there is a higher molecular mass, the particles will move at a slower rate, causing the diffusion rate to decrease.

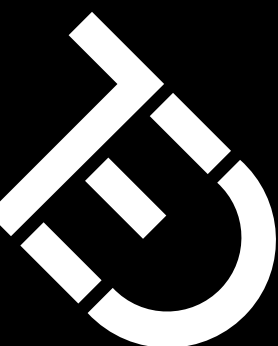
Higher $M_r \rightarrow$ particles move slower $\rightarrow$ slower diffusion rate

To calculate relative molecular mass (M_r):

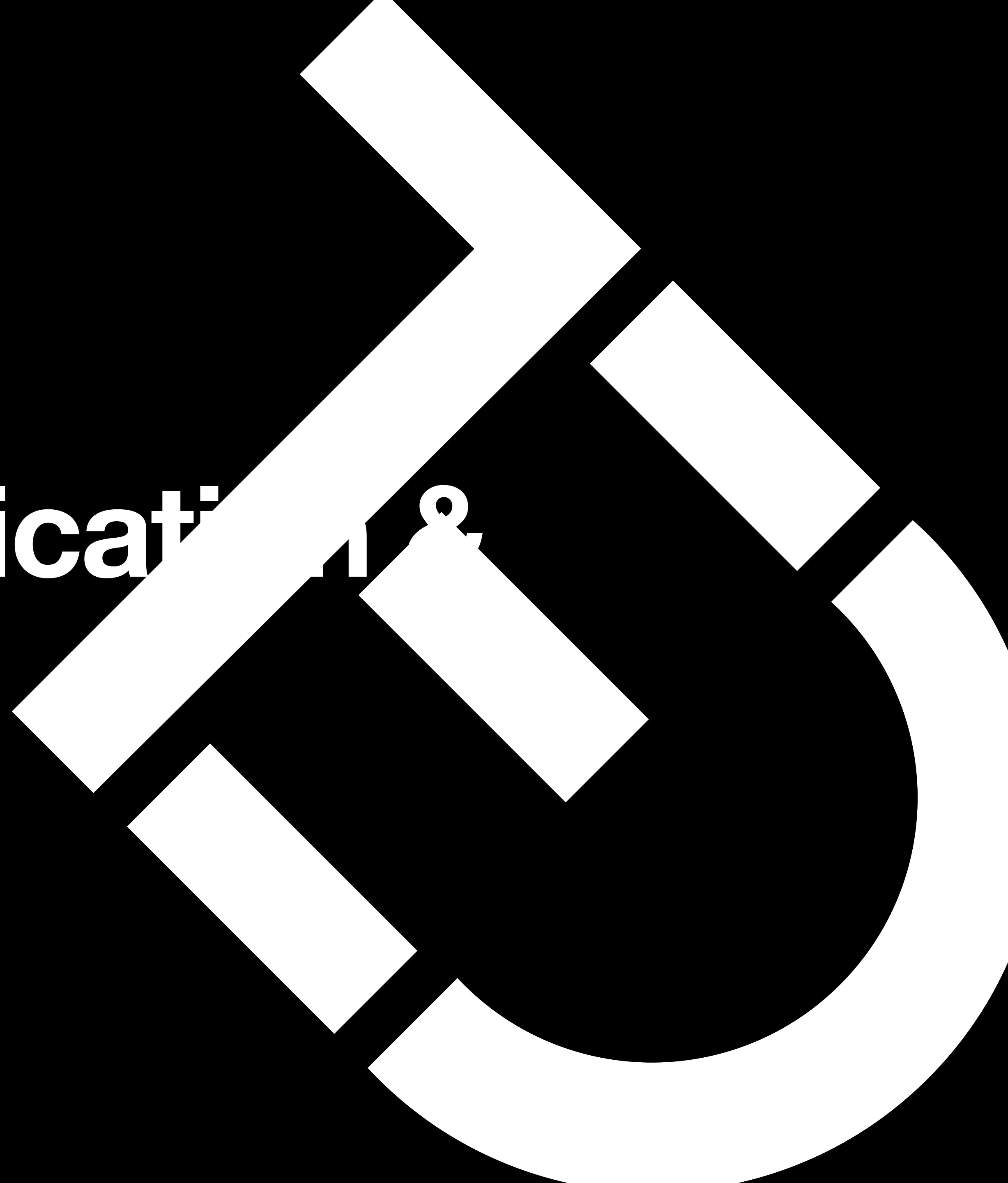
M_r = sum of all the atoms' relative atomic mass

e.g ammonia (NH_3)

$$M_r = 14 + 3(1) = 17$$

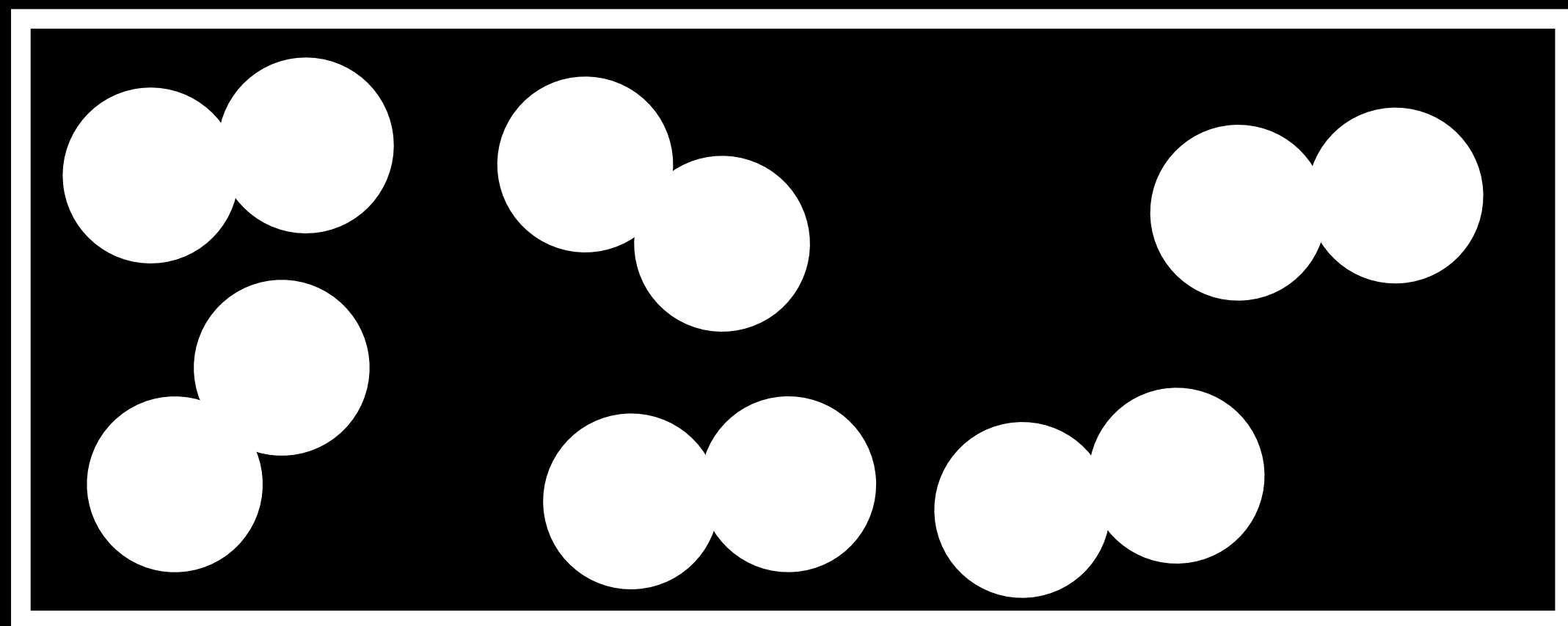
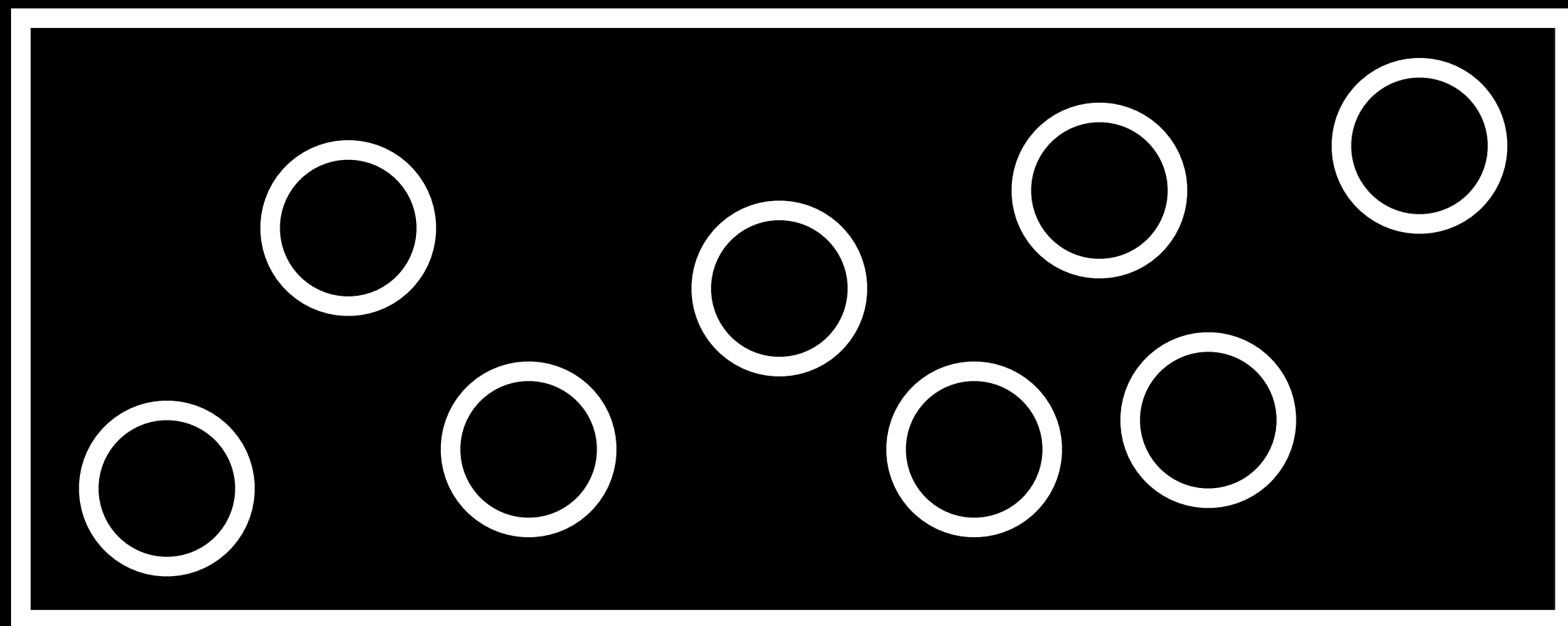


Methods of Purification & Analysis



Elements, Compounds and Mixtures

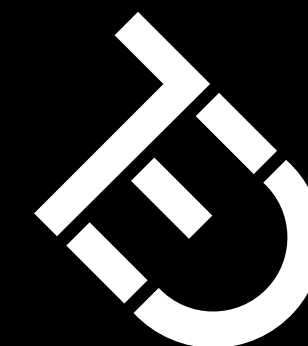
Balls (the sequel)



Elements:

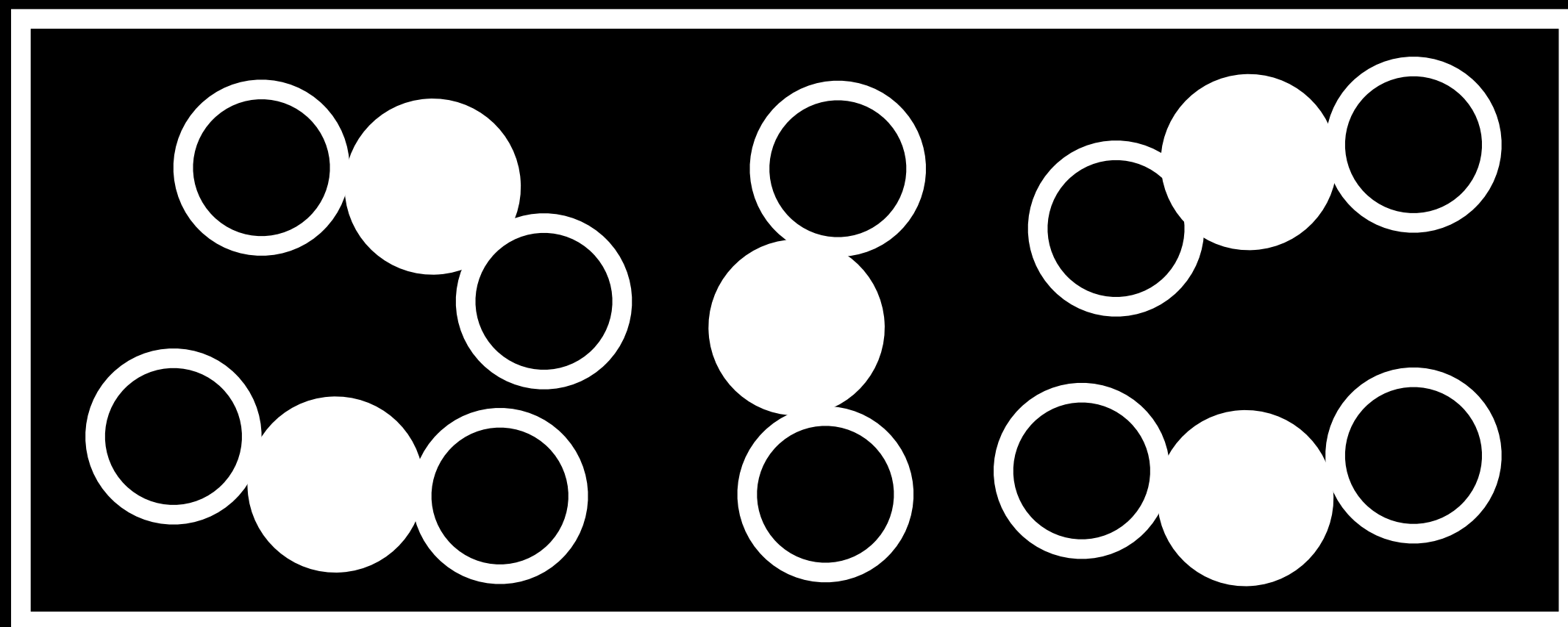
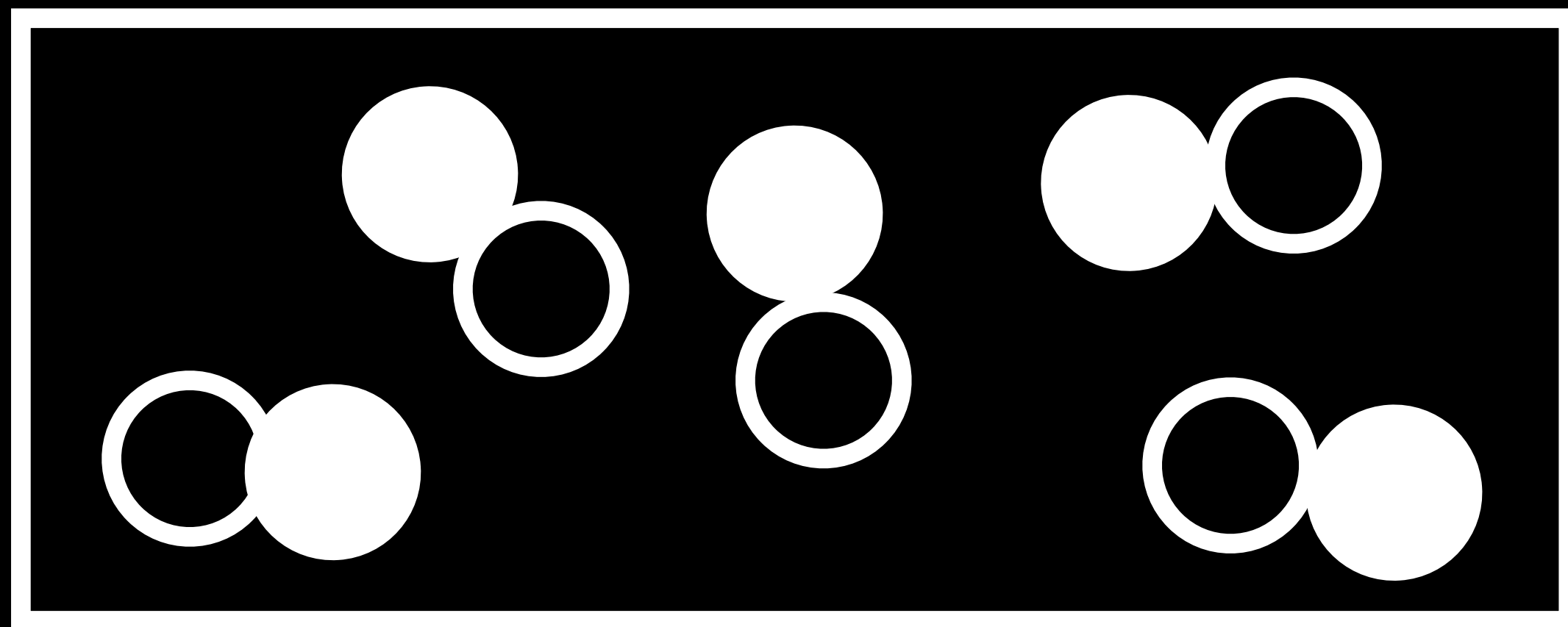
Only one type of atom than cannot be broken down into simpler substances(not necessarily monoatomic)

(Pure substance, has well defined properties like fixed m.p and b.p)



Elements, Compounds and Mixtures

Balls (the sequel)

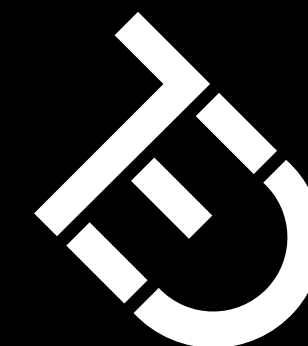


Compounds:

Two or more elements chemically combined in a fixed ratio

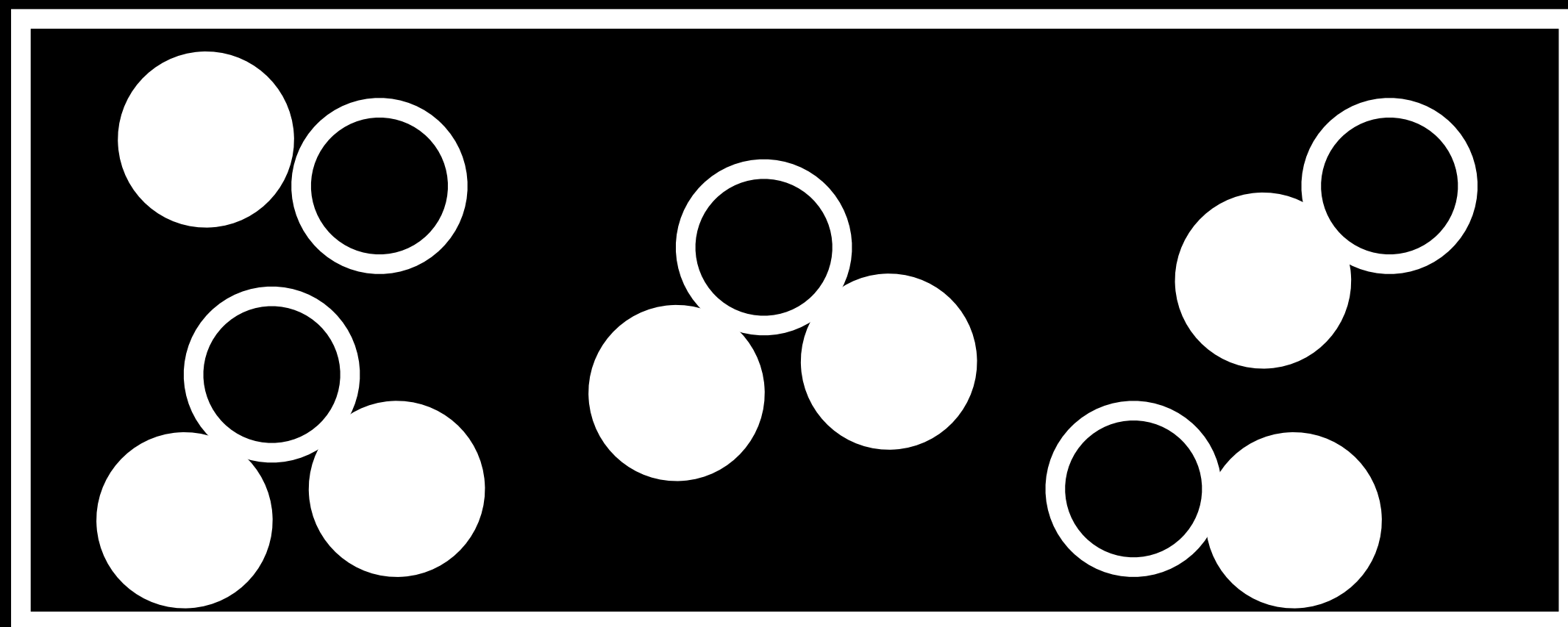
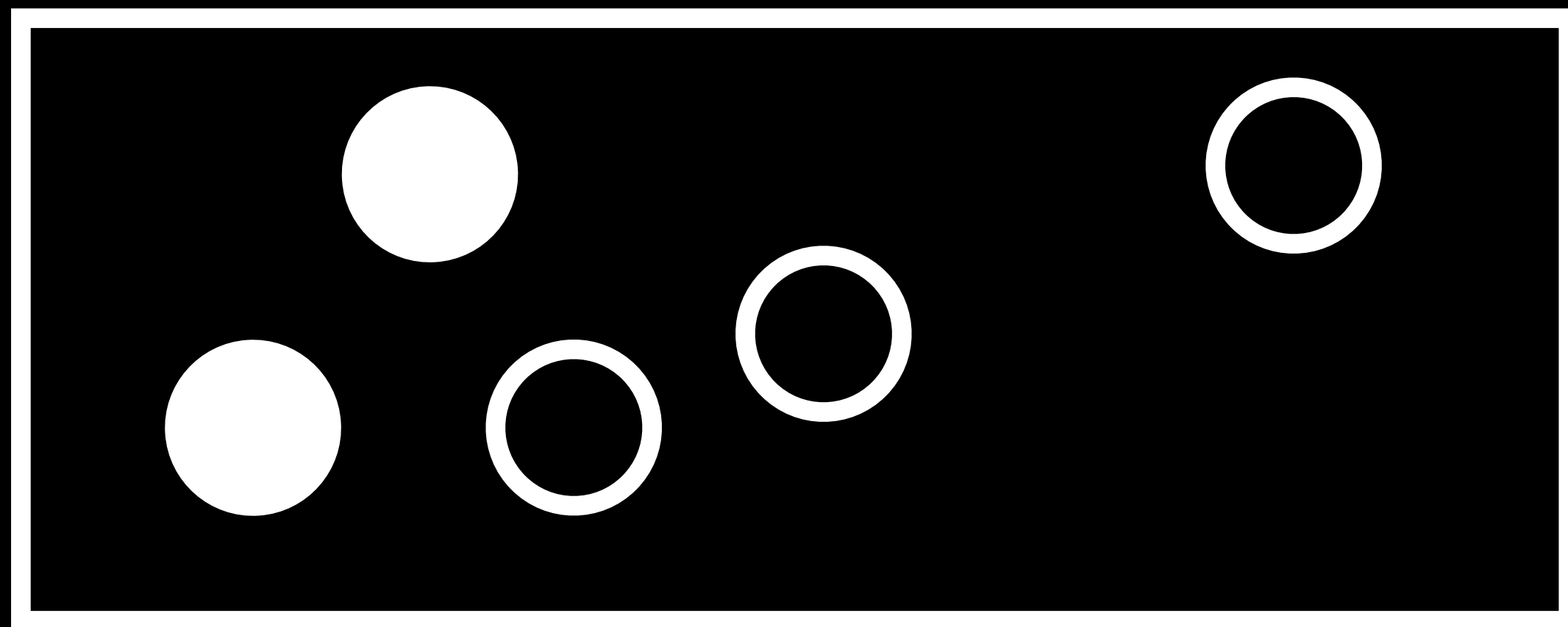
Have different properties from its constituent elements

(Pure substance)



Elements, Compounds and Mixtures

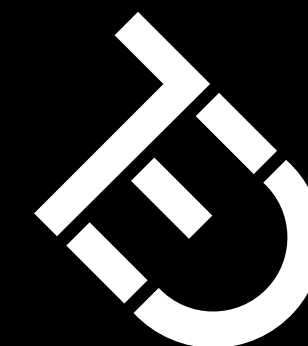
Balls (the sequel)



Mixtures:

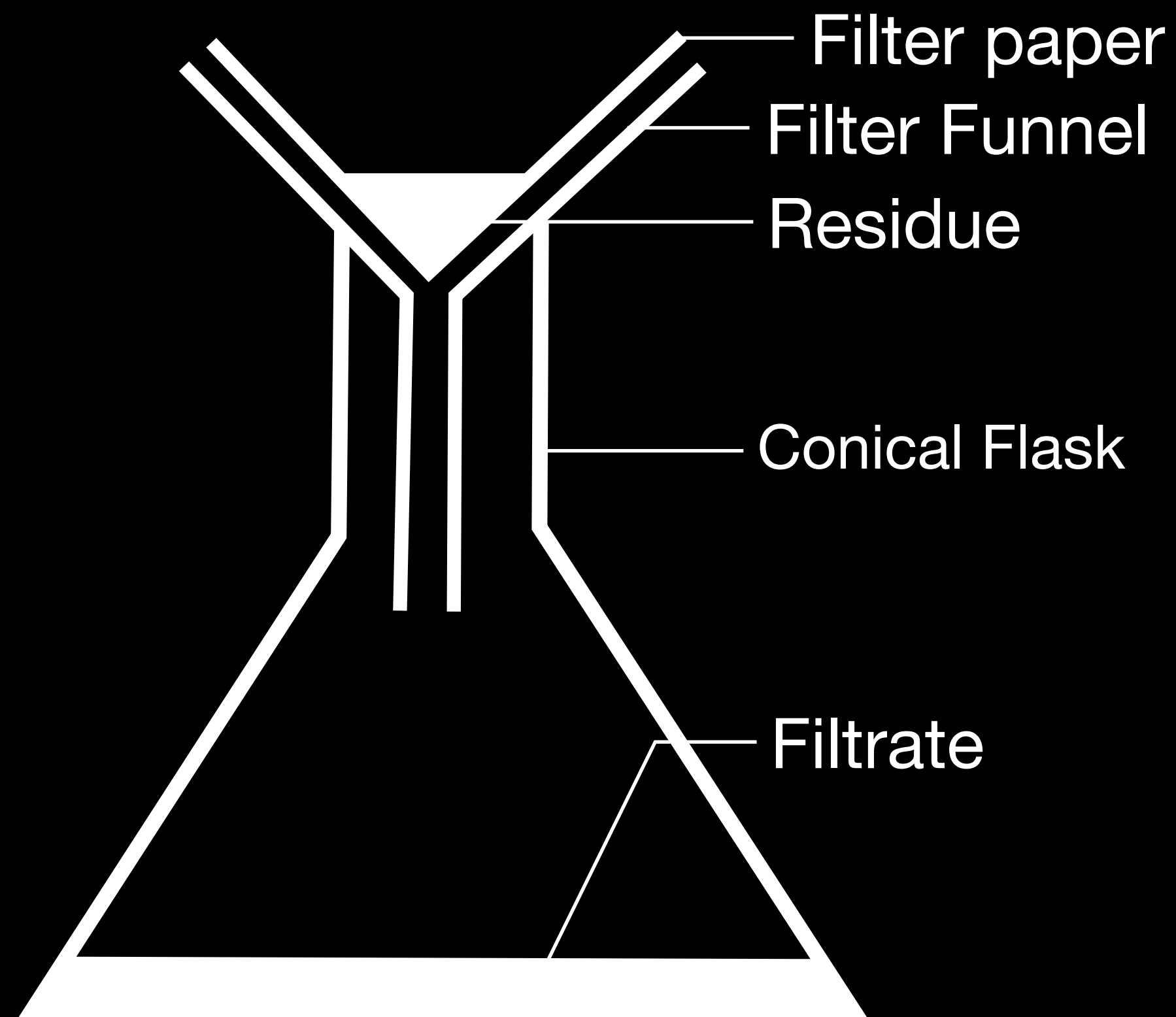
Two or more substances not chemically combined in no fixed ratio

(Impure substance, does not have well defined properties, melts/boils over a range of temperatures)



Separation Techniques - Filtration

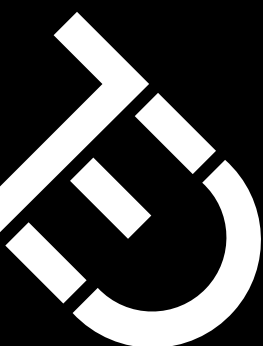
RIGHT



Separates a mixture containing an insoluble substance in a liquid

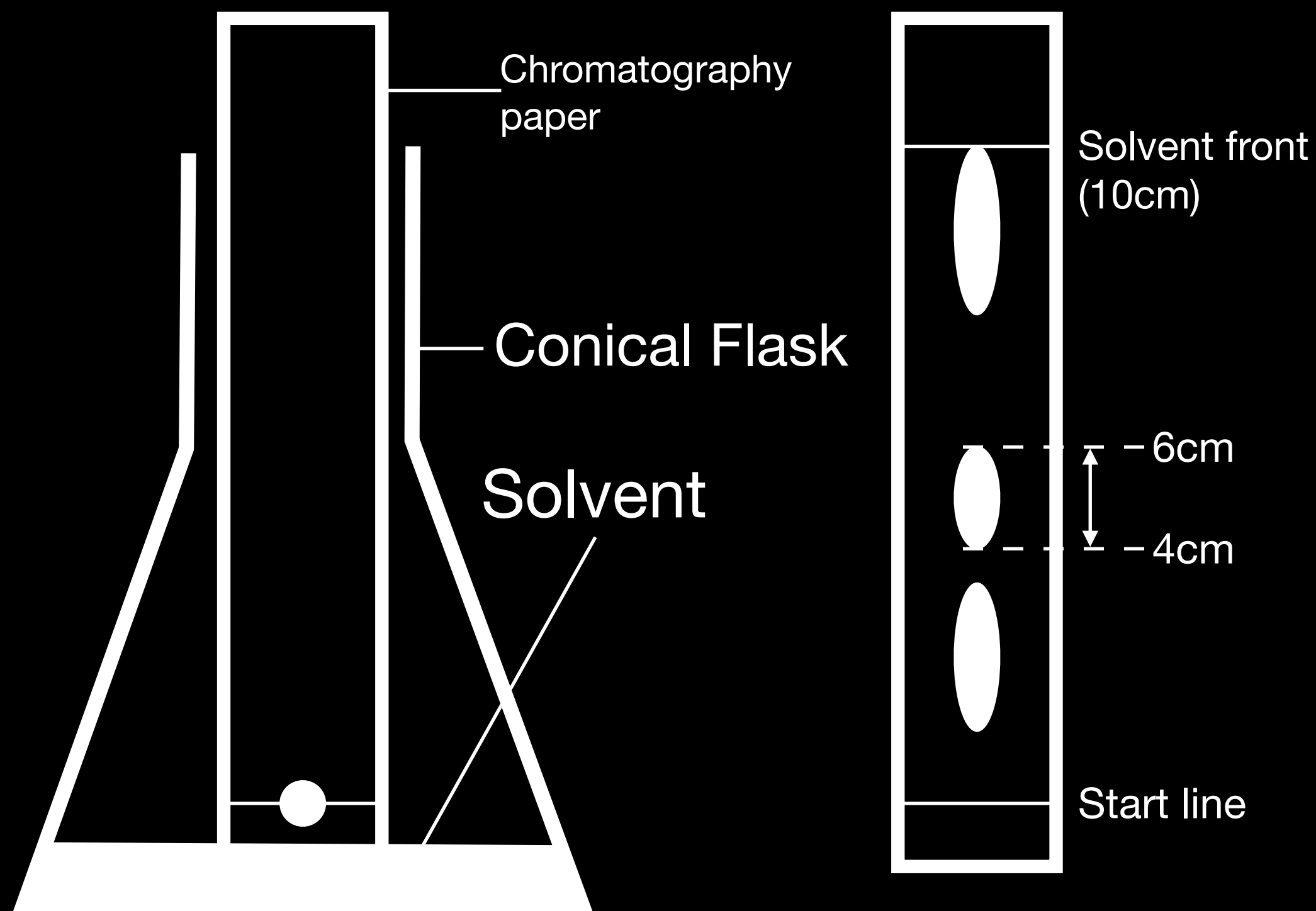
(e.g. sand and water)

Bigger insoluble particles cannot pass through the filter paper, forming as a residue, while the smaller, soluble particles can pass through the filter paper, forming as a filtrate



Separation Techniques - (Paper) Chromatography

RIGHT

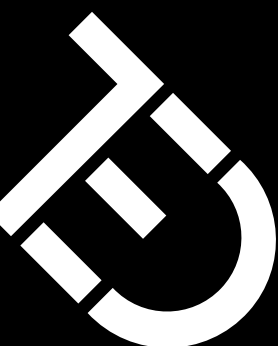


To separate a mixture of different substances of different solubilities in a solvent

Setup:

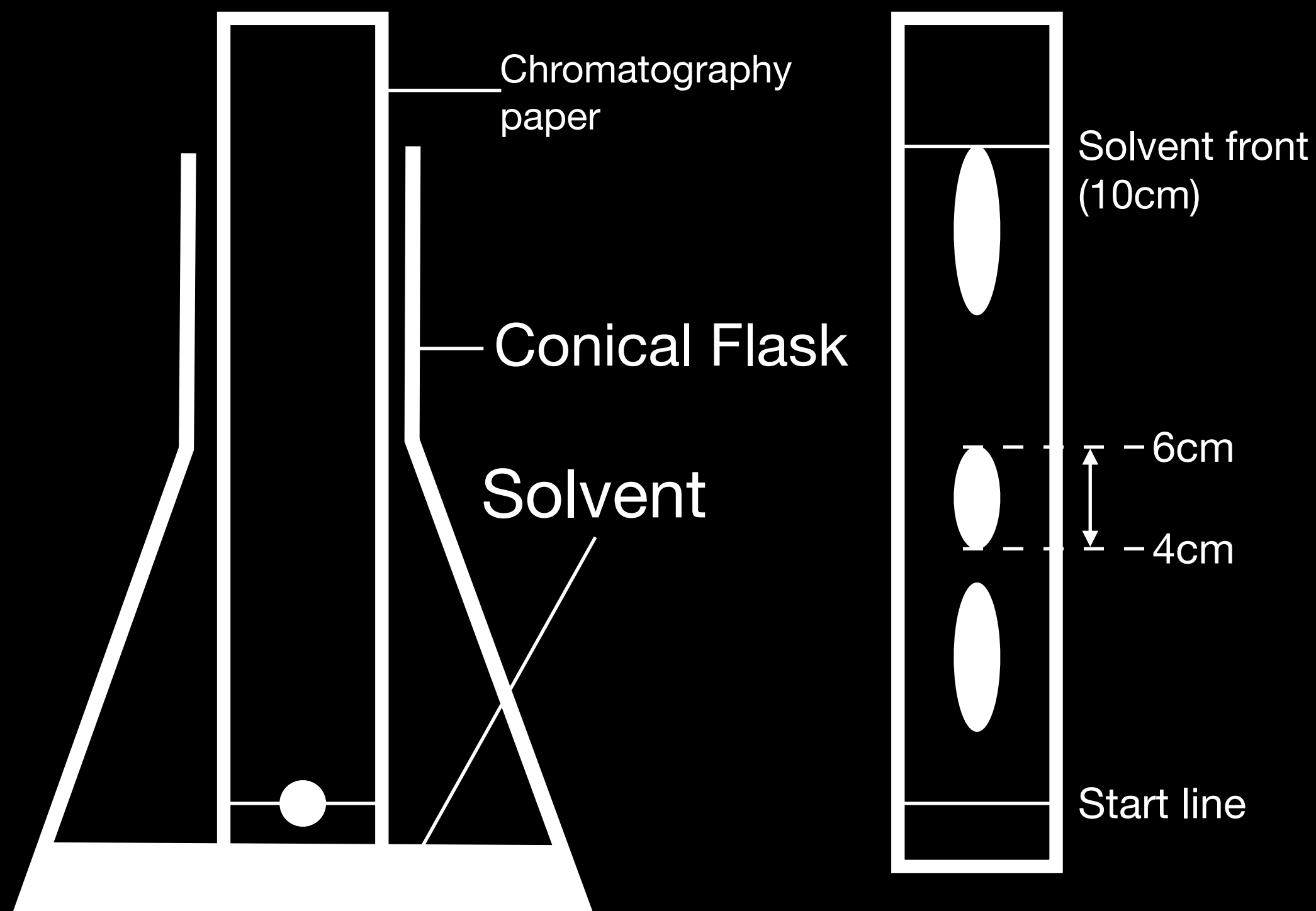
Use pencil to draw the start line (pencil graphite does not smudge and will not affect the results of the chromatography)

Sample must not be touching solvent (so that it does not dissolve before the chromatography can take place)



Separation Techniques - (Paper) Chromatography

RIGHT

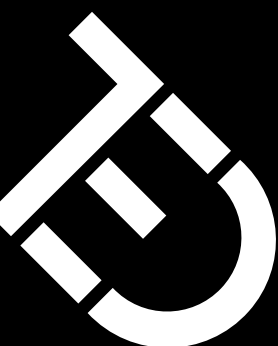


R_f Value:

$$R_f \text{ value} = \frac{\text{distance travelled by substance}}{\text{solvent front}}, 0 \leq R_f \leq 1$$

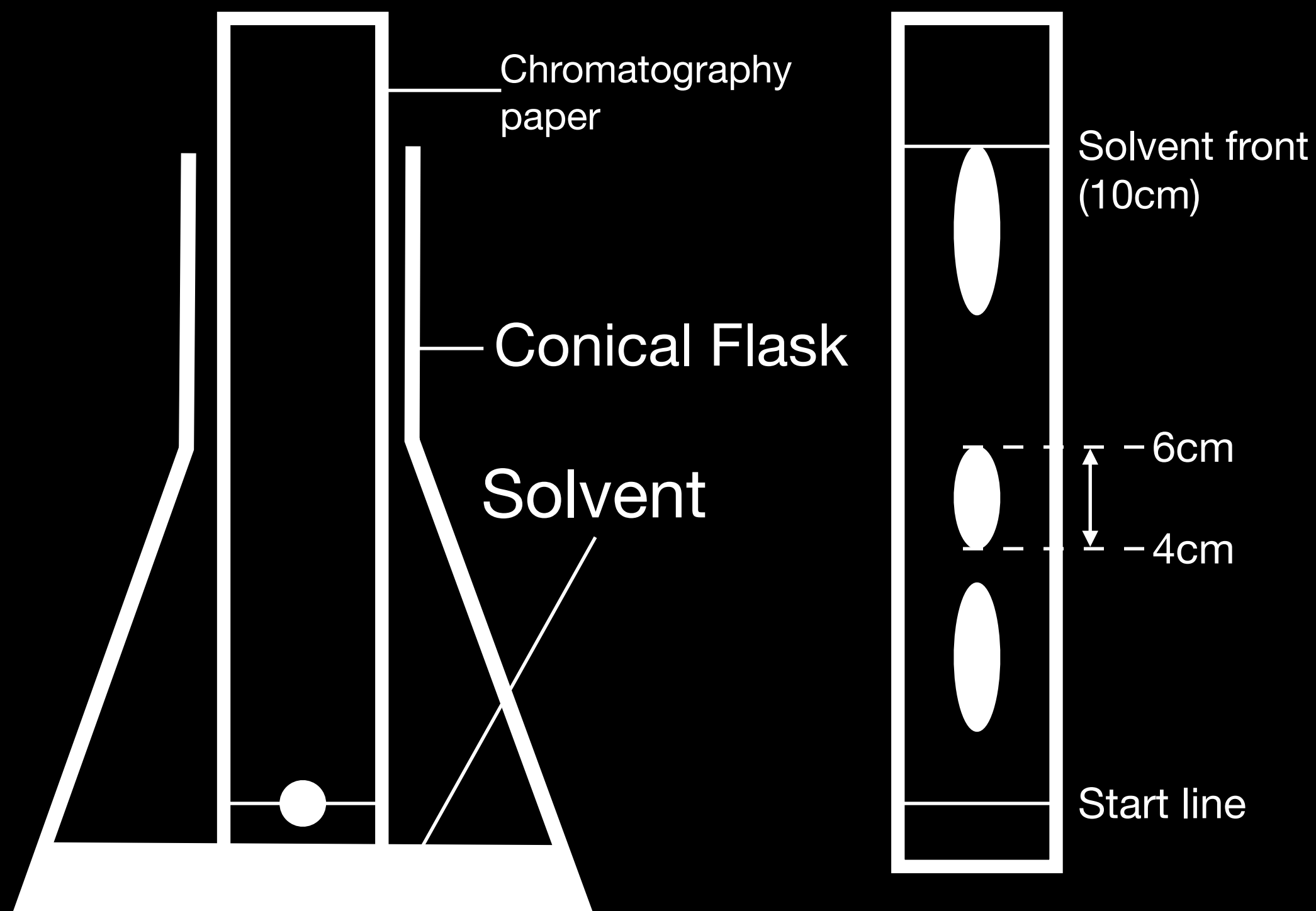
$$R_f \text{ value} = \frac{6 + 4}{10} = \frac{10}{10} = 1.0$$

R_f value is a ratio of two values, thus a substance of R_f=0.5 will always be halfway from the start line to the solvent front



Separation Techniques - (Paper) Chromatography

RIGHT

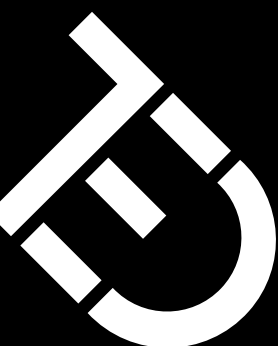


Possible scenarios:

Substance does not leave the start line
(substance is not soluble in the solvent)

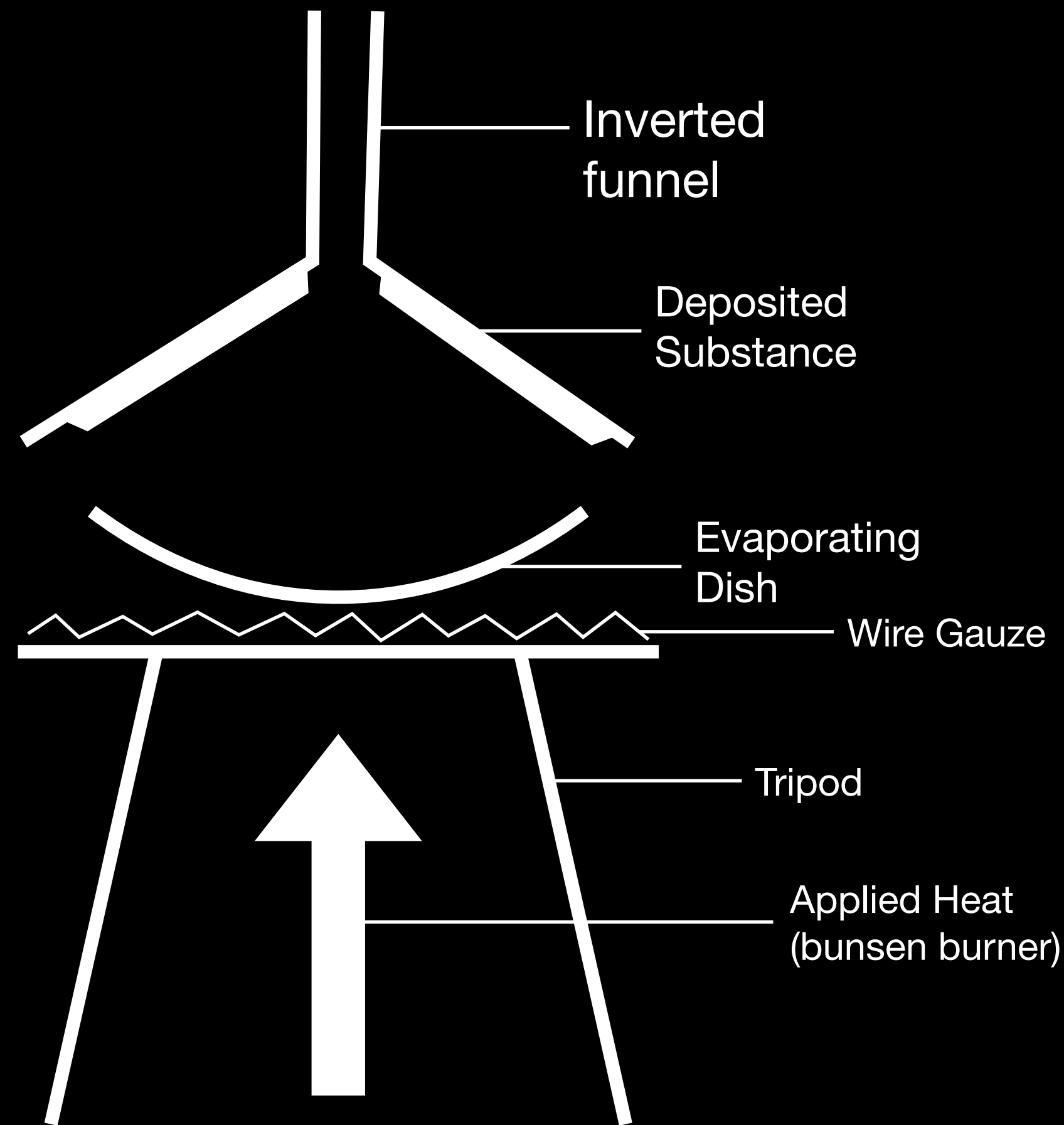
Only one substance moves up along the
chromatogram (substance is pure)

Substance seems to “disappear” after the
chromatography (a locating agent, like
ninhydrin, is needed to see the separated
substances)



Separation Techniques - Sublimation

RIGHT

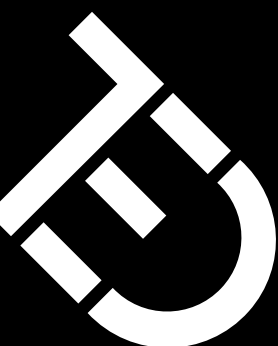


To separate sublimable substances (aka substances that can sublime) from non-sublimable substances

Sublimable substances:

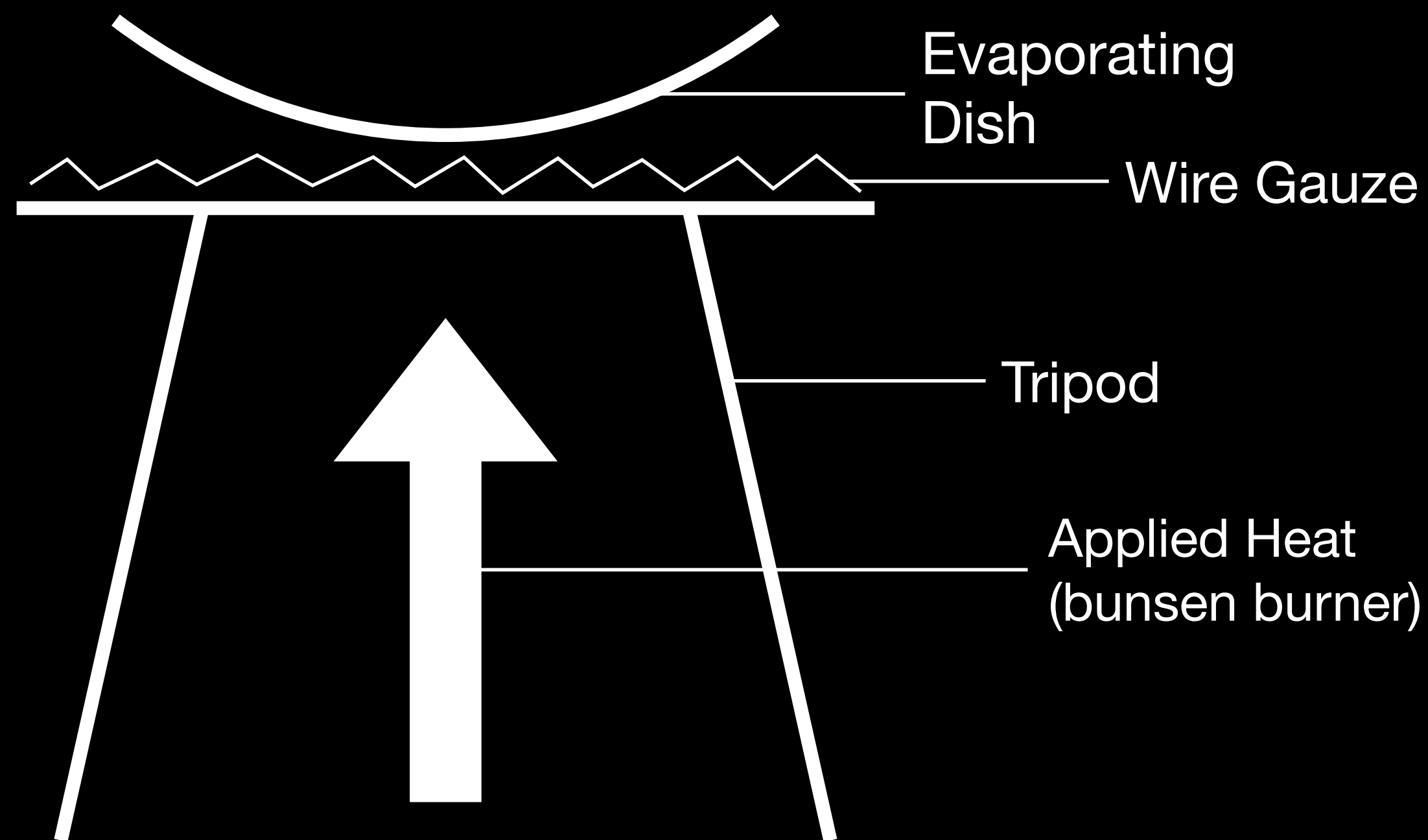
Iodine, Dry ice, Ammonium chloride

The [sublimable substance] gains heat and sublimates, until it reaches the cooler surface of the [surface it deposits on, e.g. an inverted funnel], where it cools down and deposits on [the cooler surface]



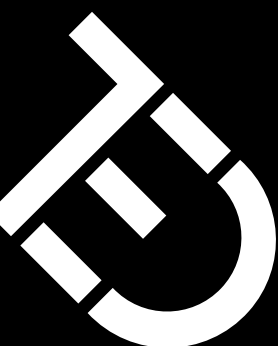
Separation Techniques - Evaporation

RIGHT



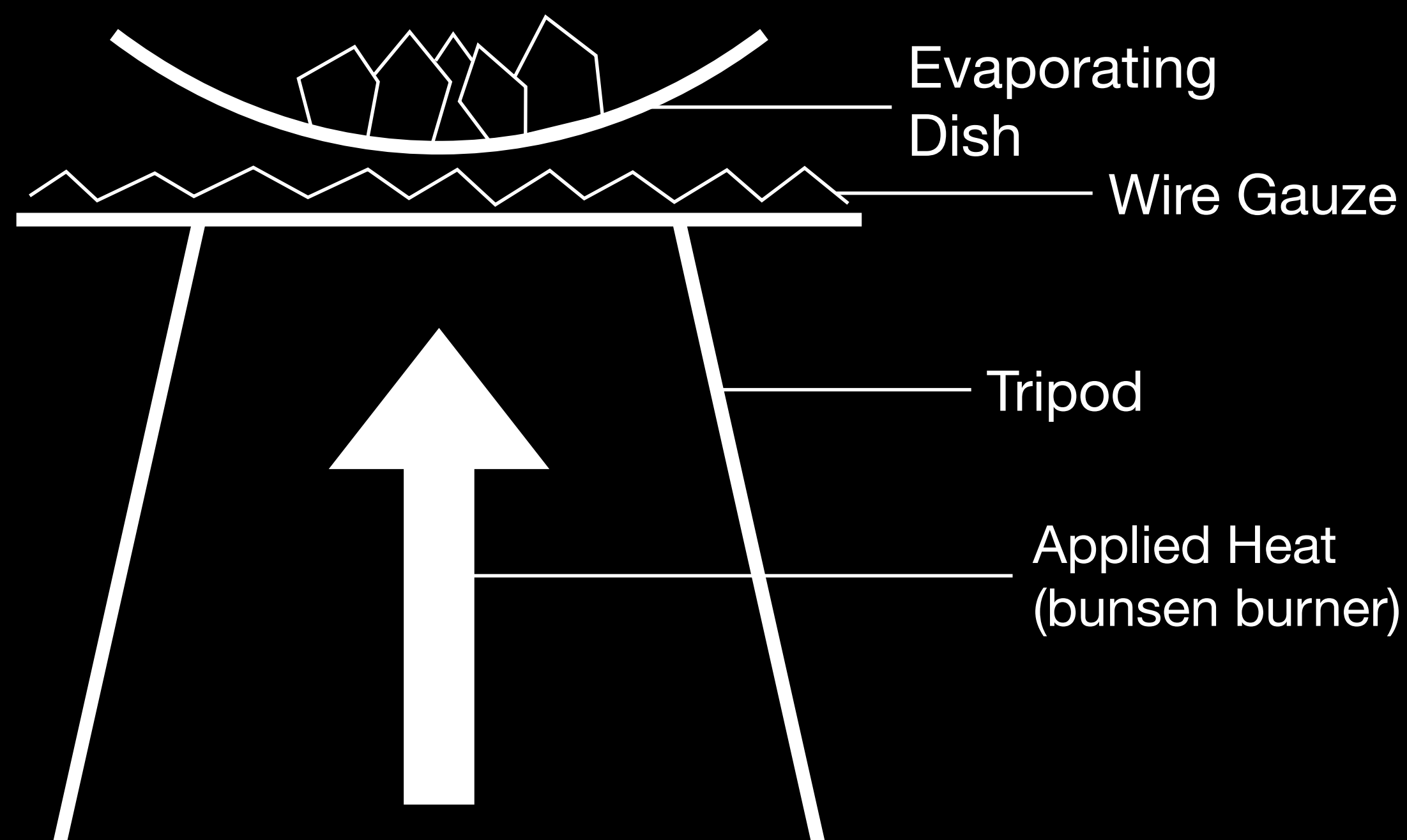
To obtain the solute(the substance that was dissolved in the solvent) dissolved in a solvent as a solid

Mixture is heated until the solvent is fully boiled away, leaving only the solute in the evaporating dish



Separation Techniques - Crystallisation

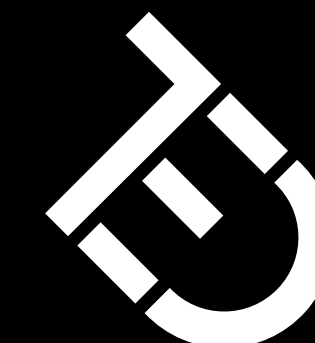
RIGHT



To obtain a solid that will decompose under strong heating/obtain a solid in the form of a crystal

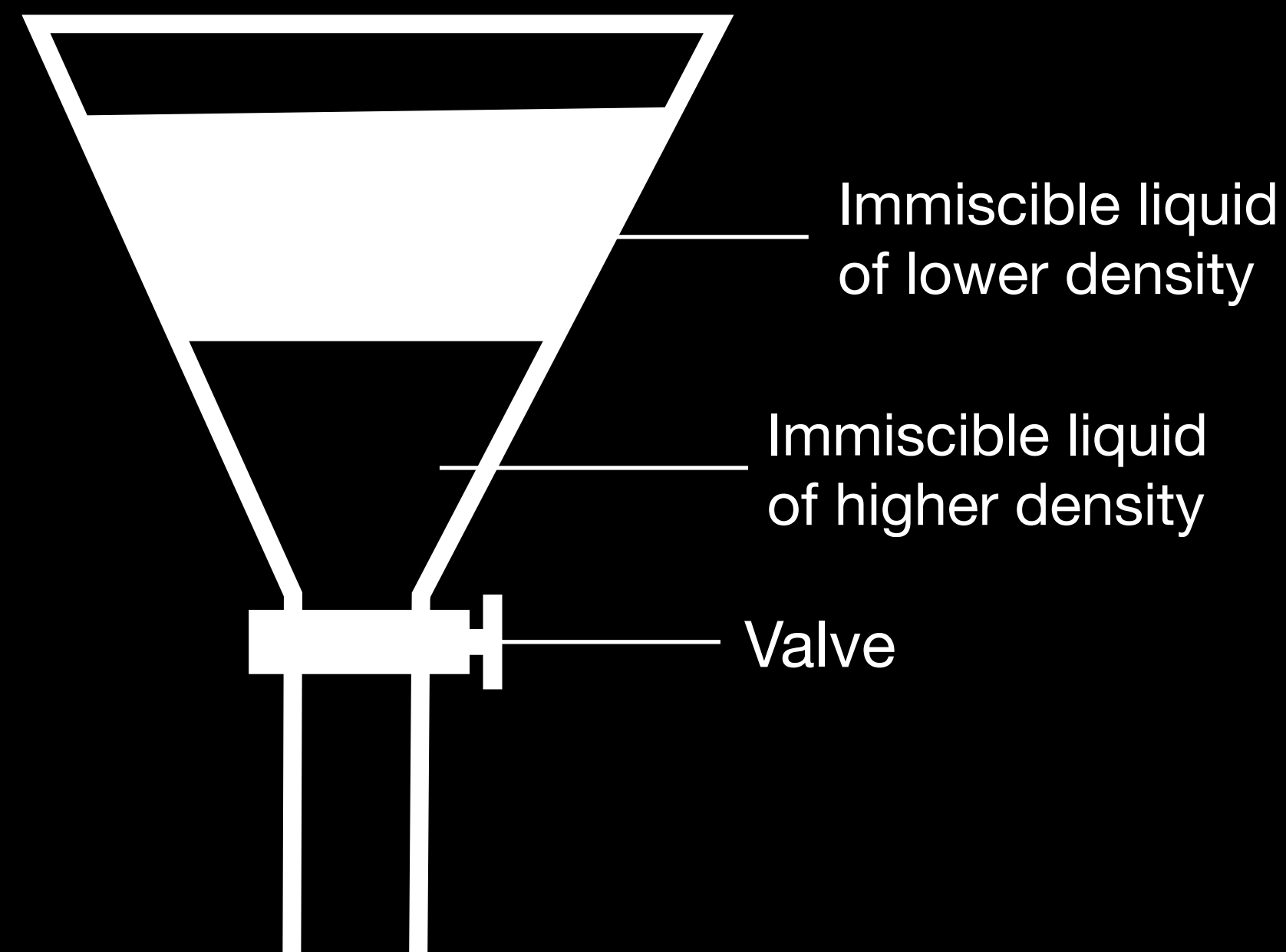
Mixture is heated until it is saturated, where it is then removed from the heat source to allow the crystals to form (Use glass rod to dip into mixture and if crystals form on glass rod, it is saturated)

Leave crystals to cool and filter to collect crystal, then rinse crystal with small amount of cold water and pat dry with filter paper



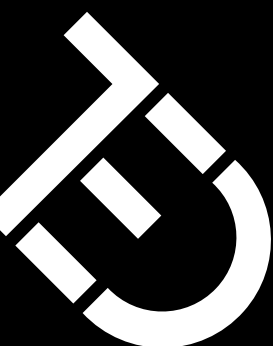
Separation Techniques - Separating Funnel

RIGHT



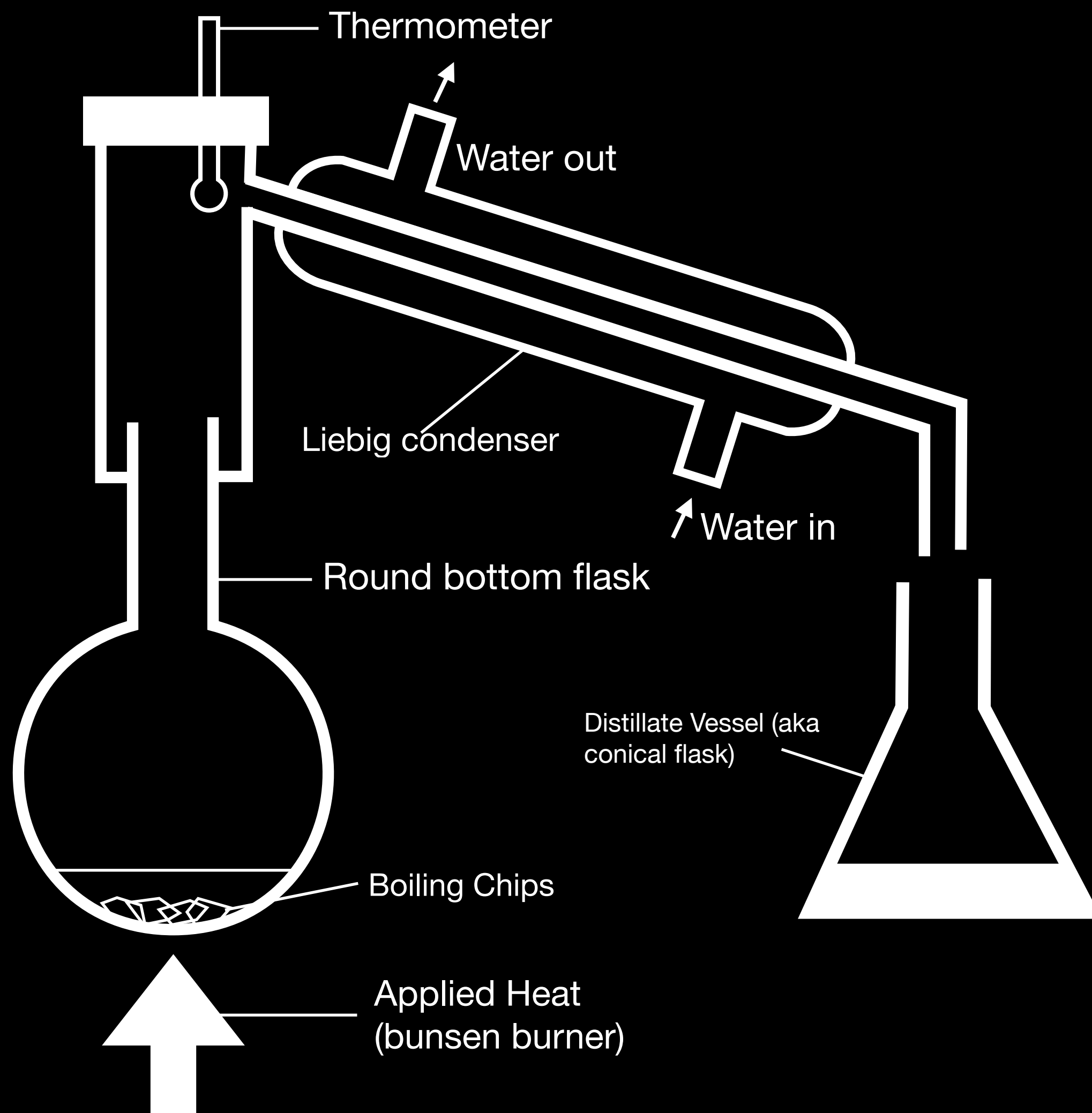
To separate two immiscible liquids (aka two liquids that do not mix together, like water and oil)

Turn valve/stopcock to allow one substance to fully flow out, turn it again to keep other substance in the separating funnel



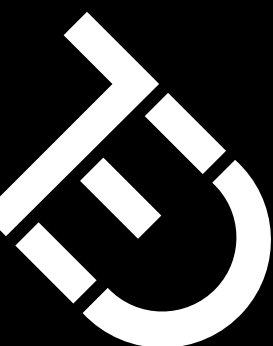
Separation Techniques - Simple Distillation

Okay



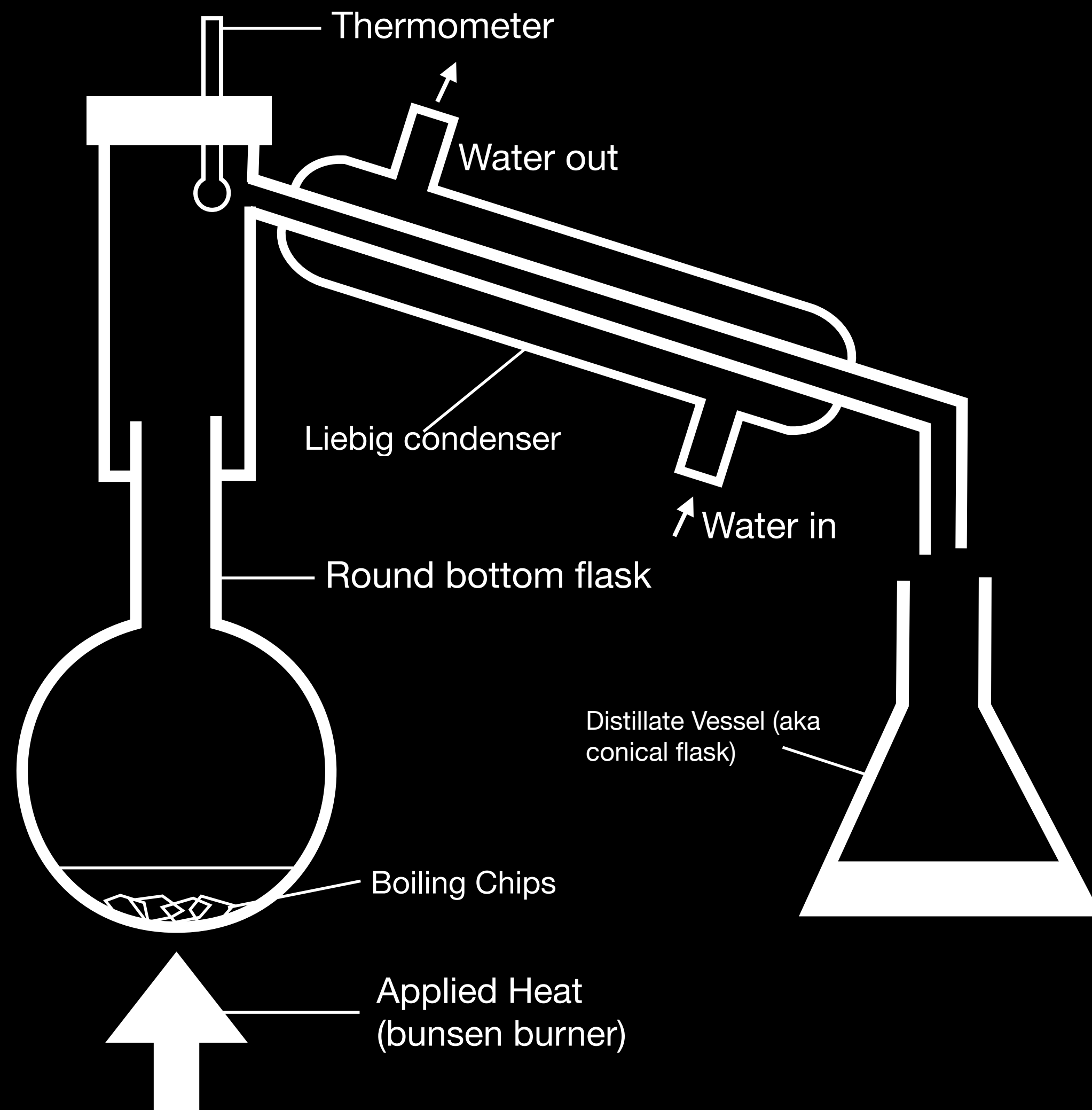
To separate a mixture of two substances with different boiling points and obtain the substance of lower boiling point

Separate a solute from a solvent that can be obtained as a liquid



Separation Techniques - Simple Distillation

Okay



Possible errors:

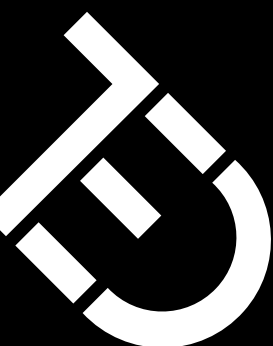
Boiling chips not present (absence of boiling chips will not allow smooth boiling)

Thermometer bulb not at side arm (Will not accurately measure the boiling point of the distillate)

Wrong flow of water (decreases efficiency of distillate condensing inside the Liebig condenser)

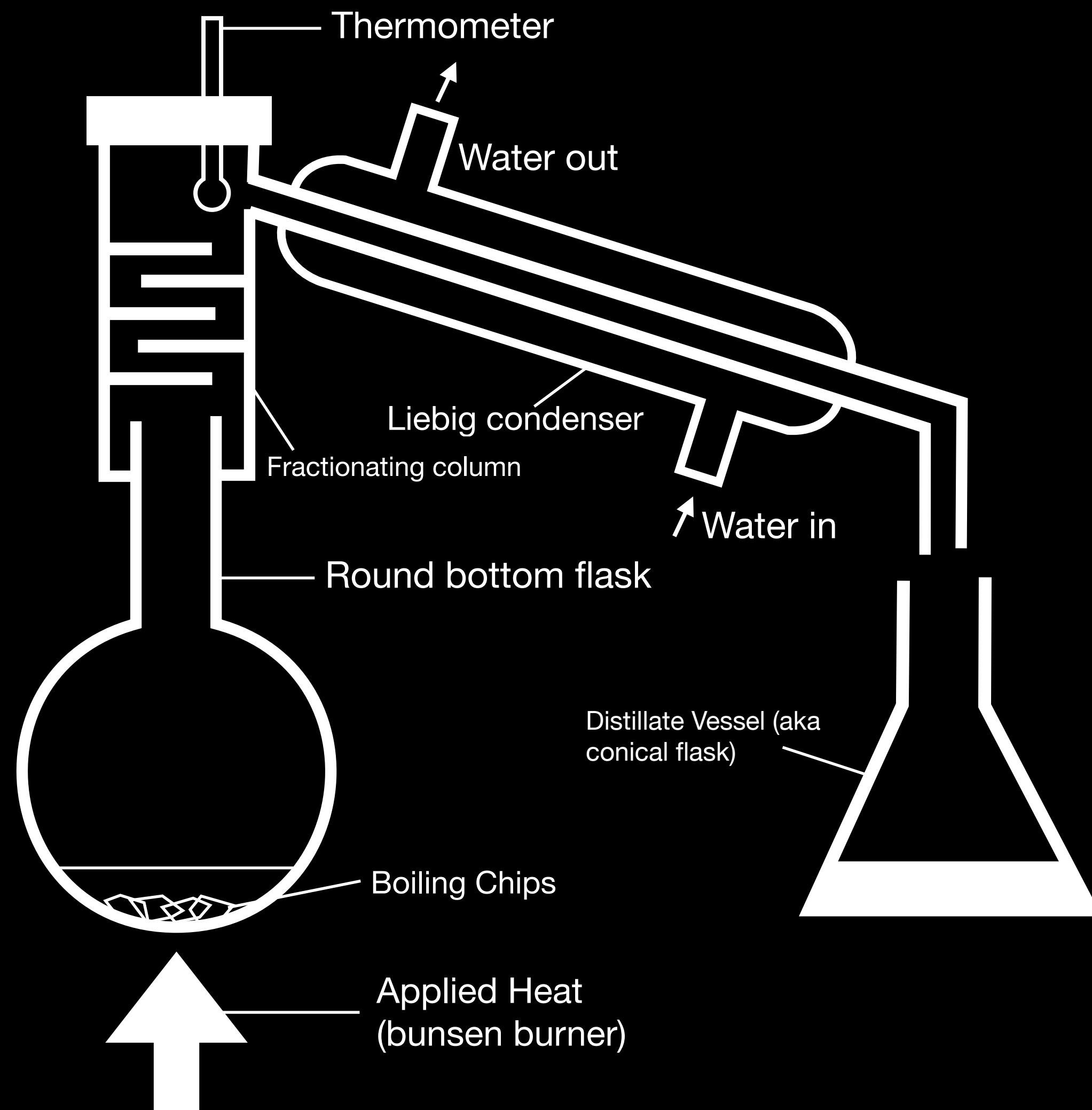
Liebig condenser not tilted towards distillate vessel (Does not ensure distillate can flow towards distillate vessel)

*The converse is also true, e.g boiling chips are present because they allow smooth boiling



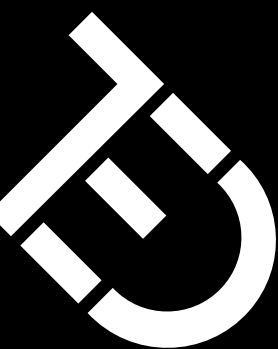
Separation Techniques - Fractional Distillation

Left



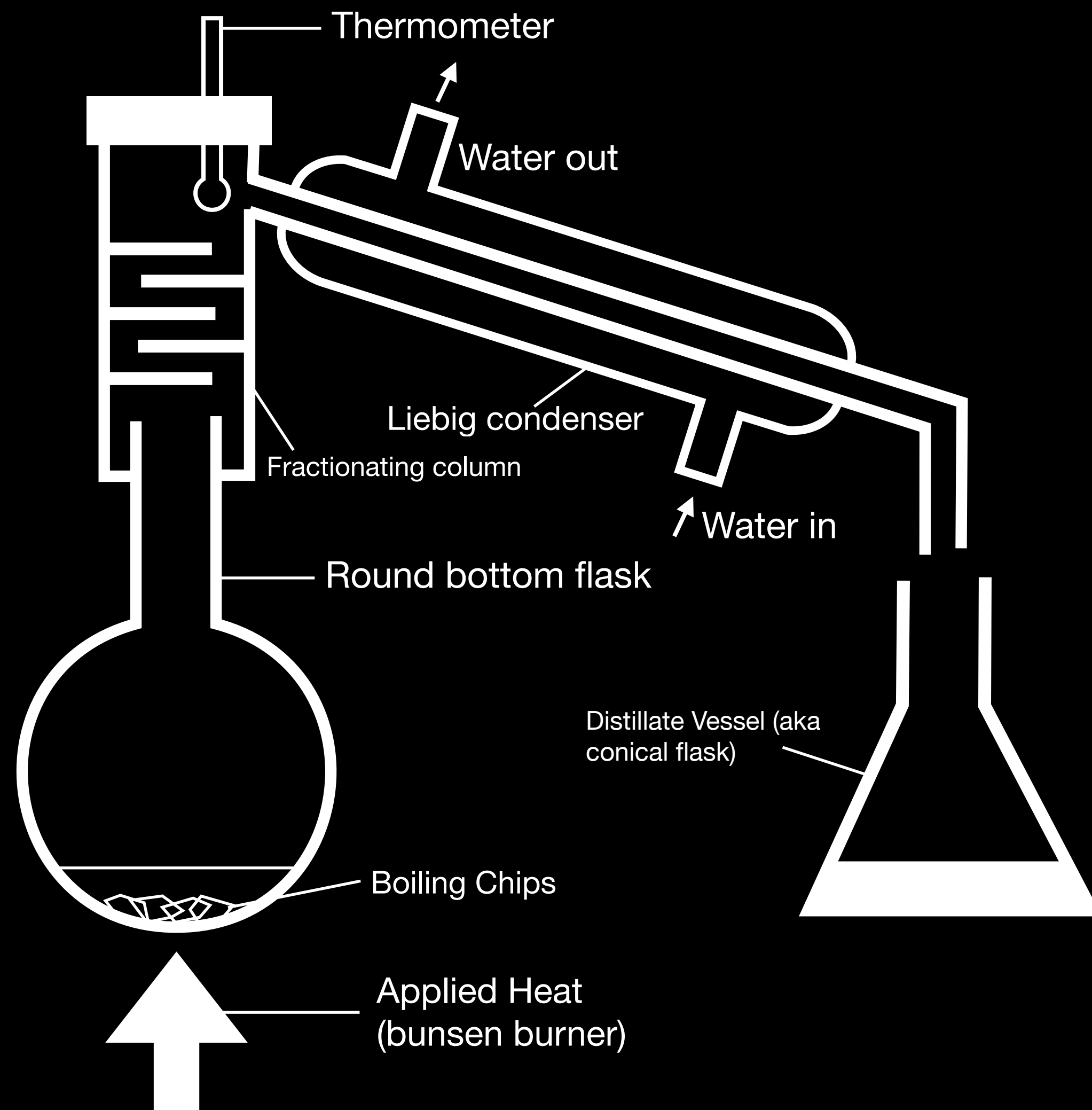
To separate a mixture of two substances with different boiling points that are too close to each other (usually dif. of less than 40°C) and obtain the substance of lower boiling point

Separate a solute from a solvent that can be obtained as a liquid



Separation Techniques - Fractional Distillation

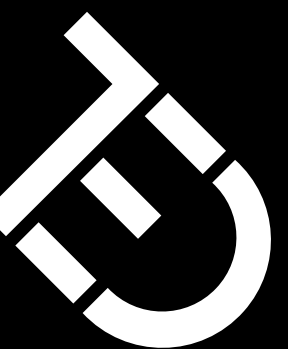
Left



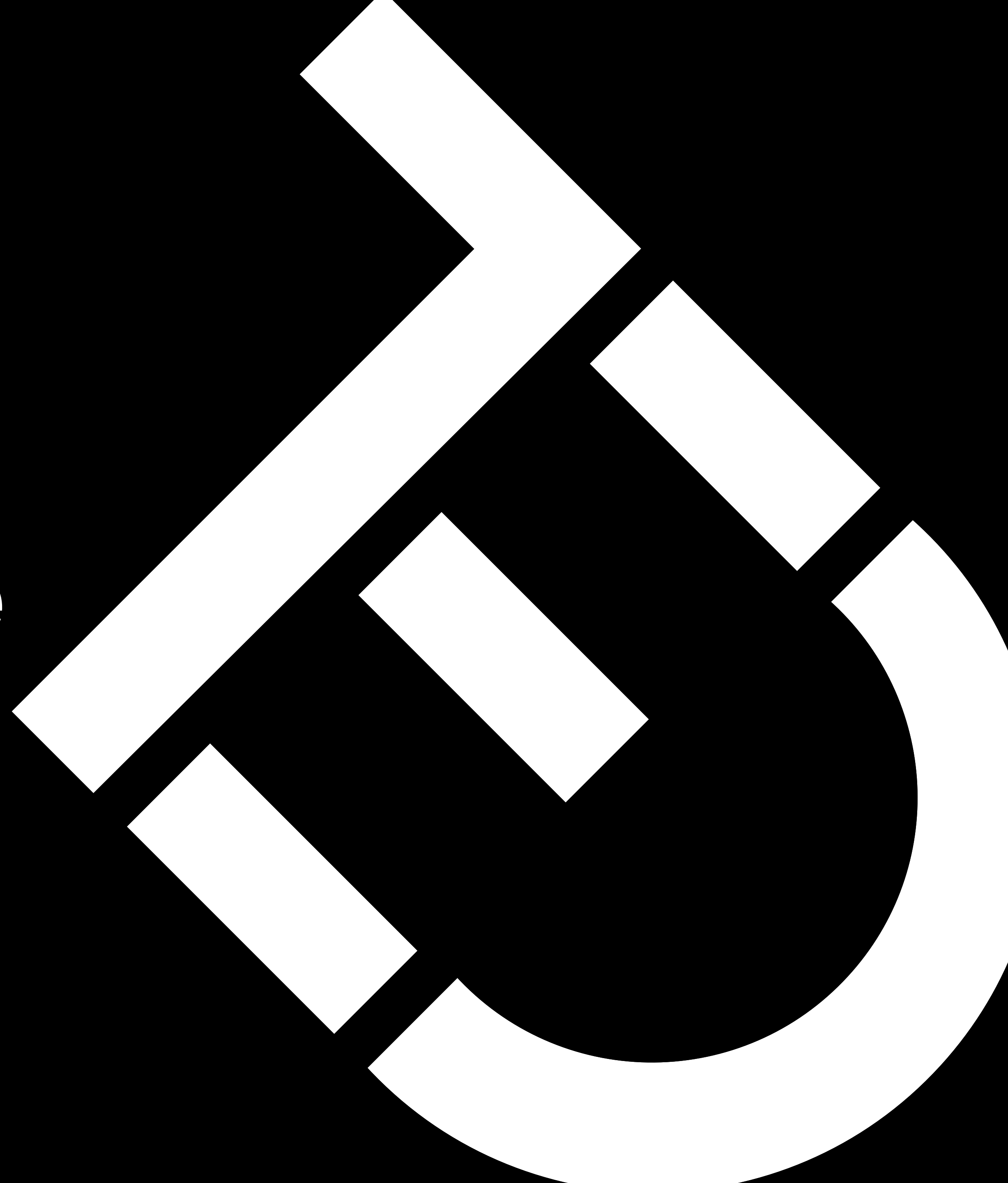
Same setup as simple distillation but with a fractionating column

Fractionating column:

Contains glass beads/plates to provide a greater surface area for the vapour of the substance of higher b.p to condense on the glass beads/plates, allowing it to drip back into the round bottom flask

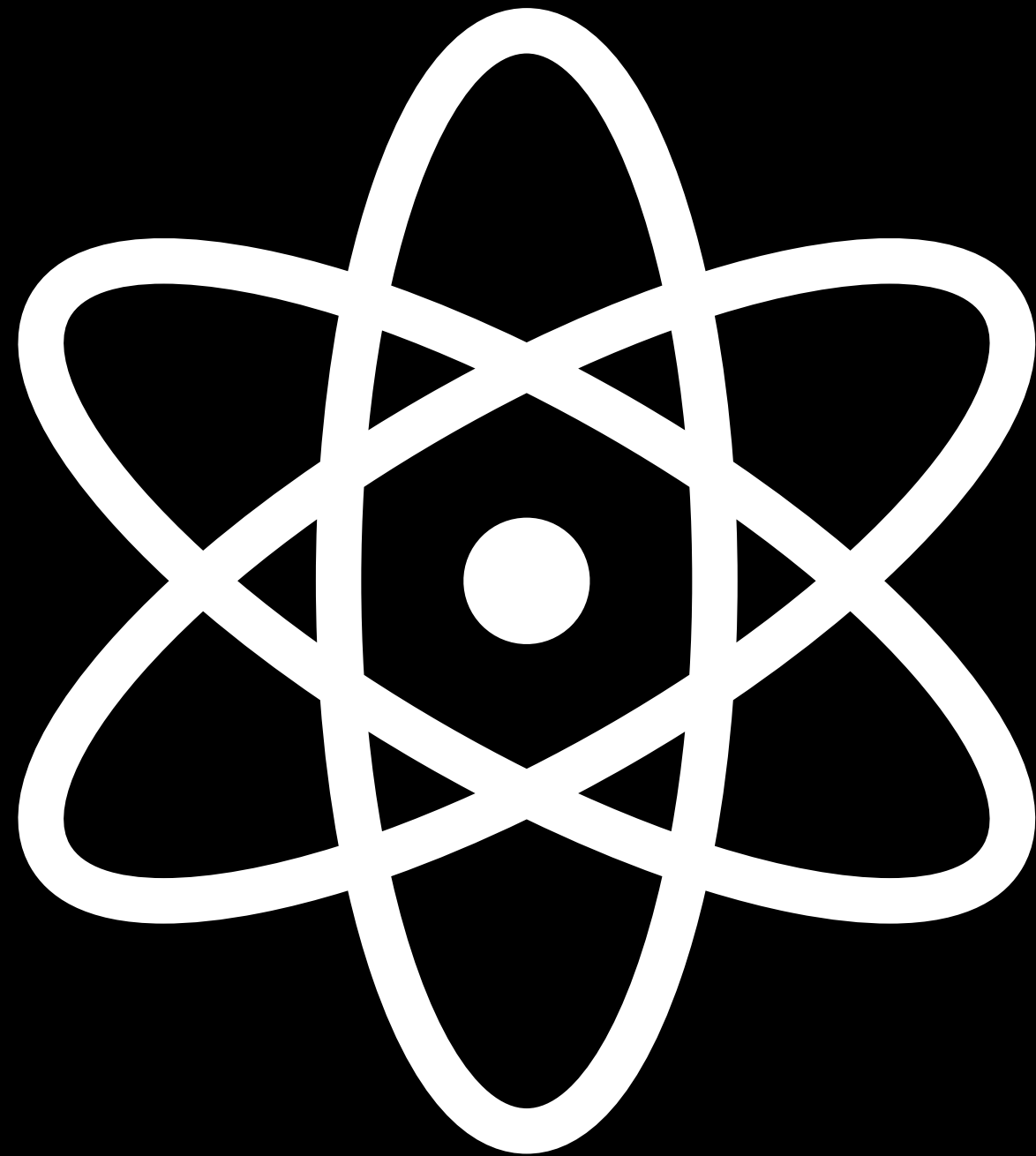


Atomic Structure

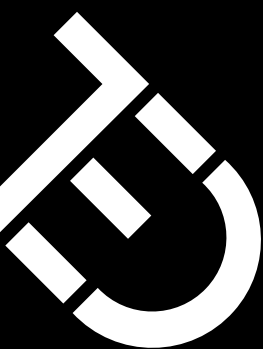


Atoms, Ions & Molecules

atomos



- things exist in 3 forms; atoms, ions, and molecules
- atoms have 0 charge
- ions have a +ve or -ve charge
- molecules are **2 or more atoms** bonded tgt



Nuclide Notation

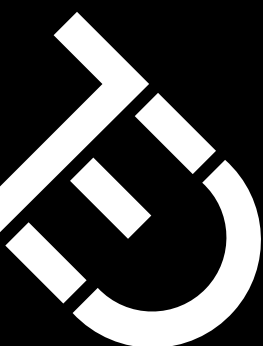
nuke

relative atomic mass



atomic number

- nuclide notation is the standard form of notation used in periodic tables and texts.
- relative atomic mass is the number of protons and neutrons in the nucleus
- atomic number is the number of protons only in the nucleus (which is also number of electrons in the atom)
- it's called the atomic number because it tells us what atom it is



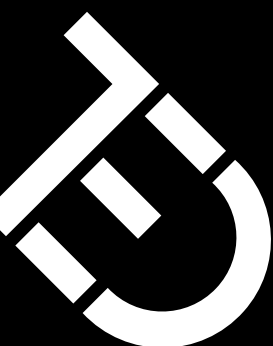
Electronic Configuration

only valid for first 20 elements !!!!!!!!!!!!!!!!



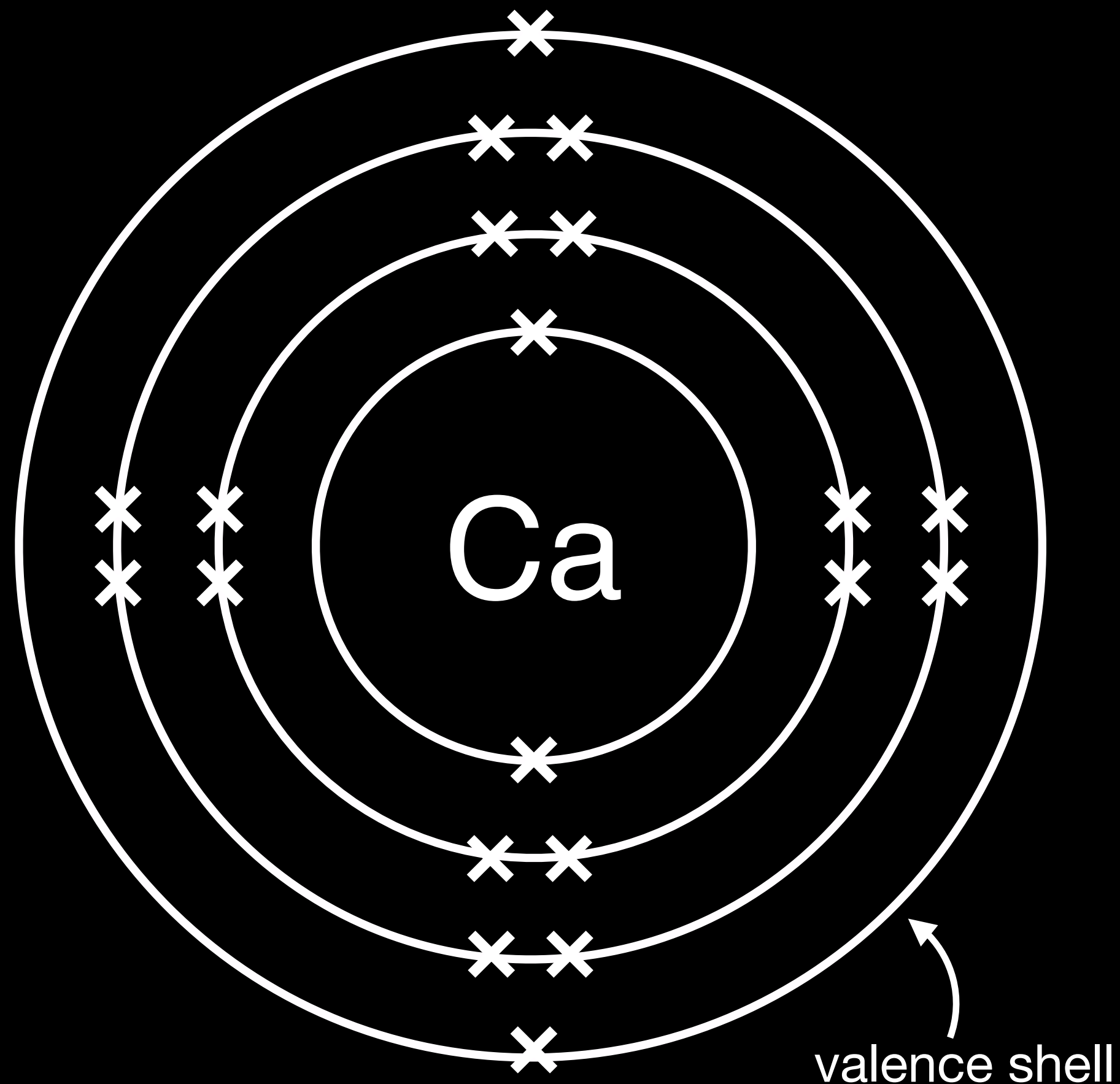
electronic configuration: 2,2

- electronic configuration shows the arrangement of electrons in the shells
- the first shell can only hold up to 2 electrons, the rest can hold 8
- In this case, there are 2 electrons in the first inner shell and 2 more in the second

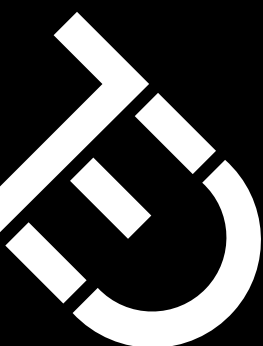


Electronic Structure

broh

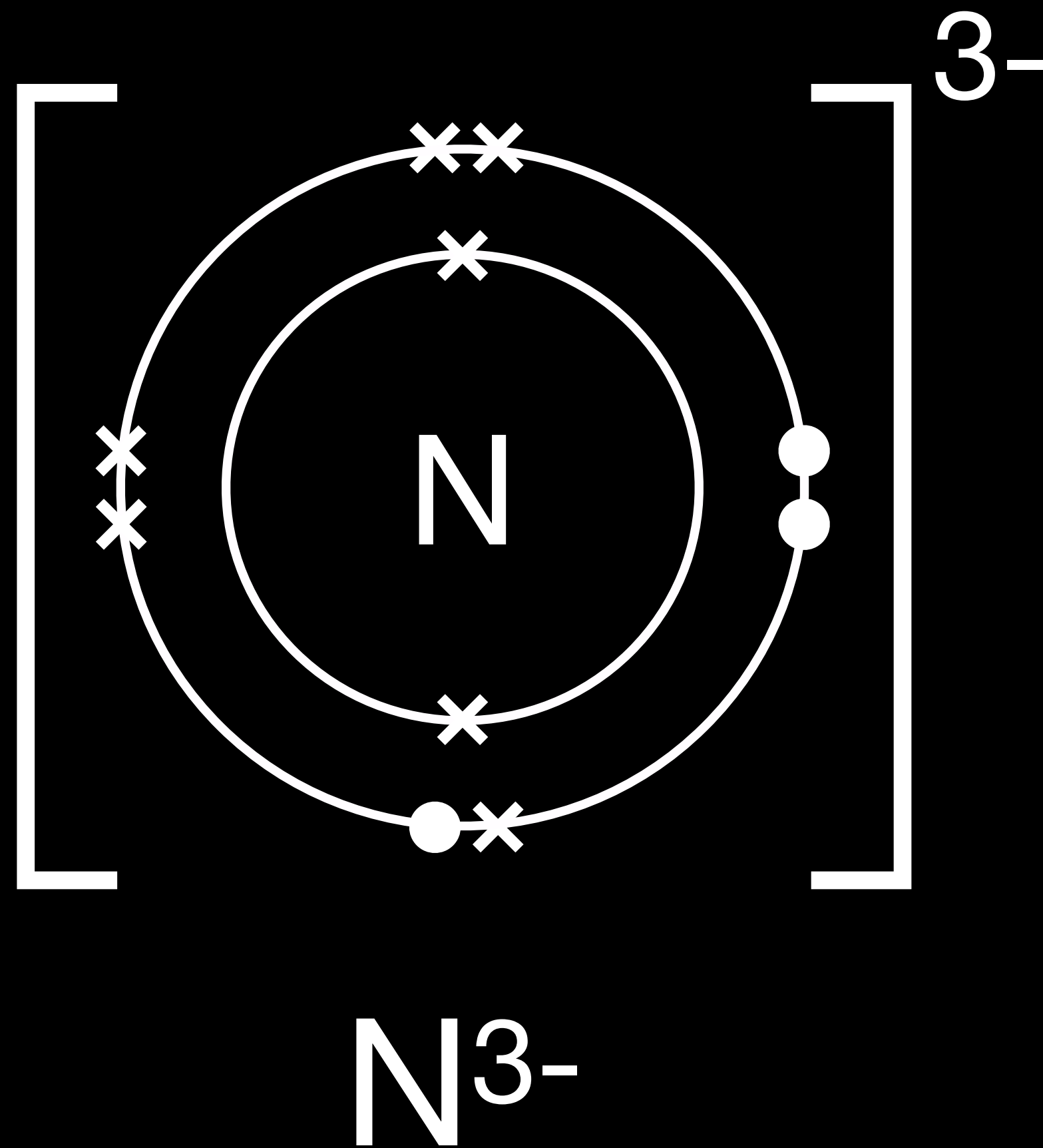


- this dot and cross diagram shows us the electron shells of an atom
- electron shells are circular orbital paths travelled by the electrons.
- the outermost shell is called the valence shell and its electrons are the valence electrons
- The number of valence electrons determines the chemical reactions and properties
- A full valence shell with 8 electrons makes the atom stable and will not react (e.g. noble gases and ions)

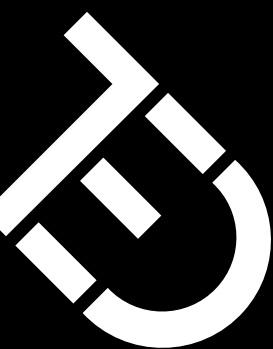


Ions

keep an i-on deez nuts

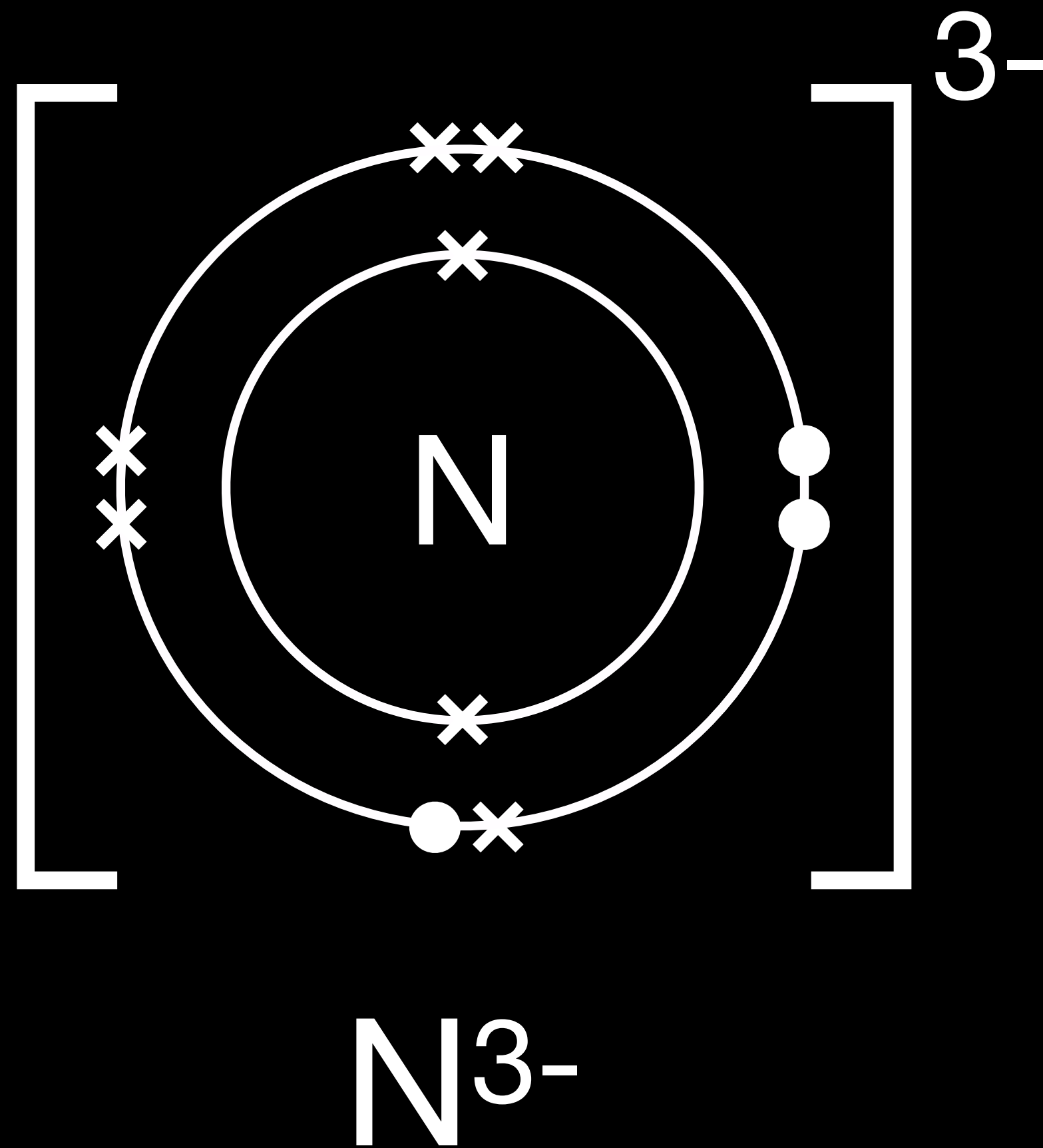


- ions are atoms with a +ve or -ve charge when they gain or lose electrons
- When electrons are lost, there is a more positive charge, thus the ion will be positive. When electrons are gained, there is a more negative charge, thus the ion is negative
- +ve ions are called cat-ions while -ve ions are called an-ions
- << this is how we draw ion model
- keep in mind the added electrons should notated differently (use dots and crosses)

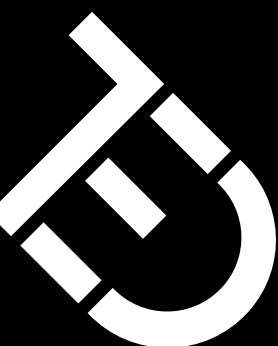


ions

property brothers

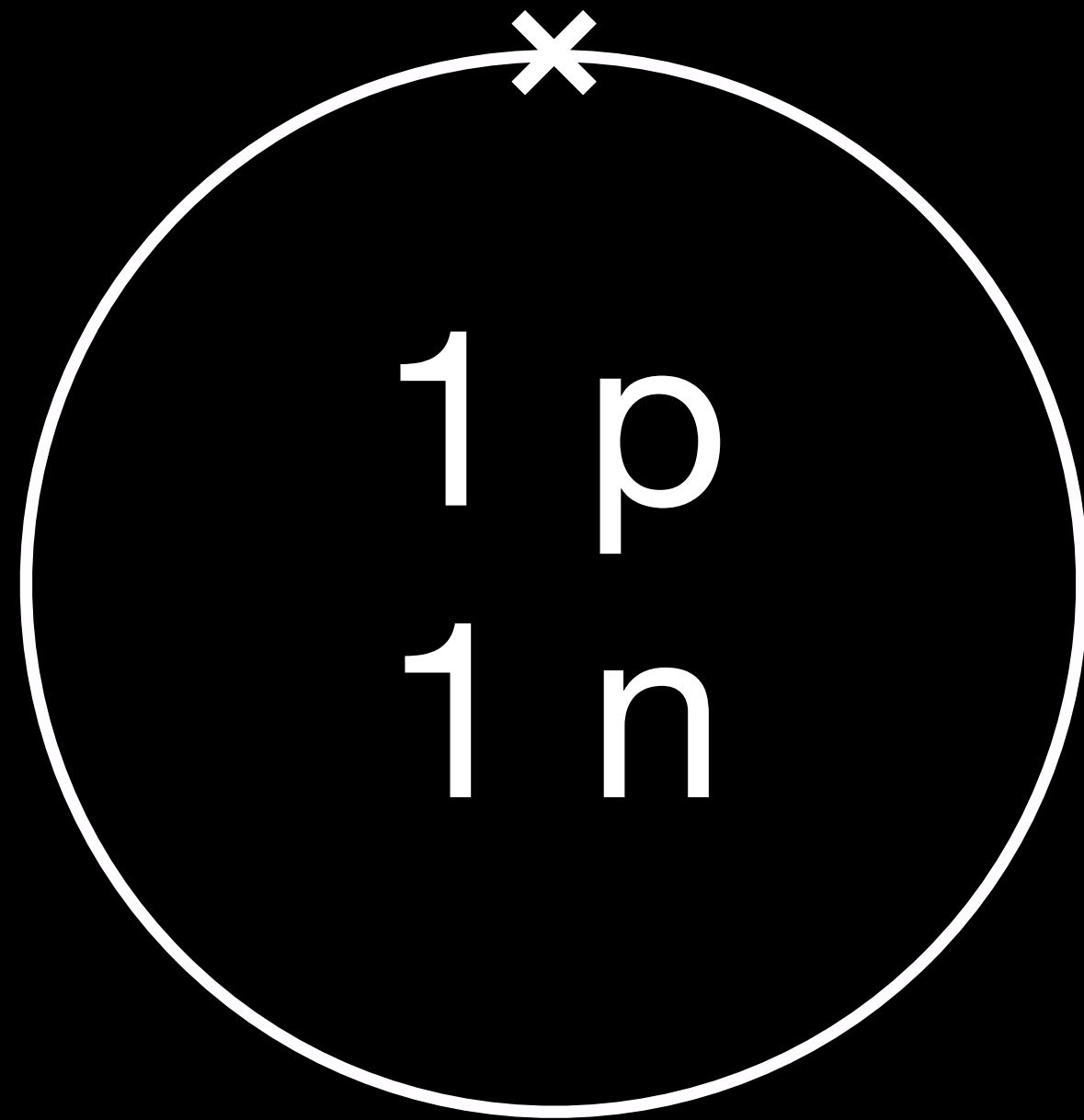


- they are stable as they gain/lose electrons to have a full valence shell, obtaining a stable duplet/octet structure (aka has 2/8 valence electrons)
- metals (grp I II III) usually become positive ion while non-metals (grp V VI VII) become negative ions
- grp IV can become both as they have 4 valence electrons (can gain/lose 4 to become stable)

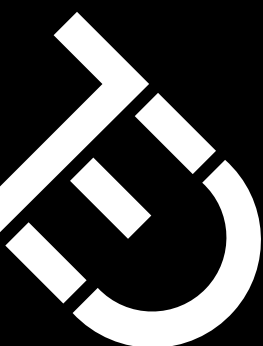


Isotopes

new tron

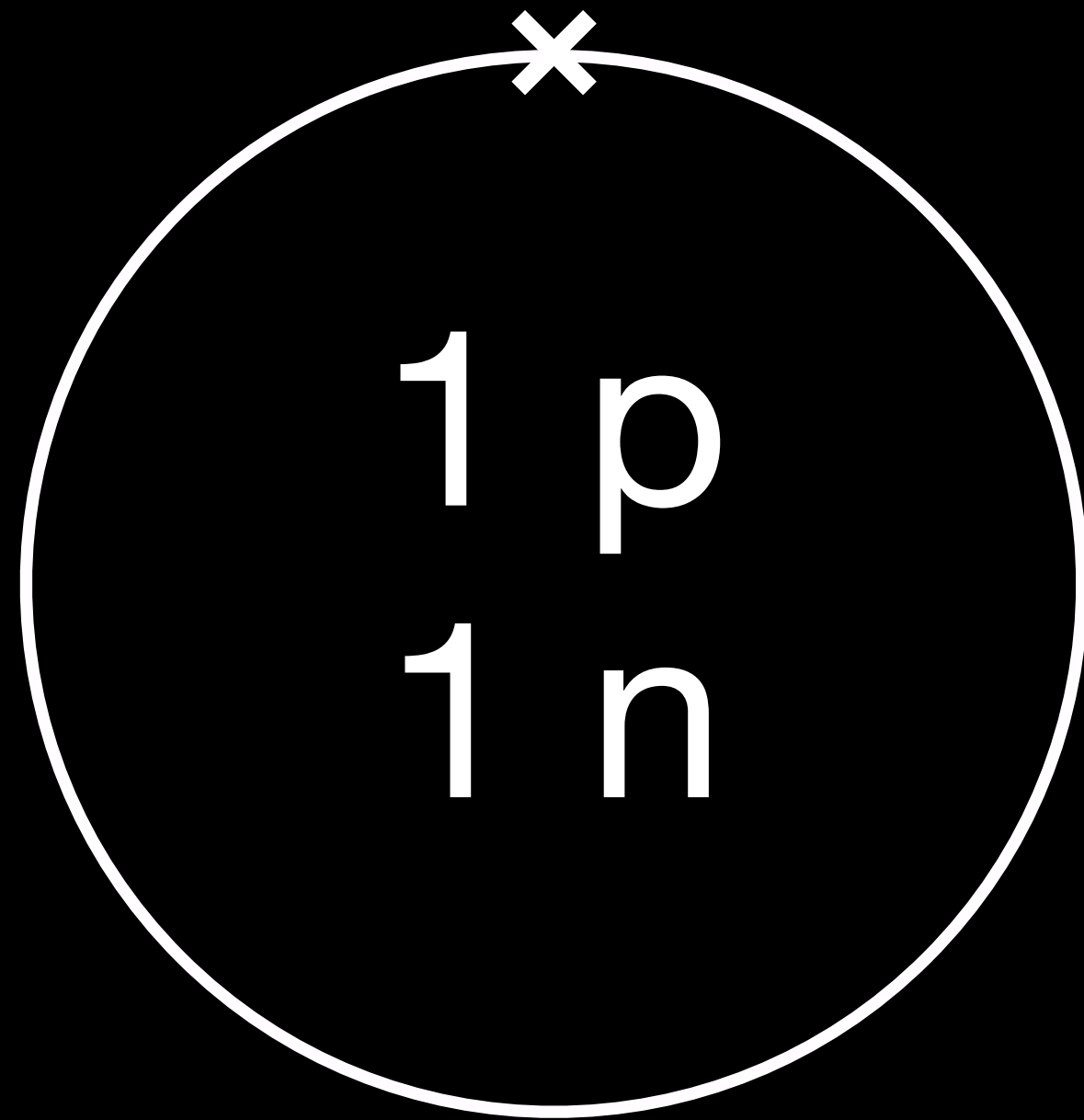


- variation of element with different number of neutrons (but with the same number of protons)
- << isotope model (hydrogen-2)
- Isotopes are named as “element-[atomic mass]” e.g. hydrogen-2

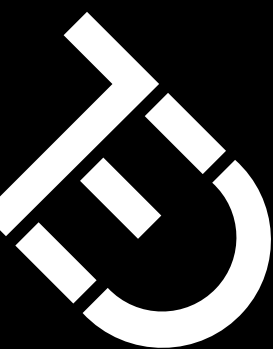


Isotopes

properties

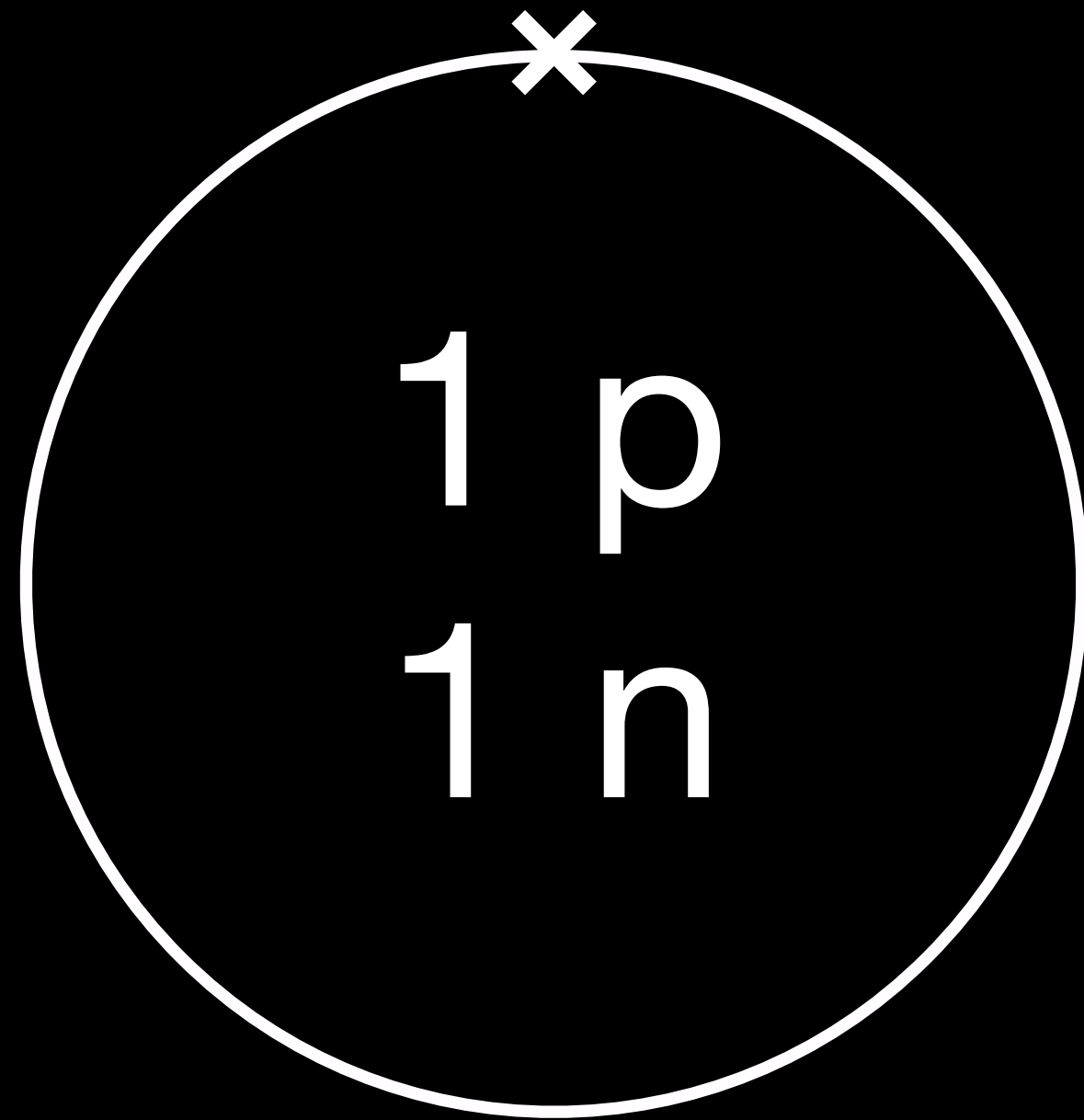


- Isotopes will have the same chemical properties as the valence shell is the same
- different physical properties as the mass is different, e.g. density, melting, and boiling points



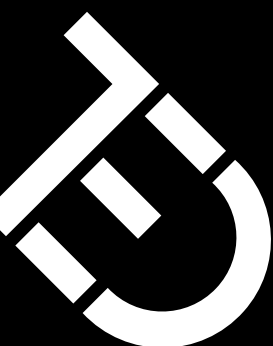
Isotopes - Relative Atomic Mass

why is chlorine 35.5??



- RELATIVE atomic mass refers to the relative atomic mass of all isotopes found naturally and their abundance percentages
- $p_1(A_r) + p_2(A_r) + \dots$, where p is the abundance percentage of the isotope, and A_r is the relative atomic mass of the isotope
- e.g. 75% of chlorine found is chlorine-35 while 25% of chlorine found is chlorine-37. Hence, the relative atom mass is

$$75 \% (35) + 25 \% (37) = 35.5$$

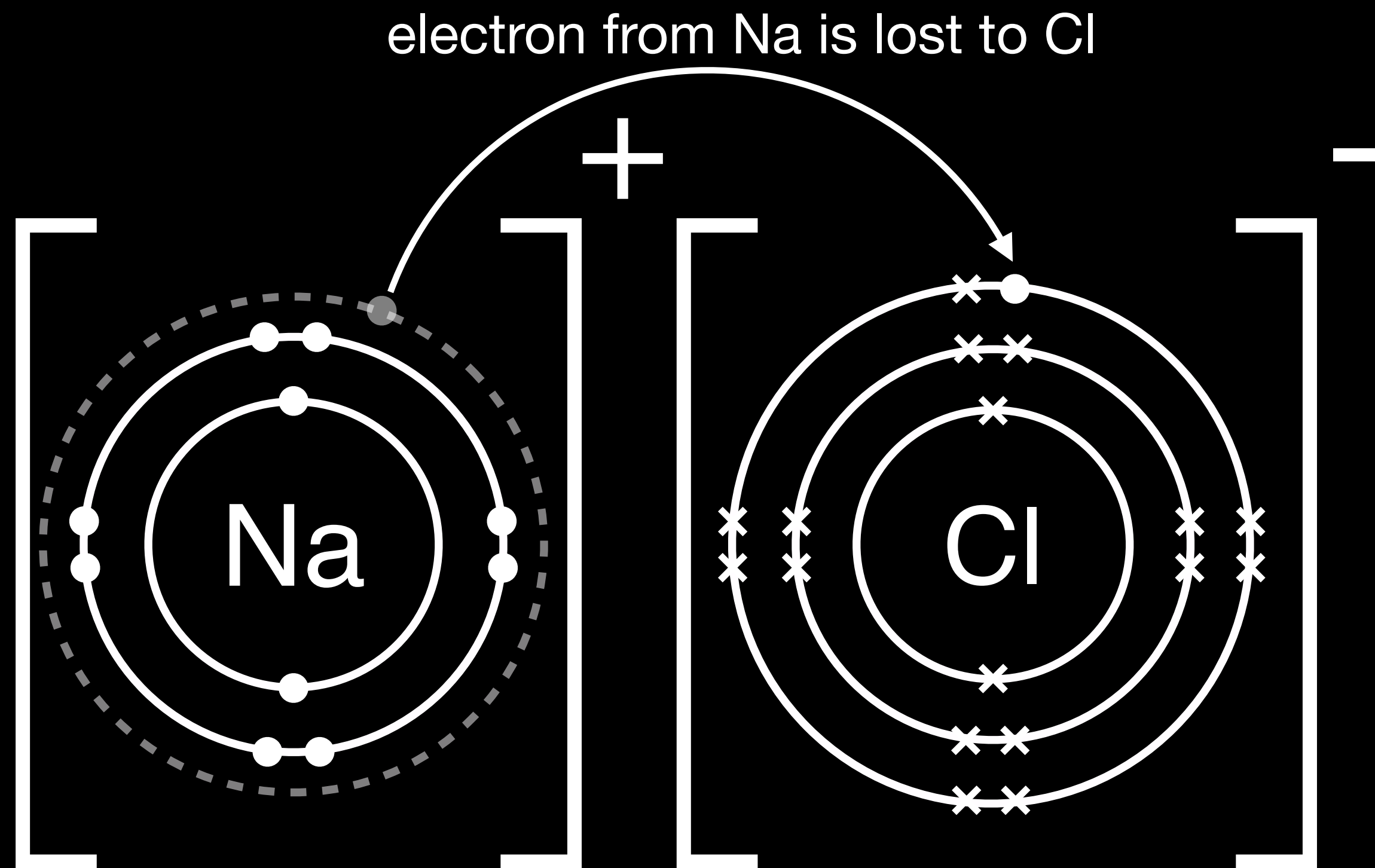


Chemical Bonding

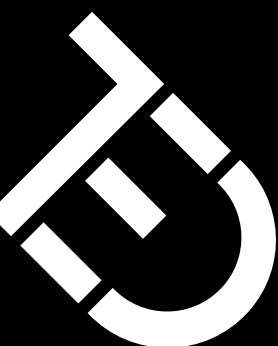


Ionic Bonding

why and how do they bond

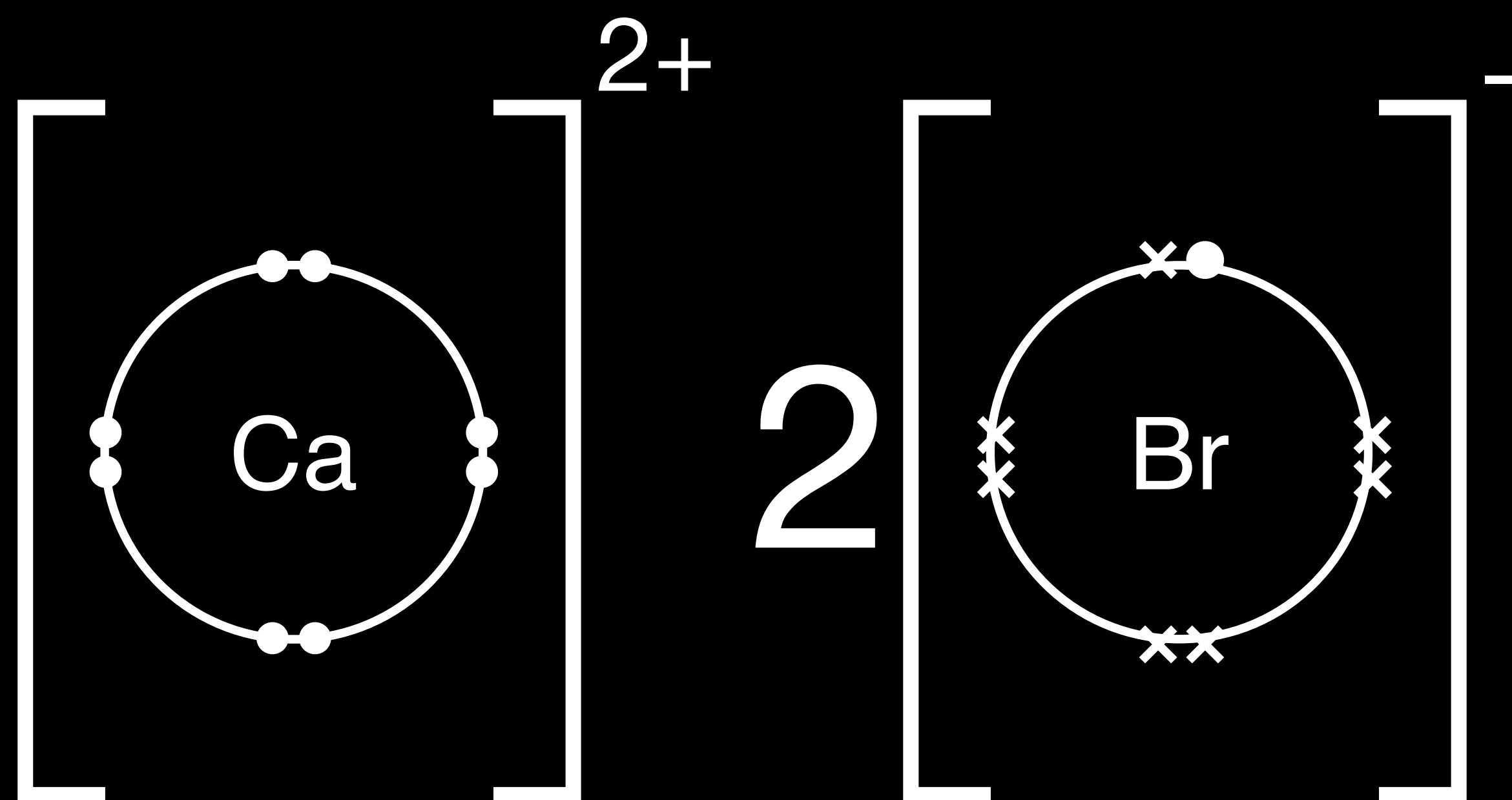


- since ions have charge, positive ions and negative ions are attracted to each other and thus form bonds
- For ionic bonding, metals usually bond with non-metals.



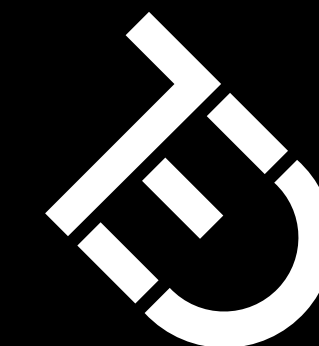
Ionic Bonding

notation and drawing



valence electrons of CaBr_2

- draw valence electrons only when asked to (or for atomic no 19 and higher)
- metal is always on the left, non metal on right
- if there is more than one ion write how many there are on the left of it e.g. CaBr_2 has 2 bromine ions, so we write 2 next to it.



Ionic Bonding

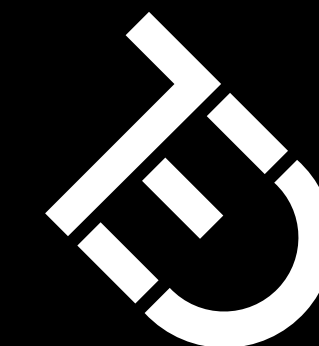
notation and drawing

Calcium Bromide: CaBr_2

Magnesium carbonate: MgCO_3

Calcium Hydroxide: Ca(OH)_2

- For naming ionic compounds, metal is still on the left without any name change while non-metal is on right and has name changed
- if there is no oxygen, add -ide to non metal.
- if theres oxygen, add -ate or -ite
 - with the exception of hydroxide



Transition Metals

built different

Iron (III) Chloride: FeCl_3

Cl_3 has a charge of 3- so Fe has a charge of 3+ (thus labelled as iron (III))

Copper(II) Phosphate: $\text{Cu}_3(\text{PO}_4)_2$

$(\text{PO}_4)_2$ has a charge of 6- so Cu has a charge of 2+

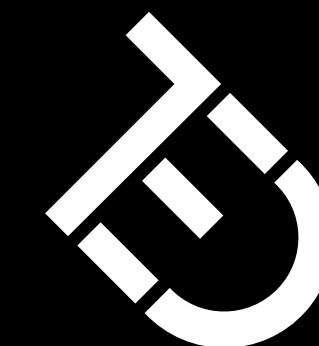
Zinc Oxide: ZnO

O has a charge of 2- so Zn has a charge of 2+

- transition metals are different as they can form ions of varying charge so roman numerals is used to indicate charge (as shown on left)

- exceptions are silver Ag^+ and zinc Zn^{2+}

(Covered more in periodic table chapter)



Polyatomic Ions

ions with more than one element

Hydroxide: OH^-

Nitrate: NO_3^-

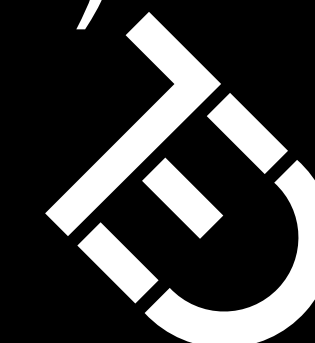
Carbonate: CO_3^{2-}

Sulfate: SO_4^{2-}

Phosphate: PO_4^{3-}

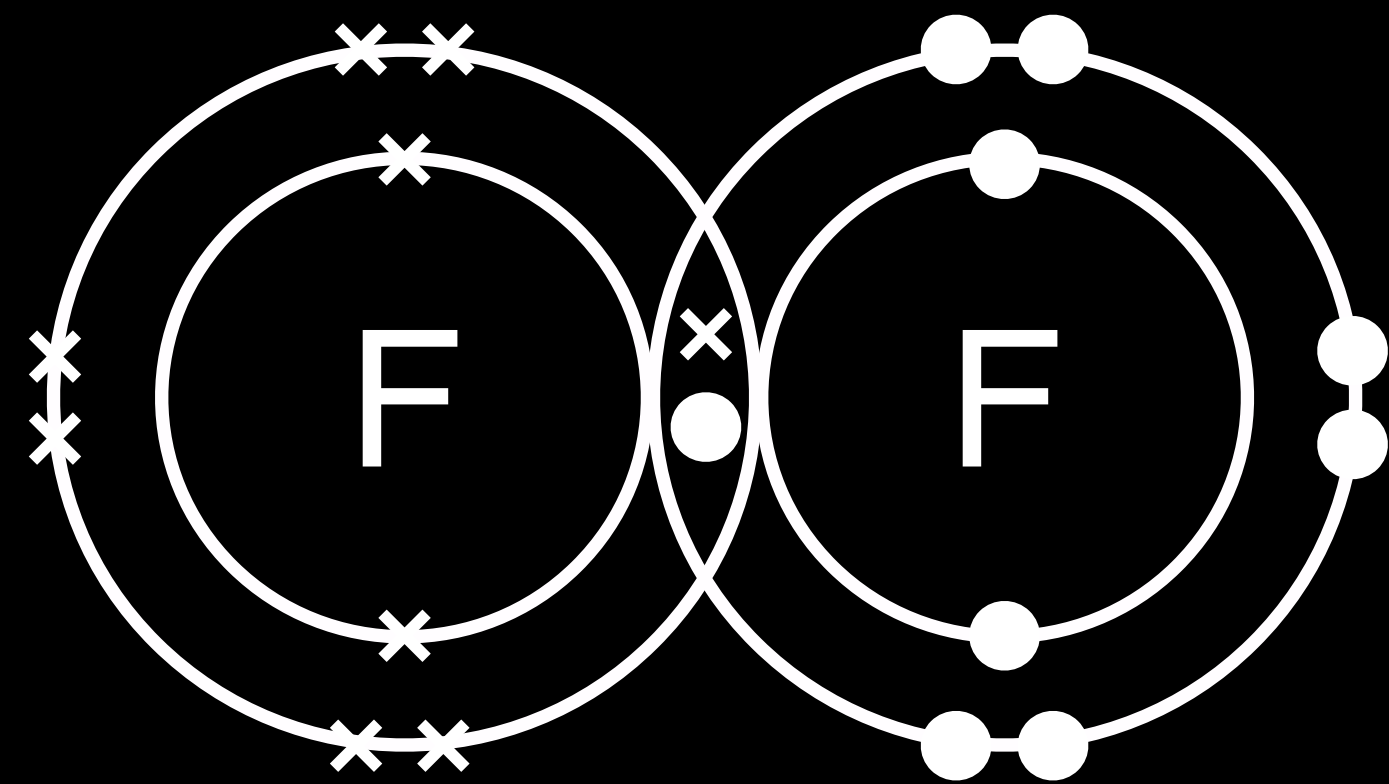
Ammonium: NH_4^+

- Ions that have more than one element
- Can be identified by suffix of -ate (exceptions of hydroxide and ammonium)
- Most have negative charge except for ammonium
- You have to memorise this (uh oh)



Covalent Bonding

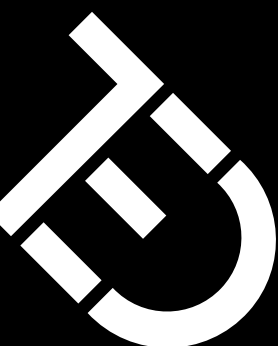
re



Bond is represented as lines, one pair of electrons representing one line

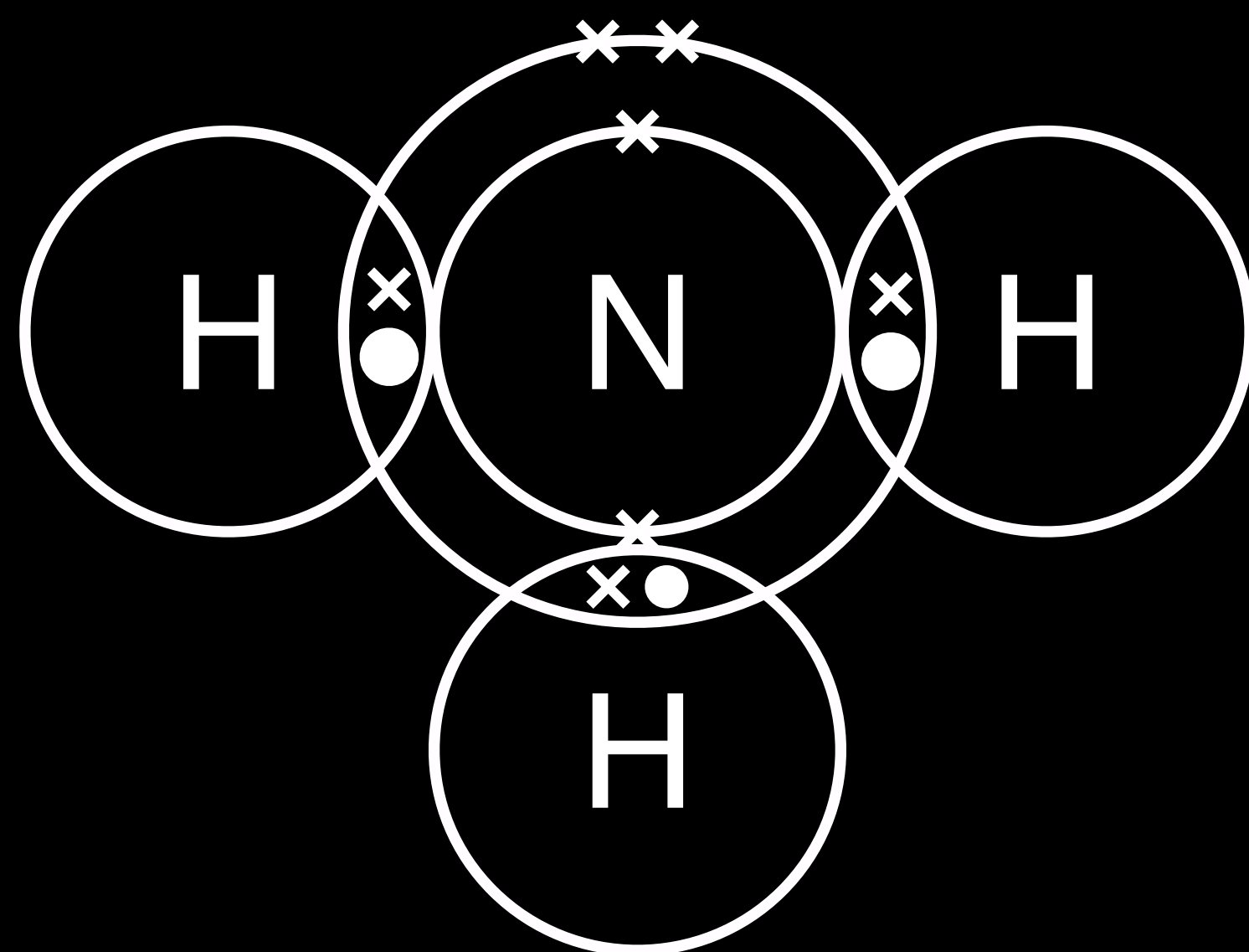
In this case, F—F

- When non metals bond together by sharing electrons
- Both elements will have duplet/octet structure aka full valence shell
- Positively charged nucleus attracts to negatively charged electrons between
- Resulting particle/molecule has no overall charge

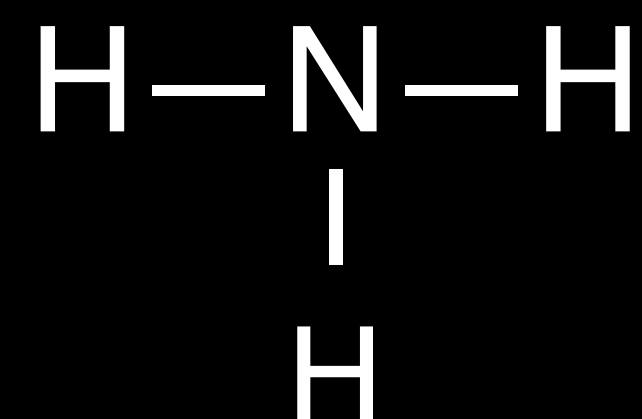


Covalent Bonding - Diagrams

Calcium does not have 16 valence electrons



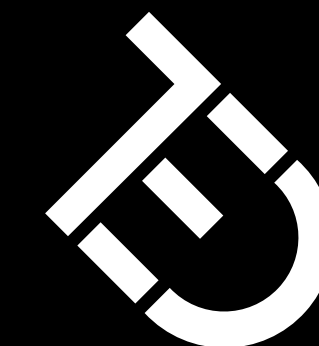
Structural formula



Chemical Formula

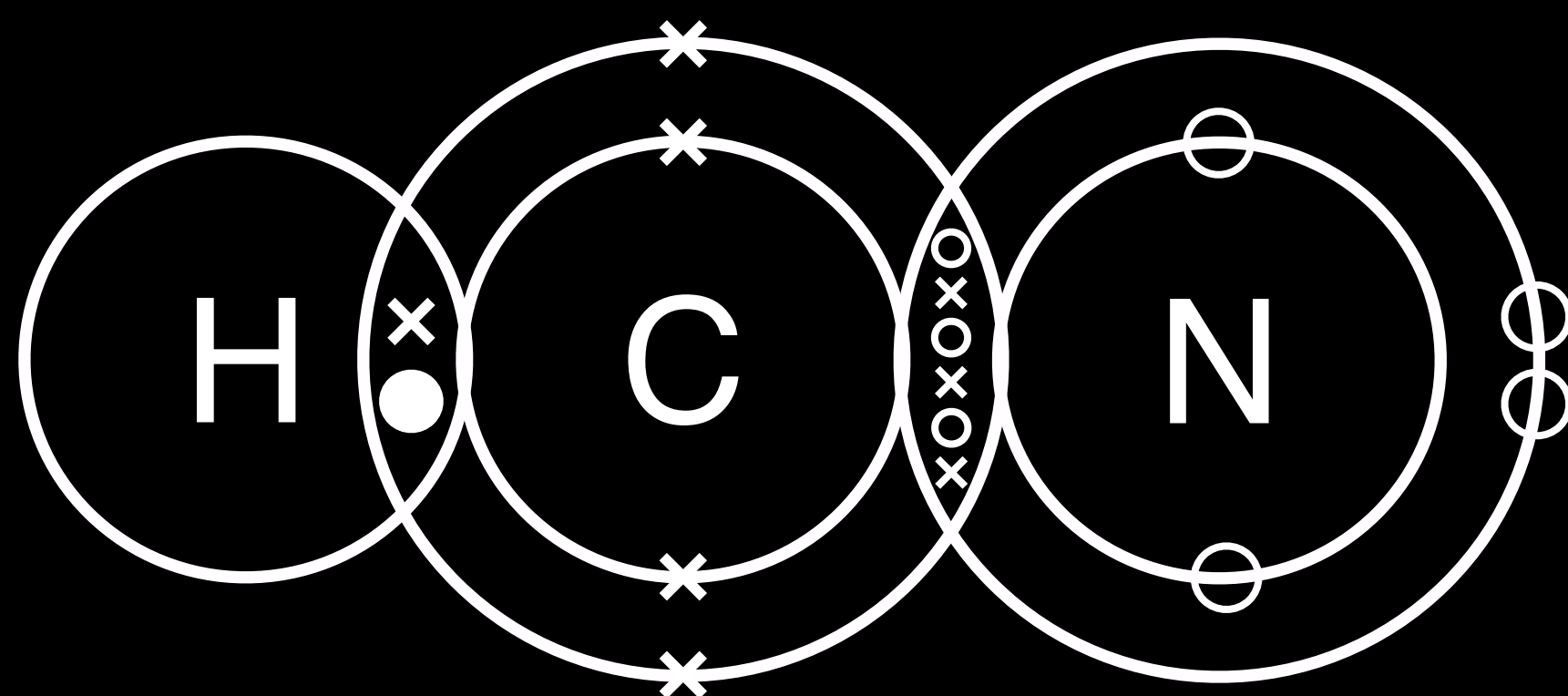


- Electrons shells should be clearly overlapping
- Different atom's electrons should be of different symbol (dot/cross)
- Shared electrons should be inside the overlapping area while non shared electrons should be on the electron shell.
- Make sure all the atoms in the molecule have a full valence shell (duplet/octet)



Covalent bonding - Diagrams

Calcium does not have 16 valence electrons



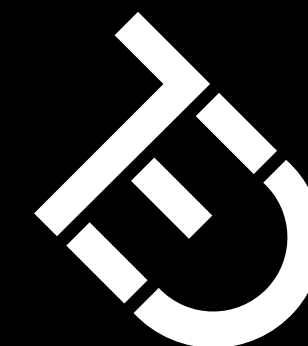
Structural formula



Chemical Formula

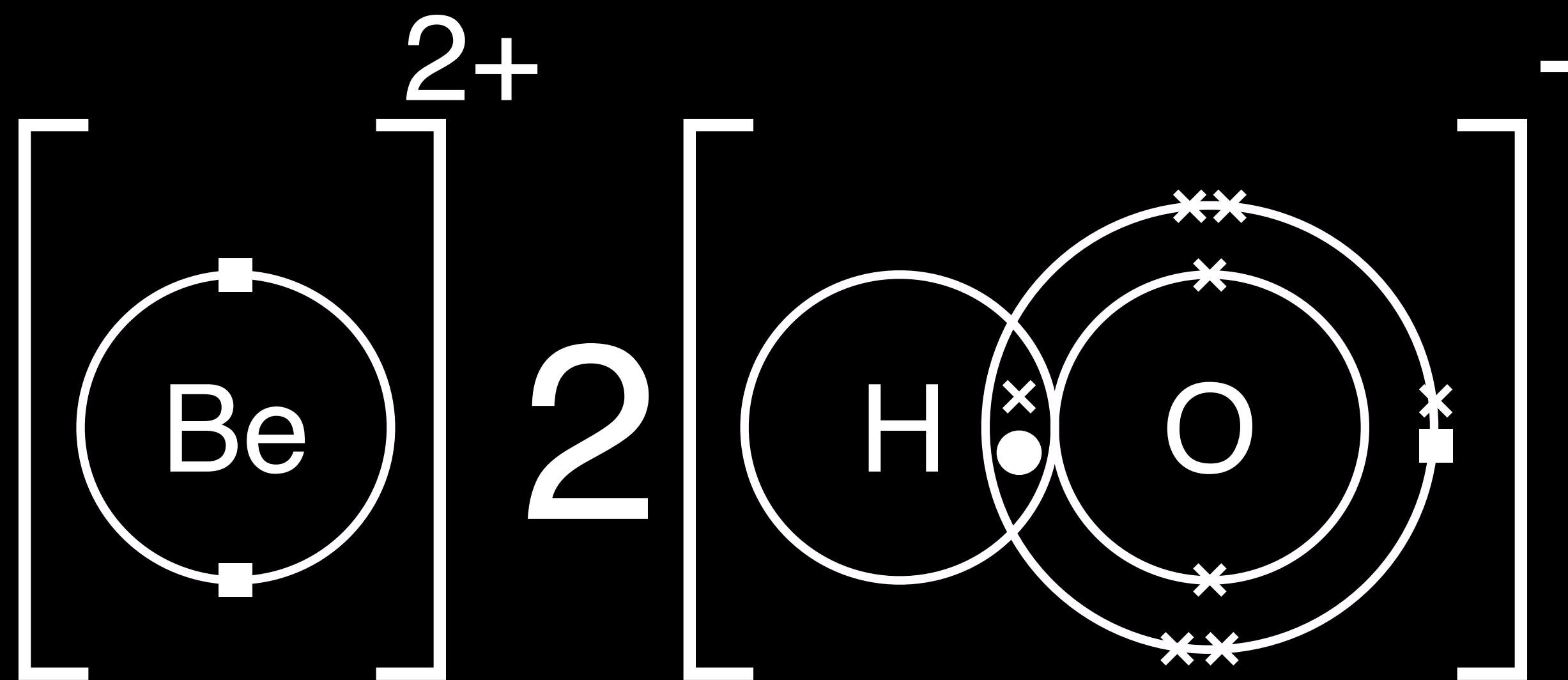


- If there are 3 or more elements (usually max 3), assign a new symbol for electron (anything simple and won't be easily confused with other symbols)
- Element with most possible number of covalent bonds should be in the center (or element with highest absolute charge). Since carbon can form 4 covalent bonds (or carbon has an absolute charge of 4), it is in the centre
- For double bonds, there will be 4 electrons between the two atoms, (and 6 for triple bonds.)



Covalent Bonding - Diagrams w/ Ionic bonding

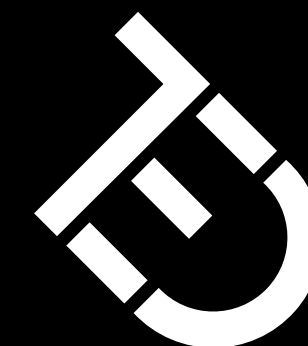
Calcium does not have 16 valence electrons



Structural formula
 $[\text{Be}]^{2+} 2[\text{O} - \text{H}]^{-}$

Chemical Formula
 $\text{Be}(\text{OH})_2$

- Similar to ionic bonding except ions are replaced with polyatomic ion (drawn as covalent bond)
- Make sure to draw different shapes for where the different electrons are from (e.g. square shape in OH represents the electron from Be)

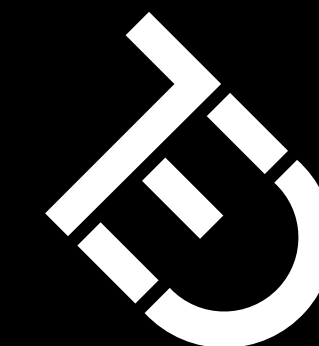


Covalent Bonding - Naming

Gee

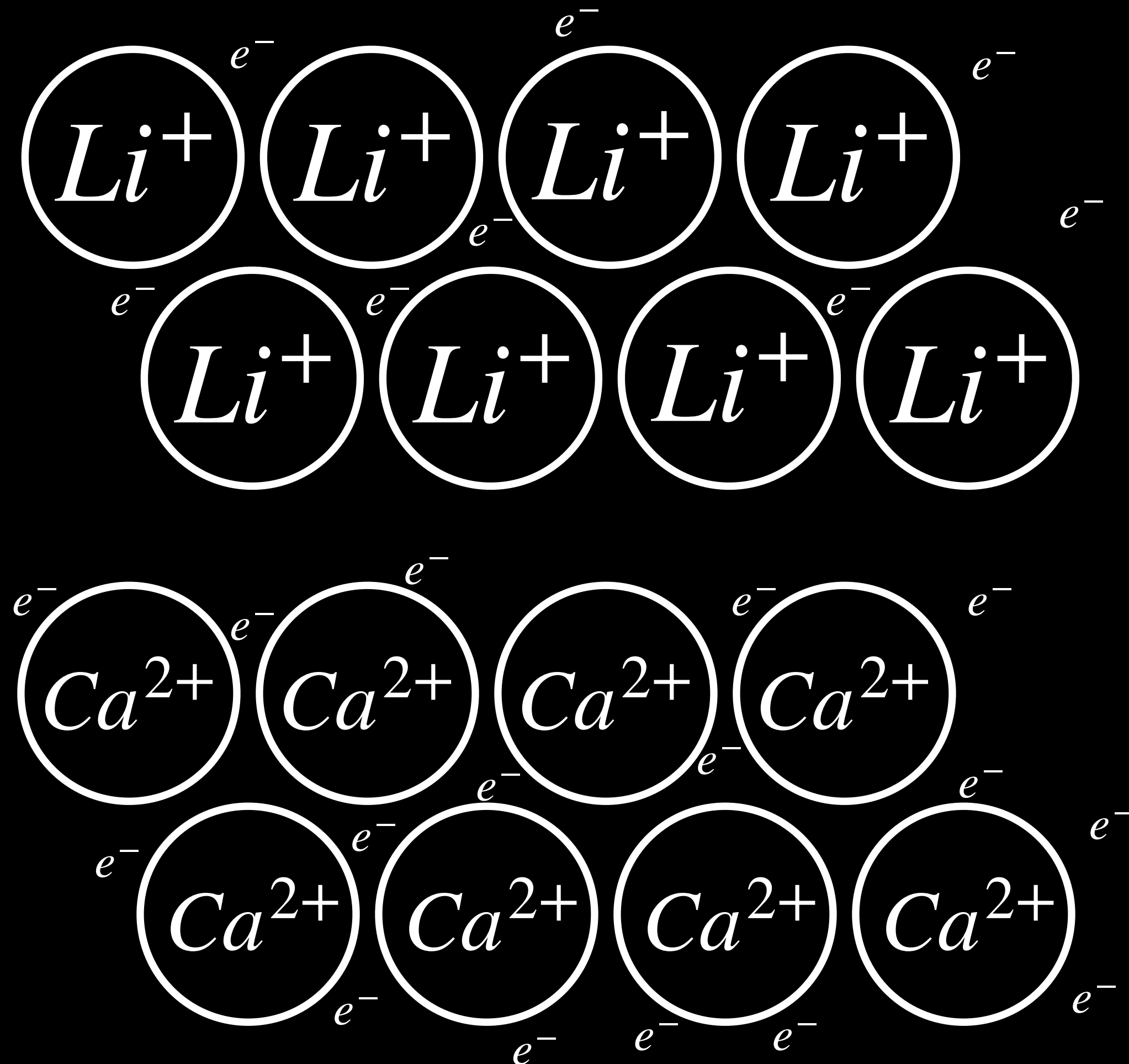
Silicon dioxide: $\overset{Si}{Si}\overset{O}{O}_2$
Sulfur hexafluoride: SF_6
Dinitrogen tetroxide: N_2O_4

- Lists its constituent elements in order and has numbers as prefixes for elements (mono, di, tri, tetra, penta, hexa)
- Exceptions apply to diatomic elements (H_2 , O_2 , Br_2 , etc) and molecules with different names/naming systems (methane, CH_4 and ammonia, NH_3)



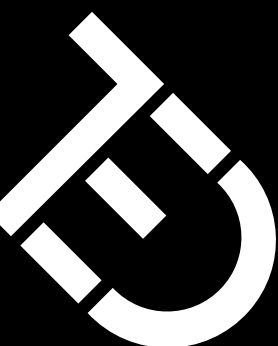
Metallic Bonding

de

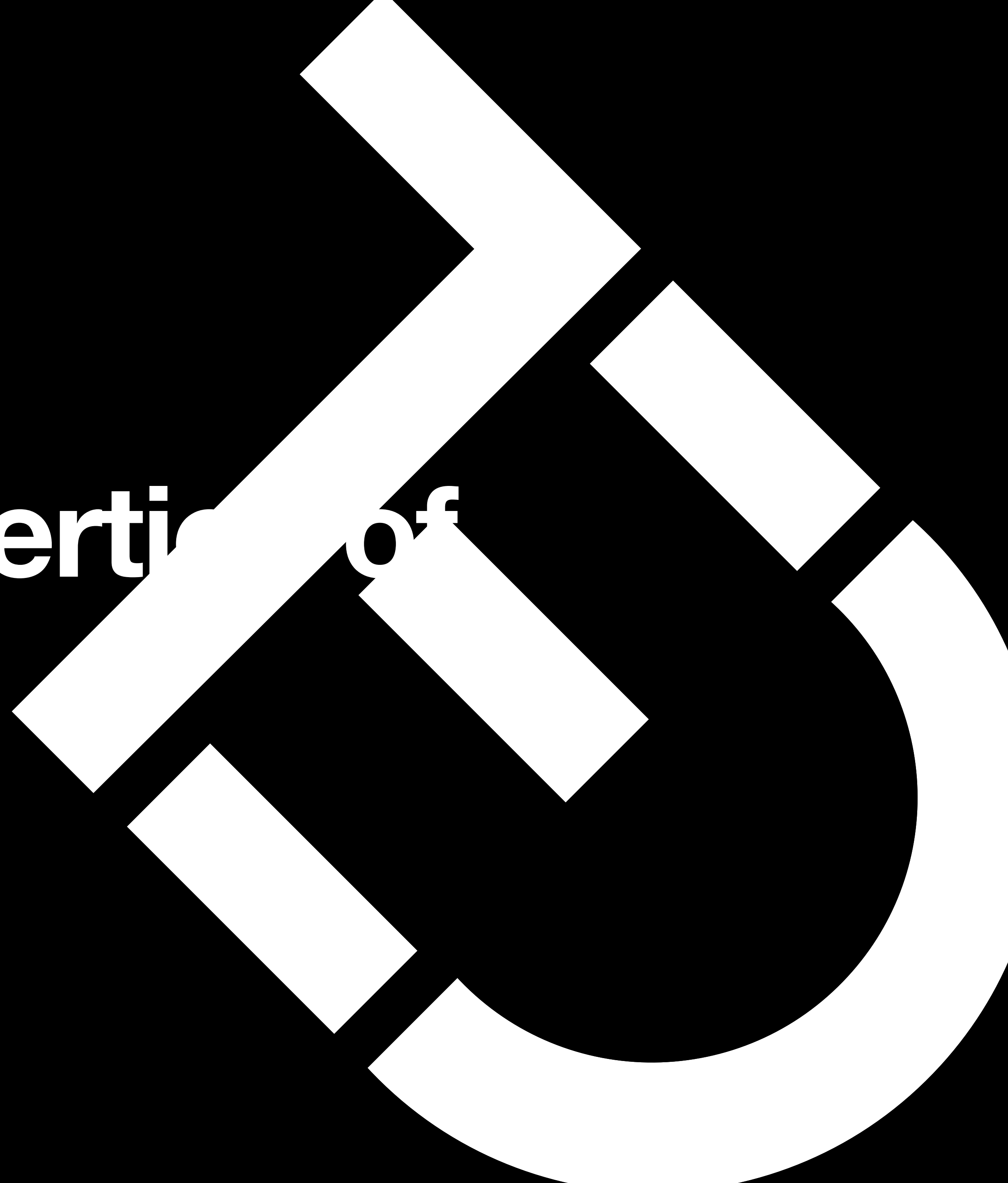


- Each lithium atom loses one valence electron to form positively charged ions to achieve duplet/octet structure
- Forms giant metal lattice structure of positive ions surrounded by a sea of mobile/delocalised electrons
- Bonds via strong forces of attraction between the positive metal ions and sea of mobile electrons

electrons should be drawn randomly, however number of electrons must match charge of cations



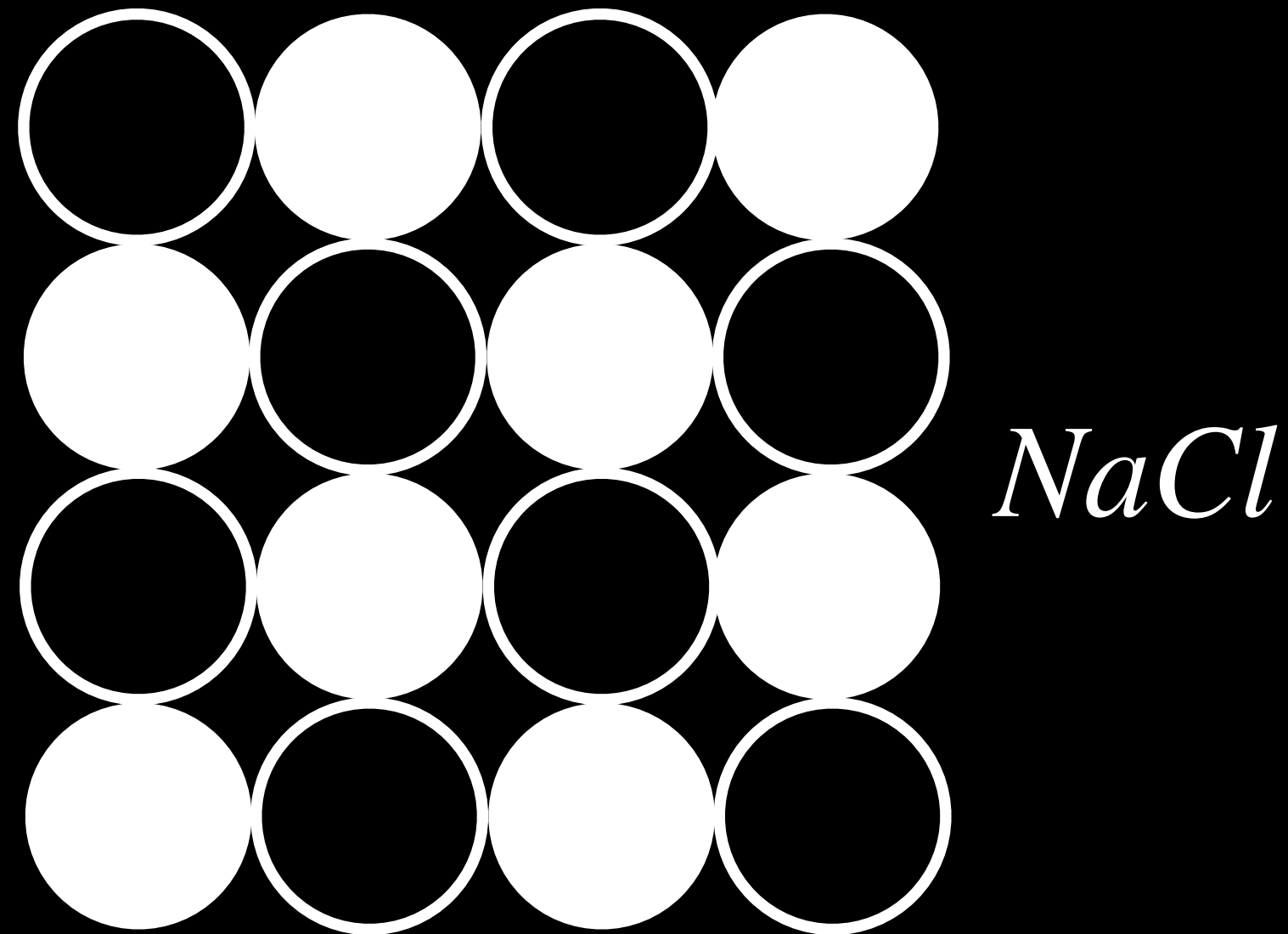
Structure & Properties of Materials



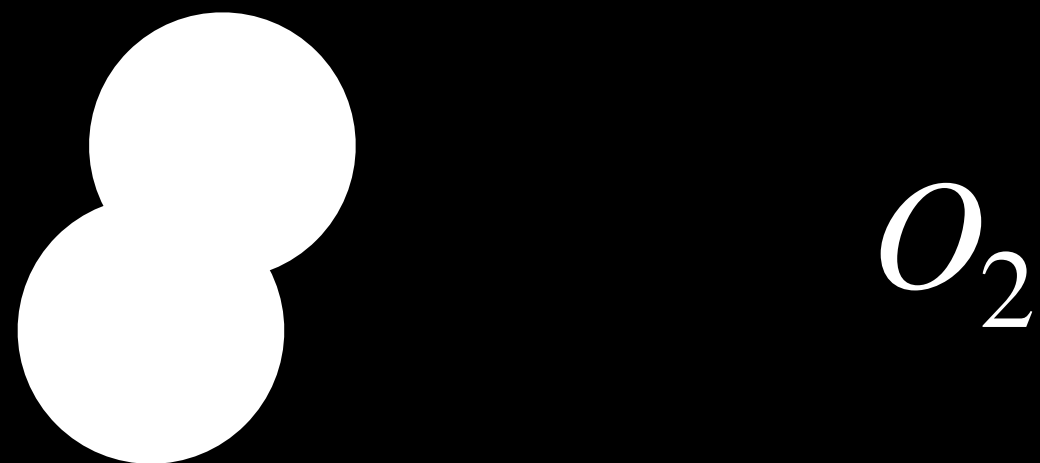
Molecular Structures

Ginseng

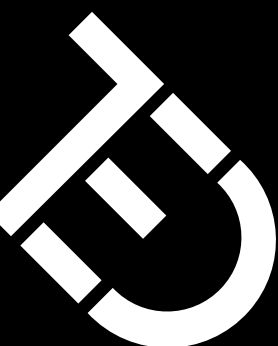
Giant molecular structure



Simple molecular structure



- Giant molecular structure
 - Strong bonds between atoms, usually in a lattice formation aka can tessellate
 - All bonding types can form this structure
- Simple molecular structure
 - Strong bonds between atoms, weak intermolecular forces between molecules
 - Only covalent can bond this way

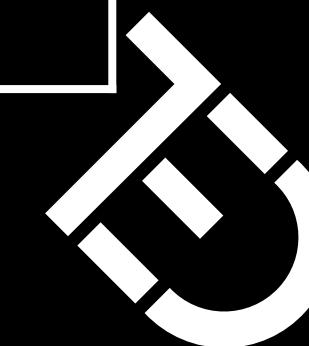


Properties - Melting/Boiling Point

Oh no

Ionic	Giant Covalent	Simple Covalent	Metallic
High mp/bp	High mp/bp	Low mp/bp	High mp/bp
Oppositely charged ions are held together by <u>very strong forces of attraction</u> —> <u>large amount of energy needed to overcome strong electrostatic forces of attraction</u>	Atoms are held together by <u>very strong covalent bonds</u> —> <u>large amount of energy needed to overcome strong covalent bonds</u>	Molecules are held together by <u>weak intermolecular forces</u> —> <u>low amount of energy needed to overcome weak intermolecular forces</u>	Ions are held together by <u>very strong metallic bonds</u> —> <u>large amount of energy needed to overcome strong metallic bonds</u>

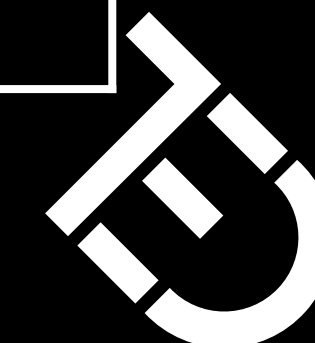
*High mp/bp is counted as greater than 500°C for ions, 1000+°C for covalent/metal



Properties - Electrical Conductivity

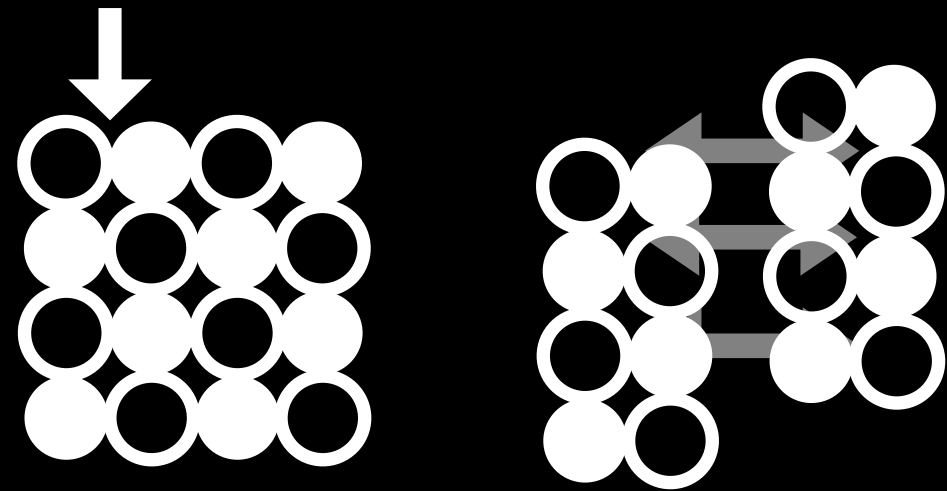
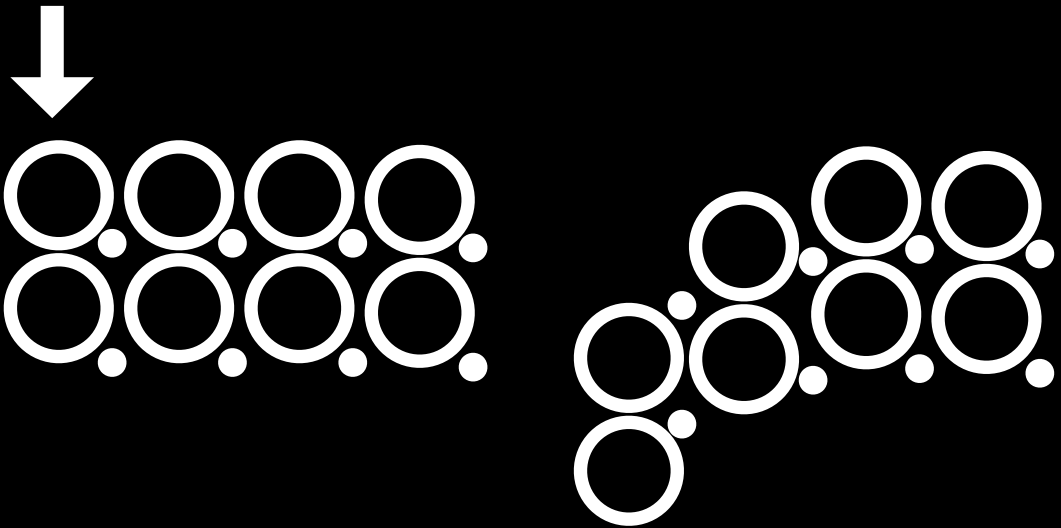
Oh no

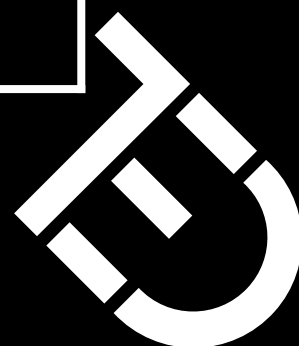
Ionic	Giant Covalent	Simple Covalent	Metallic
Only when molten/aqueous	Does not conduct electricity	Does not conduct electricity	Conducts electricity
Solid: Ions vibrate about fixed position —> unable to move towards electrode of oppositely charge —> does not conduct electricity Molten/Aqueous: Ions are mobile —> free to move to oppositely charged electrode —> conducts electricity	No mobile charged particles like electrons or ions to move to electrode of opposite charge —> does not conduct electricity Exception: Graphite (Refer to carbon allotropes section)	No mobile charged particles like electrons or ions to move to electrode of opposite charge —> does not conduct electricity	Sea of mobile electrons —> free to move to oppositely charged electrode



Properties - Hardness & Malleability

Oh no

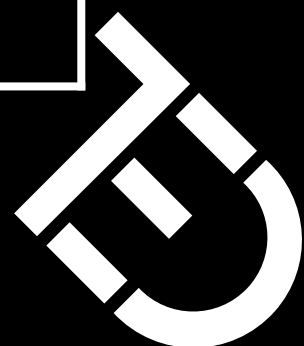
Ionic	Giant Covalent	Simple Covalent	Metallic
Hard but brittle	Refer to Carbon Allotropes		Malleable and ductile
Force applied —> similarly charged ions are forced together —> repel each other —> causes lattice to shatter 			Force applied —> layer of metal ions can slide over mobile electrons without disrupting metallic bonding throughout lattice 



Properties - Heat Conductivity

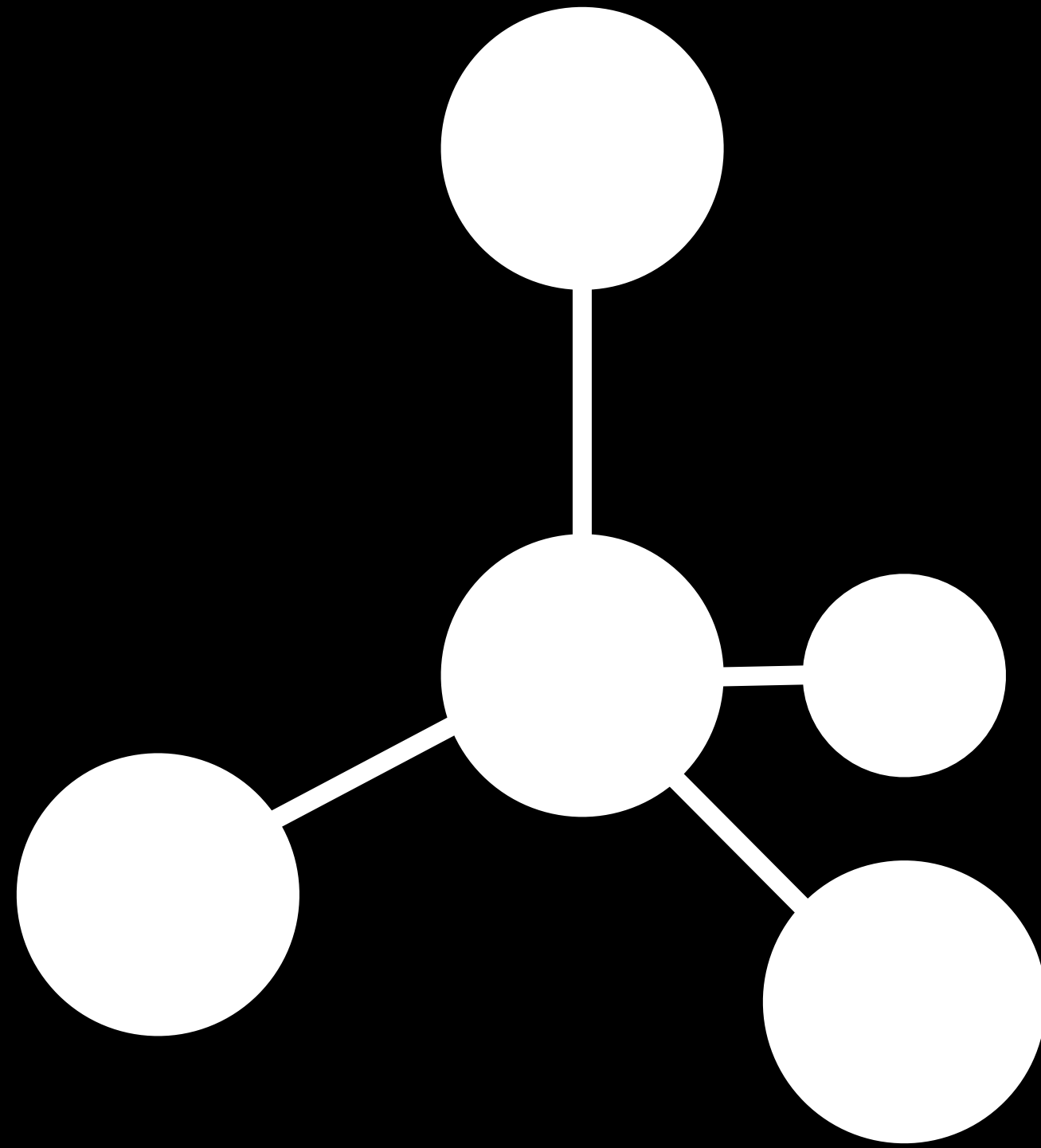
Oh no

Ionic	Giant Covalent	Simple Covalent	Metallic
Does not apply			Conducts heat well
			When heated —> delocalised electrons get more energy —> move faster and collide with neighbouring electrons —> transfers heat in collisions of electrons

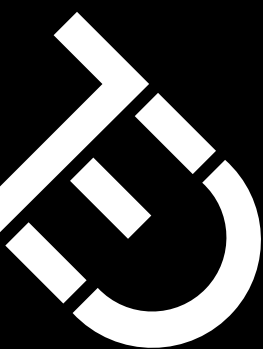


Carbon Allotropes

C

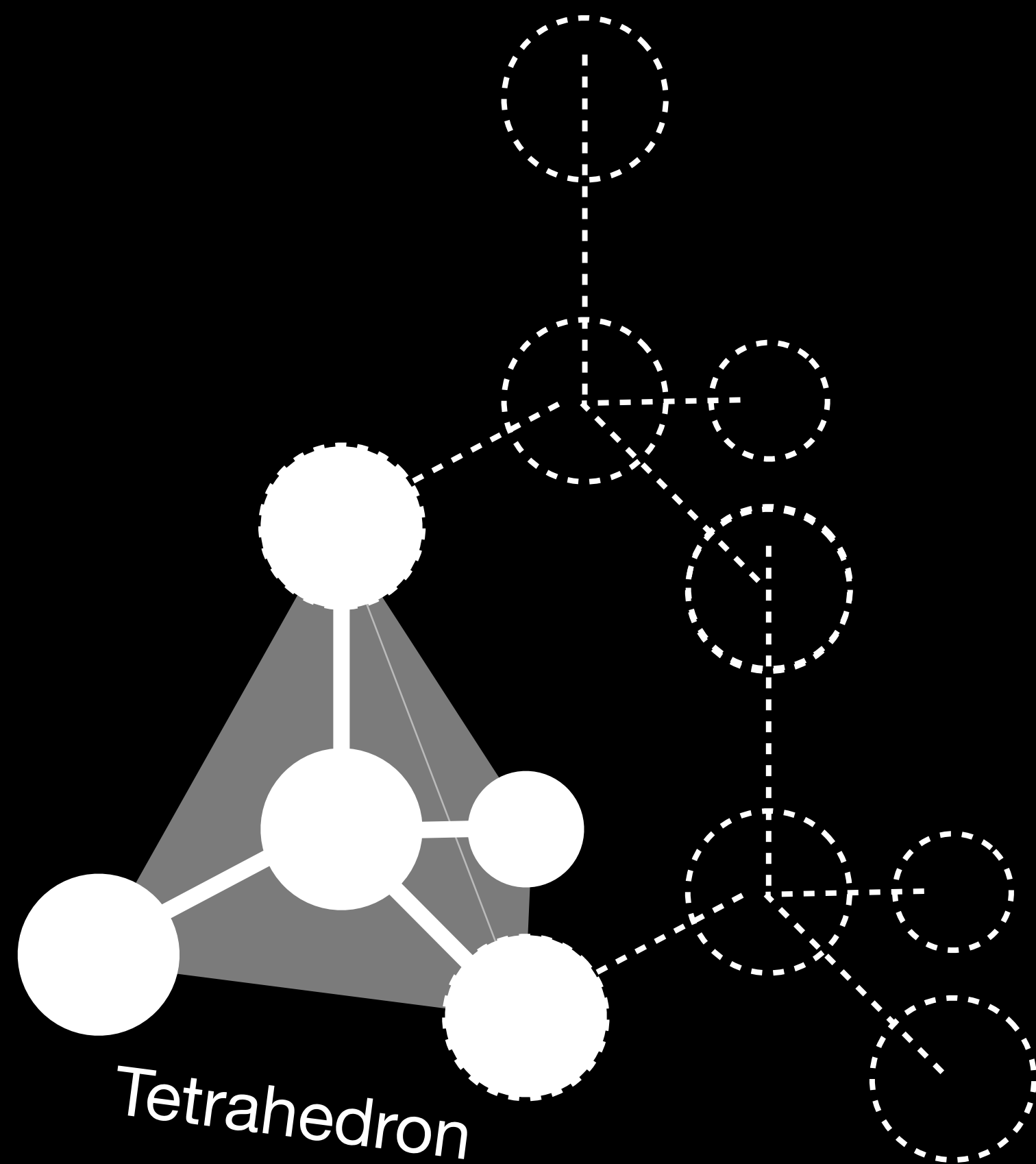


- Allotrope: physical forms of the same element in which the atoms are bonded together in different ways
- Carbon allotropes: Graphite and diamond (Most commonly tested)

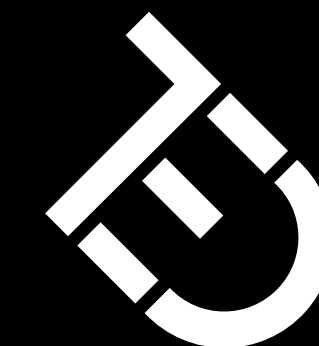


Carbon Allotropes - Diamond

Please

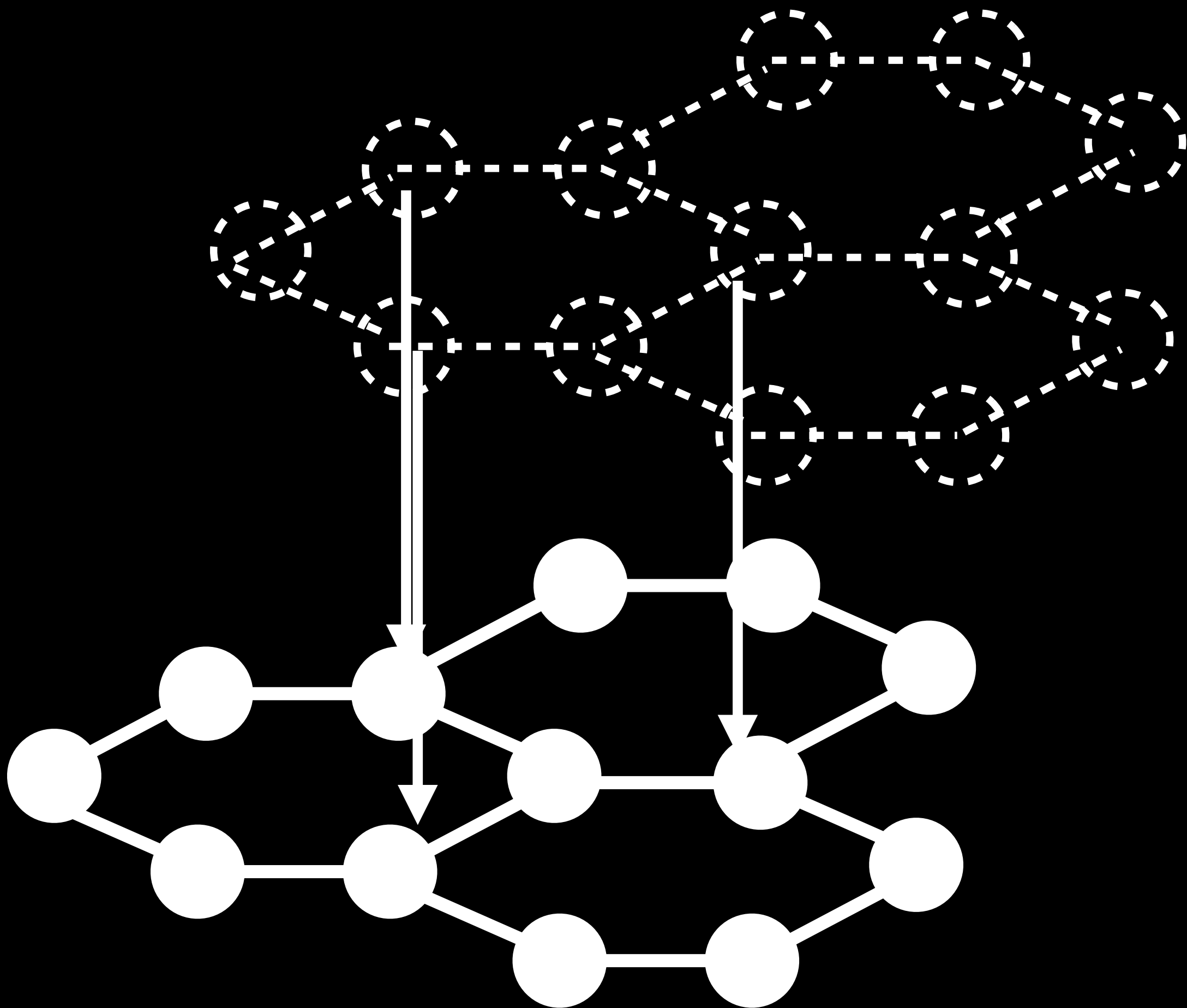


- Giant Molecular structure
- Each carbon atom uses all 4 valence electrons to covalently bond to 4 carbon atoms in a tetrahedral arrangement

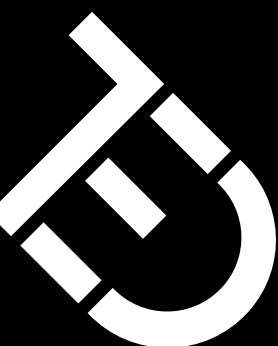


Carbon Allotropes - Graphite

Xcvhjm,..;lkjkl;'



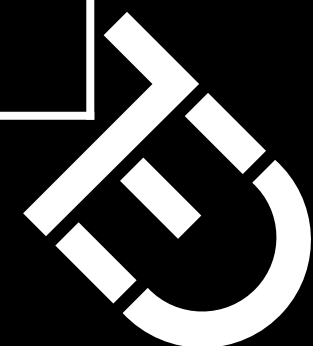
- Giant molecular structure
- Each carbon atom uses 3 valence electrons to covalently bond to 3 carbon atoms in a hexagonal layers of carbon atoms
- Hexagonal layers are held together by weak intermolecular forces (Van Der Waals forces)
- 4th valence electron is delocalised



Carbon Allotropes - Electrical Conductivity

Xcvhjm,.;lkjkl;’

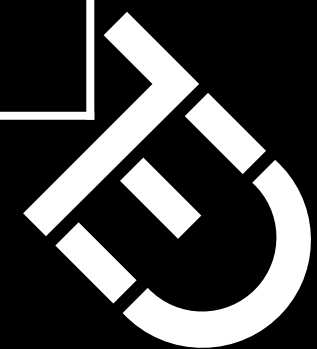
Diamond	Graphite
Does not conduct electricity	Conducts electricity
No mobile charged particles like electrons or ions to move to electrode of opposite charge —> does not conduct electricity	Presence of mobile electrons to move to electrode of opposite charge —> conducts electricity



Carbon Allotropes - Hardness

Xcvhjm,.;lkjkl;’

Diamond	Graphite
Hard	Soft & Slippery
Large amount of force needed to overcome strong covalent bonds between atoms	Low amount of force needed to overcome weak intermolecular bonds /(or Van Der Walls forces) between carbon layers



Chemical Formulae & Equations



Chemical Formula

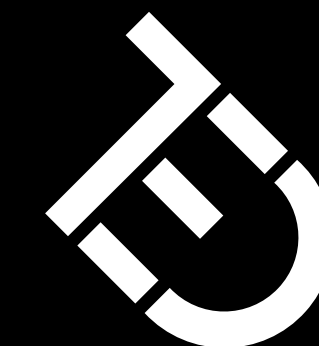
what are you

$NaCl$ ~ sodium chloride

K_2SO_4 ~ potassium sulfate

MgO ~ magnesium oxide

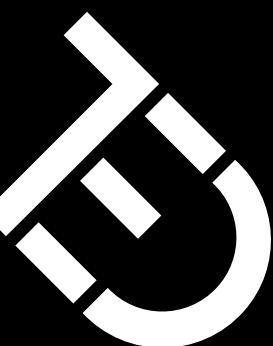
- a method of representing a chemical compound
- $<— —$ e.g
- shows the type and numebr of atoms in the compound



State Symbols

who are we

- $Fe(s)$ - solid
- $H_2O(l)$ - liquid
- $O_2(g)$ - gas
- $NaCl(aq)$ - aqueous
- remember the symbols are in lowercase
- a symbol to represent what state the chemical compound is in



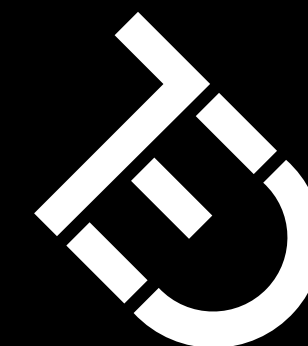
Chemical Equations

i will find you

- burning Iron metal to form Iron(III) oxide



- used to represent a chemical reaction
- contains reactants and products



Chemical Equations - Balancing Sides

wtf are these numbers



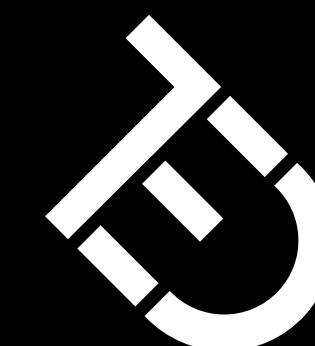
- LHS: 4 iron, 6 oxygen

- RHS: 4 iron, 6 oxygen

- there is a number before the chemical compounds (called the coefficient), picking the right coefficients balances the equation

- The equation is balanced when both sides have the same number of atoms

- For this case, the oxygen on the left side is O_2 but there is O_3 on the right hand side, so we multiply by 3 and 2 respectively to get $6O$ for both sides. As we multiplied 2 to the O_3 , we multiply 2 to Fe_2 , thus there will be $4Fe$ on the left side to balance the $2Fe_2$



Stoichiometric Ratio

L bozo + stoichiometric ratio

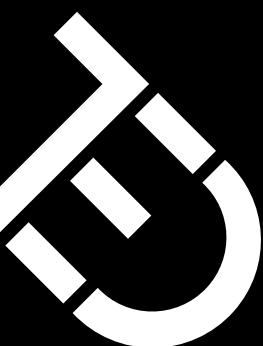


ratio = 4:3:2



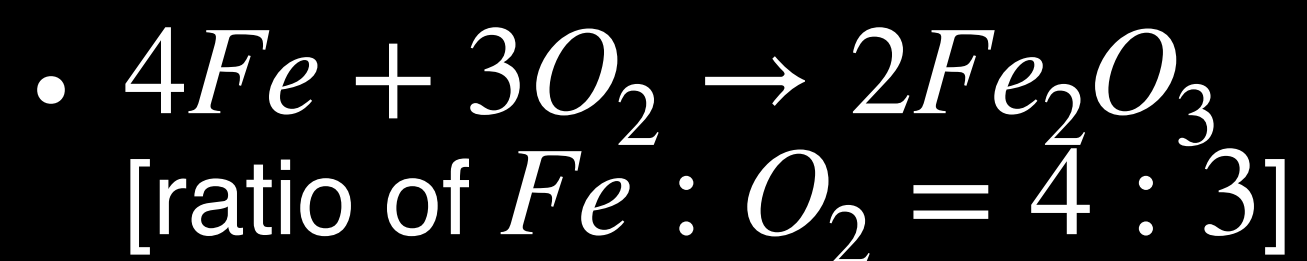
ratio = 1:1:1

- stoichiometric ratio is the ratio between all the compounds in a chemical eqn

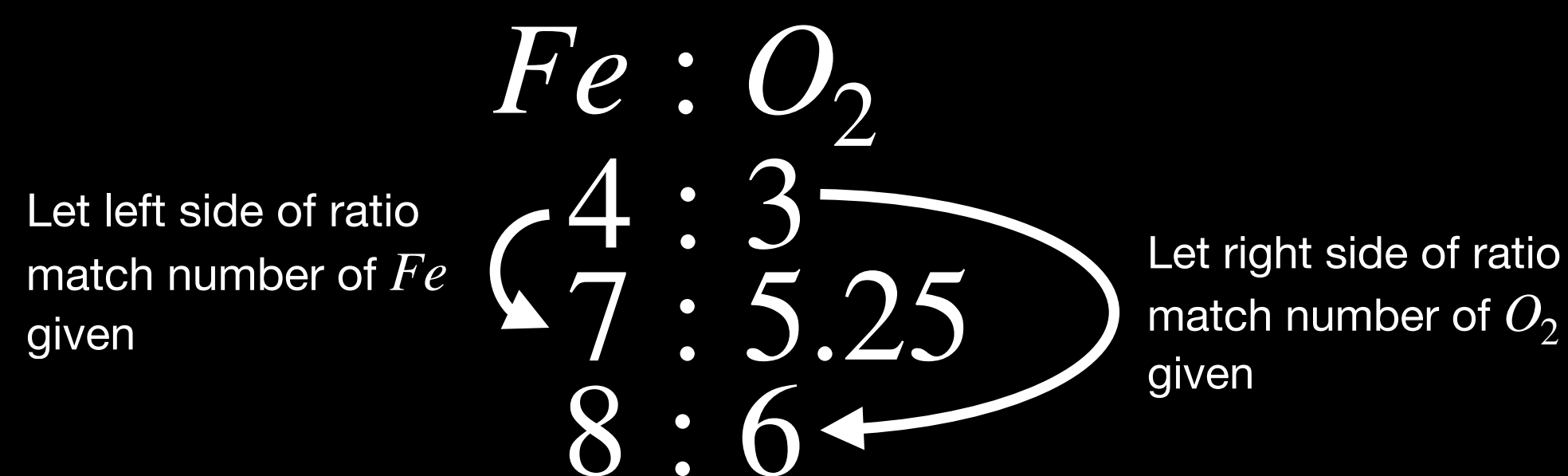


Stoichiometric Ratio - Limiting Reagents

limiting reactants

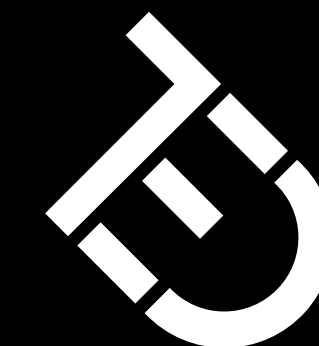


- If given $7Fe, 6O_2$:

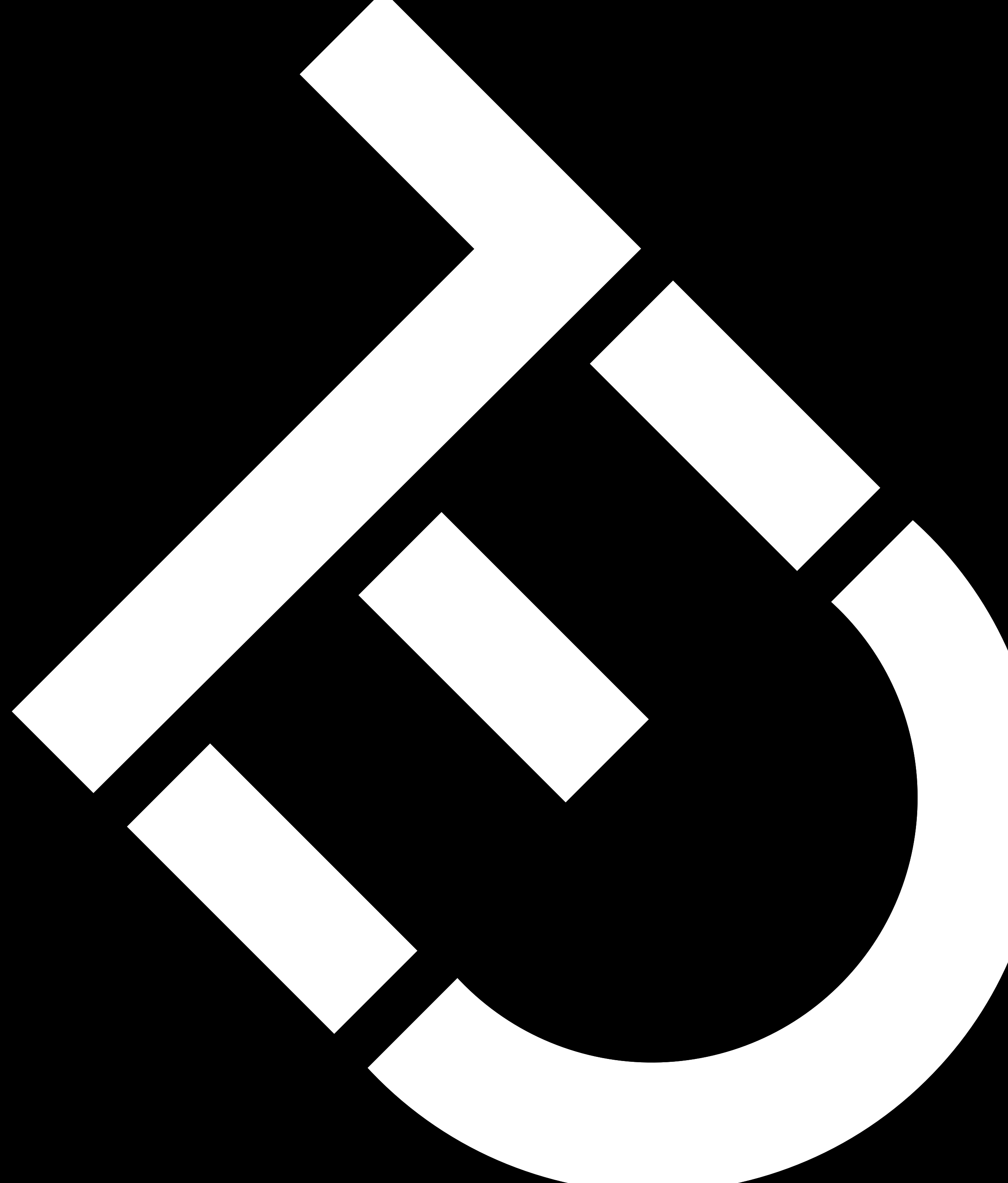


- if the ratio of the reactants given are not in the stoichiometric ratio, there will be a limiting reactant (the reactant that will be completely used up in the reaction) and the excess reactant (the reactant with some of it leftover after the reaction)

As 6 moles of O_2 needs 8 moles of Fe to react, but there are only 7 moles of Fe , Fe is a limiting reactant while O_2 is in excess.



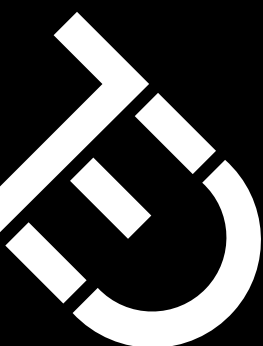
Mole Concept



Mole Definition

avogadro's number

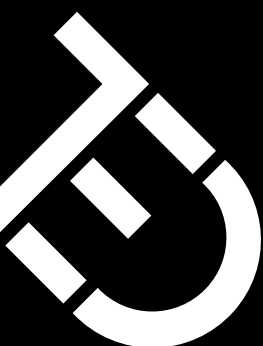
- mass of 1 carbon atom:
- $1.994 \times 10^{-23} \text{g}$
- relative atomic mass of carbon: **12**
- $1.994 \times 10^{-23} \text{g} \times 6.02 \times 10^{23} = 12 \text{g}$
- Avogadro's number is the value that when multiplied by the true mass of an element, would equal to its relative atomic mass in grams: 6×10^{23} (6.02×10^{23} is also accepted, however 6×10^{23} is the standardised version)
- you need to remember this



Molar Mass

u

- mass of 1 carbon atom:
- $1.994 \times 10^{-23} \text{g}$
- relative atomic mass of carbon: **12**
- $1.994 \times 10^{-23} \text{g} \times 6.02 \times 10^{23} = \textbf{12g}$
- molar mass = **12g/mol**
- mass of 1 mol of any substance.
- equal to relative atomic/
molecular mass in grams
- unit is g/mol



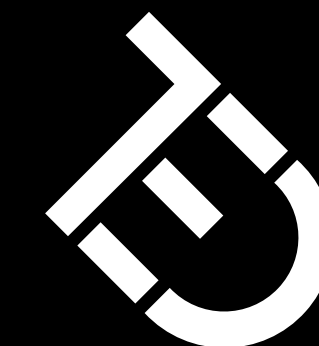
Mole Calculation: Particles/Moles

from number of particles

- Find the number of moles of 5.69×10^{24} particles.

- $$\frac{5.69 \times 10^{24}}{6 \times 10^{23}} = 9.48 \text{ moles}$$

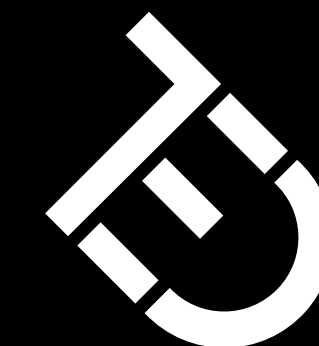
$$\text{no. of moles} = \frac{\text{no of particles}}{6 \times 10^{23}}$$



Relative Molecular Mass

what is this

- Find the M_r of MgO
- A_r of Mg = 24
- A_r of O = 16
- M_r of MgO = $24+16=40$
- relative molecular mass is the sum of the relative atomic masses of all the atoms in the molecule



Mole Calculation: Mass/Moles

using molar mass

Find the mass of 3 mol of O₂.

molar mass of O₂ = 32g/mol

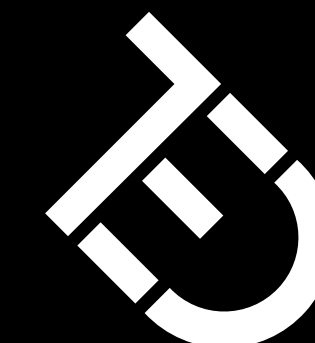
32g/mol x 3 mol = 96g

Likewise, to find moles from mass:

96g ÷ 32g/mol = 3 mol

- molar mass × moles = mass

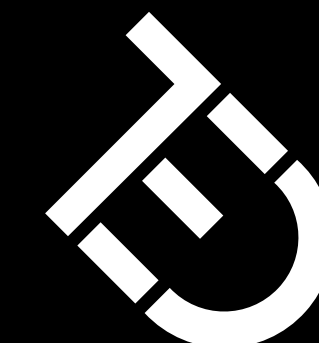
- $$\frac{\text{mass}}{\text{molar mass}} = \text{moles}$$



Mole Calculation: Empirical Formula

what the hell

- a formula that shows the simplest ratio between atoms
- Empirical formula $\neq$ Molecular formula
- E.g. empirical formula of C_4H_{10} is C_2H_5 (divide number of atoms by highest common factor, in this case 2)



Mole Calculation: Empirical Formula

tbsle

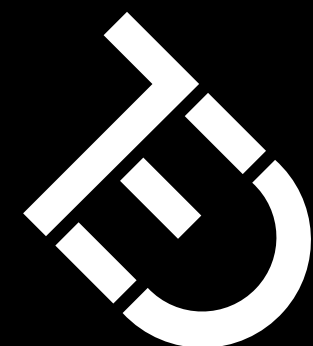
Given that a copper oxide has 89% of copper and 11% of oxygen, find the empirical formula of the compound.

Make sure to put intermediate working and round to 5.s.f for intermediate

	Cu	O
mass/g	89	11
A_r	64	16
moles/mol	$\frac{89}{64} = 1.3906 \text{ mol}(5.s.f)$	$\frac{11}{16} = 0.6875 \text{ mol}$
mole ratio (aka divide by smallest)	$\frac{1.3906}{0.6875} = 2.02(3.s.f)$	$\frac{0.6875}{0.6875} = 1$
Simplest mole ratio	2	1

The simplest mole ratio should either be rounded or both values multiplied by a common number to be a whole number ratio

$\therefore$ empirical formula is Cu_2O



Mole Calculation: Empirical Formula

what the hell

If the empirical formula of a compound is CH_2 and has a molar mass of 56g/mol, find its molecular formula

Let the molecular formula be $(CH_2)_n$

Find Mr of empirical formula

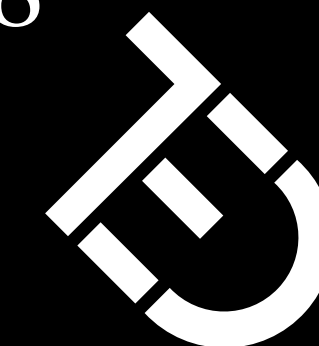
$$M_r \text{ of } CH_2 = 12 + 2 = 14$$

Divide Mr of molecule by Mr of empirical formula to find n

$$n = \frac{56}{14} = 4$$

multiply the entire empirical formula by n

$$\text{molecular formula: } (CH_2)_n = C_4H_8$$



Mole Calculation: Chemical Equations

using mole/stoichiometric ratio

Hydrochloric acid and copper carbonate react together in the following equation:



If 6.2g of copper carbonate reacts with excess hydrochloric acid, what is the mass of the copper chloride produced?

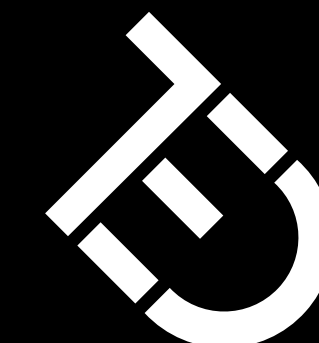
$$\begin{aligned}\text{Moles of } \text{CuCO}_3 &= 6.2\text{g} \div (64 + 12 + 3 \times 16)\text{g/mol} \\ &= 6.2\text{g} \div 124\text{g/mol} \\ &= 0.05\text{mol}\end{aligned}$$

$$\text{Ratio of } \text{CuCO}_3 : \text{CuCl}_2 = 1:1$$

$$\therefore \text{Moles of } \text{CuCl}_2 = 0.05\text{mol}$$

$$\begin{aligned}\text{Thus, mass of } \text{CuCl}_2 &= 0.05\text{mol} \times (64 + 2 \times 35.5) \\ &= 0.05\text{mol} \times 135\text{g/mol} \\ &= 6.75\text{g}\end{aligned}$$

- usually about calculating mass of unknown reactant/product when another is known.
- use stoichiometric ratio of the chemical equation to find moles of unknown



Mole Calculation: Percentage Yield

theoretical vs actual

In the reaction of $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$, when 4.6g of Na reacts with excess Cl, 10g of NaCl is formed. Find the percentage yield of NaCl.

$$\begin{aligned}\text{Moles of Na} &= 4.6\text{g} \div 23\text{g/mol} \\ &= 0.2\text{mol}\end{aligned}$$

$$\text{Ratio of Na:NaCl} = 2:2 = 1:1$$

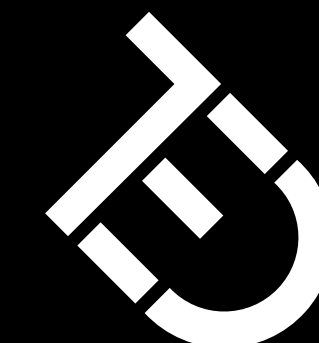
$$\therefore \text{Moles of NaCl} = 0.2\text{mol}$$

$$\begin{aligned}\text{Theoretical mass of NaCl} \\ &= 0.2\text{mol} \times (23 + 35.5)\text{g/mol} \\ &= 11.7\text{g}\end{aligned}$$

Thus, percentage yield

$$= \frac{10\text{g}}{11.7\text{g}} \times 100\% = 85.5\% (3.s.f.)$$

- $\% \text{ yield} = \frac{\text{obtained mass or moles}}{\text{theoretical mass or moles}} \times 100\%$
- Percentage yield represents how much of the product was obtained compared to the theoretical amount of product obtained (as some of the product may have been lost from transferring reagents, etc.)
- Percentage yield is always between 0% to 100%



Mole Calculation: Percentage Purity

impure

Chalk is almost pure calcium carbonate. The reaction of calcium carbonate and hydrochloric acid is as shown;
 $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$. 10g of chalk reacts with excess hydrochloric acid to produce 1.6g of water. Find the percentage purity of chalk.

$$\begin{aligned}\text{Moles of } \text{H}_2\text{O} &= 1.6\text{g} \div 18\text{g/mol} \\ &= 0.088889\text{mol} (5.s.f.)\end{aligned}$$

$$\text{Ratio of } \text{H}_2\text{O} : \text{CaCO}_3 = 1 : 1$$

$$\therefore \text{Moles of } \text{CaCO}_3 = 0.088889\text{mol}$$

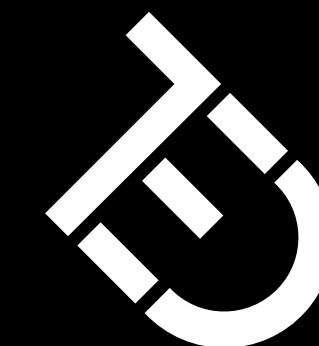
Mass of

$$\begin{aligned}\text{CaCO}_3 &= 0.088889\text{mol} \times (40 + 12 + 3 \times 16)\text{g/mol} \\ &= 8.8889\text{g}\end{aligned}$$

Thus, percentage purity

$$= \frac{8.8889\text{g}}{10\text{g}} \times 100\% = 88.9\% (3.s.f.)$$

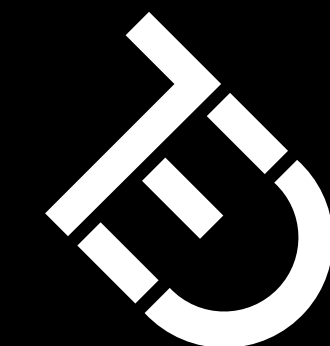
- $\% \text{ purity} = \frac{\text{mass of pure substance}}{\text{total mass}} \times 100\%$
- Percentage purity represents how pure a substance is by comparing how much of the object contains the pure substance
- The higher percentage purity is, the less impurities it has
- Percentage yield is always between 0% to 100%



Mole Calculation: Molar Volume

At room temperature and pressure reference the

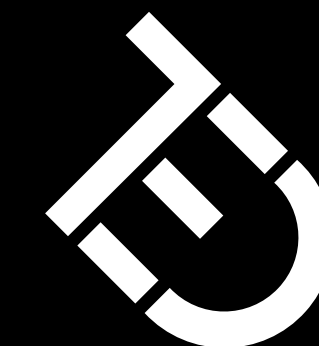
- 1 mol of a gas at room temp and pressure will take up 24dm^3
- $\text{moles} \times 24\text{dm}^3/\text{mol} = \text{volume}$
 $\text{volume} \div 24\text{dm}^3/\text{mol} = \text{moles}$
- $1\text{cm}^3 = 0.001\text{dm}^3$ (divide by 1000)
- For example:
 - 1.5 moles of O_2 gas =
 $1.5\text{mol} \times 24\text{dm}^3/\text{mol} = 36\text{dm}^3$
 - 4.8dm^3 of CO_2 gas =
 $4.8\text{dm}^3 \div 24\text{dm}^3/\text{mol} = 0.2\text{mol}$



Mole Calculation: Molar Volume

Slash/slash/slash

- Example: $N_2 + 3H_2 \rightarrow 2NH_3$
- If given a reaction with only gases, and given the volume of one substance and asked to find another, instead of converting to moles then back to volume, just use ratio directly
- e.g. given $72dm^3$ of N_2 , volume of NH_3 is $72 \times 2 = 144dm^3$
- (This works as $24dm^3$ is a constant for all gases instead of molar mass that differs for different reactants)



Mole Calculation: Concentration

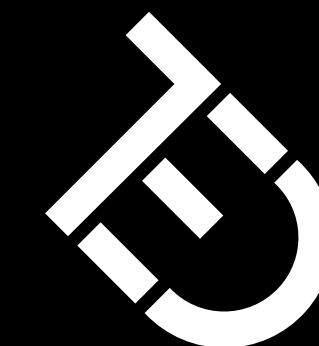
Slash/slash/slash

- For aqueous substances/
solutions,
concentration = $\frac{\text{moles or mass of substance}}{\text{total volume of solute}}$
- Thus, to find amount of
moles of substance from
concentration for
stoichiometric ratio qns,
multiply by volume

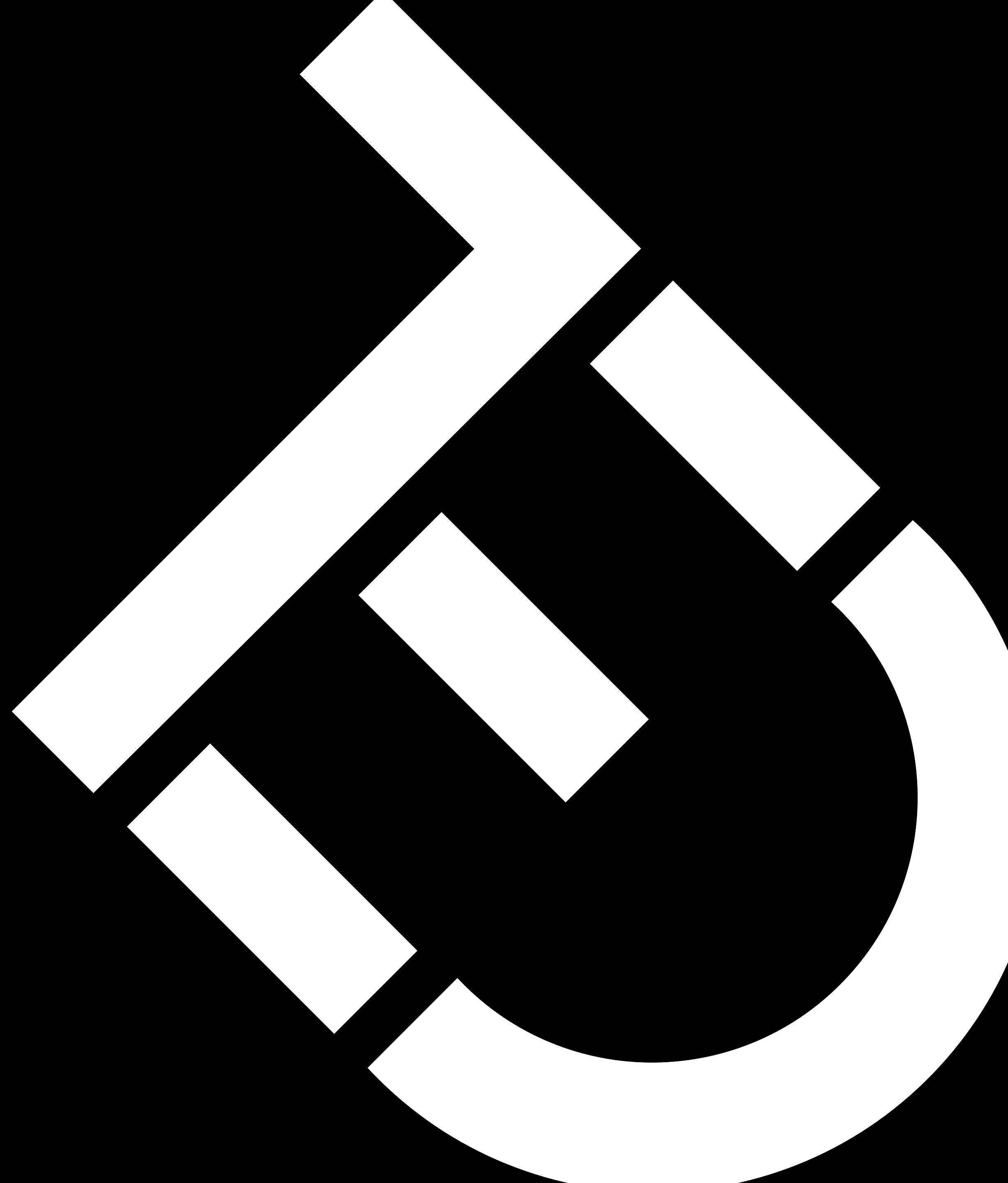
e.g. How many moles of CuSO_4 are present
in 25.0cm^3 of a 0.1mol/dm^3 solution?

$$25\text{cm}^3 = 0.025\text{dm}^3$$

$$0.025\text{dm}^3 \times 0.1\text{mol/dm}^3 = 0.00025\text{mol}$$



Acids & Bases



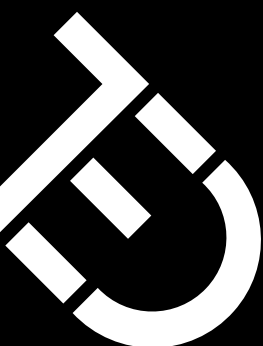
Definitions

buh

Acid: substance that dissociates(splits into its constituent atoms) in water to form H^+ ions

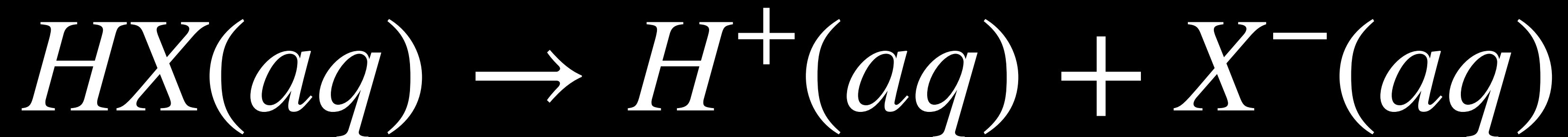
Base: metal oxide/metal hydroxide

Alkali: a base that dissociates in water to form OH^- ions. (All alkalis are bases, but not all bases are alkalis)

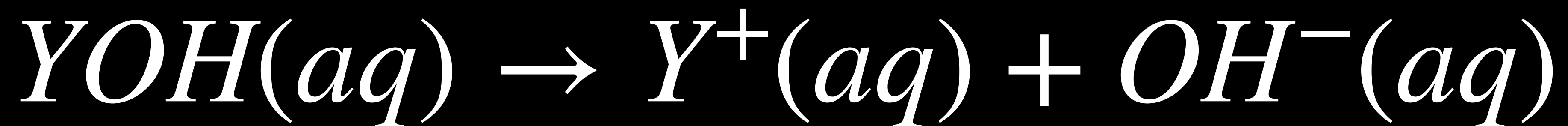


Acid/Alkali Dissociation Equation

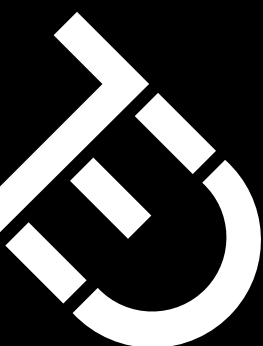
buh



acid dissociation eqn, where X is an anion like chloride

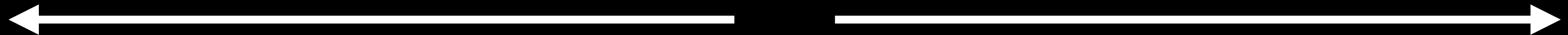


alkali dissociation eqn, where Y is an cation like sodium



pH Scale

buh



More acidic

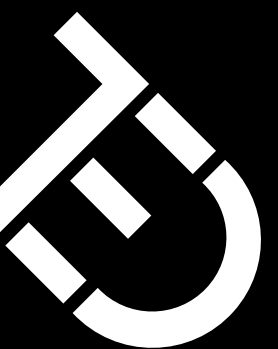
Neutral

More alkaline

More H^+ ions than OH^- ions

More OH^- ions than H^+ ions

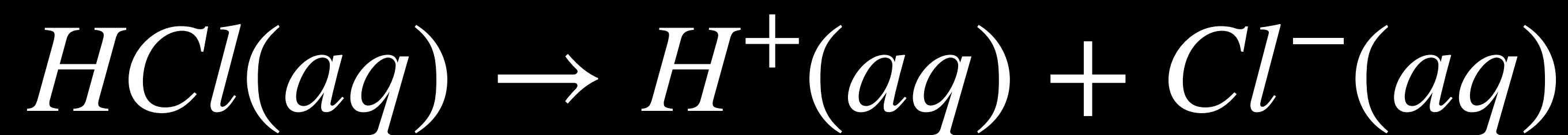
Equal OH^- and H^+ ions



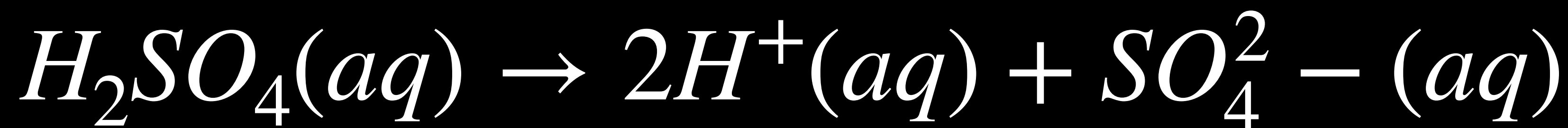
Acids - Basicity

buh

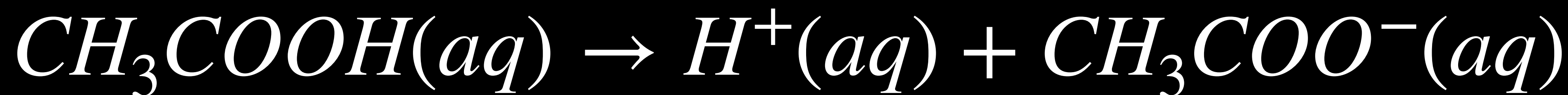
Basicity: Number of H⁺ ions dissociated from an acid



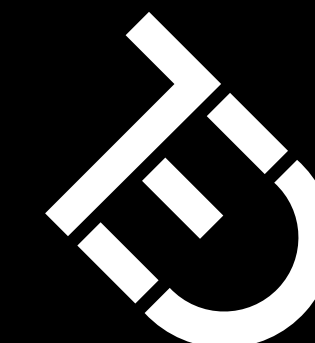
Since only one H⁺ ion dissociated, HCl is a monobasic acid



Since two H⁺ ions dissociated, H₂SO₄ is a dibasic acid



Since only one H⁺ ion dissociated, CH₃COOH is a monobasic acid



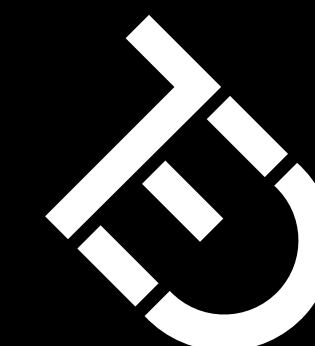
Acid/Alkali Strength

buh

Strength of acid/alkali is dependent of how it dissociates in water (dissociates fully/dissociates partially)

Substance	Acid		Alkali	
Strength	Strong	Weak	Strong	Weak
Definition	Fully dissociates	Partially dissociates	Full dissociates	Partially dissociates
Examples	Nitric acid, hydrochloric acid	ethanoic acid	Sodium hydroxide, potassium hydroxide	aq ammonia, (although has no OH, NH ₃ (aq) has alkali properties)
Example dissociation eqn	$HNO_3 \rightarrow H^+ + NO_3^-$	$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$	$NaOH \rightarrow Na^+ + OH^-$	$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

Acid/Alkali Strength is also known as % ionisation or degree of ionisation

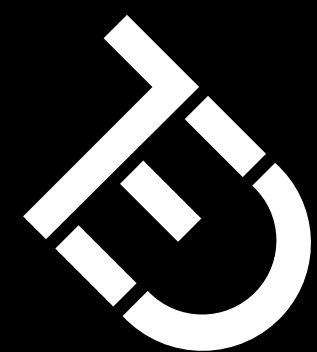


Indicators

buh

	More acidic							Neutral		More alkaline					
pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Methyl Orange	Red				Orange	Yellow									
Screened Methyl orange	Violet				Grey	Green									
Thymolphthalein	Colourless										Blue				
Red Litmus paper	No change (stays red)								Turns Blue						
Blue Litmus paper	Turns Red							No change (stays blue)							

You should know the colour change of both litmus papers, and at least one other indicator (for titration planning)



Reactions

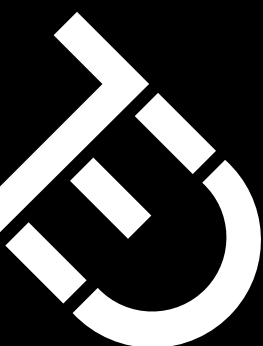
buh

acid + base $\rightarrow$ salt + water

acid + metal $\rightarrow$ salt + hydrogen gas

acid + carbonate $\rightarrow$ salt + water + carbon dioxide gas

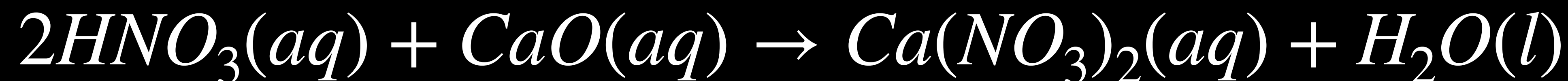
base + ammonium salt $\rightarrow$ salt + water + ammonia gas



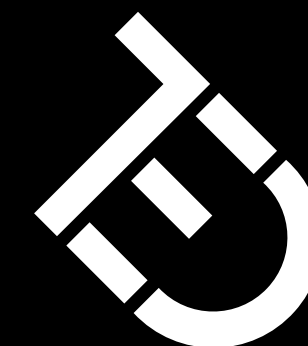
Reactions

buh

acid + base $\rightarrow$ salt + water



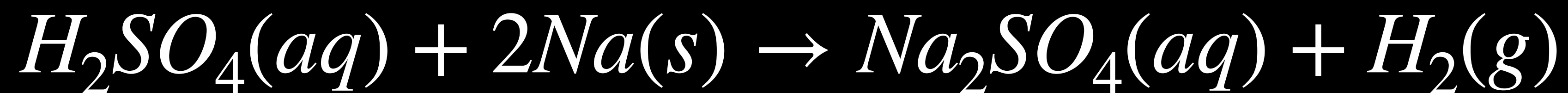
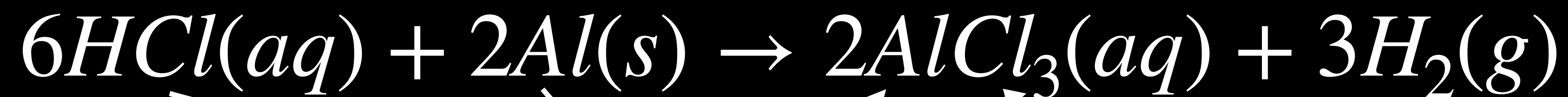
this process is also known as neutralisation
if balanced reaction with no excess, reactants are neutral



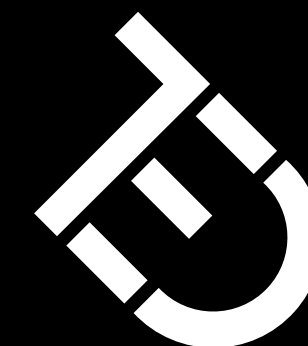
Reactions

buh

acid + metal $\rightarrow$ salt + hydrogen gas



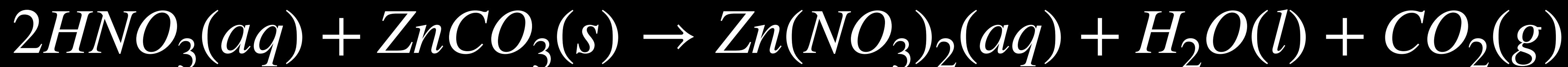
Metal has to be reactive enough to work
i.e. metal cannot be copper, silver, or gold



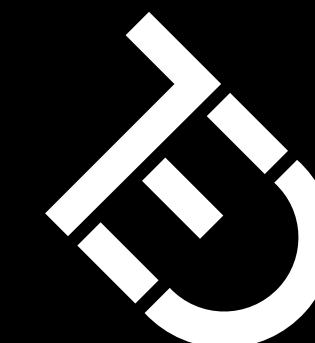
Reactions

buh

acid + carbonate $\rightarrow$ salt + water + carbon dioxide gas



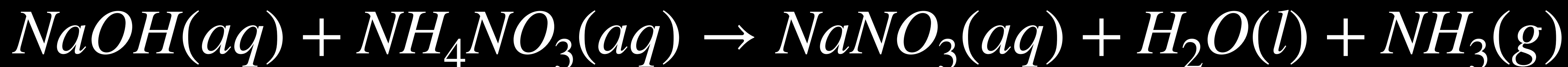
Carbonate anion and H^+ ion from acid forms CO_2 and water



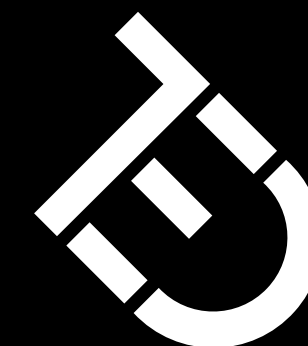
Reactions

buh

base + ammonium salt $\rightarrow$ salt + water + ammonia gas



NH₄ cation and oxide/hydroxide ion forms NH₃ and H₂O



Oxides

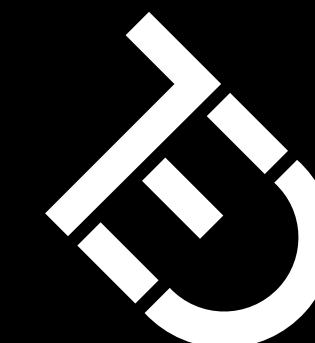
Sssssssssssssssssssssss

Generally,

Metal oxides are basic (K_2O , BaO , etc.), thus reacts with acids

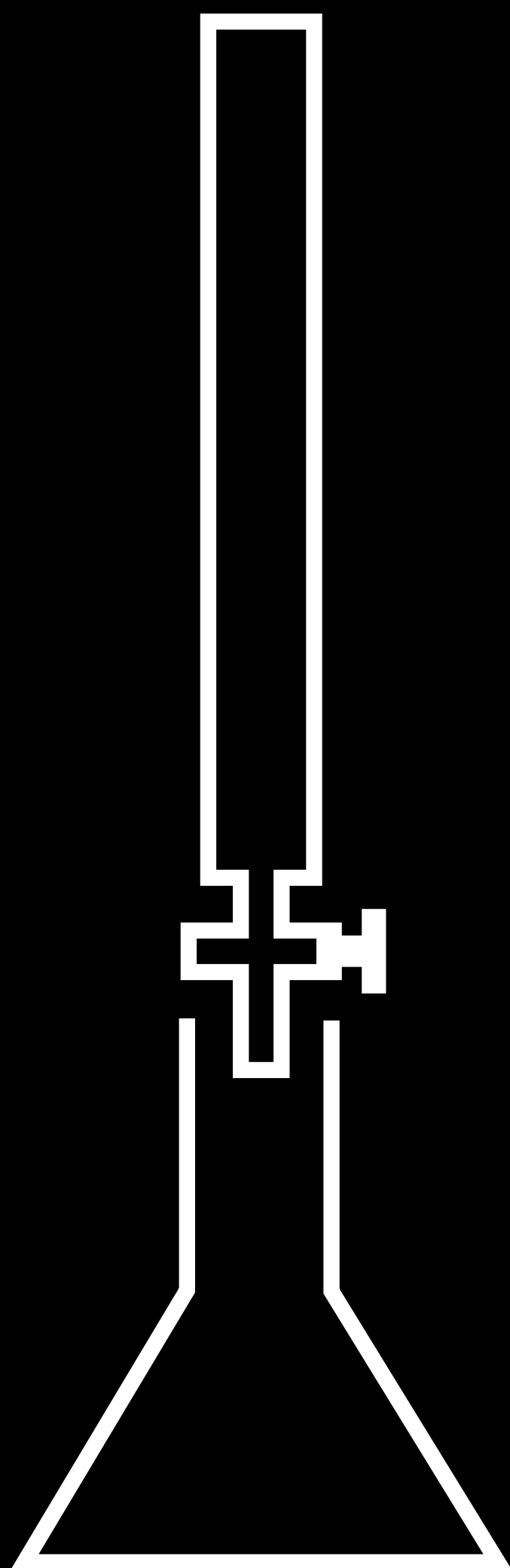
Non-metal oxides are acidic (SO_2 , CO_2 , etc.), thus reacts with bases

Other oxides can be neutral(react with neither, like H_2O , CO) or amphoteric (reacts with both, like ZnO , AlO , PbO) [memorise as ZAP]



Titration

slurps the burette

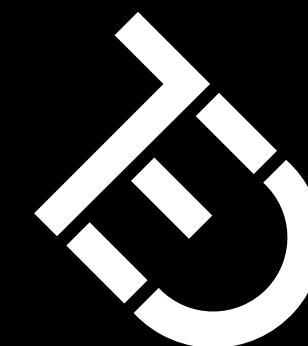


For an acid X and an alkali Y, where either one has unknown concentration

1. Fill burette 50 cm³ with X, making sure tip of burette is full of solution and does not have any air bubbles
2. Pipette 25 cm³ Y into a conical flask
3. Add indicator to flask
4. Place flask on white tile and titrate X with Y, swirling until solution reaches end-point(aka changes colour to neutral, depends on indicator used), then record volume of substance used
5. Repeat step 4 thrice and take the average of two best values for calculations

(Substance titrated is not particular, i.e. Y can be titrated as well)

The indicator endpoints depends on the indicator itself; for example, screened methyl orange is grey at the endpoint, while methyl orange is orange at the endpoint



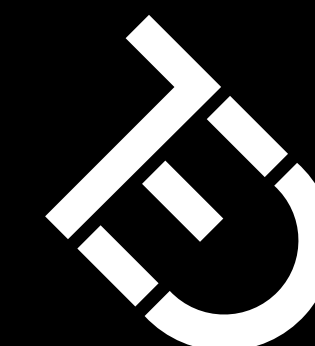
Titration

slurps the burette

Titration number (Just to show is this your 1st/2nd/etc titration)	1	2	3
Initial Burette Reading/ cm ³	Numbers	Numbers	Numbers
Final Burette Reading/ cm ³	Numbers	Numbers	Numbers
Volume of [substance] used/cm ³	Numbers	Numbers	Numbers
Best of 2 (tick the two closest values)	[tick here]	[tick here]	[tick here]

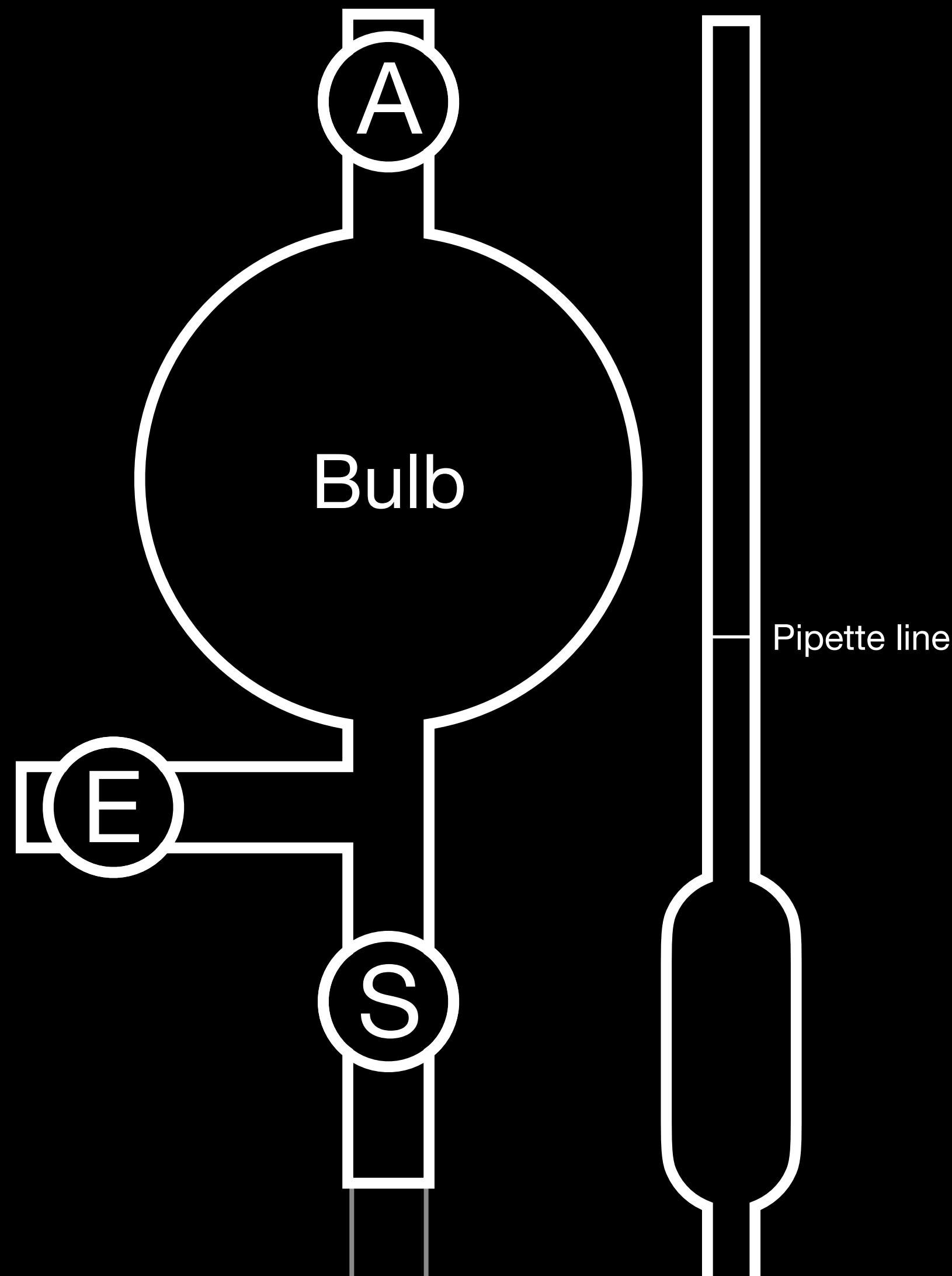
1st titration can be a rough titration,
aka you do not need to be very exact/
meticulous for this one

After finding the two closest values, take
the mean of the two to find the average
value of [substance] used, which is what
you use for later mole calculations



Titration

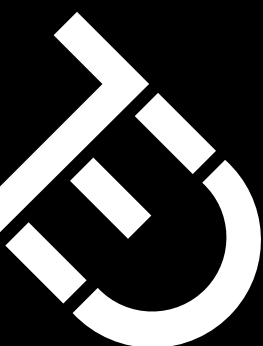
Pipette moment



To use a pipette filler:

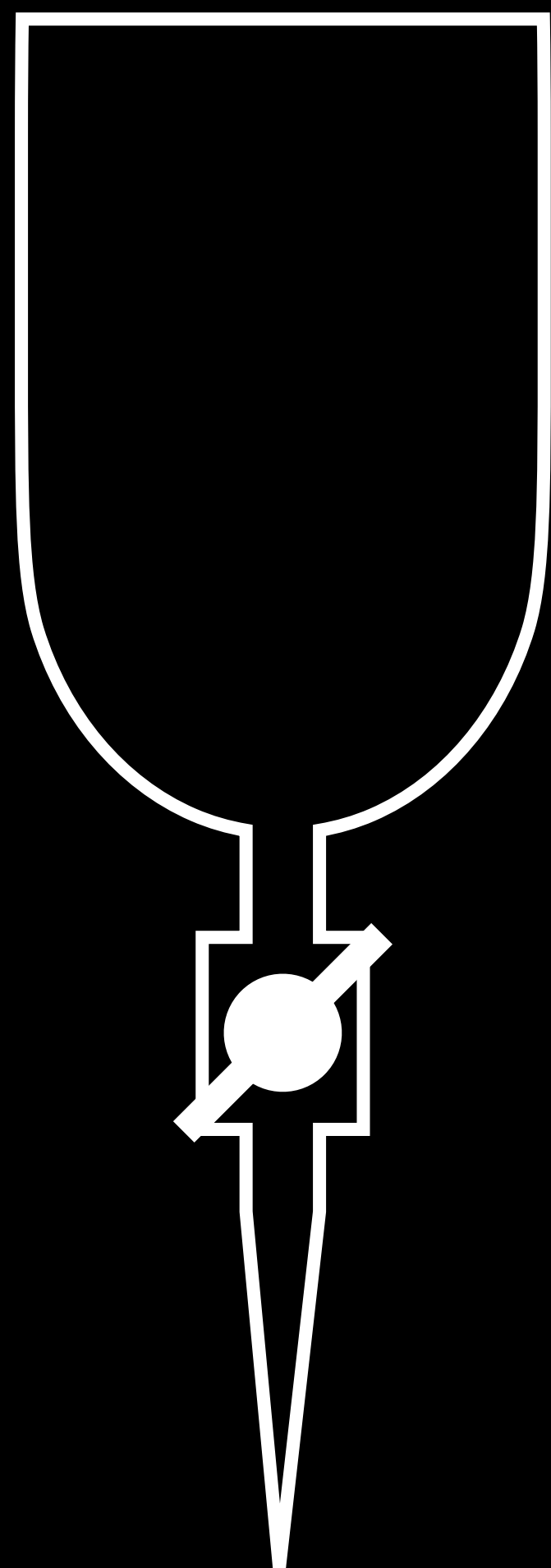
1. Press onto air button[A] & squeeze bulb to empty out air from filler
2. Gently twist filler onto pipette and immerse tip of pipette into liquid
3. Press onto suction button [S] to suck liquid in to the line on pipette
4. If liquid meniscus is above pipette, use exit button [E] to release liquid
5. Once liquid is measured out, place pipette into other vessel and twist off pipette to quickly release all liquid(this allows you to also prep other stuff without needing to hold onto pipette holder)

if small bits of liquid are caught at bottom, do not attempt to remove, pipette has been calibrated to account for that

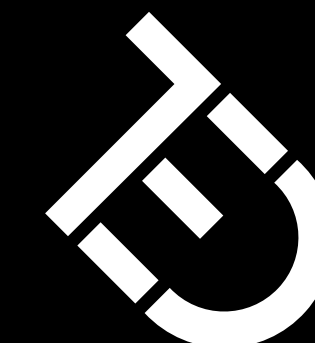


Titration

not a Pipette moment



1. Make sure the burette's valve is horizontal before filling burette with liquid with funnel (make sure to use thumb to have gap between burette mouth and funnel for faster filling)
2. Record where the meniscus is at (need not be at 0 mark)
3. Turn the valve slowly to adjust the rate of flow (closer to vertical —> faster flow, closer to horizontal —> slower flow)



Titration

slurps the burette

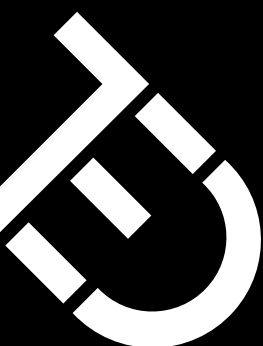
Average value taken is amount of substance needed for neutralisation.

To find concentration of substance:

Find no. of moles of substance of known concentration

Use stoichiometric ratio to find no of moles of other substance needed

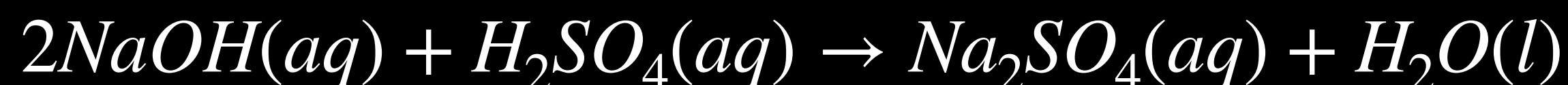
Take moles/volume to find unknown concentration



Titration

Sssssssssssssssssssssss

10.0cm^3 of sulfuric acid of an unknown concentration is titrated into 25.0cm^3 of sodium hydroxide with a concentration of 0.1mol/dm^3 for neutralisation. Find the concentration of the sulfuric acid.



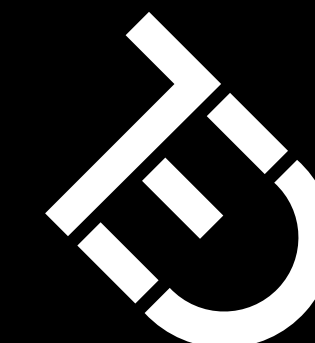
$$25.0\text{cm}^3 = 0.025\text{dm}^3$$

$$\begin{aligned}\text{Moles of NaOH} &= 0.025\text{dm}^3 \times 0.1\text{mol/dm}^3 \\ &= 0.0025\text{mol}\end{aligned}$$

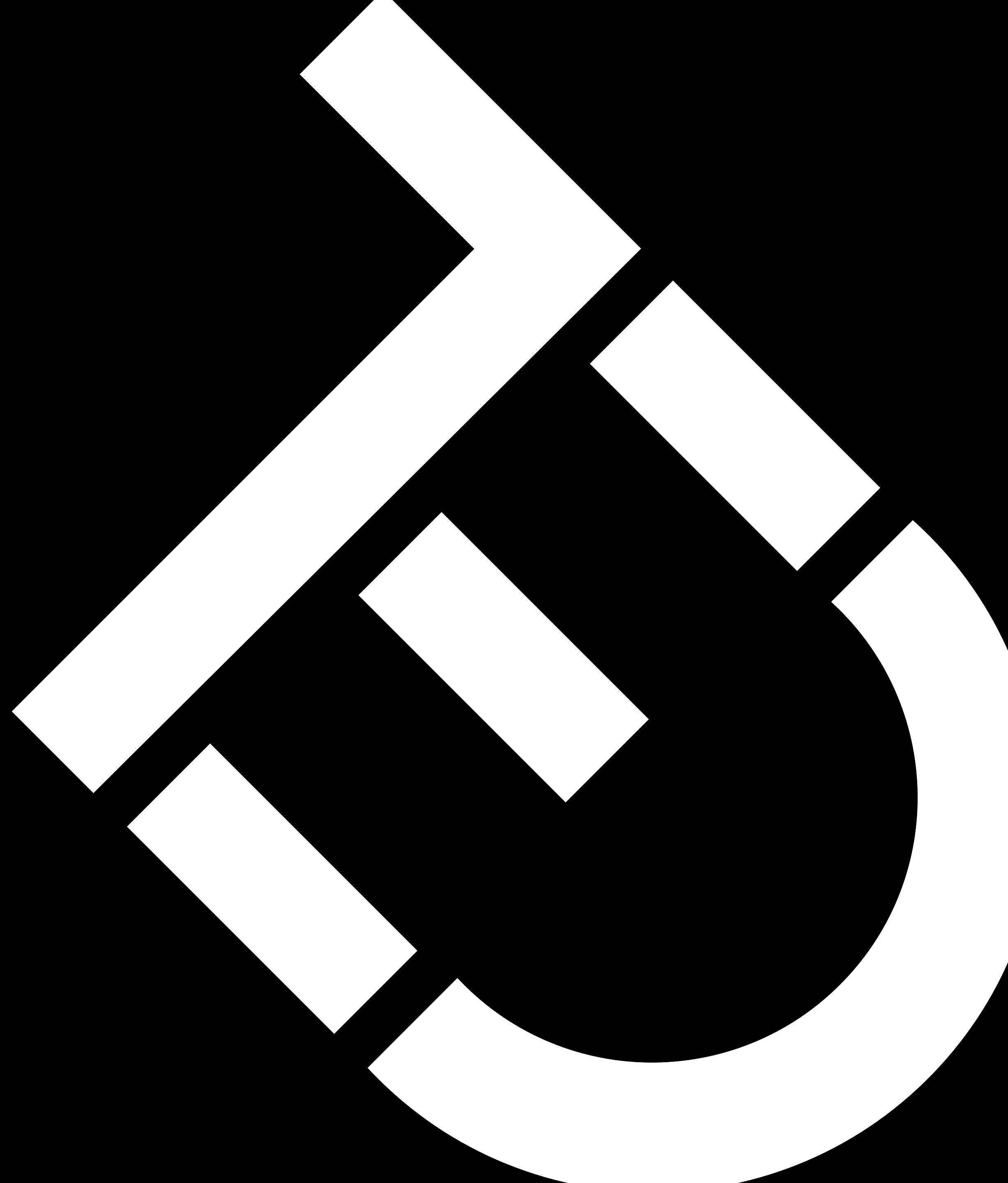
$$\text{Ratio of NaOH} : \text{H}_2\text{SO}_4 = 2 : 1$$

$$\begin{aligned}\therefore \text{Moles of H}_2\text{SO}_4 &= 0.0025\text{mol}/2 \\ &= 0.00125\text{mol}\end{aligned}$$

$$\begin{aligned}\text{concentration of H}_2\text{SO}_4 &= 0.00125\text{mol}/10.0\text{cm}^3 \\ &= 0.00125\text{mol}/0.01\text{dm}^3 \\ &= 0.125\text{mol/dm}^3\end{aligned}$$



Salts



Salt Solubility (in water)

Salts

All SPAN salts are soluble (Sodium, Potassium, Ammonium, Nitrate)

Chloride salts are generally soluble (except for silver and lead(II))

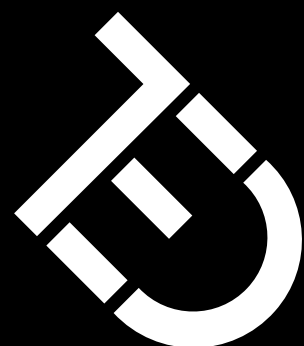
Sulfate salts are generally soluble (except for barium, calcium, and lead(II))

Carbonate salts are generally insoluble unless contains SPA

Oxide/Hydroxide salts are generally insoluble unless contains SPA or barium

(Have to remember(lmao))

Soluble		Insoluble	
	Exceptions		Exceptions
All SPAN salts (sodium, potassium, ammonium, nitrate)	Nil	Carbonate(CO3)	SPA
Chlorine (Cl)	Ag, Pb	Oxides/ Hydroxides (O/OH)	SPA, Ba
Sulfate (SO4)	Ba, Ca, Pb	Pb is common	



Salt Preparation - Titration

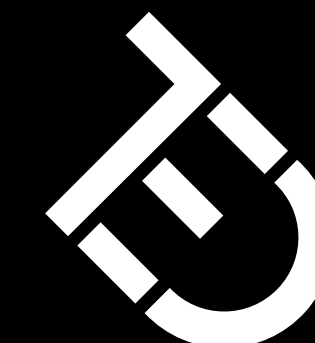
Salts

For preparing soluble Group 1 salts and ammonium salts

Refer to Acid and bases on how to titrate

After using acid bases titration method to find amount of substance needed to form a neutral solution, repeat titration without indicator

Take neutral solution and crystallise it to obtain the salt and dry between filter paper to obtain salt



Salt Preparation - Ionic Precipitation

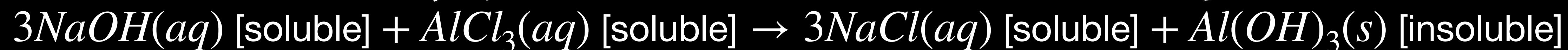
stlas

For preparing insoluble salts

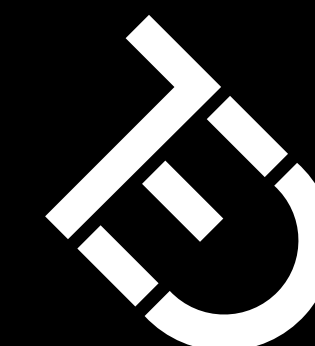
For two salt solutions AB and CD (where each letter is an ion), mix the two solutions together (forms two new salts, AC and BD, where one of them is insoluble)

Filter out soluble salt solution via filtration and wash with distilled water

e.g



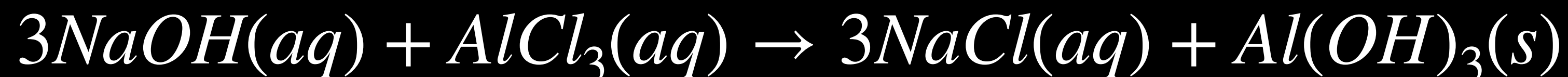
QA tests also use this precipitation technique



Ionic Equation Writing

there

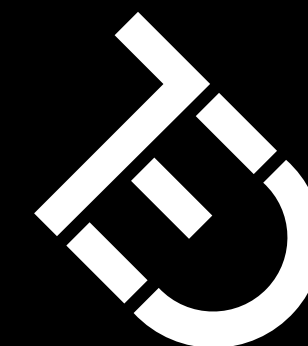
Start with balanced chemical equation



Split each aqueous salt into its constituent ions

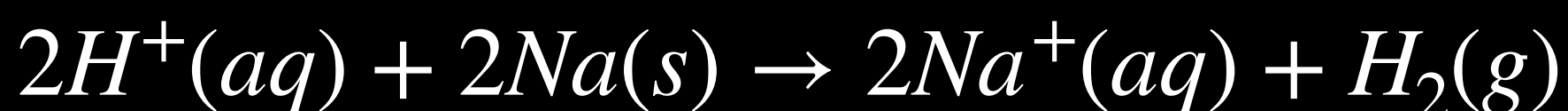
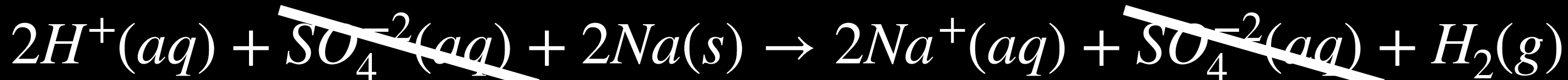
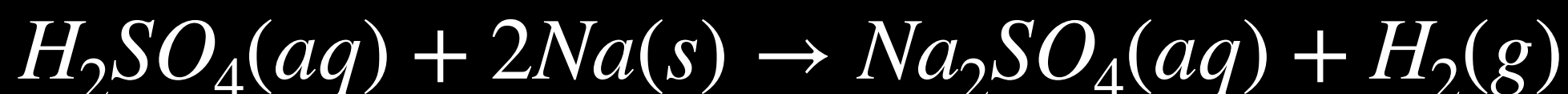


Cancel out common ions from both sides

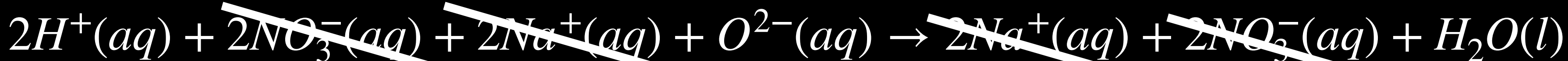
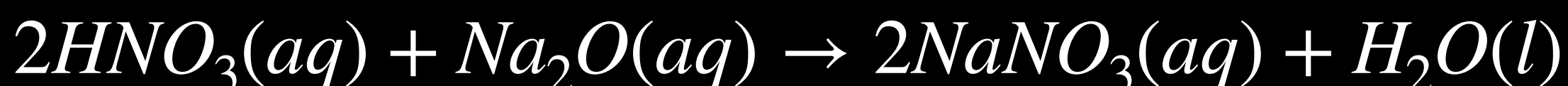


Ionic Equation Writing

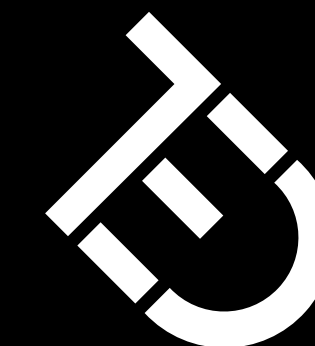
there



Why not just find solid and see constituent ions? Some reactions do not have precipitate, therefore this method is most reliable



note that for all neutralisation reactions, even with hydroxides, the ionic equation is as shown



Salt Preparation - Adding Excess Insoluble Substance

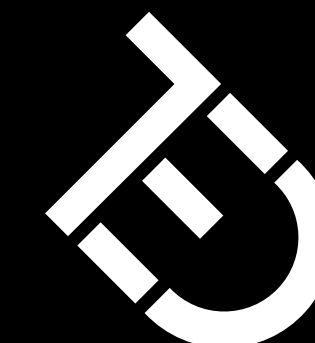
stlas

For preparing soluble salts that are not group 1 or ammonium salts or soluble salts that are prepared with an acid and an insoluble base or carbonate

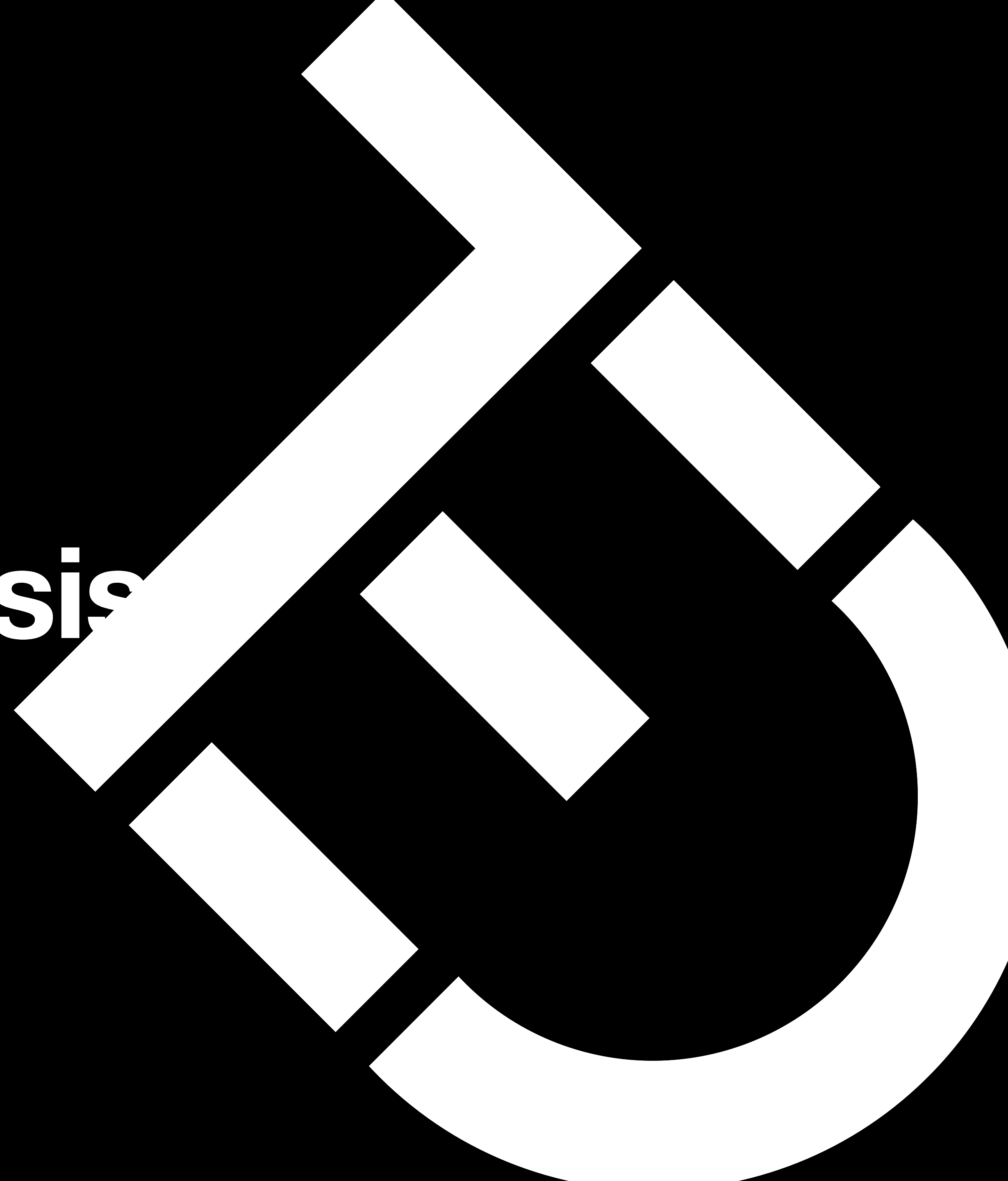
Keep adding base/carbonate to a volume of acid until an excess is formed(i.e. the base/carbonate starts to form at the bottom of the vessel)

Filter out excess then extract salt from filtrate via crystallisation

Filter crystals out then dry between pieces of filter paper to achieve crystals



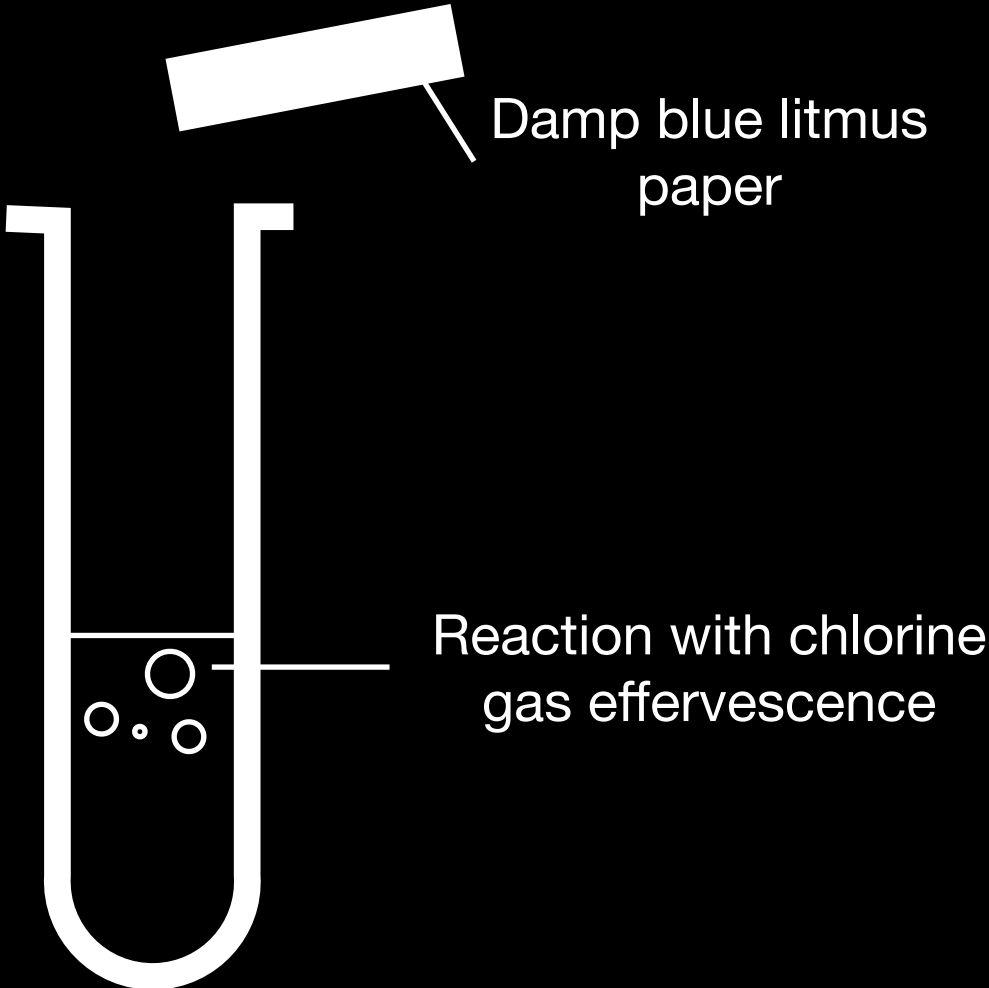
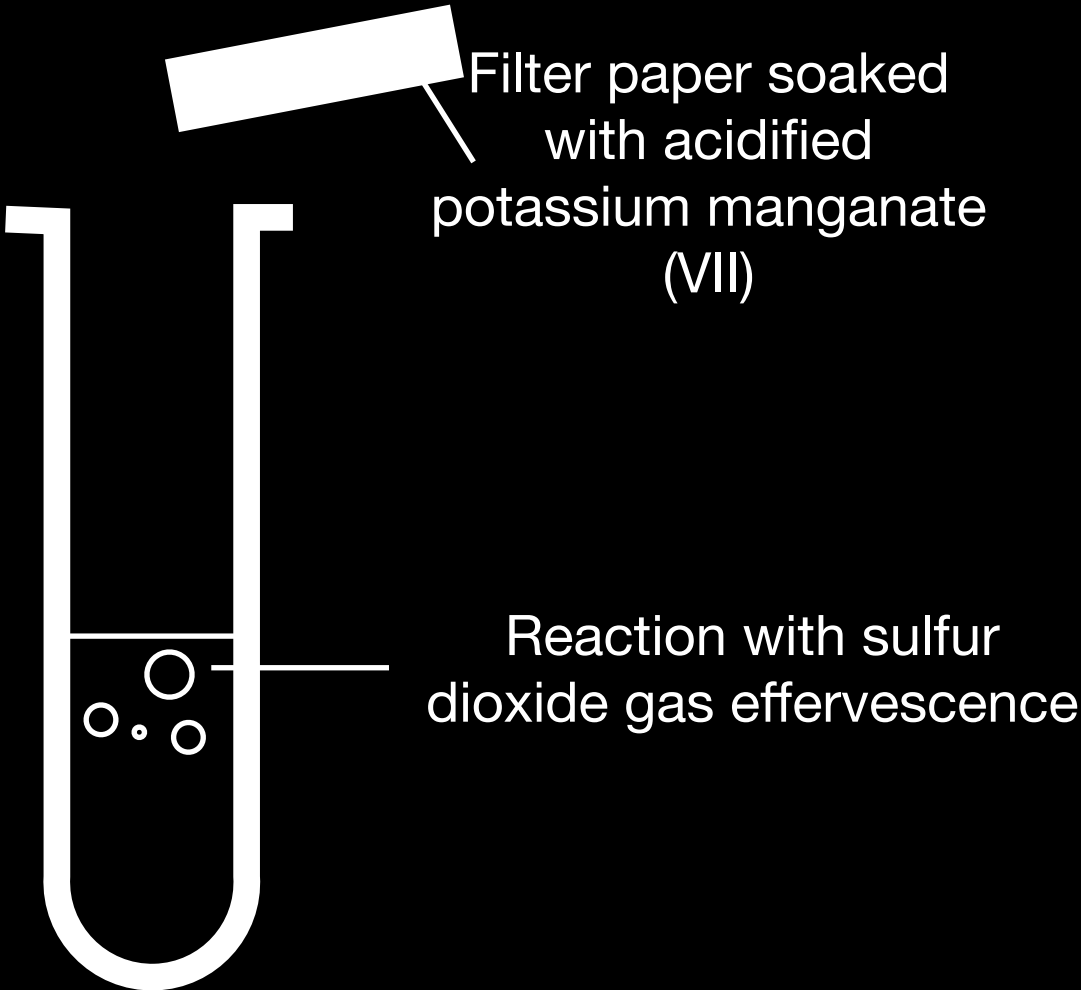
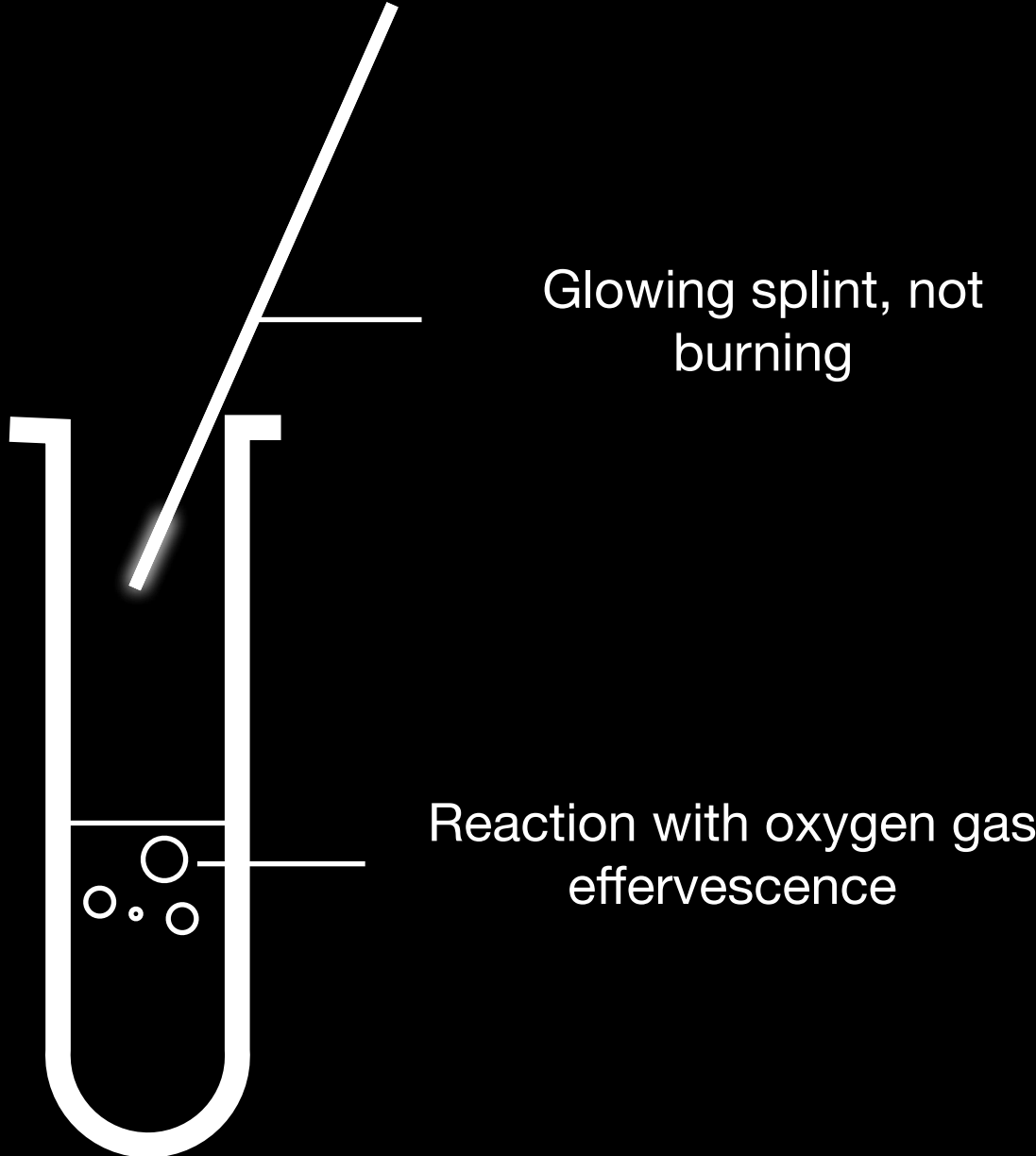
Qualitative Analysis

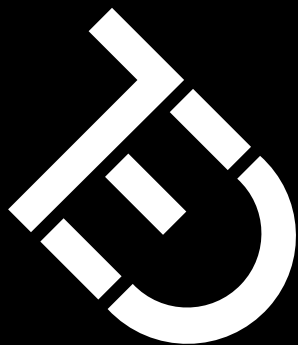


Gas Tests

Snort gas

You do not need to do this, just know in theory

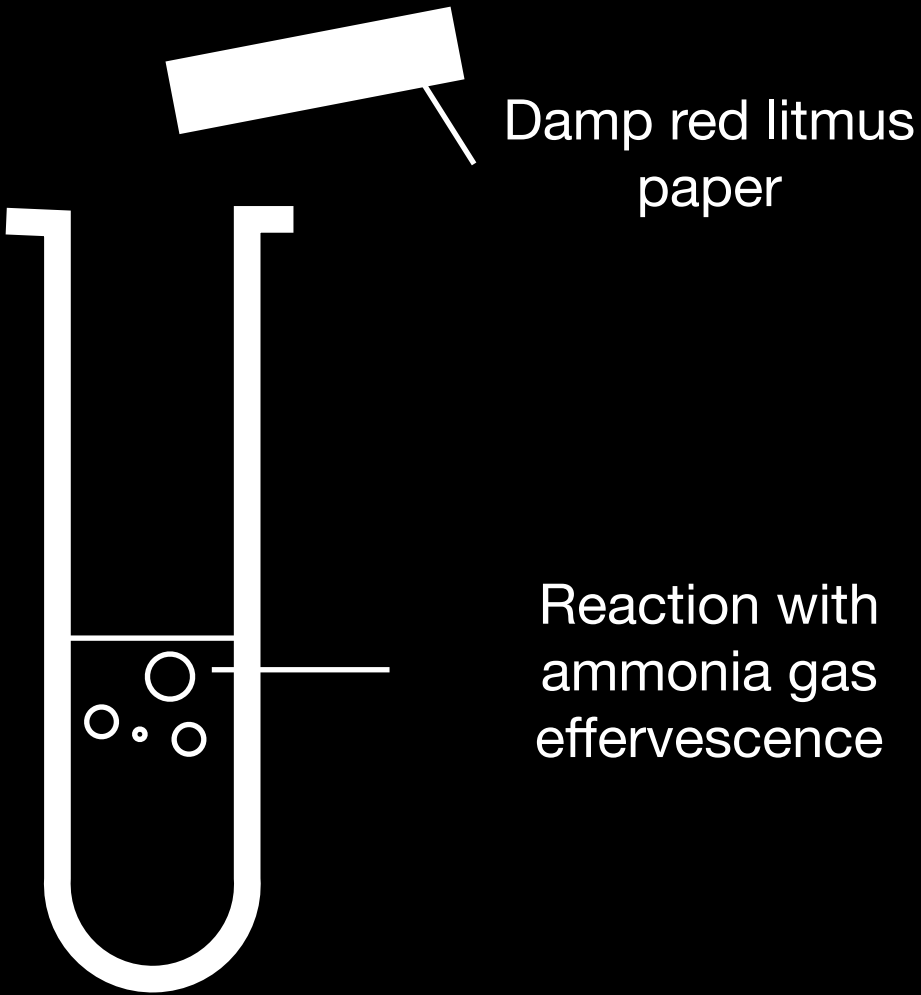
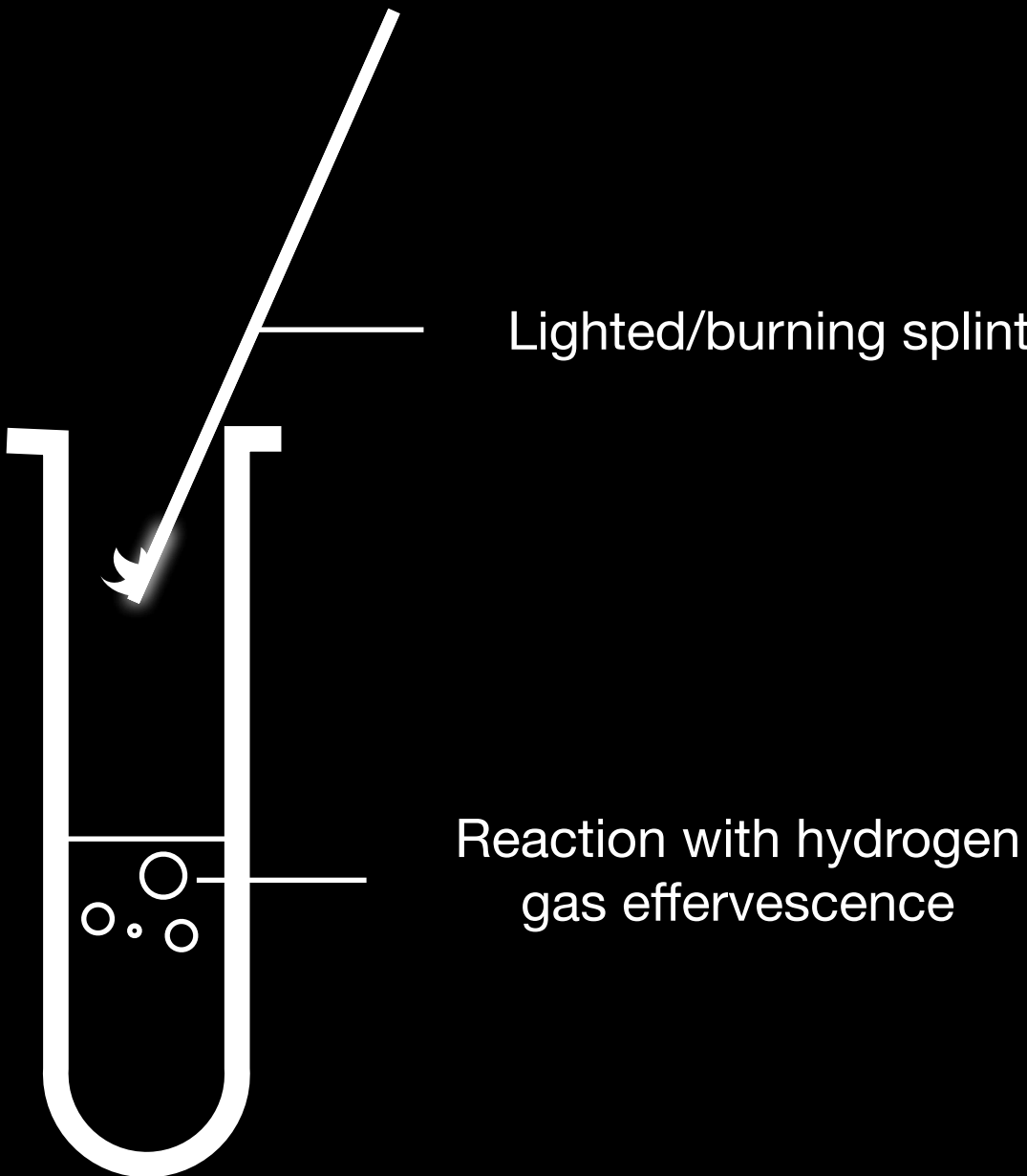
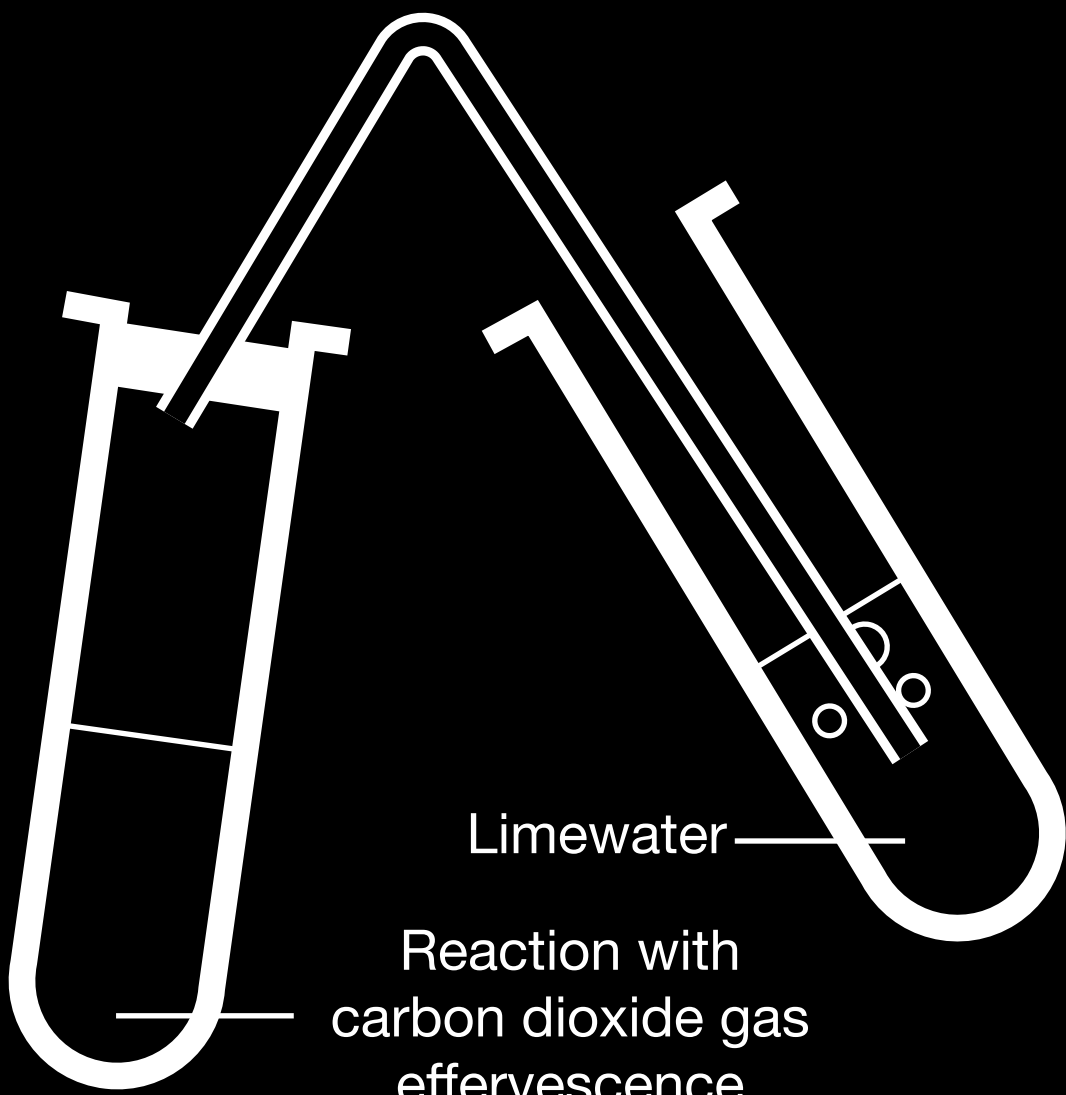
Gas	Chlorine	Sulfur dioxide	Oxygen
Test	 Damp blue litmus paper Reaction with chlorine gas effervescence	 Filter paper soaked with acidified potassium manganate (VII) Reaction with sulfur dioxide gas effervescence	 Glowing splint, not burning Reaction with oxygen gas effervescence
If present	Litmus paper turns red, then white (bleaches litmus paper)	Turns filter paper from purple to colourless	Wooden splint reignites

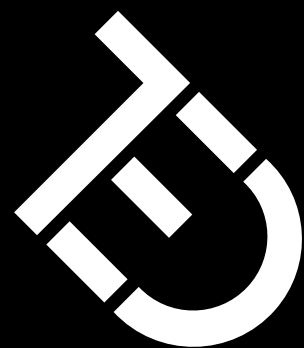


Gas Tests

Snort gas

After starting the reaction, recommended to use thumb to cover test tube while preparing tube/splint to prevent gas from escaping

Gas	Ammonia	Hydrogen	Carbon dioxide
Test			
If present	Red Litmus paper turns blue	Extinguishes splint with a pop sound	White precipitate forms in limewater



Ion Tests - Cation

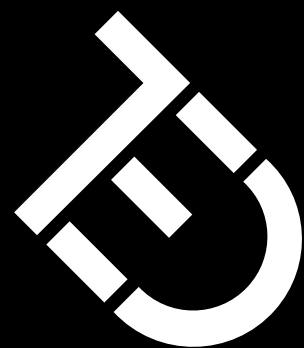
capz(it has a z at the end so I am cool)

Identity of precipitate for both NaOH and NH3 is [cation]OH

CAPZ

Cation	Effect under NaOH(aq)		Effect under NH3(aq)	
	Precipitate	Solubility in excess	Precipitate	Solubility in excess
Ca2+	White	Insoluble	No ppt	
Al3+	White	Soluble	White	Insoluble
Pb2+	White	Soluble	White	Insoluble
Zn2+	White	Soluble	White	Soluble
Cu2+	Light Blue	Insoluble	Light Blue	Soluble to form dark blue sol.
Fe2+	Green	Insoluble	Green	Insoluble
Fe3+	Red-brown	Insoluble	Red-brown	Insoluble
NH4+	After adding NaOH, warm to test ammonia gas			

Fe2+ when left to stand (aka when left alone) oxidises to become Fe3+, thus it will change colour from green to red-brown



Ion Tests - Anion

Slubide subksutiies

Nitric acid is for testing for any excess carbonates

If they ask you to only add nitric acid, make sure to prepare limewater first

Anion	Anion Test	If present
CO3 2- (carbonate)	Add acid(e.g nitric acid), then use carbon dioxide test	White precipitate formed
Cl- (chloride)	Add nitric acid, then add silver nitrate(aq)	White precipitate
I- (iodide)		Yellow precipitate
SO4 2- (sulfate)	Add nitric acid, then add barium nitrate	White precipitate
NO3 - (nitrate)	Add NaOH+aluminium foil, then warm gently and test for ammonia	Effervescence + ammonia gas formed

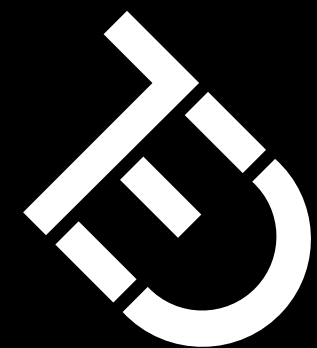
Precipitate identity is *AgCl*

Precipitate identity is *AgI*

Precipitate identity is *BaSO₄*

To warm gently, leave air hole half-open

Prepare damp red litmus paper before this!!!!!!



Energy from Chemicals

An abstract graphic design featuring a black background. On the right side, there is a large, thick white circle. Overlaid on this circle and extending across the frame are several thick white diagonal lines. These lines are arranged in a way that they appear to be part of a larger, repeating geometric pattern, possibly a stylized 'X' or a series of parallel lines intersecting. The overall effect is a high-contrast, modern, and geometric composition.

Endothermic & Exothermic Reactions

hot and cold reaction

Endothermic - surrounding feels cold

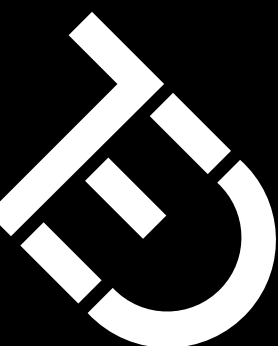
in endothermic reactions, the amount of energy absorbed is more than the amount of energy released

absorbs energy from surroundings -> surroundings feel cold

Exothermic - surrounding feels warm

in exothermic reactions, the amount of energy absorbed is less than the amount of energy released

release energy to surroundings -> surroundings warm



Enthalpy Change

energy

Enthalpy change is the overall energy change of a substance after a reaction.

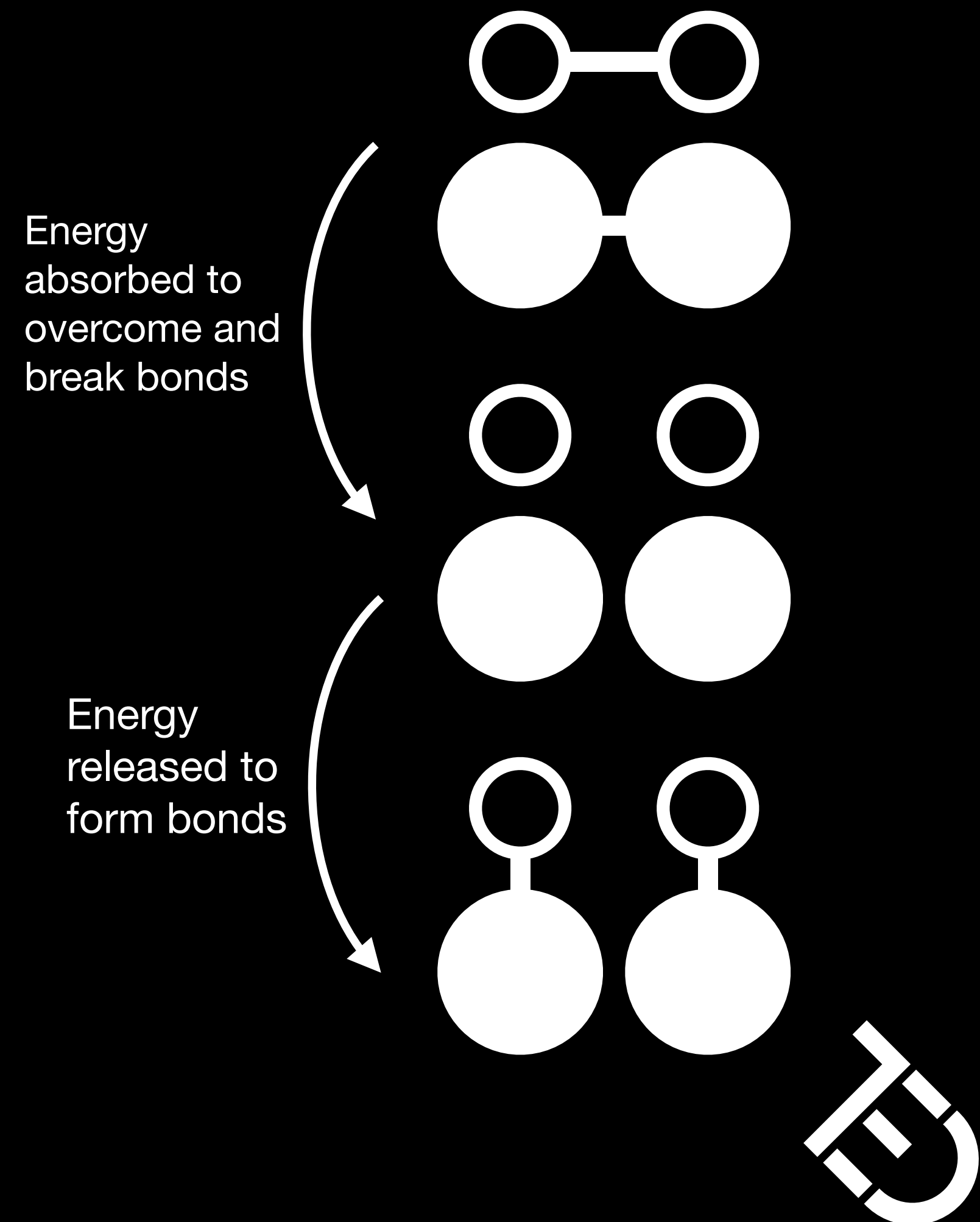
In a reaction, bonds of the reactants are first broken, which requires energy to be absorbed to overcome the bonds (recall structure & properties of materials).

Afterwards, new bonds are formed to form the products, requiring energy to be lost to allow the bonds to form again.

Thus, the enthalpy change can be calculated by finding the difference between the energy gained (when bonds break) and energy released (when bonds form)

enthalpy is represented by ΔH

unit for enthalpy change is usually kJ/mol or is kJ/g



Enthalpy Change - Calculations

calculating

Finding enthalpy change questions will give a table with the energy between each type of bond (called bond energies).

using the chemical equation, you can use this info to calculate enthalpy change as so:

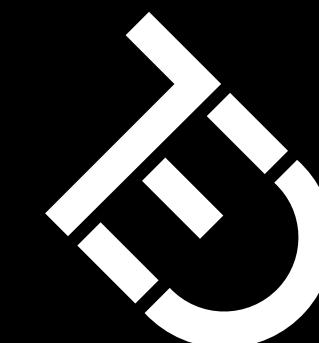
$\Delta H = \text{sum of bond energies in reactants (energy gained)} - \text{sum of bond energies in products (energy lost)}$

If ΔH is negative $[-\Delta H]$, the reaction is exothermic

If ΔH is positive $[\Delta H]$, the reaction is endothermic

eg. $\text{HNO}_3(aq) + \text{NaOH}(aq) \rightarrow \text{NaNO}_3(aq) + \text{H}_2\text{O}(l)$ is exothermic $\therefore -\Delta H$

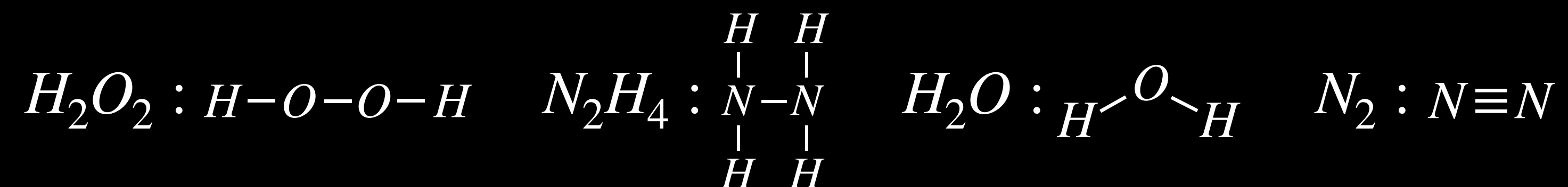
eg. $\text{NaHCO}_3(s) + \text{HCl}(aq) \rightarrow \text{NaCl}(aq) + \text{H}_2\text{O}(l) + \text{CO}_2(g)$ is endothermic $\therefore +\Delta H$



Enthalpy Change - Calculations

calculating

calculate enthalpy change of $2H_2O_2(g) + N_2H_4(g) \rightarrow 4H_2O(g) + N_2(g)$



energy gained:

$$2(2(H-O) + O-O) + N-N + 4(N-H) \\ = 2(2(460) + 150) + 390 + 4(390) = 4090 \text{ kJ/mol}$$

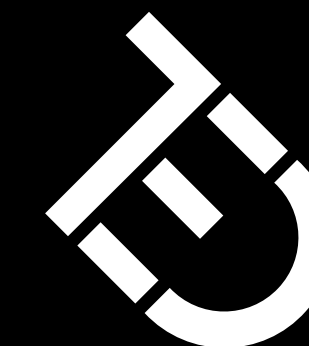
energy released:

$$4(2(H-O)) + (N \equiv N) \\ = 4[2(460)] + 994 = 4674 \text{ kJ/mol}$$

$$\Delta H = 4090 - 4674 = -584 \text{ kJ/mol (thus the reaction is exothermic)}$$

You need to know the structure of the reactants and products to do these qns !!!!

Bond	Bond energy in kJ/mol
O-O	150
N-N	160
N-H	390
O-H	460
N≡N	994



Energy Profile Diagrams

omg its prolife #canceled

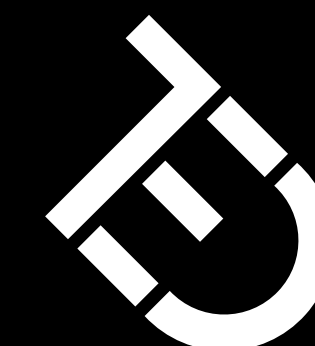
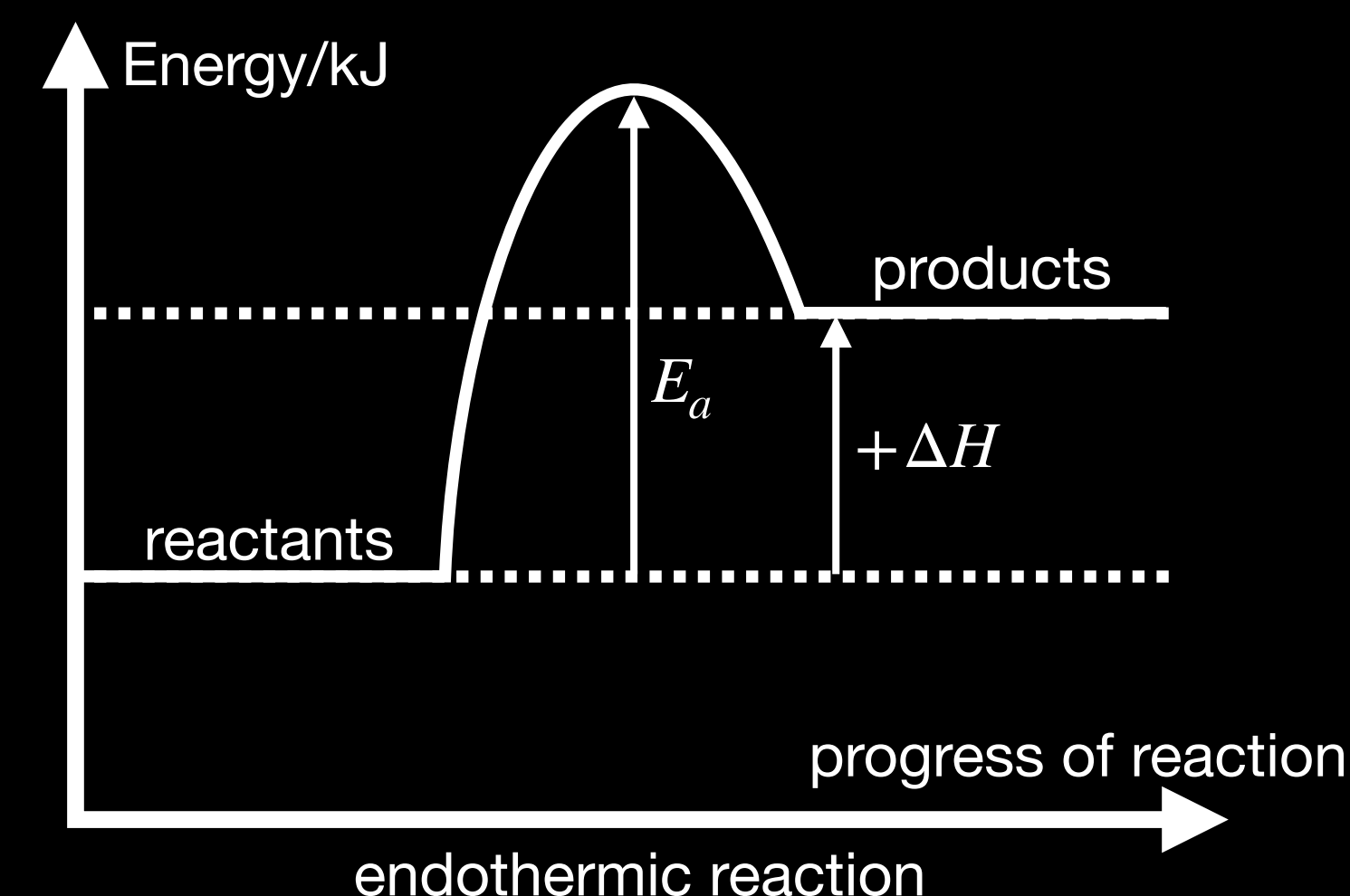
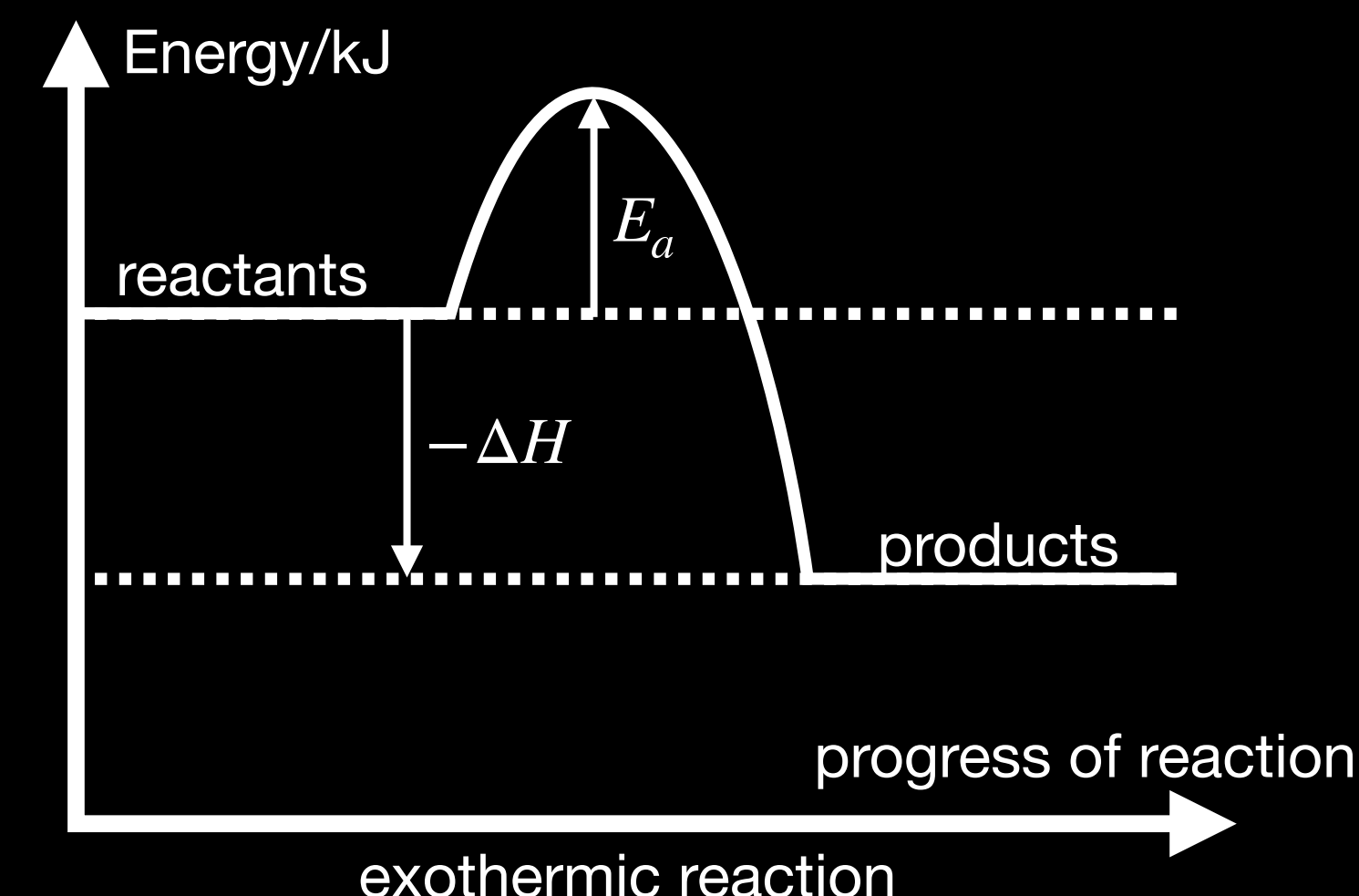
energy profile diagram is a graph to show the change in energy of a substance with the process of reaction.

E_a : activation energy; minimum energy required to start a reaction

ΔH : enthalpy change

arrows are meant to show the change in energy

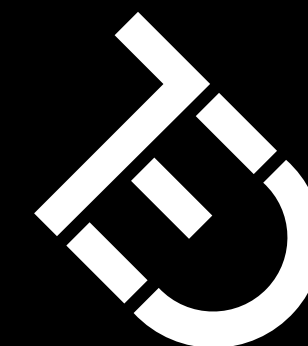
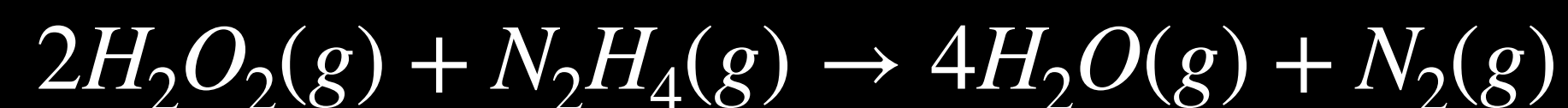
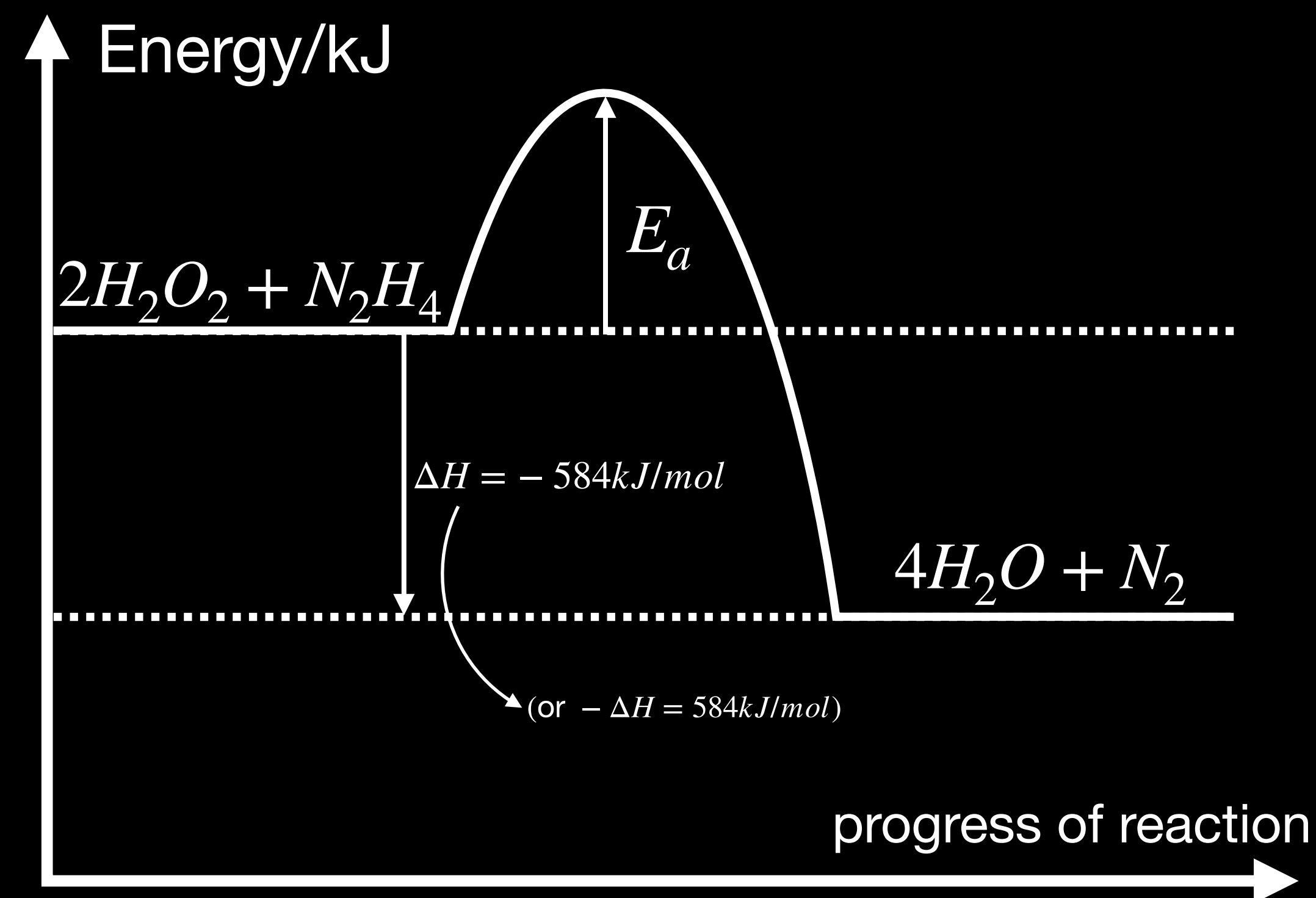
up for increasing, down for decreasing



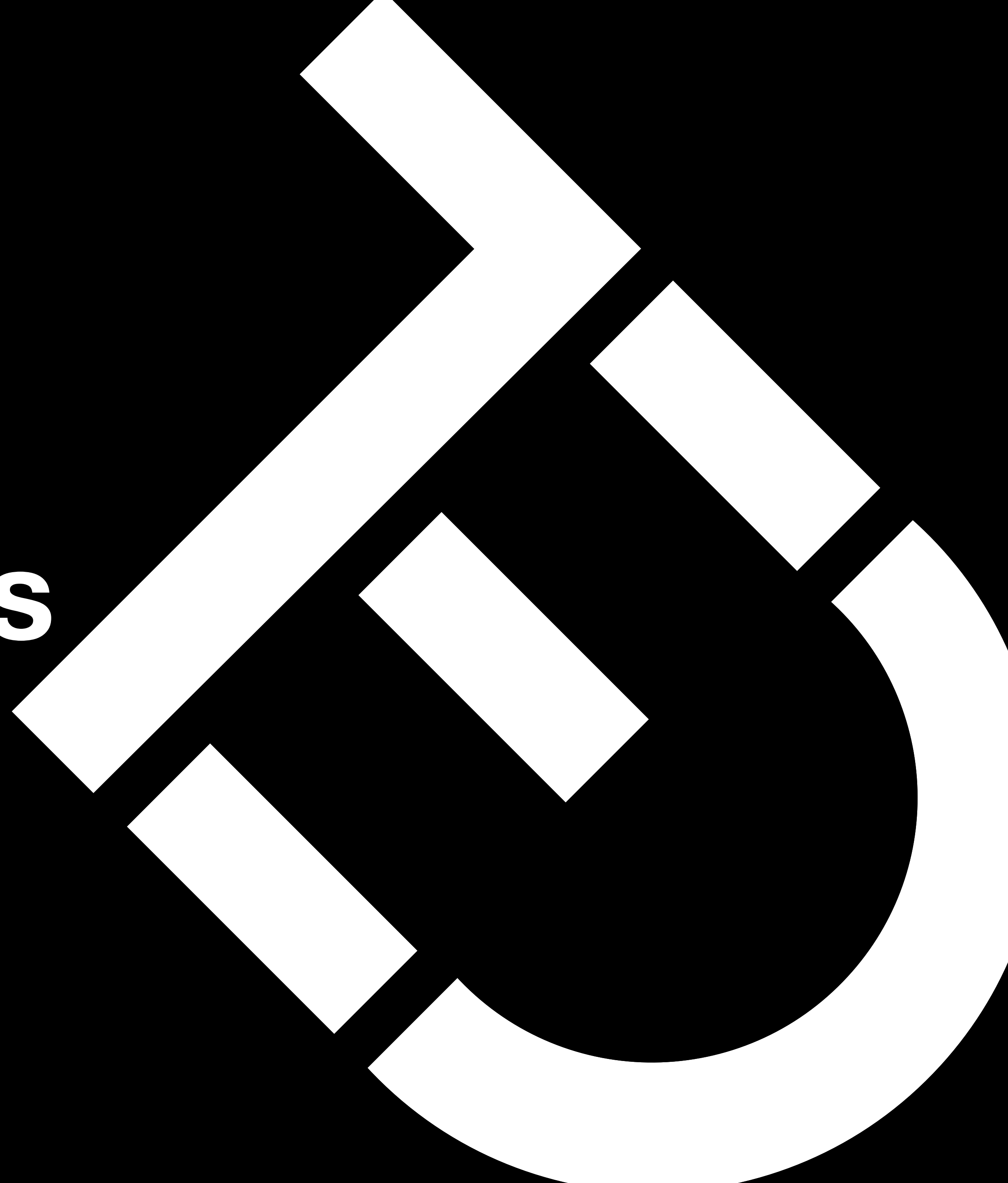
Energy Profile Diagrams

how to draw

1. 2 horizontal lines for the reactant and product at start and end respectively
product is above reactant if endo and below if exo
2. curve from reactant going up to activation energy and down to product
3. draw dotted horizontal lines following the reactant line and product lines
4. draw the E_a arrow from the reactants to the peak of the curve and the ΔH arrow from the reactants to the products (BOTH arrows are drawn from the reactants line)
5. Label the axes as shown (energy/kJ for vertical axis, and progress of reaction for horizontal axis, not time), the reactants and products with the reaction's reactants and products, and the enthalpy change with the sign.



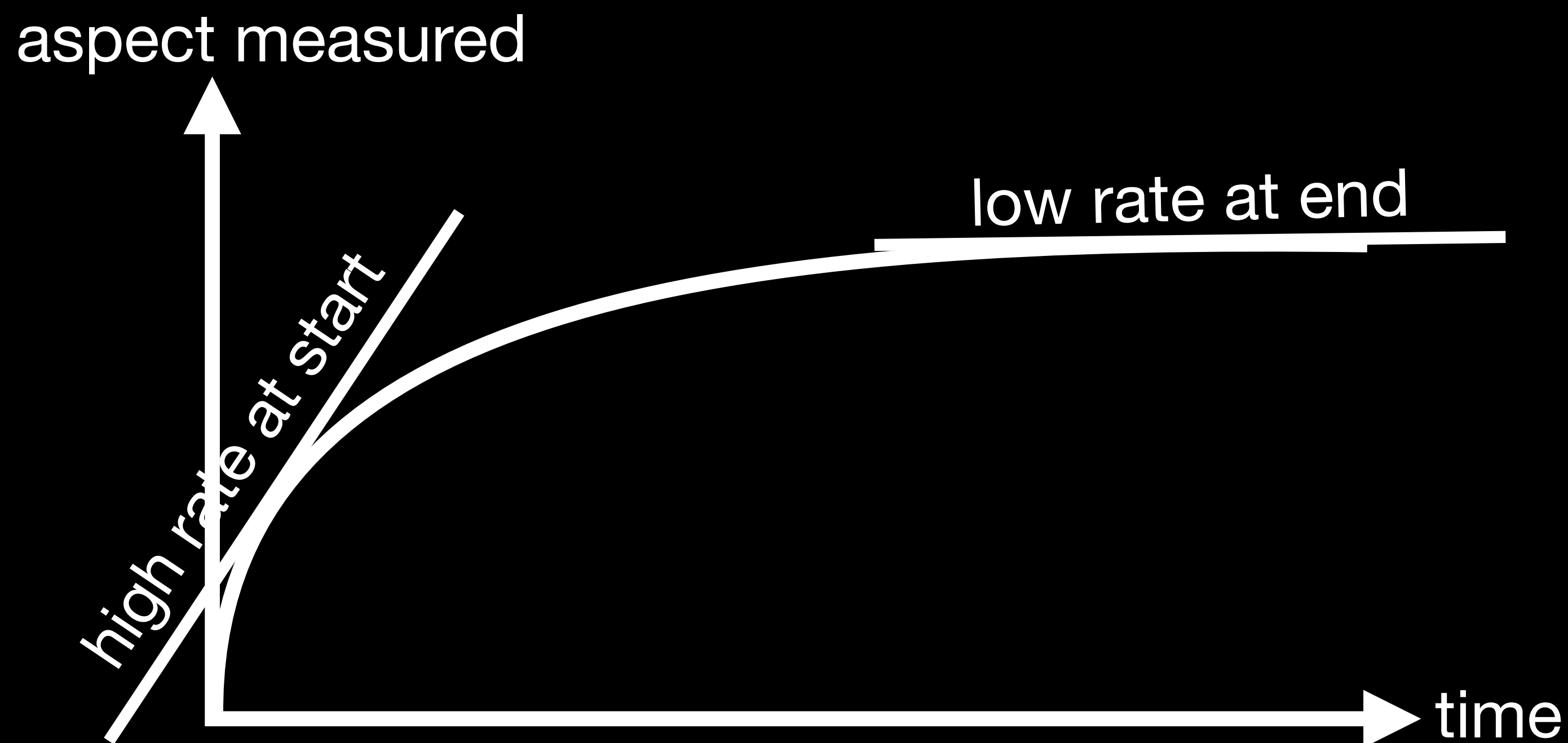
Rate of Reactions



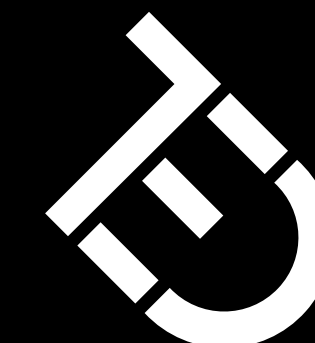
Measuring Rate of Reaction

or

To measure the speed of a reaction, we monitor an aspect of the reaction respect to time.
e.g. volume of gas produced, mass of contents in reaction vessel, temperature change



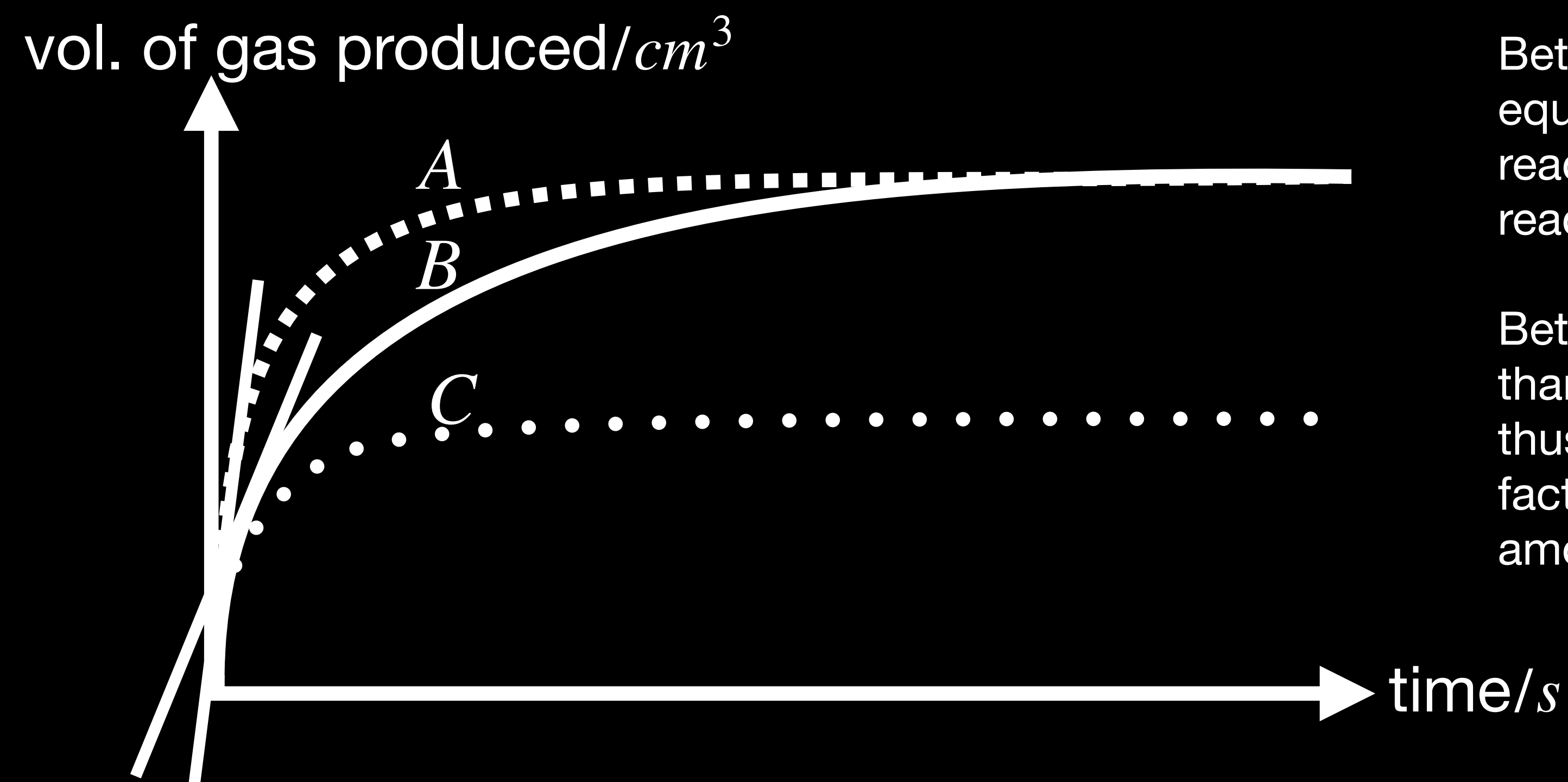
Depending on which aspect is monitored, the graph is either increasing (e.g volume of gas produced) or decreasing (mass of reactant vessel)



Measuring Rate of Reaction

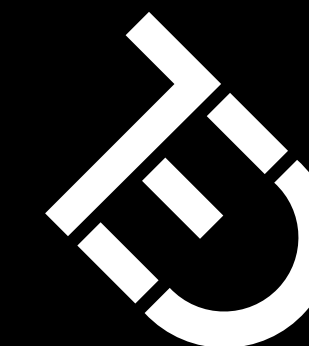
ror

For example, if we were to measure the volume of gas produced in an acid carbonate reaction, we can analyse the rate of reaction and the amount of products used.



Between B and C, the rate of reaction (at the start) are equal, however C has less gas produced, thus less reactants (in the same conditions) were used in reaction C.

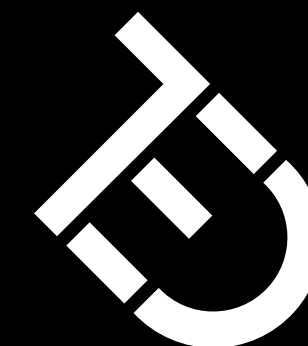
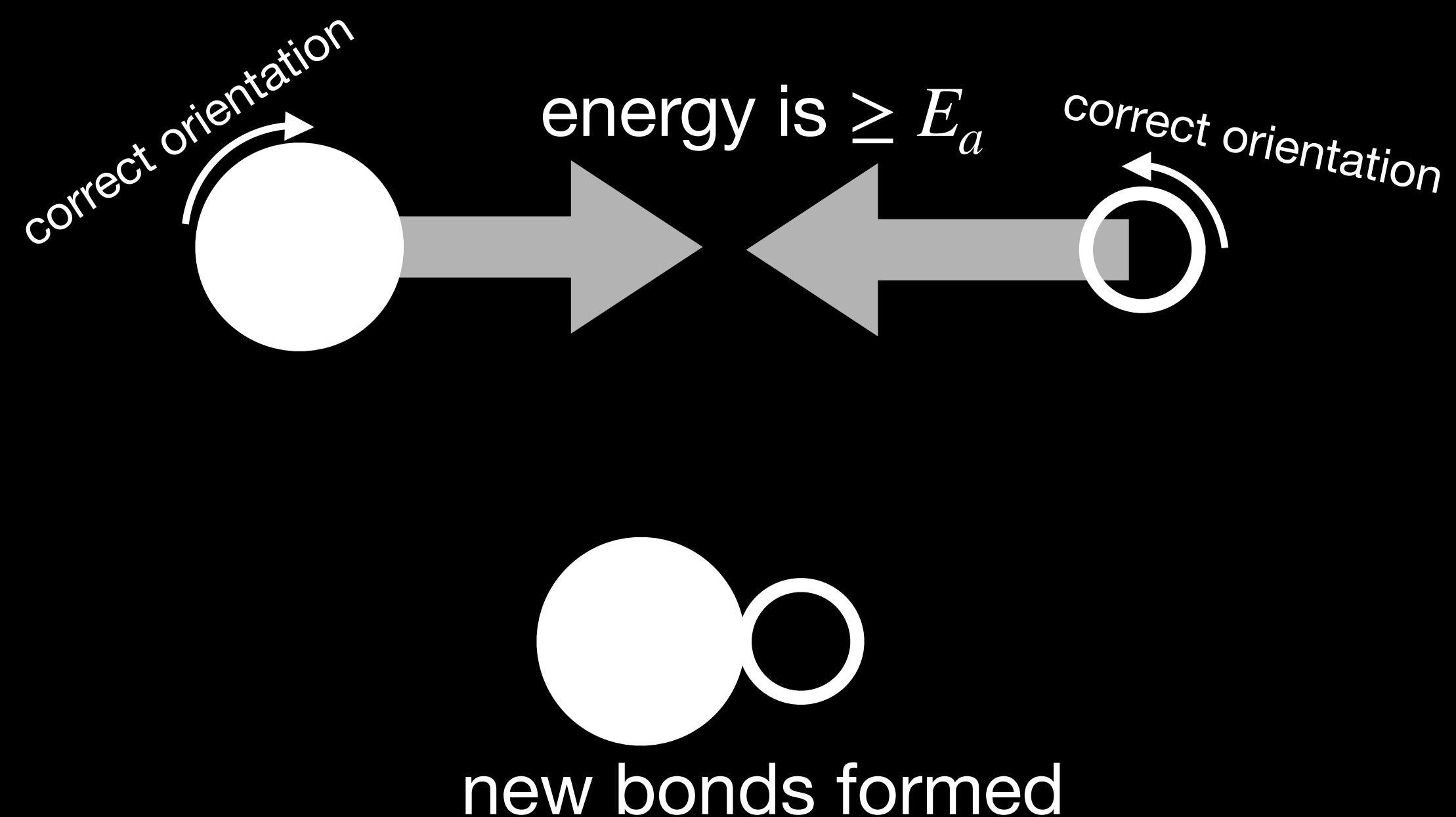
Between A and B, the rate of reaction of A is faster than B, while the same amount of gas was produced, thus the same amount of reactants was used, while a factor affecting rate of reaction (that does not affect amount of products produced) was changed



Collision Theory

Col

For a reaction to occur, the reactant particles must undergo effective collisions, where:
particles must have enough energy ($\geq$ activation energy) and must collide in the right orientation
**not all collisions are effective collisions

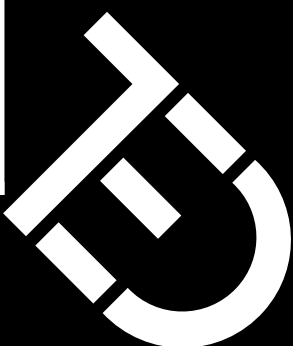


Factors Affecting Rate of Reaction

Fffff

Factor	Trend	Explanation
Temperature	↑ temp, ↑ rate of reaction	As temp ↑, avg KE of particles ↑, more particles with energy $\geq E_a$, ↑ frequency of (normal) collisions, ↑ frequency of effective collisions, ↑ RoR
Concentration (only for solutions)	↑ conc., ↑ rate of reaction	As conc. ↑, no. of particles per volume ↑, ↑ frequency of (normal) collisions, ↑ frequency of effective collisions, ↑ RoR (note that concentration will also increase the products formed, unless if volume of solution used counters that)
Pressure/Volume (Only for gases)	↑ pressure or ↓ volume, ↑ rate of reaction	As pressure ↑/vol ↓, no. of particles per volume <u>of gas</u> ↑, ↑ frequency of (normal) collisions, ↑ frequency of effective collisions, ↑ RoR
Surface Area/Particle Size (only for solids)	↑ surface area or ↓ particle size, ↑ rate of reaction	As surface area ↑/particle size ↓, Surface area:volume ratio ↑, ↑ frequency of (normal) collisions, ↑ frequency of effective collisions, ↑ RoR
Catalyst presence	Catalyst present, ↑ rate of reaction	When catalyst is present, alternate reaction pathway of lower E_a is provided, more particles with energy $\geq E_a$, ↑ frequency of effective collisions, ↑ RoR (catalysts do NOT increase normal reactions)

All factors' explanation must include ↑ frequency of collisions and effective collisions, ↑ RoR (except for catalyst, do not include normal reactions)



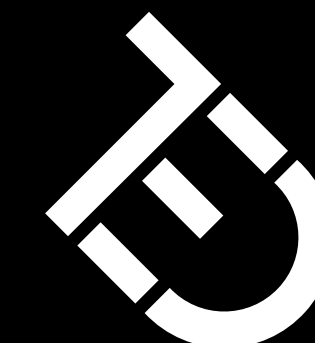
Catalysts

Cccccccc

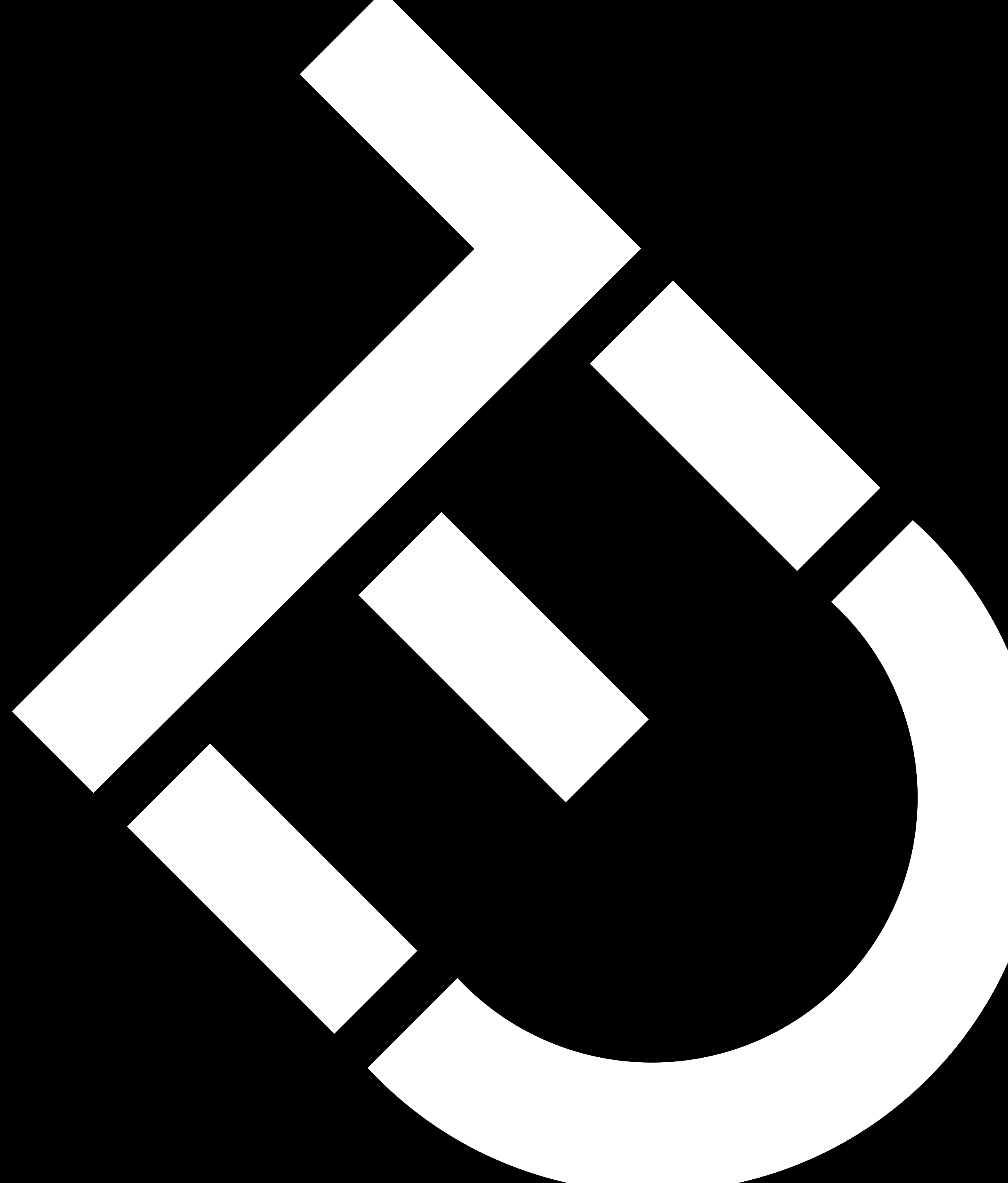
As said, a catalyst speeds up a reaction by providing an alternate pathway of lower activation energy

Catalysts also have these characteristics:

- only a small amount is needed to speed up a reaction
- Catalysts do not increase the reaction yield
- The chemical property will remain unchanged, even if the physical properties does
- ****Catalysts are not used up in the reaction (so do not write in chemical equation)**
- Impurities can prevent catalysts from working
- Catalysts are selective; one catalyst only works for one purpose, there is no catalyst that speeds up all reactions



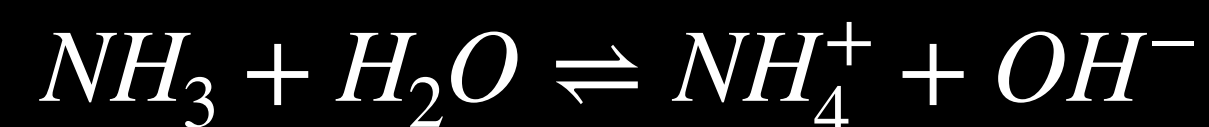
Ammonia



Properties of Ammonia

Get real

Ammonia partially dissociates in water to form a weak alkali



aqueous ammonia can conduct electricity (due to ions being charge carriers)

Ammonia gas is an alkaline

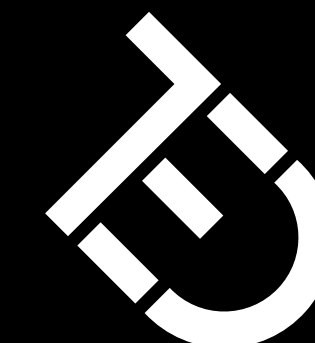
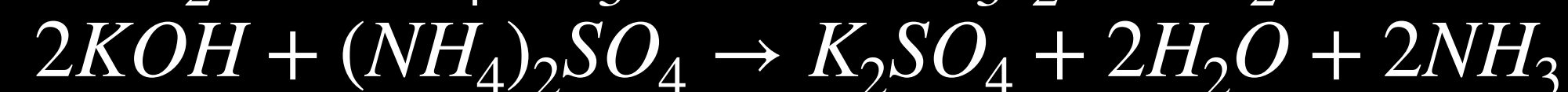
is also a colourless and pungent gas

causes damp red litmus paper to turn blue

thus can react with acids to form ammonium salts, e.g. $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

Ammonium salts react with bases (laboratory production of ammonia)

base + ammonium salt $\rightarrow$ salt + water + ammonia gas (also known as ammonia displacement)



Uses of Ammonia

Get rel

Ammonia is used in:

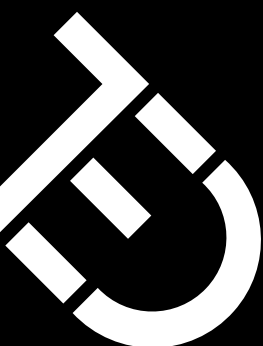
Household cleaning products

Nitric acid production

Fertiliser production

common artificial fertilisers: ammonium sulfate $[(NH_4)_2SO_4]$ and ammonium nitrate $[NH_4NO_3]$
(produced by ammonia and acid)

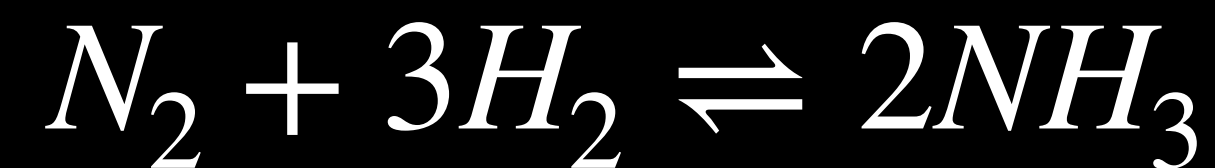
allows plants to take in nitrogen through roots as nitrogen cannot be absorbed through the air directly



Haber Process - Reaction

ammm

The Haber Process is a method to commercially produce ammonia at large scales using the reaction:



N_2 is obtained by fractional distillation of liquified air

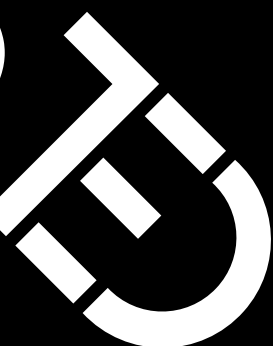
H_2 is obtained by cracking of crude oil (or steam and methane reforming, however less used nowadays, and better to write cracking instead)

The molar ratio of the two gases $N_2 : H_2$ is 1 : 3, thus we use that ratio of gases in the Haber process

The symbol $[\rightleftharpoons]$ represents that the reaction is reversible, aka the ammonia can decompose back into its original reagents while the reagents form into ammonia simultaneously

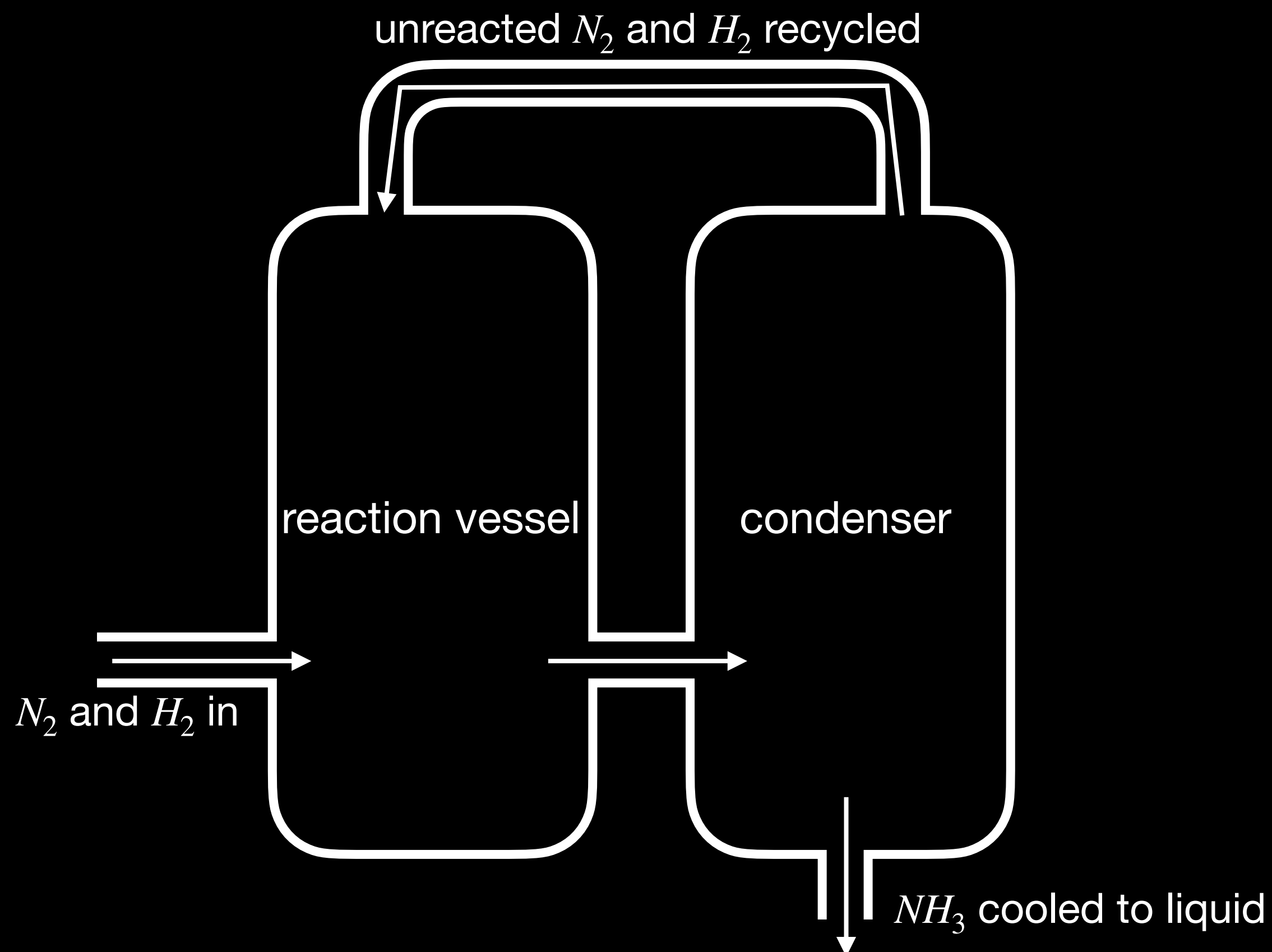
After an amount of time, the reaction will be at an equilibrium; which means the rate of the forward reaction(formation) is equal to rate of the backward reaction(decomposition).

This is also the reason why all of the reactants will never be fully reacted or reach completion, the reaction will force the two forward and backward reactions to have the same rate (aka will always reach equilibrium)

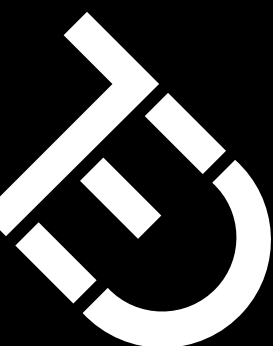


Haber Process - Setup & Conditions

ammm

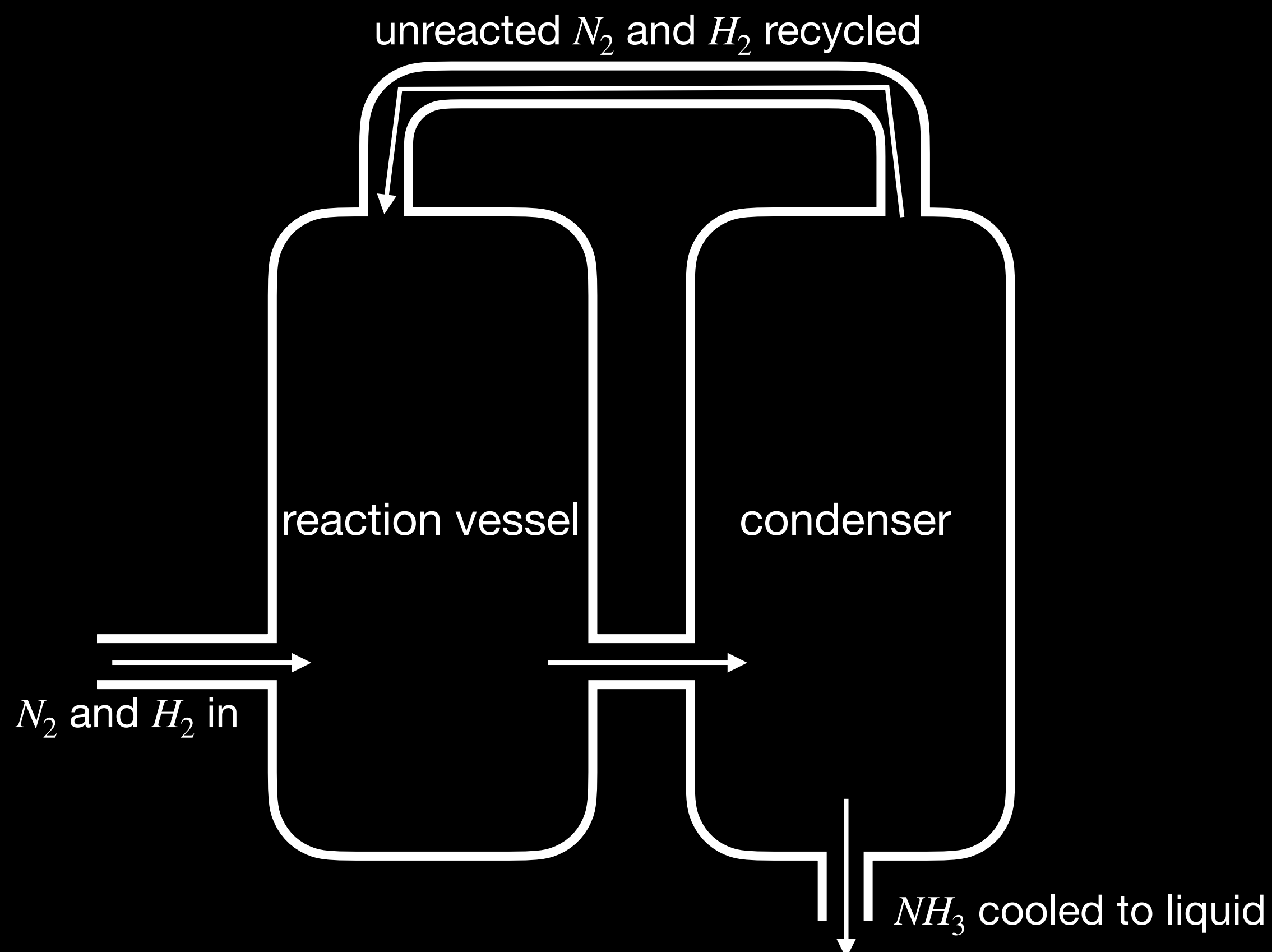


The Haber process takes in N_2 and H_2 into a 1:3 ratio (because of molar ratio) where they react in the reaction vessel, followed by being cooled in the condenser, where the reacted ammonia is cooled into a liquid, while the unreacted reagents are recycled back into the reaction vessel.



Haber Process - Setup & Conditions

ammm



The conditions for the Haber process is:

Temp has to be 450°C

Pressure has to be 200 or 250 atm (accept both)

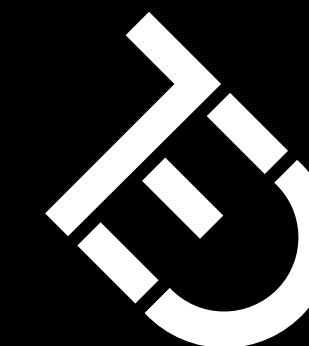
Finely divided iron (catalyst)

Ratio of $N_2 : H_2$ must be mixed in 1:3 ratio

NH_3 has to be continuously removed from system

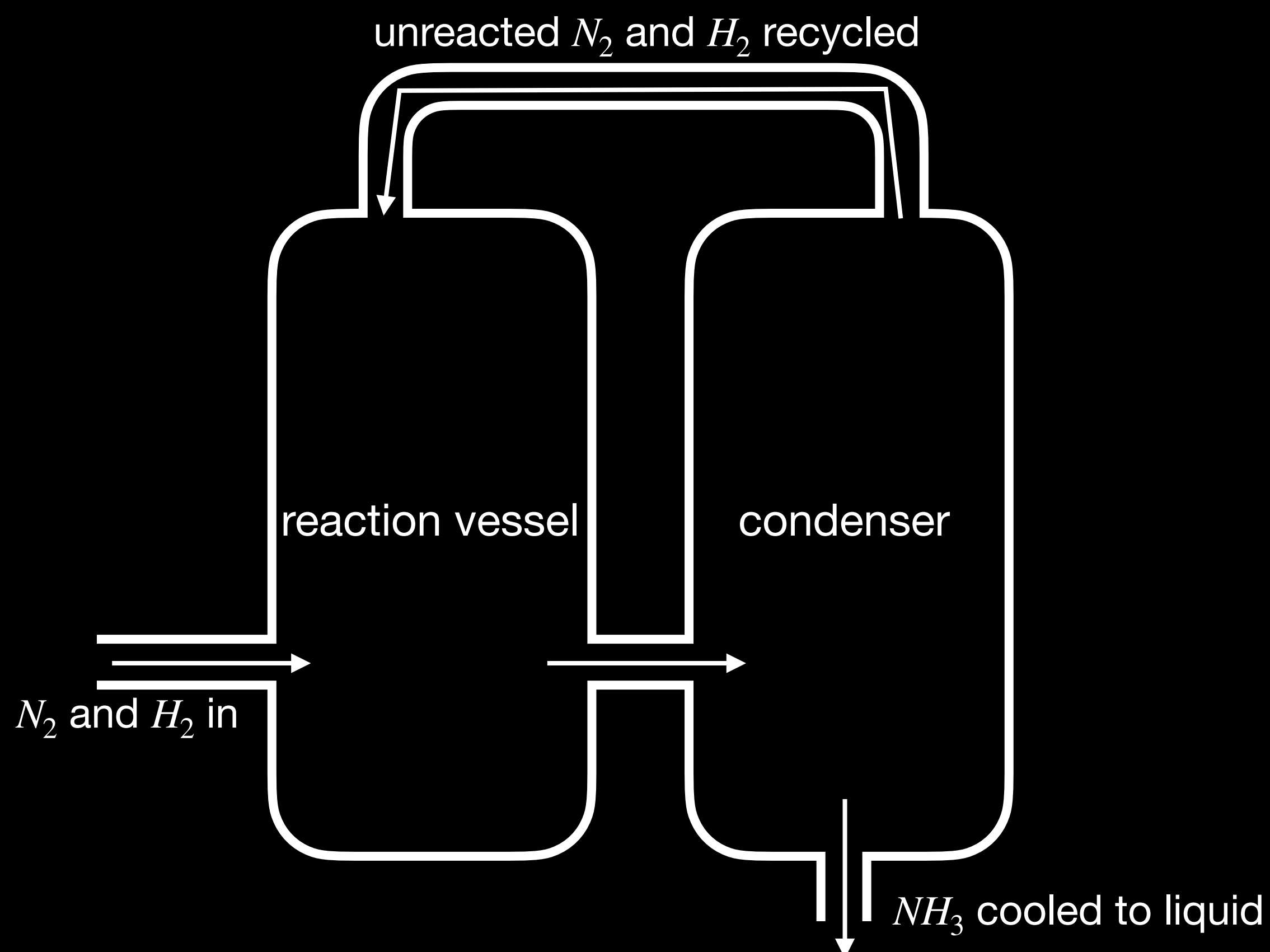
Unreacted N_2 and H_2 has to be recycled.

(NEED TO MEMORISE CONDITIONS)



Haber Process - Setup & Conditions

ammm



Trends of yield:

As temp decreases, yield increases

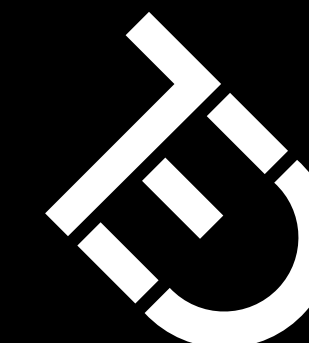
As pressure increases, yield increases

**note that yield $\neq$ rate of reaction, so increasing temp will increase rate of reaction but decrease yield

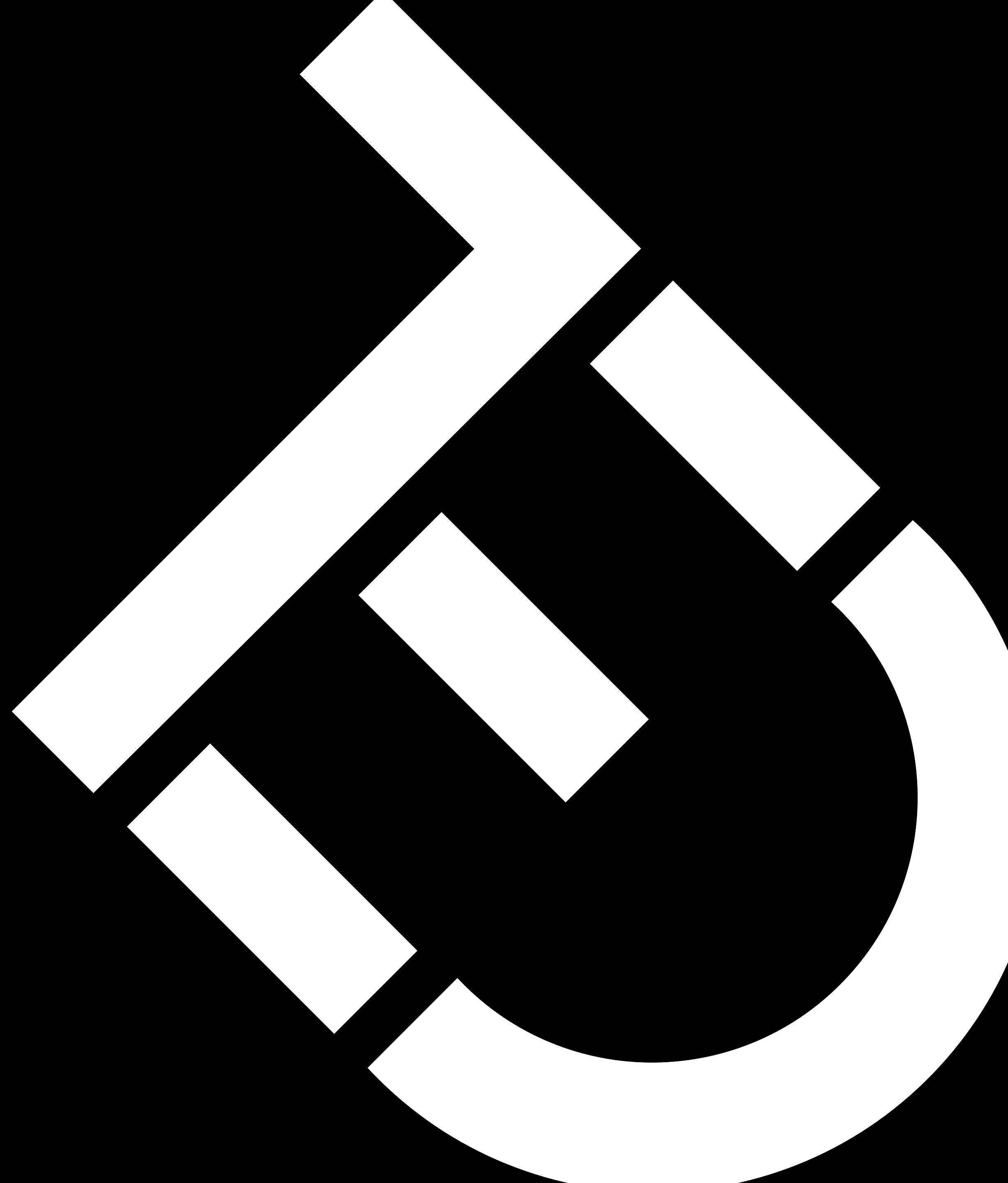
This is due to something called Le Chatlier's principle; basically when we change the initial conditions of the (reversible) reaction, the reaction will want to counteract the conditions set.

For example, as the forward reaction of producing ammonia is exothermic while the backwards reaction of decomposition is endothermic; when we set the temperature of the reaction to be lower, the reaction will want to counteract the imbalance in temp by releasing energy, thus more of the exothermic reaction will occur, increasing yield (likewise for pressure)

**this is NOT TESTED, however it is useful to know in application qns where they may give background information to help you answer qns relating to this



Redox

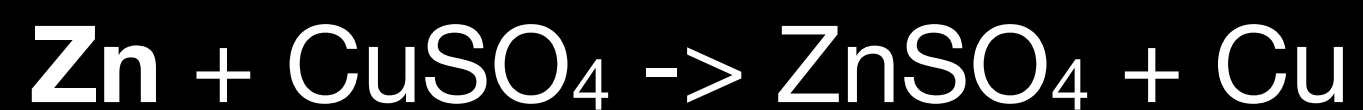
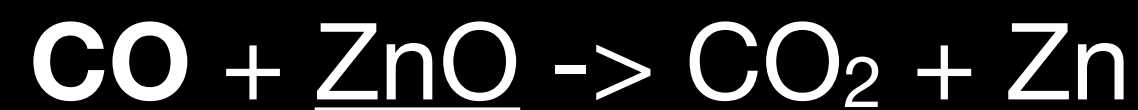


Redox

reduction and oxidation

REDuction + OXidation REDOX

examples:



bolded reactants are oxidised

underlined reactants are reduced

redox reactions are comprised of 2 processes that happen at the same time; reduction and Oxidation

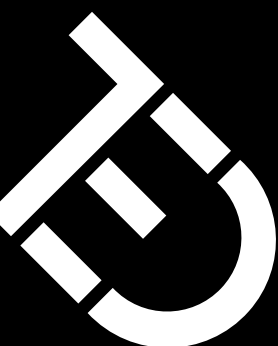
There are 4 definitions of reduction and oxidation

gain/loss of oxygen

gain/loss of hydrogen

gain/loss of electrons

change in oxidation state



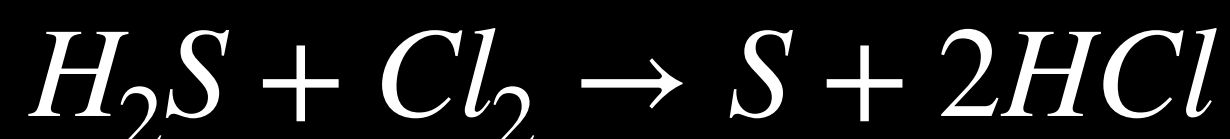
Redox Definitions: Oxygen and Hydrogen

gain lose

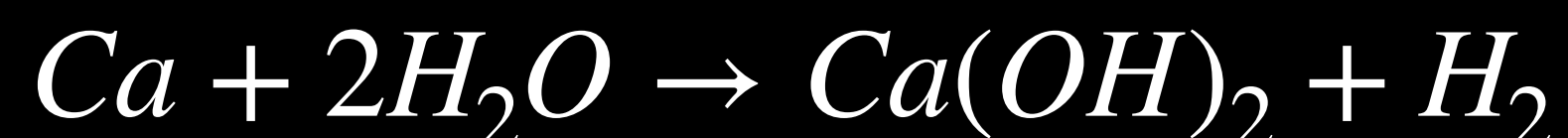
examples:



C gains 2 O and O₂ loses 2 O



H₂S loses 2 H and Cl gains H



Ca gains both O and H????

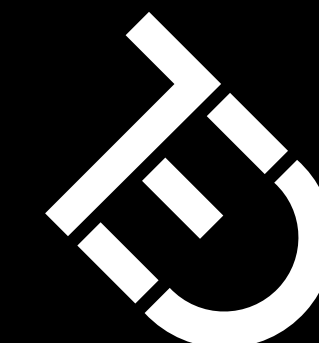
when a reactant is oxidised, it gains oxygen

when a reactant is reduced, it loses oxygen

when a reactant is oxidised, it loses hydrogen

when a reactant is reduced, it gains hydrogen

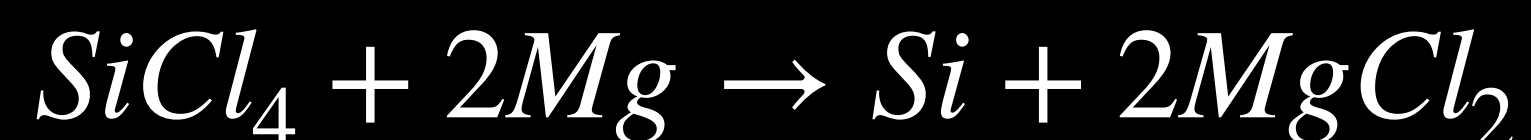
what happens when the redox reaction does not contain gain/loss of oxygen or hydrogen?



Redox Definitions: Electrons

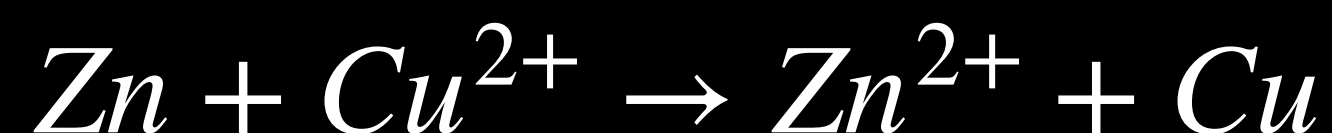
transfer

examples:



Si^{4+} ion in SiCl_4 gains 4e^- to form Si

Mg loses 2e^- to form Mg^{2+} in MgCl_2



Zn loses 2e^- to form Zn^{2+}

Cu^{2+} gains 2e^- to form Cu

Redox reactions can also be defined as a process in which electrons are transferred from one reactant to another

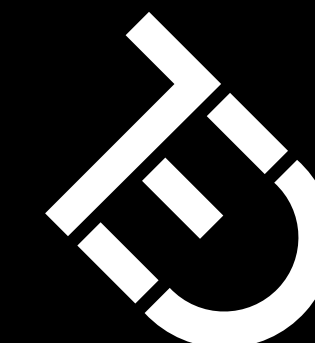
when a reactant is oxidised, they lose electrons

when a reactant is reduced, they gain electrons

For questions that ask about redox in terms of electron transfer, write as so:

As [element] gains/loses electrons to form [product], it is reduced/oxidised

however, forming ionic equations to determine whether a reaction is a redox reaction can be a long and tedious process, thus there is another way



Redox Definitions: Oxidation States

Slide subtitle

oxidation state is a way to keep track of the electrons in redox reactions

there are 8 rules for oxidation state:

1. elements that are uncombined with any other element has O.S of 0

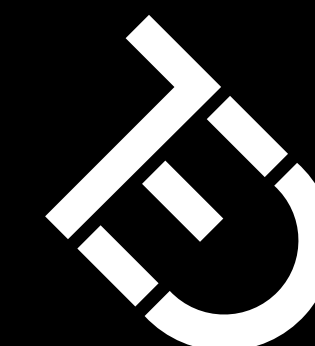
e.g Mg, Br₂, O₂, H₂ all have O.S of 0

2. in simple ions, O.S is the charge

e.g Cl⁻, Na⁺, Al³⁺ have O.S of -1, +1, 3 respectively

3. in complex ions, sum of O.S of each individual element is the charge of the ion

e.g in CO₃²⁻, O.S of C is +4, O.S of O is -2. Sum of the O.S is 4+3(-2)=-2



Redox Definitions: Oxidation States

Slide subtitle

4. O.S of hydrogen in all compounds is +1 but it is -1 in metal hydrides

e.g O.S of H in H_2O is +1 but in NaH it is -1

5. O.S of oxygen in all compounds is -2 but it is -1 in peroxides

e.g O.S of O in H_2O is -2 but in H_2O_2 it is -1

6. group I metals have O.S of +1, group II has +2, and Al has +3, in all their compounds

e.g O.S of Na in NaCl is +1

7. group VII usually have O.S of -1

8. sum of O.S of elements in compound = 0

e.g in NaOH , $+1+1-2=0$

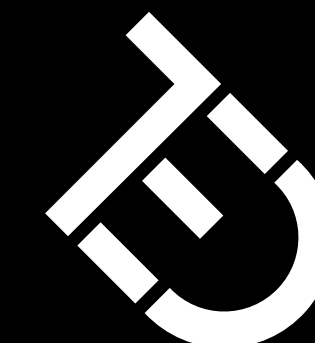
when a reactant is oxidised, O.S increases
when a reactant is reduced, O.S decreases

****three reactions that are NEVER redox:**

Neutralisation

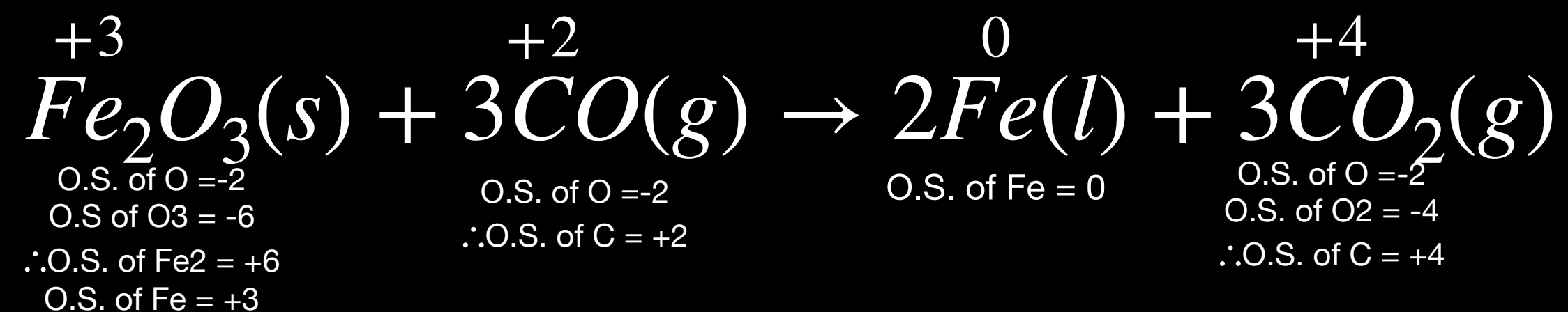
Ionic precipitation

Thermal Decomposition



Redox Definitions: Oxidation States

Slide subtitle



Show that $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$ is a redox reaction.

As the oxidation state of Fe decreases from +3 in Fe_2O_3 to 0 in Fe , it is reduced.

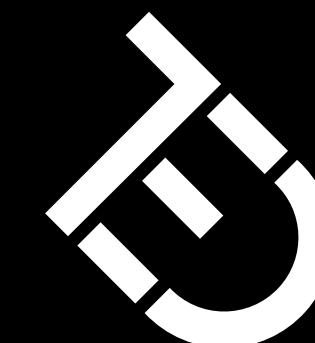
As the oxidation state of C increases from +2 in CO to +4 in CO_2 , it is oxidised

Thus, as reduction and oxidation takes place, the reaction is a redox reaction.

Write qns about O.S change as

As the oxidation state of [element] increases/decreases from [original O.S.] in [reactant] to [final O.S.] in [product], it is oxidised/reduced.

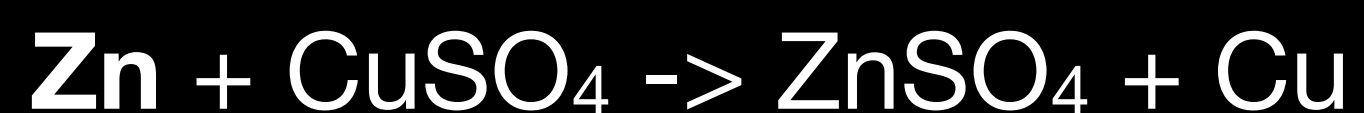
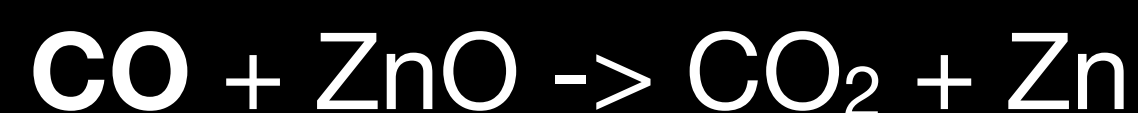
For qns asking about showing a redox reaction, show both oxidation and reduction.



Oxidising/Reducing Agents

Slide subtitle

examples:



bolded are reducing agents

underlined are oxidising agents

Acidified KMnO_4 is a commonly used oxidising agent (changes from purple to colourless)

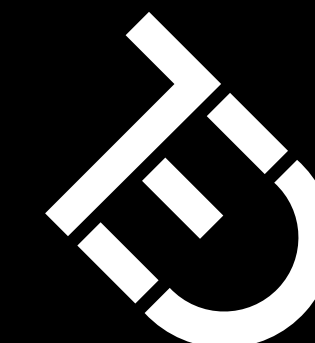
KI is a commonly used reducing agent (changes from colourless to brown)

oxidising agents are reactants that oxidise another reactant

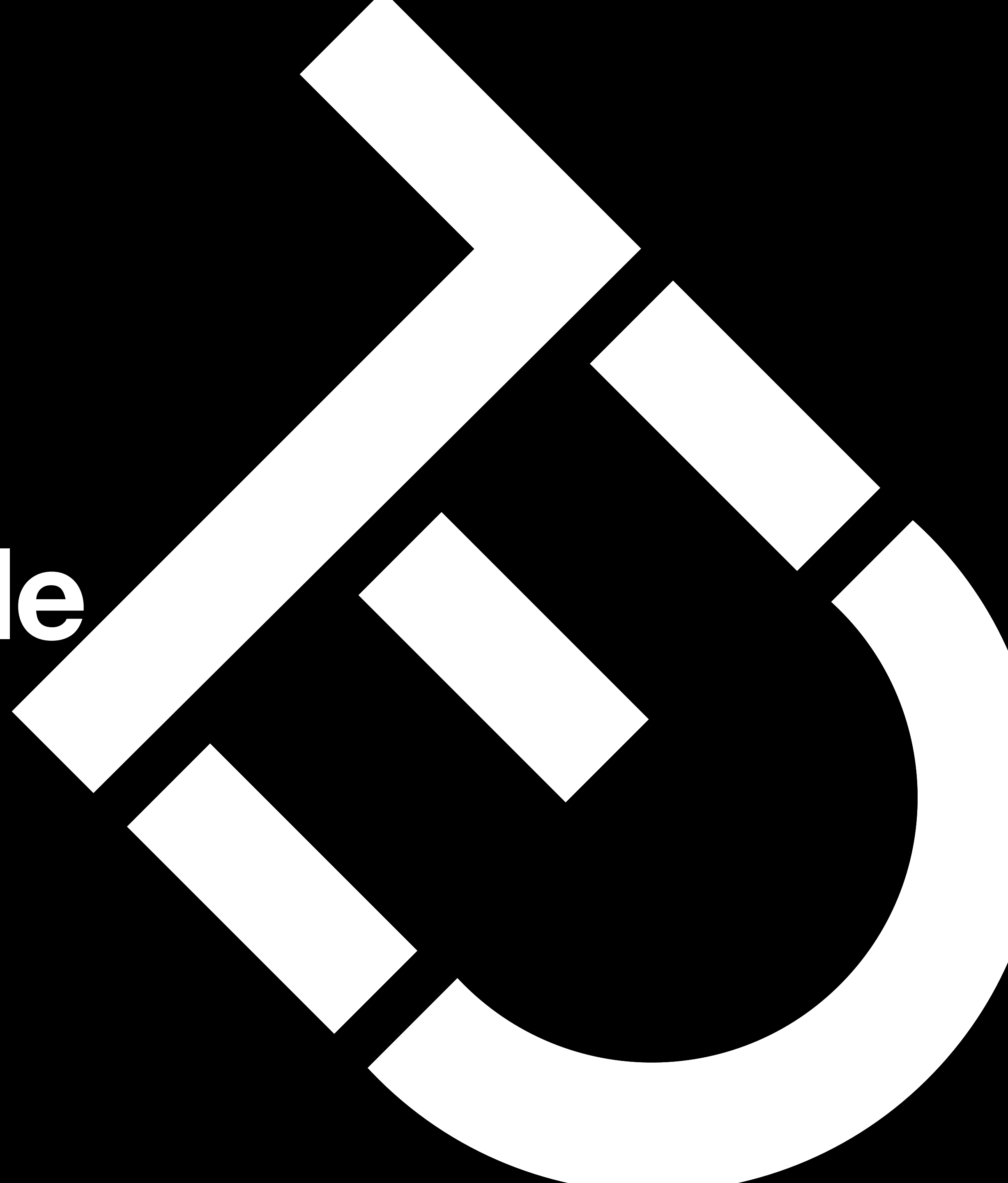
oxidising agents get reduced when oxidising another reactant

reducing agents are reactants that reduce another reactant

reducing agents get oxidised when reducing another reactant



The Periodic Table



General Periodic Trends

tabel

The Periodic Table of Elements																	
Group																	
I	II											III	IV	V	VI	VII	0
Key proton (atomic) number atomic symbol name relative atomic mass							1										2
							H										He
							hydrogen										4
3	4											5	6	7	8	9	10
Li	Be											B	C	N	O	F	Ne
lithium	beryllium											boron	carbon	nitrogen	oxygen	fluorine	neon
7	9											11	12	14	16	19	20
Na	Mg											Al	Si	P	S	Cl	Ar
sodium	magnesium											aluminium	silicon	phosphorus	sulfur	chlorine	argon
23	24											27	28	31	32	35.5	40
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
potassium	calcium	scandium	titanium	vanadium	chromium	manganese	iron	cobalt	nickel	copper	zinc	gallium	germanium	arsenic	selenium	bromine	krypton
39	40	45	48	51	52	55	56	59	59	64	65	70	73	75	79	80	84
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
rubidium	strontium	yttrium	zirconium	niobium	molybdenum	technetium	ruthenium	rhodium	palladium	silver	cadmium	indium	tin	antimony	tellurium	iodine	xenon
85	88	89	91	93	96	—	101	103	106	108	112	115	119	122	128	127	131
55	56	57 – 71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
Cs	Ba	lanthanoids	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
caesium	barium		hafnium	tantalum	tungsten	rhenium	osmium	iridium	platinum	gold	mercury	thallium	lead	bismuth	polonium	astatine	radon
133	137		178	181	184	186	190	192	195	197	201	204	207	209	—	—	—
87	88	89 – 103	104	105	106	107	108	109	110	111	112		114		116		
Fr	Ra	actinoids	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn		F		L		
francium	radium		rutherfordium	dubnium	seaborgium	bohrium	hassium	meitnerium	darmstadtium	roentgenium	copernicium		flerovium		livermorium		
—	—		—	—	—	—	—	—	—	—	—		—		—		

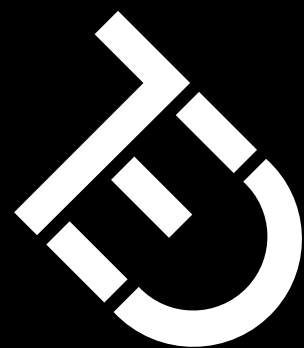
lanthanoids

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
lanthanum	cerium	praseodymium	neodymium	promethium	samarium	europtium	gadolinium	terbium	dysprosium	holmium	erbium	thulium	ytterbium	lutetium
139	140	141	144	—	150	152	157	159	163	165	167	169	173	175
89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
actinium	thorium	protactinium	uranium	neptunium	plutonium	americium	curium	berkelium	californium	einsteinium	fermium	mendelevium	nobelium	lawrencium
—	232	231	238	—	—	—	—	—	—	—	—	—	—	—

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.)

Elements are arranged in increasing atomic number, not relative atomic mass
Group number = Number of valence electrons
Period number = Number of electron shells
Group I to IV form positive charges (1+ to 3+) while Group V to VII forms negative charges (3- to 1-)



General Periodic Trends

tabel

Metallic property decreases along period (no. of valence e⁻ increases, drop in tendency to lose e⁻ for stable octet config.)

recall that metals lose electrons (which is what gives metals their properties)

Metallic property increases along Grp I (atom size increases, valence e⁻ is further from nucleus, attractive force on e⁻ decreases, less energy needed to overcome electrostatic forces, easier to lose e⁻, more metallic properties)

Non-metallic property decreases along Grp VII (atom size increases, valence e⁻ is further from nucleus, electrostatic attractive force on e⁻ decreases, harder to gain e⁻ and achieve stable octet structure)

The Periodic Table of Elements																			
Group																			
I	II											III	IV	V	VI	VII	0		
							1 H hydrogen 1											2 He helium 4	
							non-metals →												
3 Li lithium 7	4 Be beryllium 9	Key proton (atomic) number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20		
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57 – 71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89 – 103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —				

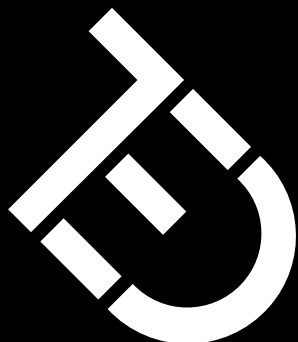
lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
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actinoids

89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —
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The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.)



Group I Elements - Alkali Metals

tabel

3 Li lithium 7
11 Na sodium 23
19 K potassium 39
37 Rb rubidium 85
55 Cs caesium 133
87 Fr francium —

Group I elements are also known as Alkali metals, as they form alkalis when reacting with water

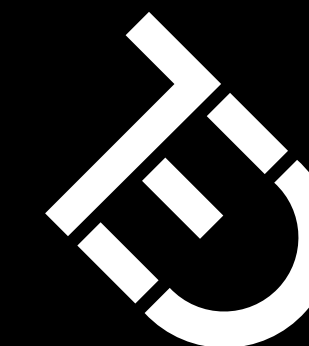
e.g $2\text{Na}(s) + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH}(aq) + \text{H}_2(g)$, or in general for a alkali metal M,
 $2\text{M}(s) + 2\text{H}_2\text{O} \rightarrow 2\text{MOH}(aq) + \text{H}_2(g)$

Physical properties

- Low densities than other metals, density increases down group (thus floats on water)
- Low melting points, melting point decreases down group
- Softer than other metals (can be cut by knife)
- Other metallic properties (electrical conductivity, ductility, malleability, etc.)

Chemical properties

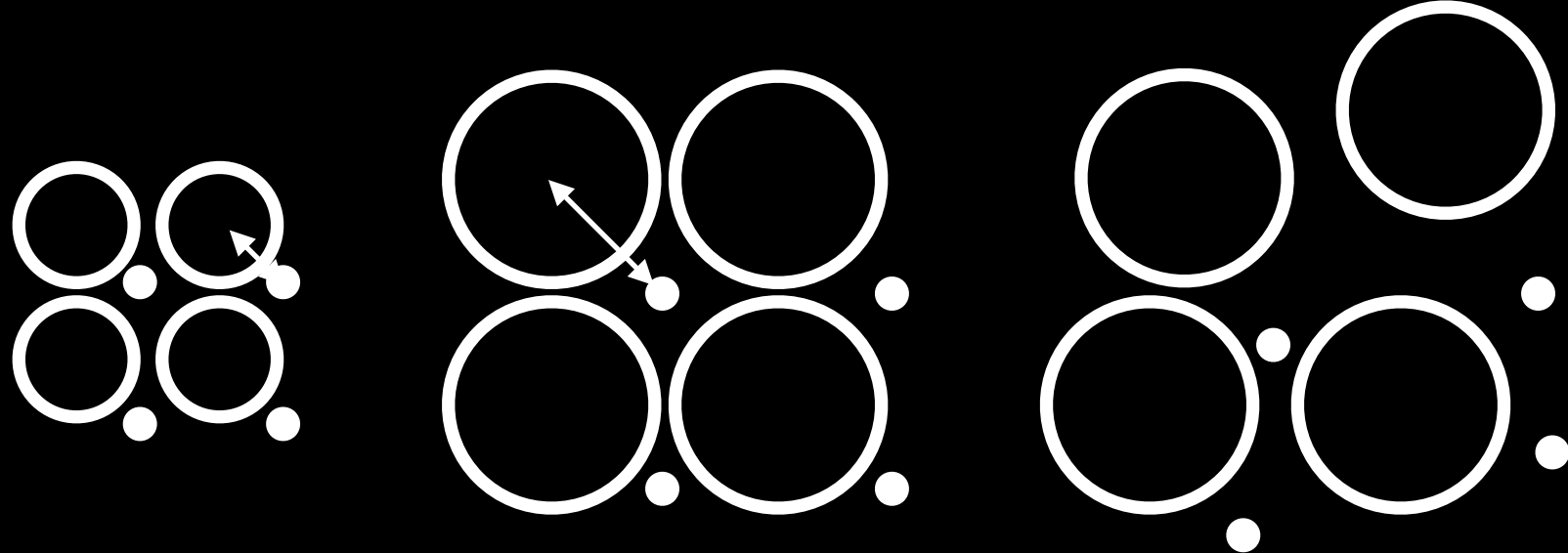
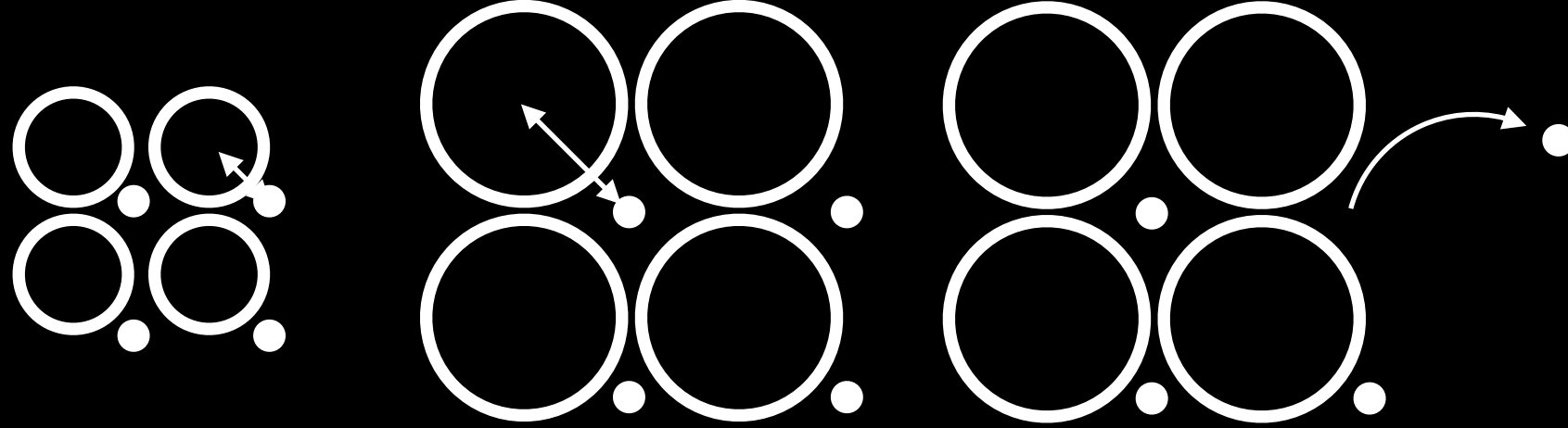
- Highly reactive, reactivity increases down group

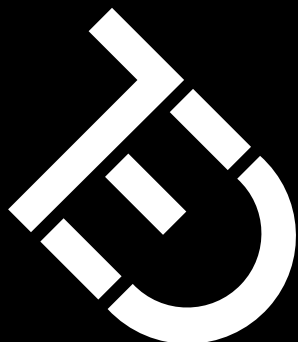


Group I Elements - Alkali Metals

tabel

3 Li lithium 7
11 Na sodium 23
19 K potassium 39
37 Rb rubidium 85
55 Cs caesium 133
87 Fr francium —

m.p decreases down group	Reactivity increases down group
As size of atom increases down group, distance between cation & e⁻ increases, electrostatic force decreases, less energy needed to overcome weak electrostatic forces, thus lower m.p 	As size of atom increases down group, distance between cation & e⁻ increases, electrostatic force decreases, less energy needed to overcome weak electrostatic forces, easier for e⁻ to be lost, more reactive 



Group VII Elements - Halogens

tabel

9 F fluorine 19
17 Cl chlorine 35.5
35 Br bromine 80
53 I iodine 127
85 At astatine —

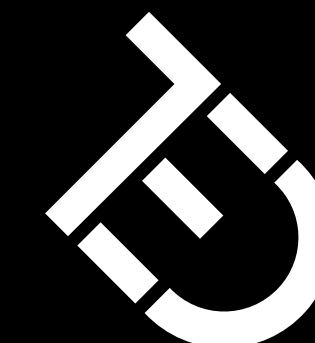
Group VII elements are known as halogens and they exist as diatomic elements.

Physical properties

- The colour of halogens darkens down the group
- Boiling point of halogens increases down group

Chemical properties

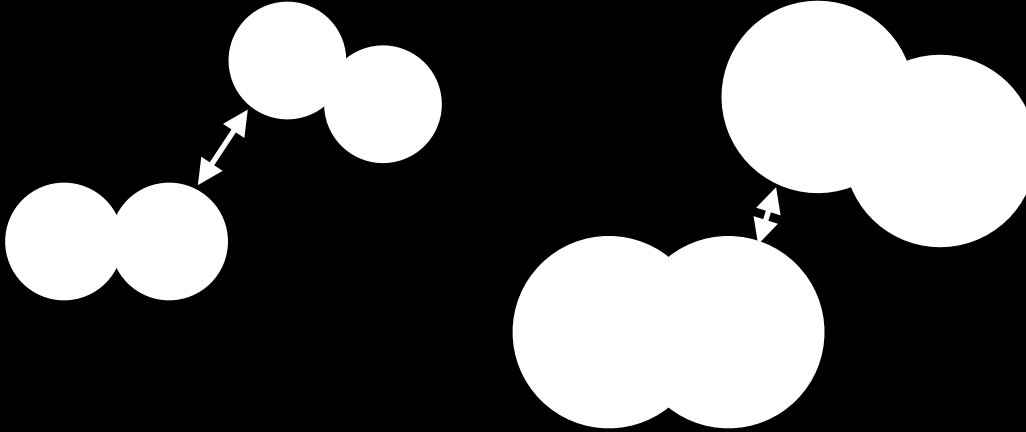
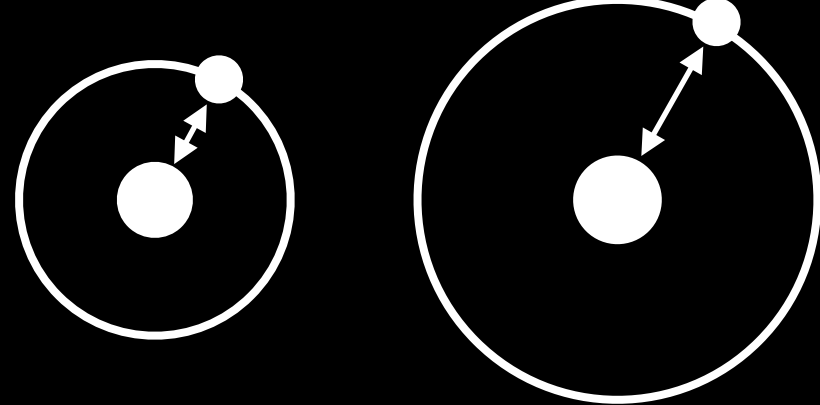
- Reacts with reactive metals $[Na + Cl \rightarrow NaCl]$ and halide solutions (reactivity decreases down group)
- (generally) acts as oxidising agents as they tend to gain electrons, thus being reduced and an oxidising agent $[Cl_2 + 2e^- \rightarrow 2Cl^-]$

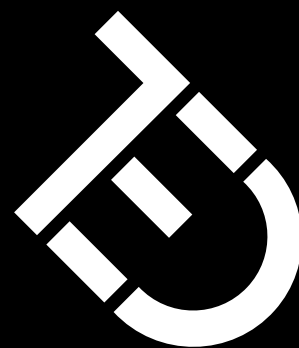


Group VII Elements - Halogens

tabel

9 F fluorine 19
17 Cl chlorine 35.5
35 Br bromine 80
53 I iodine 127
85 At astatine —

b.p increases down group	Reactivity decreases down group
As size of atom increases down group, atomic radius increases, inter-molecular forces between molecules increases, more energy needed to overcome forces, thus higher b.p. 	As size of atom increases down group, distance between nucleus & e⁻ increases, electrostatic force decreases, harder to gain e⁻, less reactive 



Group VII Elements - Halogen Displacement

tabel

less reactive ↓

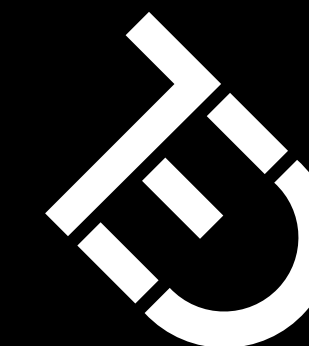
9 F fluorine 19
17 Cl chlorine 35.5
35 Br bromine 80
53 I iodine 127
85 At astatine —

A displacement reaction refers to when a more reactive element replaces a less reactive element in a compound. For example, if A is more reactive than B, then $A + BC \rightarrow AC + B$.

Halogen displacement is a displacement reaction between a halogen (usually called [halogen] water, e.g. chlorine water, except for iodine which is just iodine) and a halide solution (solution of a halide salt, e.g. KCl)

For example, as Br is less reactive than Cl, when Cl and KBr react, $[Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)]$, the Cl displaces the Br out of the solution. However, if we were to combine KCl and Br, no reaction would occur.

From the ionic equation, we get $Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$. As the oxidation state of Cl increase from 0 in Cl_2 to -1 in Cl^- , Cl is oxidised, while Br is reduced. Thus, the reaction is a redox reaction (and all displacement reactions are redox)



Group VII Elements - Halogen Displacement

tabel

most reactive halogen *F* can displace all except *KF*

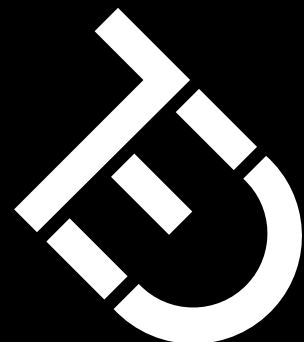
least reactive halogen *I* cannot displace any

	F_2	Cl_2	Br_2	I_2
<i>KF</i>	No reaction	No reaction	No reaction	No reaction
<i>KCl</i>	Reacts	No reaction	No reaction	No reaction
<i>KBr</i>	Reacts	Reacts	No reaction	No reaction
<i>KI</i>	Reacts	Reacts	Reacts	No reaction

Other cations will work for displacement

less reactive

9 F fluorine 19
17 Cl chlorine 35.5
35 Br bromine 80
53 I iodine 127
85 At astatine —



Group 0 Elements - Noble Gases

tabel

2 He helium 4
10 Ne neon 20
18 Ar argon 40
36 Kr krypton 84
54 Xe xenon 131
86 Rn radon —

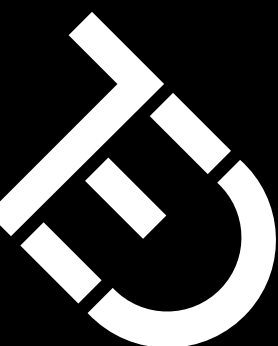
Group 0 Elements are known as Noble Gases and they exist as monoatomic elements

Physical properties

- Low boiling points (increases down the group)
- Density increases down the group

Chemical properties

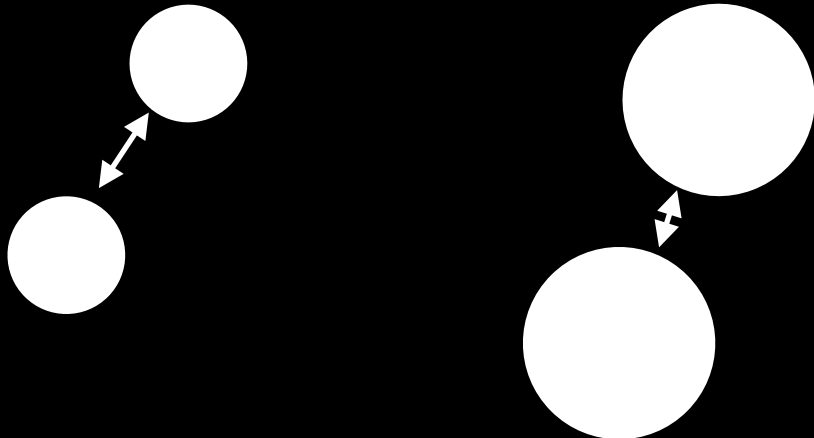
- Chemically inert due to full octet structure

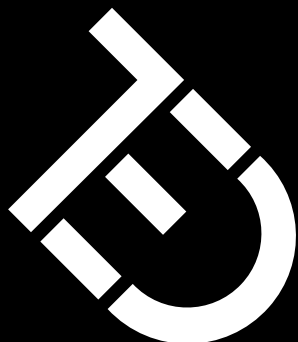


Group 0 Elements - Noble Gases

tabel

2 He helium 4
10 Ne neon 20
18 Ar argon 40
36 Kr krypton 84
54 Xe xenon 131
86 Rn radon —

b.p increases down group	Density increases down group
As size of atom increases down group, atomic radius increases, inter-molecular forces between molecules increases, more energy needed to overcome forces, thus higher b.p. (same as halogens) 	At room temp & pressure, 1 mol of gas = 24dm³. Thus, as atomic mass increases down group while volume stays the same, density increases.

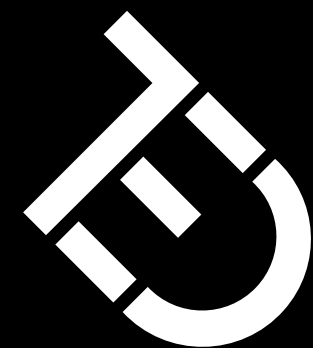


Group 0 Elements - Noble Gases

tabel

2 He helium 4
10 Ne neon 20
18 Ar argon 40
36 Kr krypton 84
54 Xe xenon 131
86 Rn radon —

Noble Gas	Uses
<i>He</i>	Balloons/weather balloons
<i>Ne</i>	Neon signs
<i>Ar</i>	Gas in light bulbs
<i>Kr</i>	Lasers for eye surgery and car headlamps
<i>Xe</i>	Lighthouse lamps, Car headlamps



Transition Elements

tabel

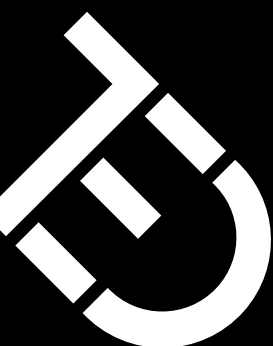
Transition elements are elements in the middle block of the Periodic Table between group II and III. All transition metals are also metals, thus they will also have characteristics of metals

Physical properties

- Hard unlike group I metals
- High melting points except for mercury
- High densities
- Other metallic properties (electrical conductivity, ductility, malleability, etc.)

Chemical properties

- Less reactive (but not inert)
- Have cations of variable charge and oxidation state
- Forms coloured compounds
- Tend to act as catalysts (e.g. iron in Haber Process) as able to hold more than 8 e^- in their valence shells, making it easier to take in and give away e^- from other molecules



Metals



Metal Properties

yskt

High m.p. and b.p. (except for Grp I metals, Gallium and Mercury)

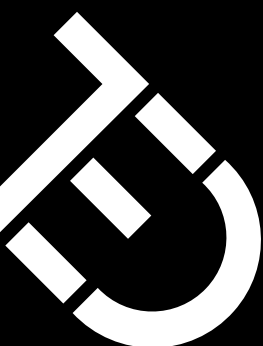
Malleable & ductile

Good conductors of heat & electricity

High density (except for Grp I metals)

Shiny

Refer to structure &
properties of materials



Alloys

aloy

Alloys are a mixture of a metal and one or more elements (that may or may not be a metal)

Alloys are generally stronger than their constituent metals as differently sized particles of different elements —> disrupts regular layers of metals —> more difficult for layers to slide over one another —> stronger

examples of alloys:

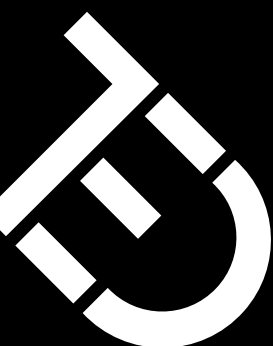
rose gold - 75% gold, 25% copper

sterling silver - 92.5% silver, 7.5% copper

brass - 60% copper, 40% zinc

bronze - 88% copper, 12% tin

Mainly need to know brass and bronze, but just the constituent metals, NOT the percentages



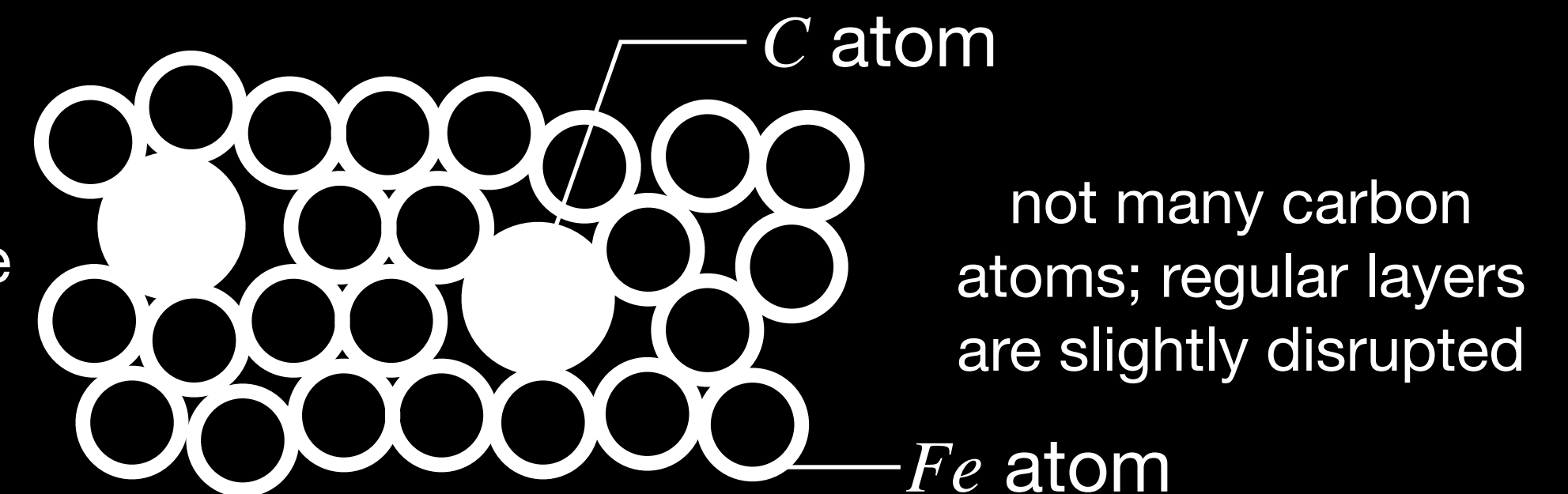
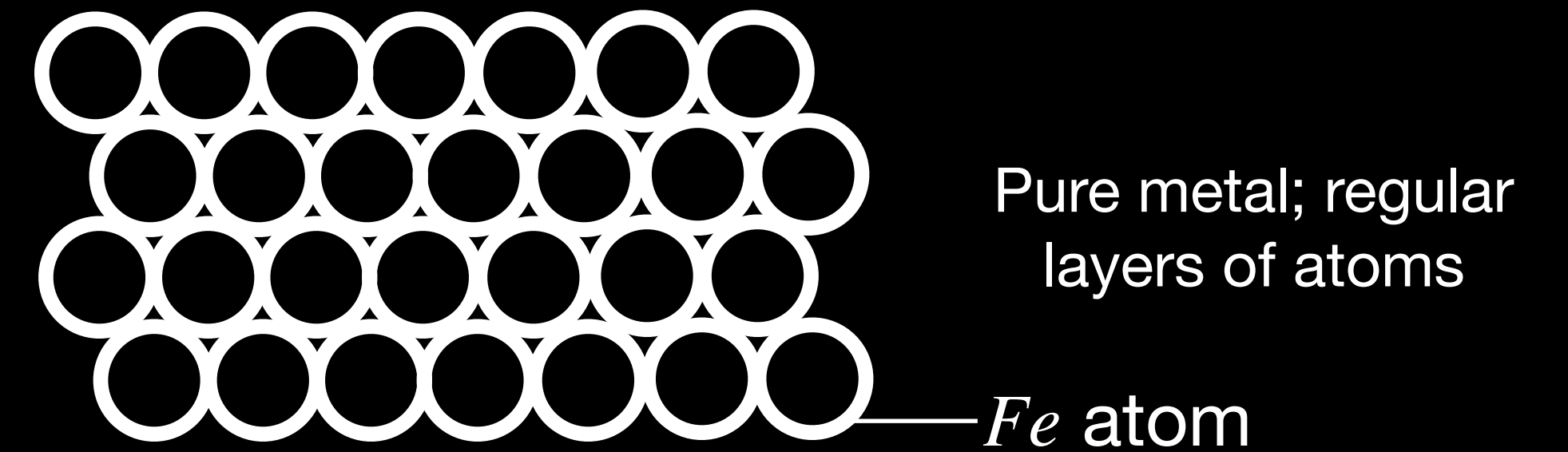
Alloys - Steel

iron maiden reference

Steel is an alloy of iron, combining mainly iron and carbon together in different proportions for different properties.

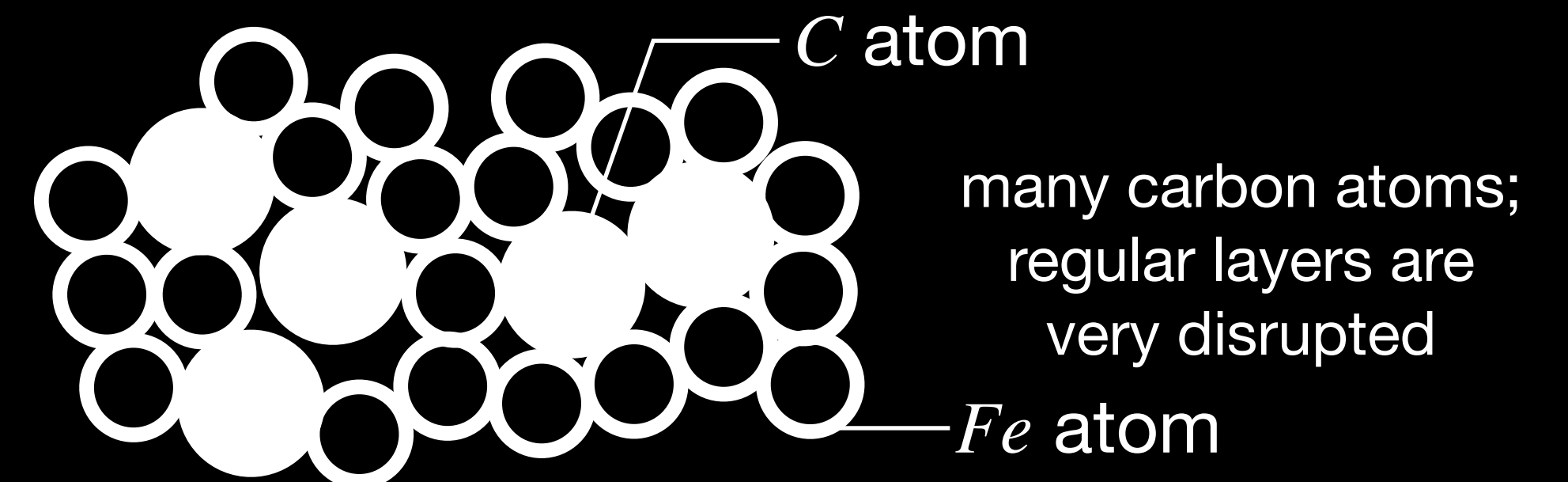
low carbon:

low percentage of carbon: ~0.30% (no need to memorise percentages)
ductile and malleable, soft (to memorise: recall how metals are malleable and ductile, so a small amount of carbon does not affect its ability for layers to slide too much)



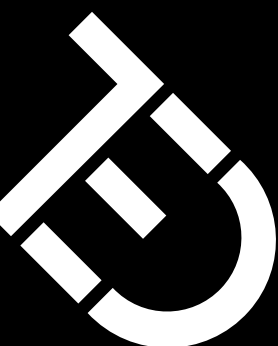
high carbon:

high percentage of carbon: ~1.5% (no need to memorise percentages)
strong but brittle (to memorise: remember that more carbon will disrupt the layers which makes it stronger, but more force is needed to slide the layers, making it more brittle)



stainless:

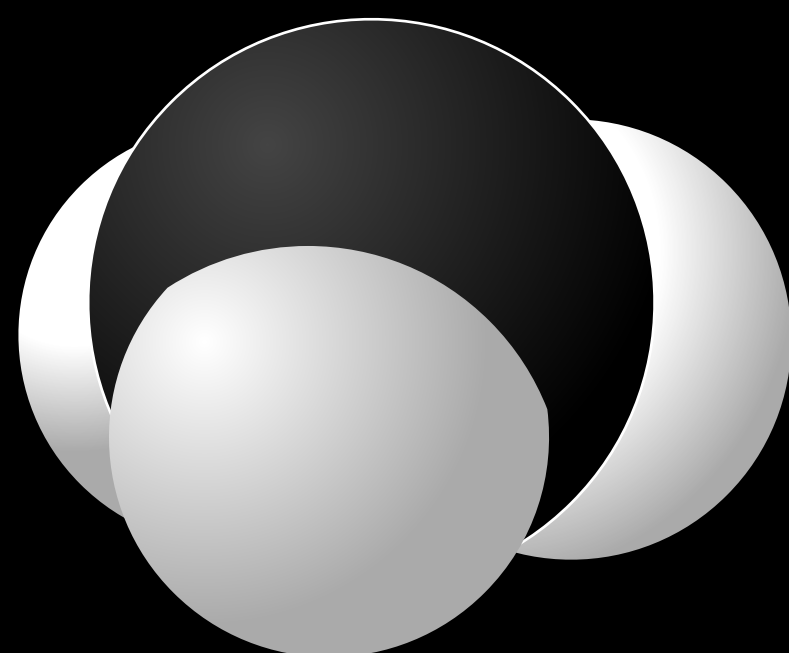
contains other metals such as chromium, nickel, and molybdenum
resistant to rusting



Models

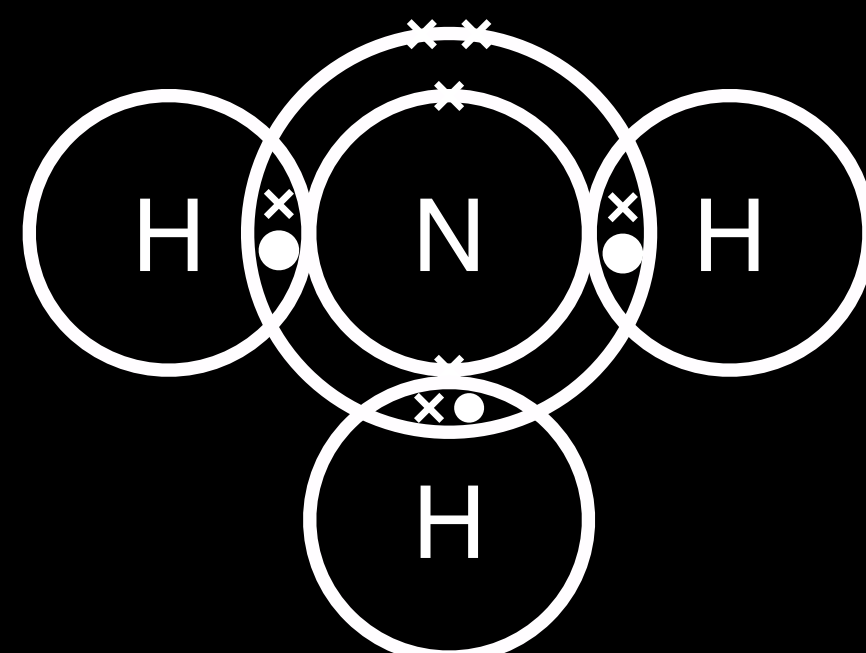
steal models

Some questions may ask about the accuracy of molecular models, or about why a molecular model is inaccurate. Questions may include actual data of molecules for you to compare with the model, or may ask you to simply state the inaccuracies. Shown are common inaccuracies of models:

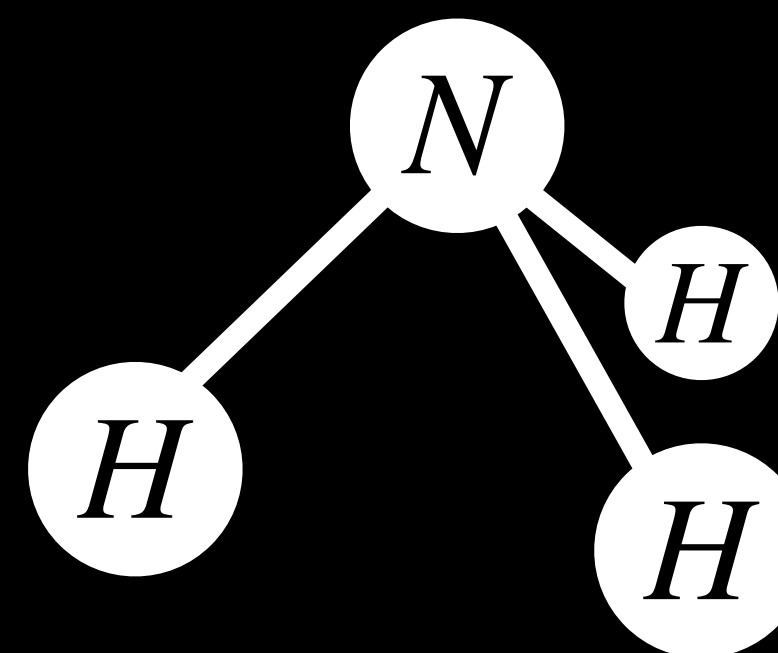


space filling model

Does not show length of bonds

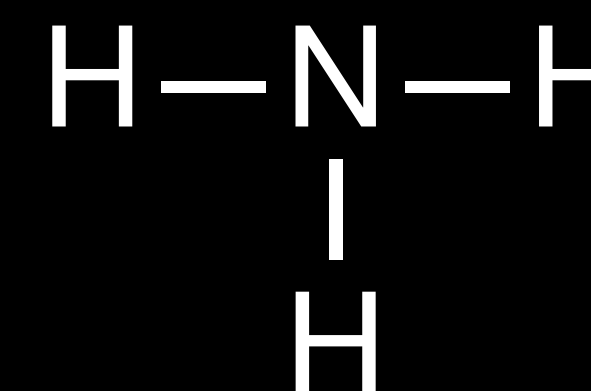


dot and cross model



ball and stick model

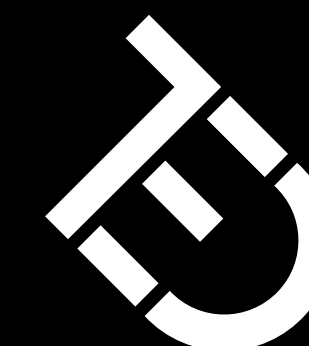
Does not show accurate atomic radius (maybe all atoms are same sized)/ does not show accurate bond length (bonds may be inaccurate)



skeletal model

Does not accurate size of atoms

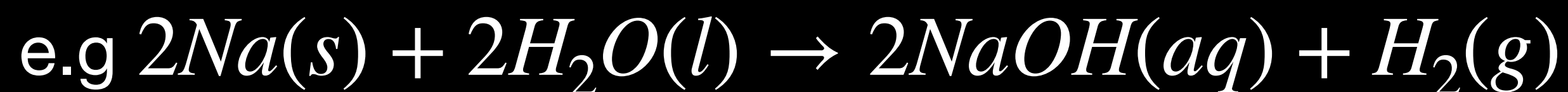
Note that you do not need to memorise the name of these models, they will usually provide labels if they test this, just use logic to see what data is not present/misrepresented in the model (usually inaccurate atom size/bond length)



Metal Reactions

retraction

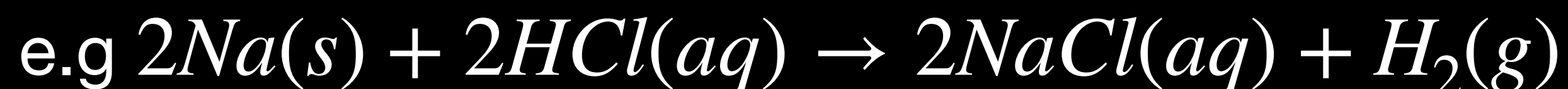
metal + cold water $\rightarrow$ metal hydroxide + hydrogen gas



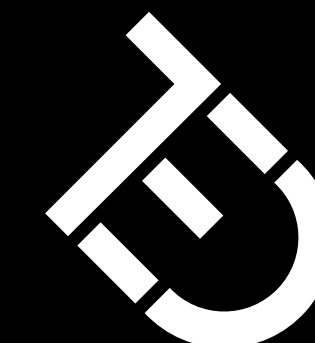
Metal + steam $\rightarrow$ metal oxide + hydrogen gas



Metal + acid $\rightarrow$ salt + hydrogen gas



All main metal reactions have hydrogen gas produced, which for reactive enough metals, may cause the hydrogen gas to ignite and cause flames to occur.

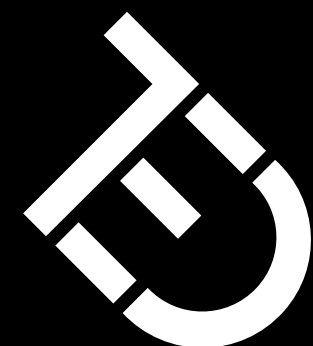


Metal Reactivity Series

toomany

Potassium	Please	Please
Sodium	Stop	Stop
Calcium	Calling	Calling
Magnesium	Me	Me
Aluminium	A	A
Carbon	Crazy	Cute
Zinc	Zebra	Zebra
Iron	I	I
Tin	Truly	Too
Lead	Love	Like
Hydrogen	Hairy	Having
Copper	Cats	Coloured
Silver	Sitting	Stripes
Gold	Gently	Gerald

You MUST memorise the metal reactivity series, which you can memorise by mnemonics or with logic



Metal Reactivity Series

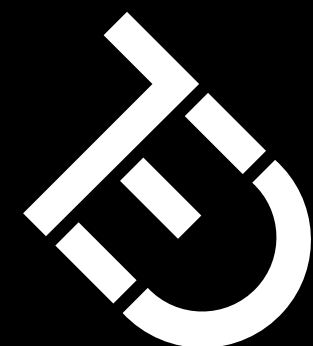
toomany

Metal	Reacts w/ cold water?	Reacts w/ steam?	Reacts w/ acid?
Potassium	✓	✓	✓
Sodium	✓	✓	✓
Calcium	✓	✓	✓
Magnesium	✓	✓	✓
Aluminium	✓, however Al ₂ O ₃ layer from oxidation usually forms on Al which prevents reaction	✓	✓
Carbon			
Zinc		✓	✓
Iron		✓	✓
Tin		✓	✓
Lead		✓	✓ (very slowly)
Hydrogen			
Copper			
Silver			
Gold			

You should also know which metals take part in the metal reactions

Steam and acids react with more metals than cold water as they either are gases which allows it to react with the metal or dissociate H⁺ ions which are easier to react with than cold water.

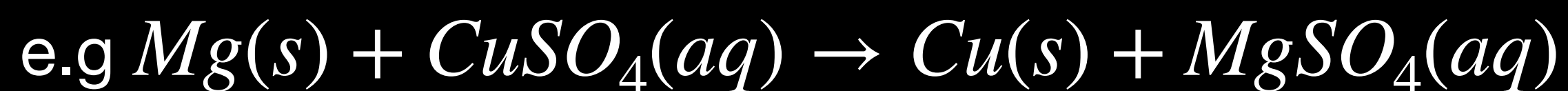
Copper, Silver and Gold do NOT partake in any reactions



Metal Displacement

md

a process where a more reactive metal replaces a less reactive metal in its compound



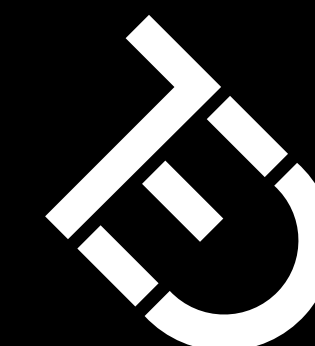
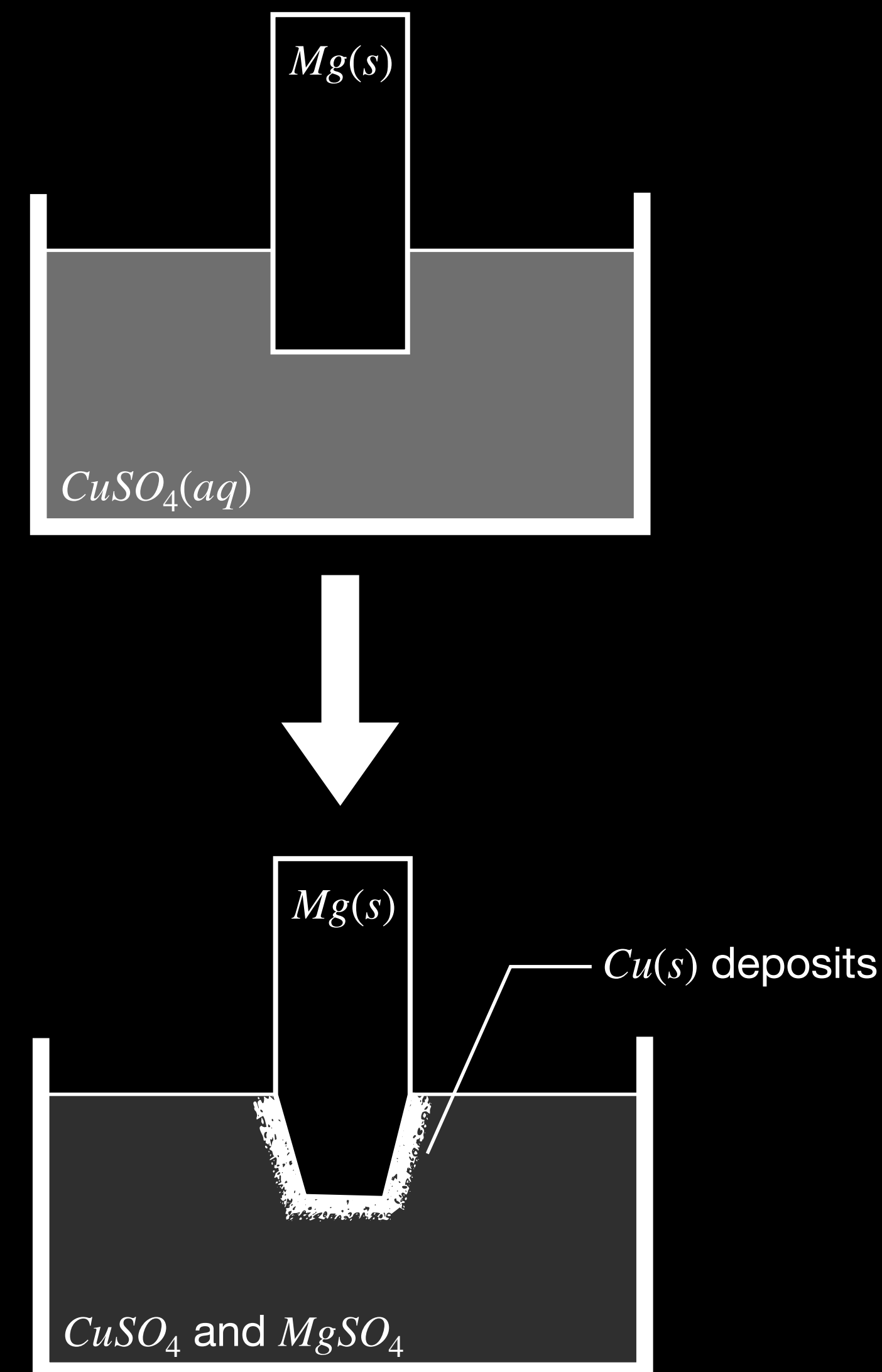
In a metal displacement reaction, you will observe:

Size of metal decreasing

New solid deposits on more reactive metal (describe colour of solid if asked about observations, e.g. copper deposits are red-brown)

Colour of solution may change (e.g. for CuSO_4 , the blue colour will fade.)

Effervescence may be observed (but this is because of the metal reacting with the water in the solution, and NOT due to the metal displacement)



Metal Carbonate Decomposition

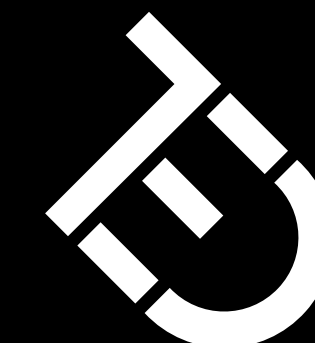
thermal decomposition of metal compound

When exposed to heat, metal carbonates will be broken down into metal oxides and carbon dioxide: [metal carbonate $\rightarrow$ metal oxide + $CO_2(g)$], e.g. $MgCO_3 \rightarrow MgO + CO_2$]

compounds with more reactive metals tend to thermally decompose only at very high temperatures and vice versa for less reactive metals

this is because more reactive metals form more stable compounds which form stronger bonds which require more energy to break apart and decompose

**note that potassium and sodium carbonate do not decompose as their constituent metals are too reactive



Metal Extraction

lo

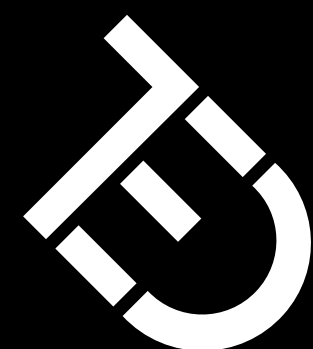
metals usually react with water/oxygen in their surroundings
therefore when we mine them, the pure metals need to be extracted
differently depending on the reactivity of the metal

Metal	Extraction method
Potassium	Electrolysis of molten metal electrolyte
Sodium	
Calcium	
Magnesium	
Aluminium	
Carbon	
Zinc	Reduction of ore under heat by carbon
Iron	
Tin	
Lead	
Hydrogen	
Copper	Reduction of ore under heat by hydrogen
Silver	
Gold	Usually found as pure

can be used for all metals but only
used for metals more reactive than
carbon because expensive

Reduction under heat by carbon:
used for metals less reactive than carbon
metal oxide + carbon -> metal + carbon
dioxide
e.g $FeO(s) + C(s) \rightarrow Fe(s) + CO(g)$

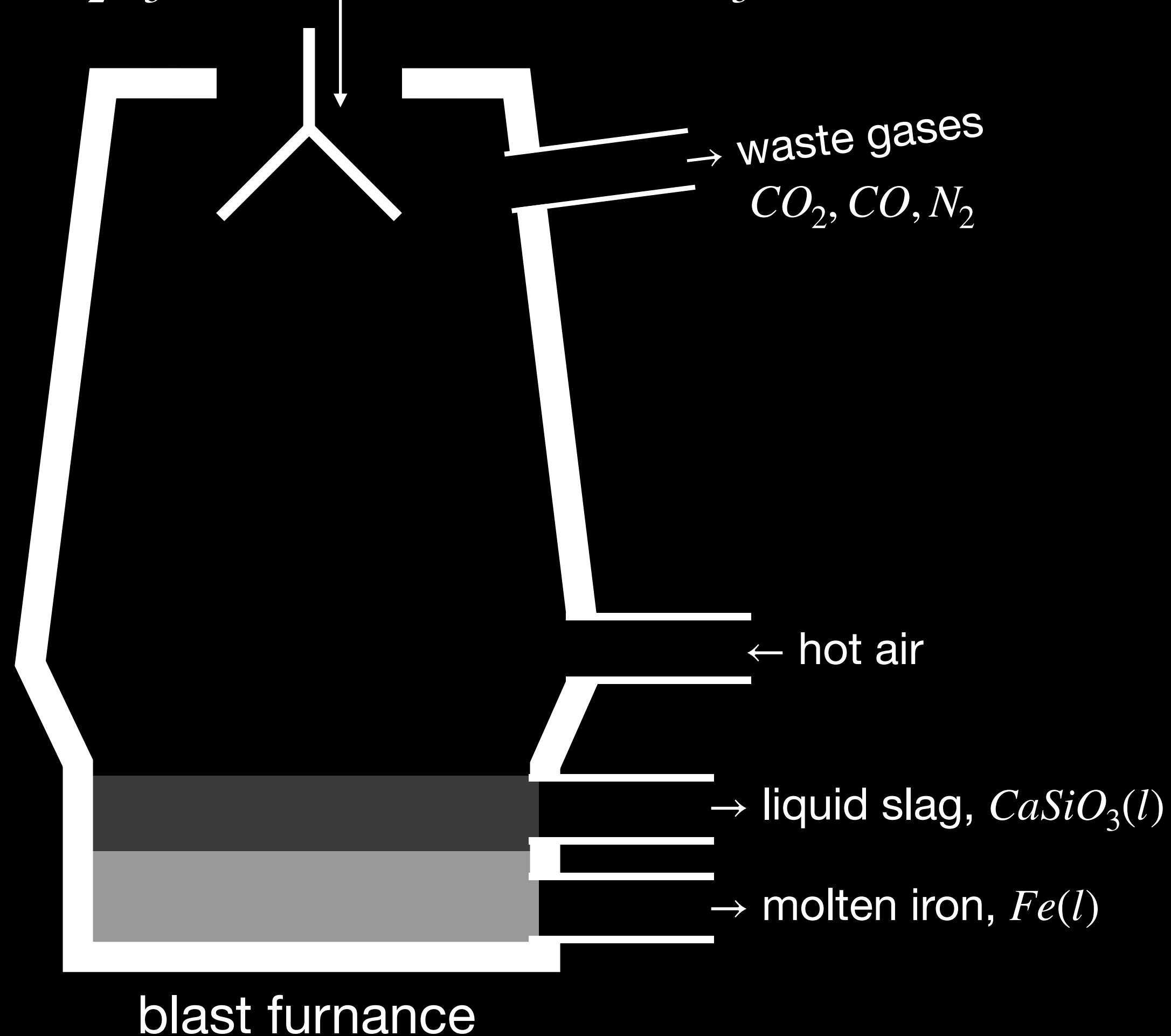
Reduction under heat by hydrogen:
used for metals less reactive than hydrogen
metal oxide + hydrogen -> metal + water
e.g $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(g)$, C is reduced.
**NOTE: Lead can also be reduced by hydrogen to obtain lead
even though it is more reactive than hydrogen



Metal Extraction - Blast Furnace

me

Iron ore (Fe_2O_3), coke (C), limestone ($CaCO_3$)



The blast furnace is an industrial process for producing iron, which uses the following substances:

Intake:

Coke [C]

Limestone [$CaCO_3$]

Iron Ore/Haematite ore [Fe_2O_3] with mainly sand impurities [SiO_2]

Hot air, which contains nitrogen [N_2] and oxygen [O_2]

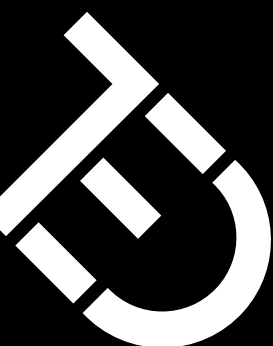
Outtake:

Molten iron [$Fe(l)$]

Liquid Slag [$CaSiO_3(l)$]

Waste Gases of carbon dioxide, carbon monoxide, and nitrogen. [CO_2, CO, N_2]

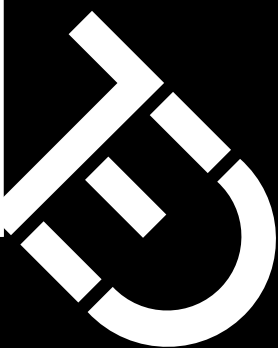
(CO_2 and CO are excess gases from the reactions and N_2 is a gas from the hot air that does not partake in the reaction)



Metal Extraction - Blast Furnace

me

Reaction	Type of reaction	Purpose & Description of reaction	Exo/Endo
$C(s) + O_2(g) \rightarrow CO_2(g)$	Redox	To produce CO from coke [carbon] and oxygen from the hot air by oxidising coke	EXO
$\rightarrow CO_2(g) + C(s) \rightarrow 2CO(g)$			ENDO
$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$	Redox	Iron ore is reduced by carbon in CO to form molten iron, where the CO2 produced can be reused in forming more CO.	EXO
$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$	Decomposition	To decompose limestone into CO2 (to form more CO) and to form CaO	ENDO
$SiO_2(s) + CaO(s) \rightarrow CaSiO_3(l)$ acidic oxide as non-metal basic oxide as metal	Neutralisation (basic+acidic oxide)	CaO formed from decomp of CaCO3 reacts with sand impurities in the molten iron to form slag, which can be removed easily	EXO



Corrosion & Rust

no way rust like rust in peace

Corrosion is a redox reaction in which any metal is oxidised by surroundings
water and oxygen are required for corrosion

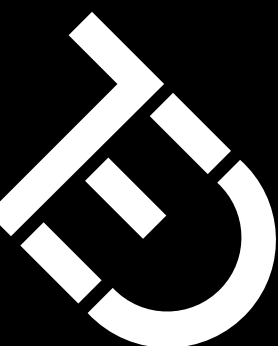
rusting refers to the corrosion of ONLY IRON in which hydrated iron(III) oxide is formed

equation for rusting:



where x is an arbitrary number of H₂O

salts and acids are catalysts for rusting as they provide mobile ions to increase rate of electron transfer



Corrosion & Rust - Prevention

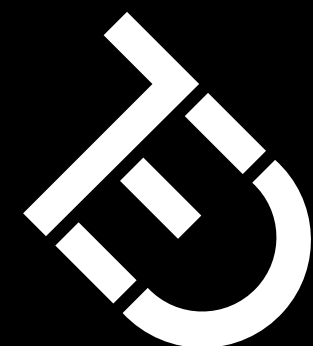
no way rust like rust in peace

Shown are the methods to prevent the rusting of iron

general method	examples	how does it prevent	uses
surface protection	coat with a layer of paint, less reactive metal, oil, plastic tube etc	prevent contact with oxygen and water	cars, machines, bridges, ships
sacrificial protection	coat with layer of more reactive metal e.g zinc	zinc is more reactive and thus will corrode preferentially than iron	kitchen sink
alloys	use alloys like stainless steel	stainless steel contains Cr, Ni, Mn which forms protective oxide layer around iron to prevent contact with water and oxygen	cutlery, surgical equipment, sink

Coating iron with zinc is called galvanisation, which has both benefits of sacrificial protection and surface protection

Note that sacrificial protection will work as long as the two metals are touching/ if there is a wire connecting the two metals as when Zn corrodes, the electrons lost allows the corroded Fe³⁺ to reduce back to Fe



Recycling Metals

using old metal instead of new metal

Advantages:

conserve limited resources of metals

less land use

land used for landfill and mining

more environmentally friendly than mining

less pollution and habitat destruction due to mining

less waste in landfills

lower cost than extracting new metal

saves energy

conserve fossil fuel

produce less waste

Disadvantages:

possible air pollution

extracting lead from car battery releases pollutants during smelting

if waste product from extraction is not disposed properly, it can pollute land and water

cost of recycling cheap metals is higher than just mining more

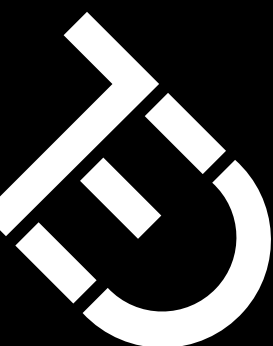
businesses may not think it's worthwhile

high transport cost for scrap metal

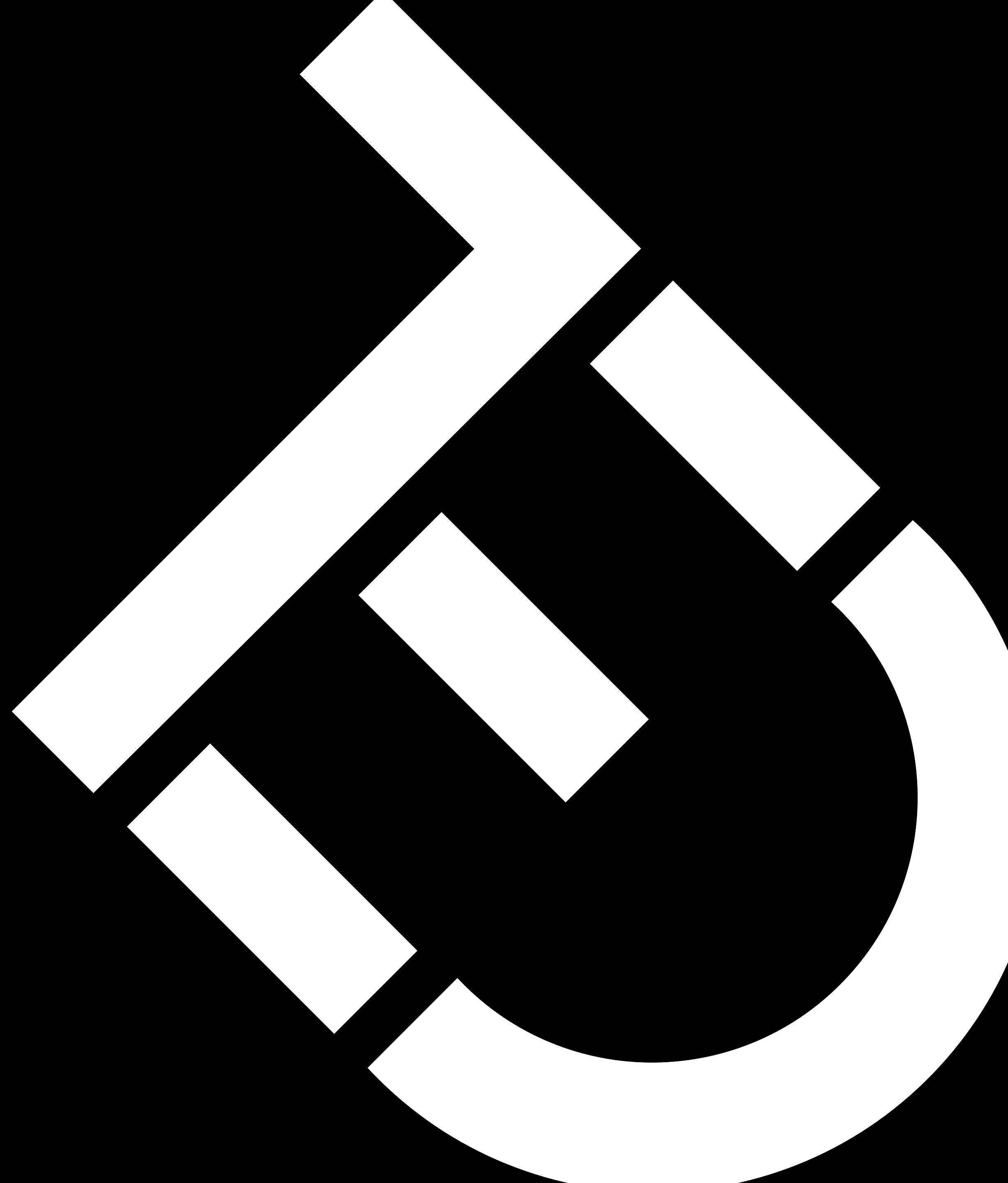
difficult to persuade people to deposit used material into recycling container

time and effort to practice recycling as a way of life

Remember at least 2 advantages/disadvantages for qns when they ask you to state about why recycling is good/bad (most of the reasons are quite intuitive though)



Electrolysis



Simple Cells

bio?????

Made up of two electrodes (one more reactive metal and one less reactive metal) and electrolyte (ionic solution, e.g. $\text{NaCl}(aq)$ or $\text{HCl}(aq)$)

Electrons flow from more reactive metal to less reactive metal

At the more reactive metal:

Electrons are lost to form metal cations (**from the electrode**)

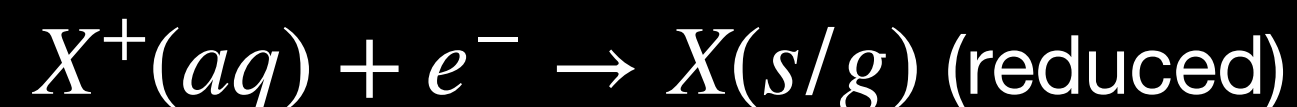


As oxidised, more reactive electrode called ANODE [AN OX]

More reactive metal is also negative electrode

At the less reactive metal:

Electrons are gained to reduce cations **in the electrolyte** (e.g. H^+)

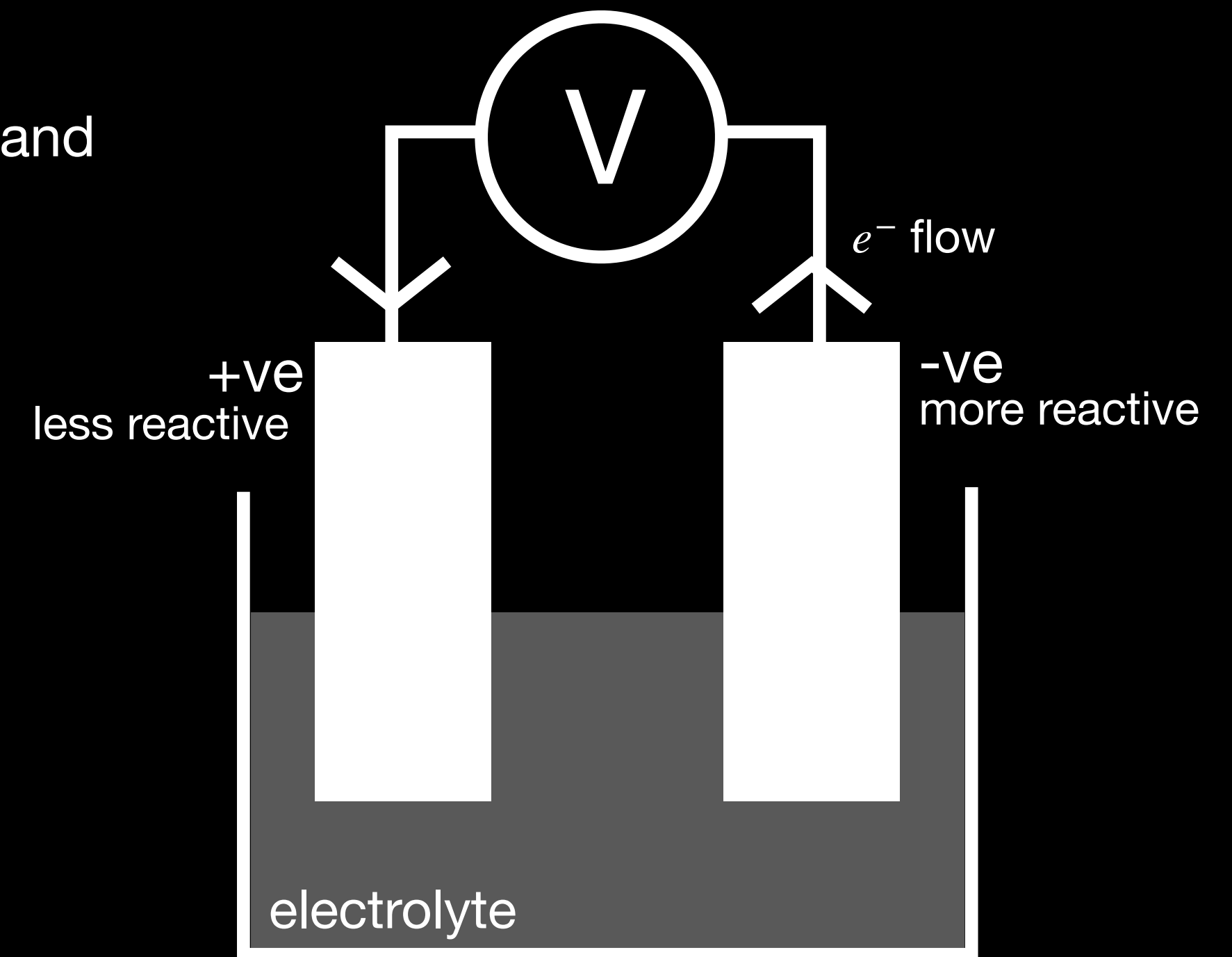


As reduced, less reactive electrode called CATHODE [RED CAT]

Less reactive metal is also positive electrode

Current is generated by flow of electrons $\rightarrow$ voltage produced

Voltage is dependent of difference in metal's reactivity (higher difference, higher voltage)

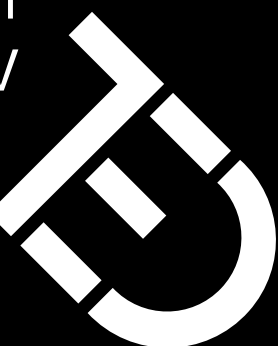


magnesium	—	—	—	
aluminium				
zinc	—	—	—	
iron				
copper	—	—	—	

0.78V

2.72V

If voltage between Mg and Cu is 2.72, and voltage between Zn and Cu is 0.78V, voltage between Mg and Zn is $2.72 - 0.78 = 1.94\text{V}$



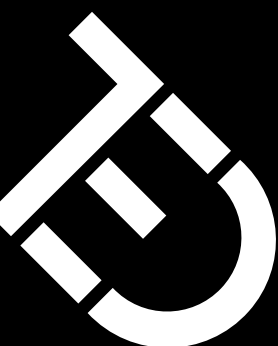
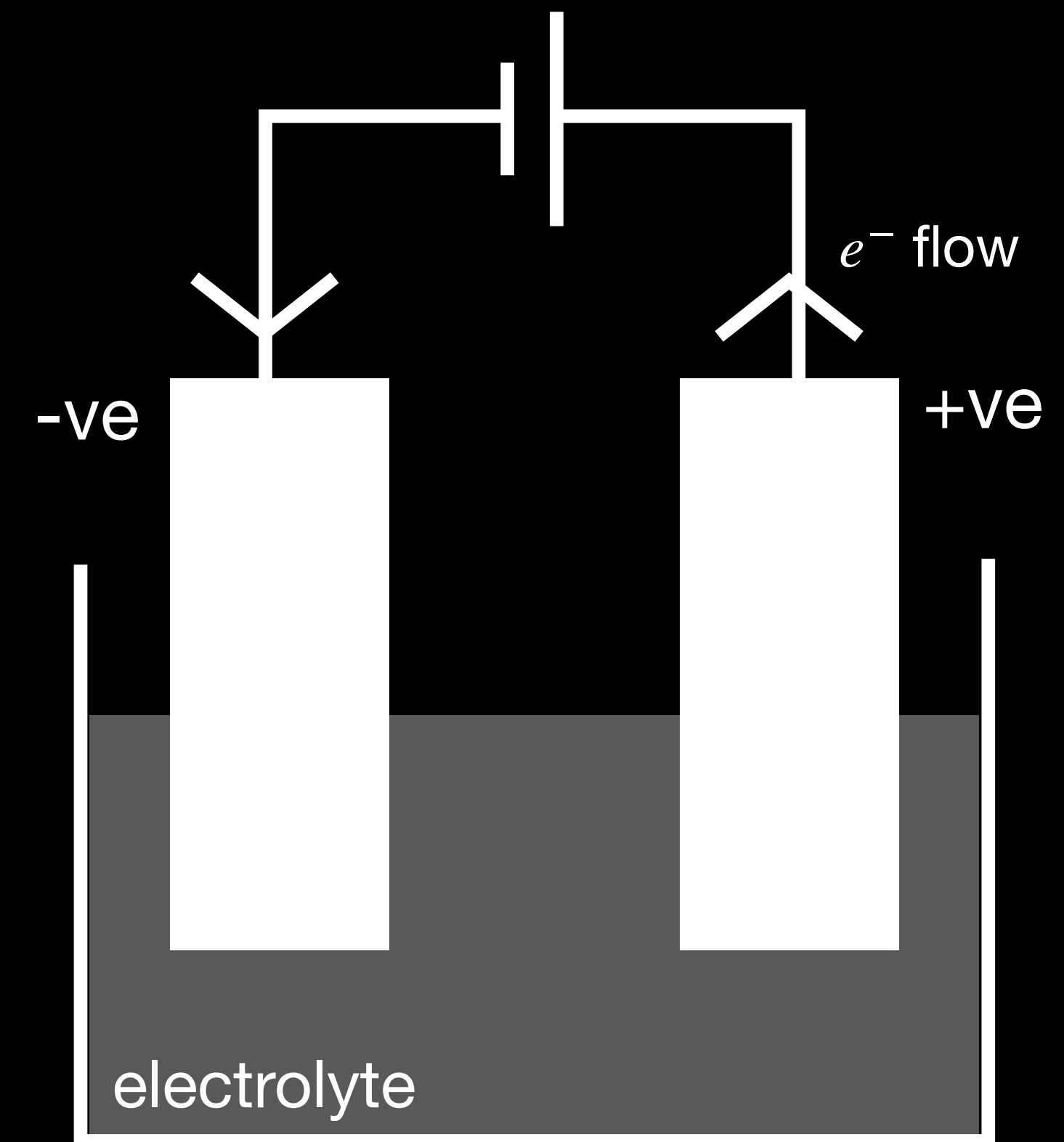
Electrolytic Cells

e

Made up of two electrodes and electrolyte with a power source

Usually used to decompose a compound(the electrolyte)

Identity of electrolyte and electrodes affects type of reaction



Molten Ionic Substances

m

Setup: graphite electrodes with molten $PbBr_2$ as electrolyte

As molten, ions present are Pb^{2+} and Br^- .

At the positive electrode (connected to positive end of battery):

Br^- ions are attracted to the electrode; discharged to form Br_2 gas

$2Br^-(l) \rightarrow Br_2(g) + 2e^-$ (lose electrons, oxidised)

As oxidised, positive electrode is ANODE

At the negative electrode (connected to negative end of battery): Pb^{2+}

ions are attracted to the electrode; discharged to form liquid Pb

metal(heat applied melts Pb metal)

$Pb^{2+}(l) + 2e^- \rightarrow Pb(l)$ (gain electrons, reduced)

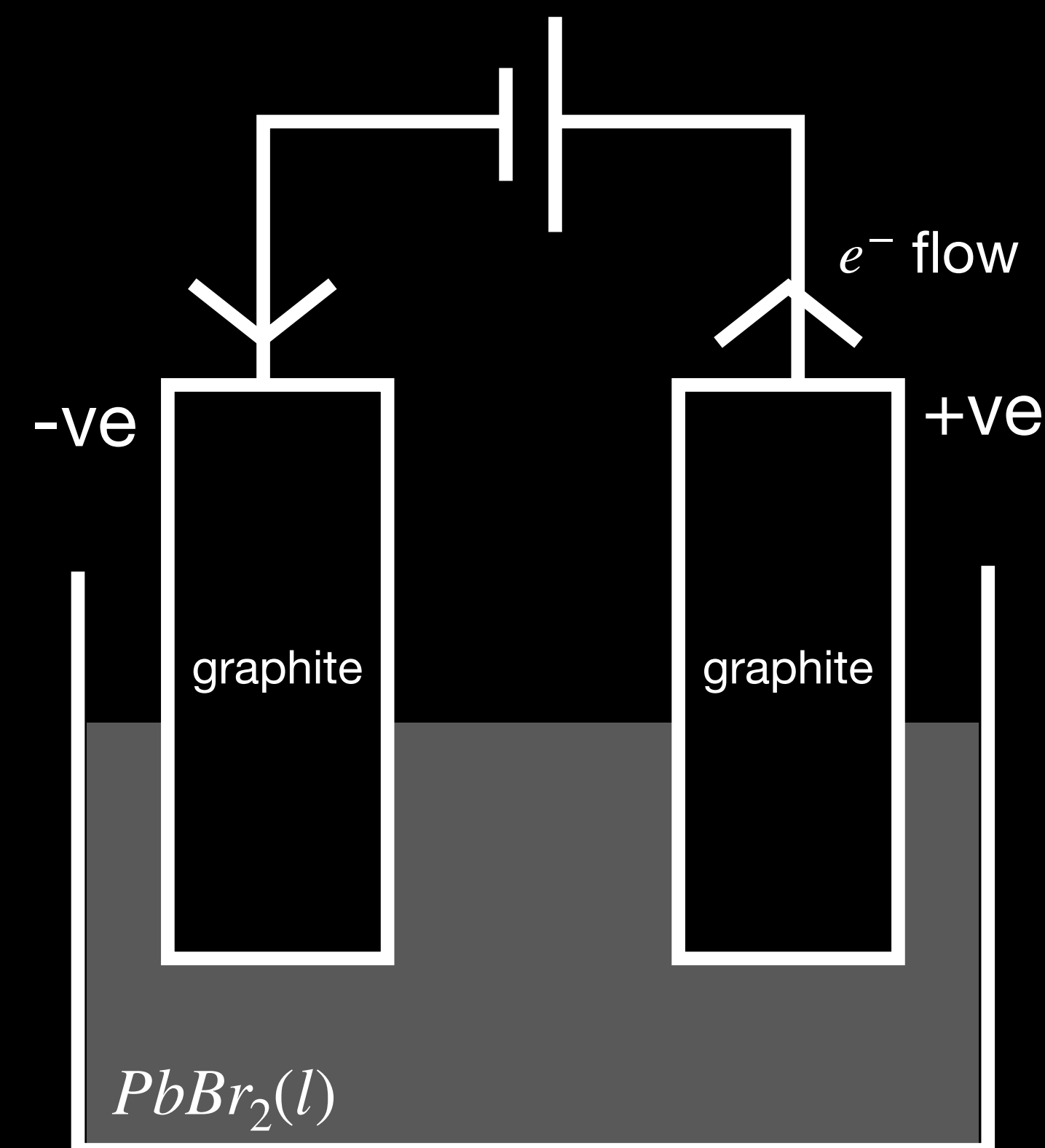
As reduced, negative electrode is CATHODE

Full ionic eqn: $Pb^{2+}(l) + 2Br^-(l) \rightarrow Br_2(g) + Pb(l)$

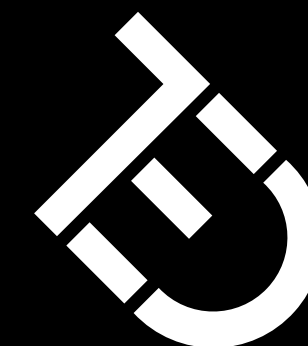
PRODUCTS FORMED: Bromine gas and liquid Pb metal

All ions are liquid, not aqueous

graphite used as inert and can still conduct electricity
does not react with electrolyte
(another example is platinum)



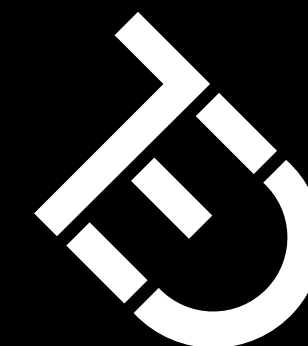
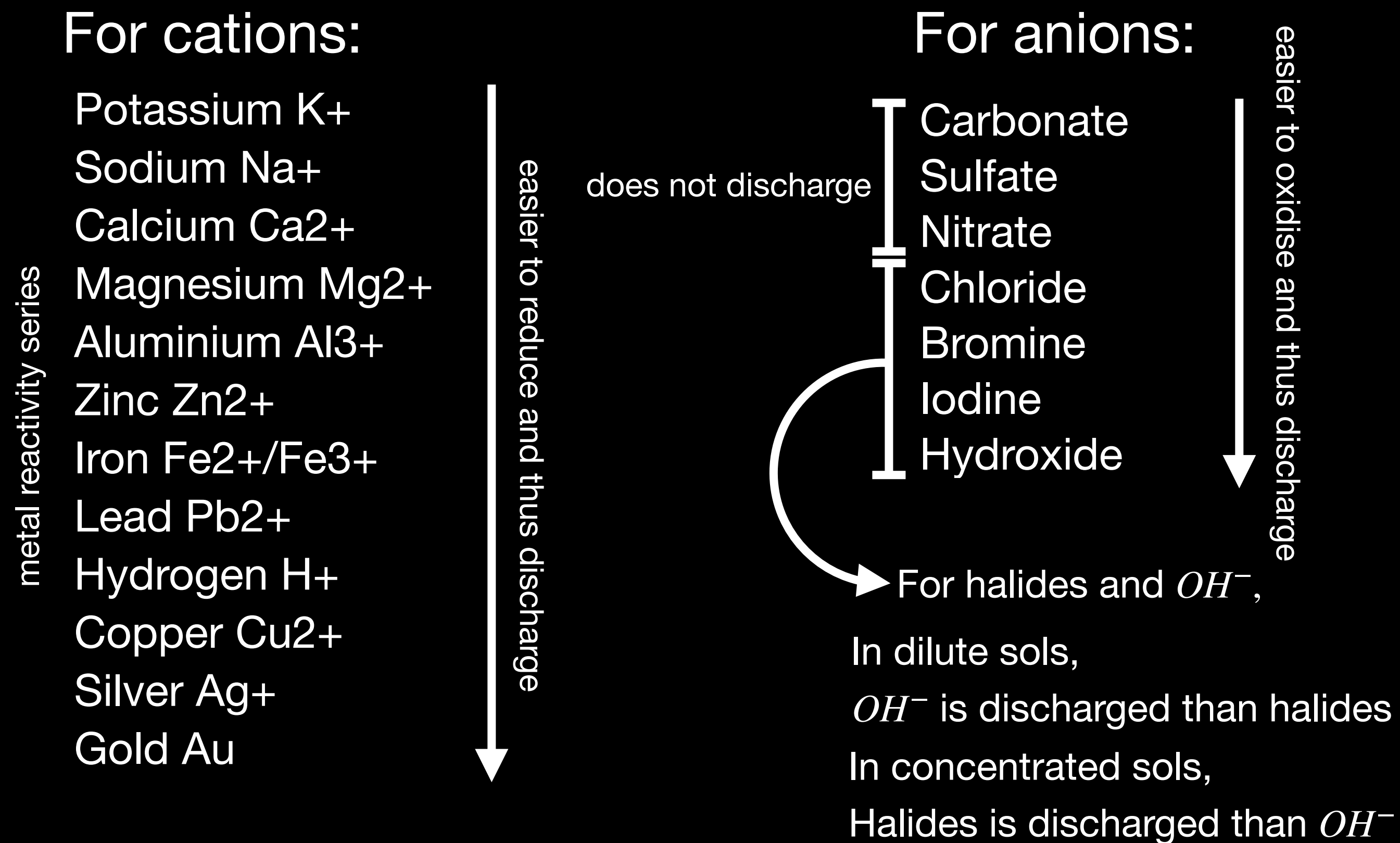
heat applied, as ionic has high melting point



Aqueous Ionic Substances - Selective Discharge

S

As aqueous solutions contain ions from the compound and the water, not all of the ions will be oxidised/reduced. Thus, some ions will be selectively discharged:



Aqueous Ionic Substances

aq

Setup: graphite electrodes with dilute aqueous NaCl(aq) as electrolyte

Ions present: Na^+ , Cl^- , H^+ , OH^-

At the positive electrode/anode:

Both Cl^- and OH^- are attracted to electrode, however as dilute, OH^- is discharged.

$4\text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{aq}) + 4\text{e}^-$ (loses electrons, oxidised) **take note of eqn as commonly tested

As OH^- ions lost, higher concentration of H^+ ions near anode, thus more acidic at electrode

At the negative electrode/cathode:

Both Na^+ and H^+ are attracted, however only H^+ is discharged

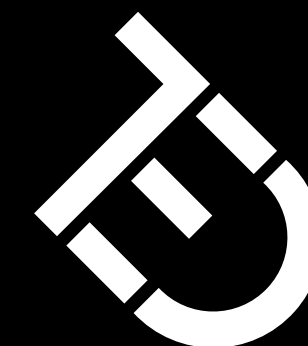
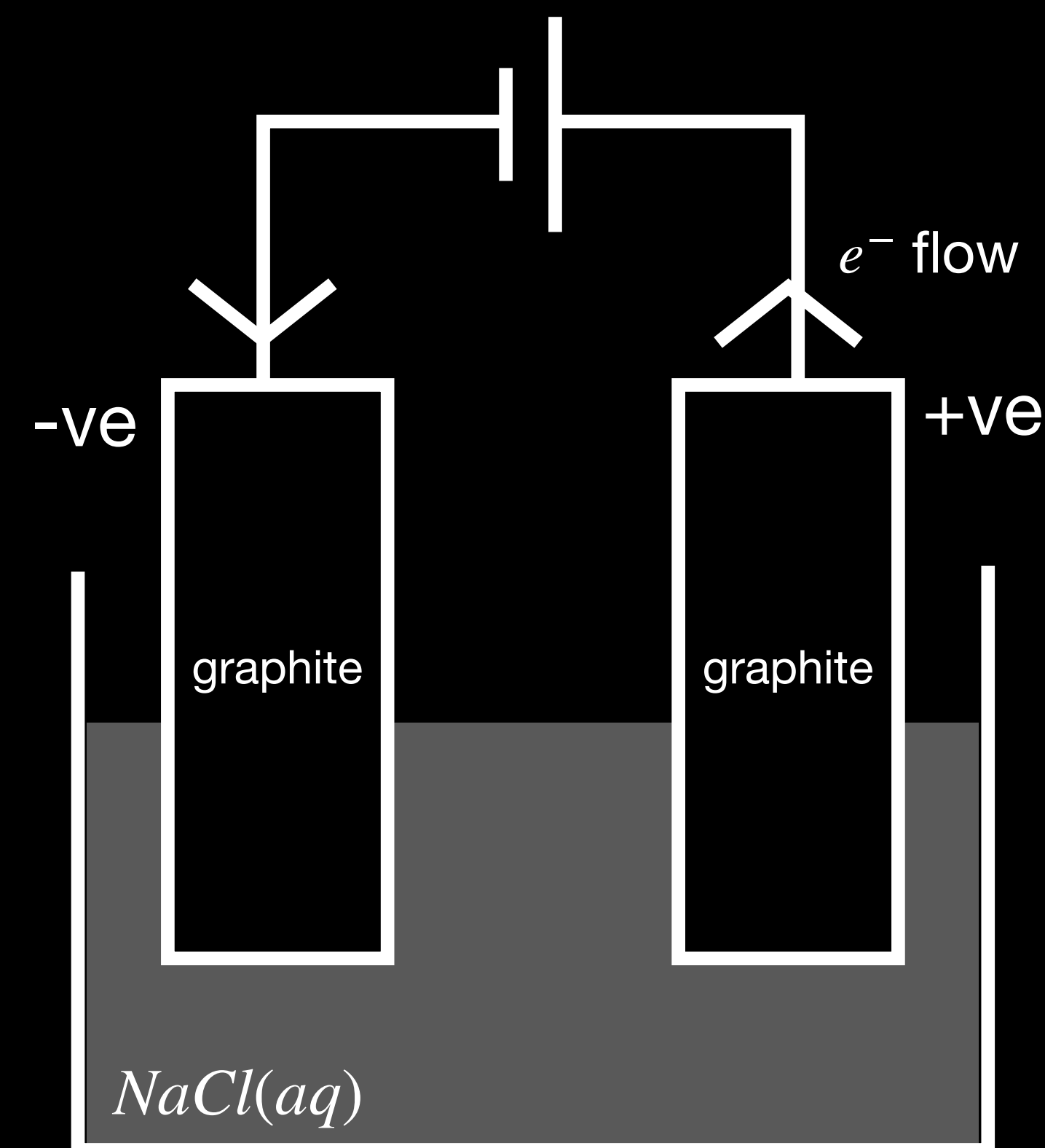
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$ (gains electrons, reduced)

As H^+ ions lost, higher concentration of OH^- ions near cathode, thus more alkaline at electrode

Full eqn: $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2(\text{g})$

PRODUCTS FORMED: O_2 and H_2 gas

pH of solution remains same as roughly equal amount of H^+ and OH^- ions present



Aqueous Ionic Substances

aq

Setup: graphite electrodes with concentrated aqueous NaCl(aq) as electrolyte

Ions present: Na^+ , Cl^- , H^+ , OH^-

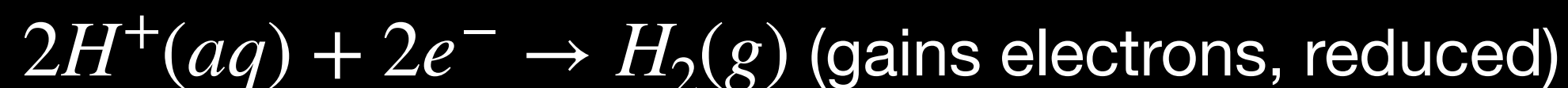
At the positive electrode/anode:

Both Cl^- and OH^- are attracted to electrode, however as concentrated, Cl^- is discharged.



At the negative electrode/cathode:

Both Na^+ and H^+ are attracted, however only H^+ is discharged

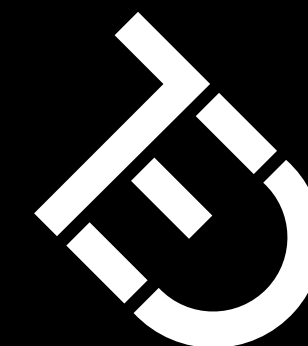
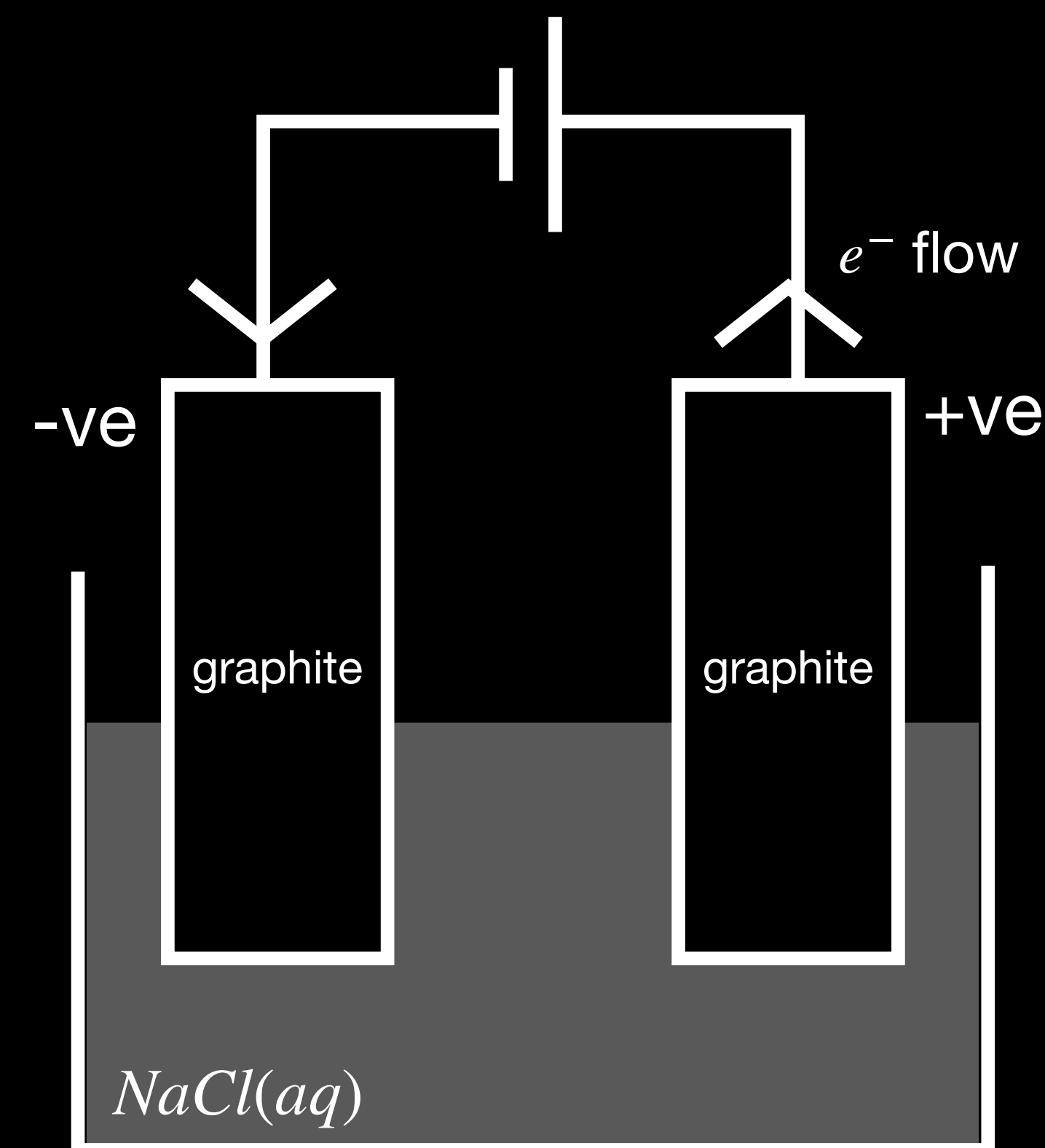


As H^+ ions lost, higher concentration of OH^- ions near cathode, thus more alkaline at electrode



PRODUCTS FORMED: H_2 and Cl_2 gas

Solution will be more alkaline



Aqueous Ionic Substances

aq

Setup: graphite electrodes with dilute aqueous $\text{CuSO}_4(\text{aq})$ as electrolyte
Ions present: Cu^{2+} , SO_4^{2-} , H^+ , OH^-

At the positive electrode/anode:

Both SO_4^{2-} and OH^- are attracted to electrode, however only OH^- is discharged



As OH^- ions lost, higher concentration of H^+ ions near anode, thus more acidic at electrode

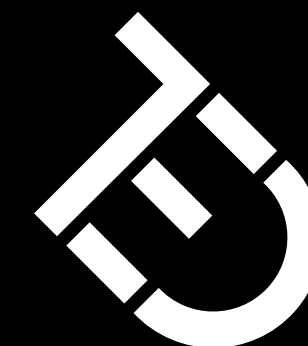
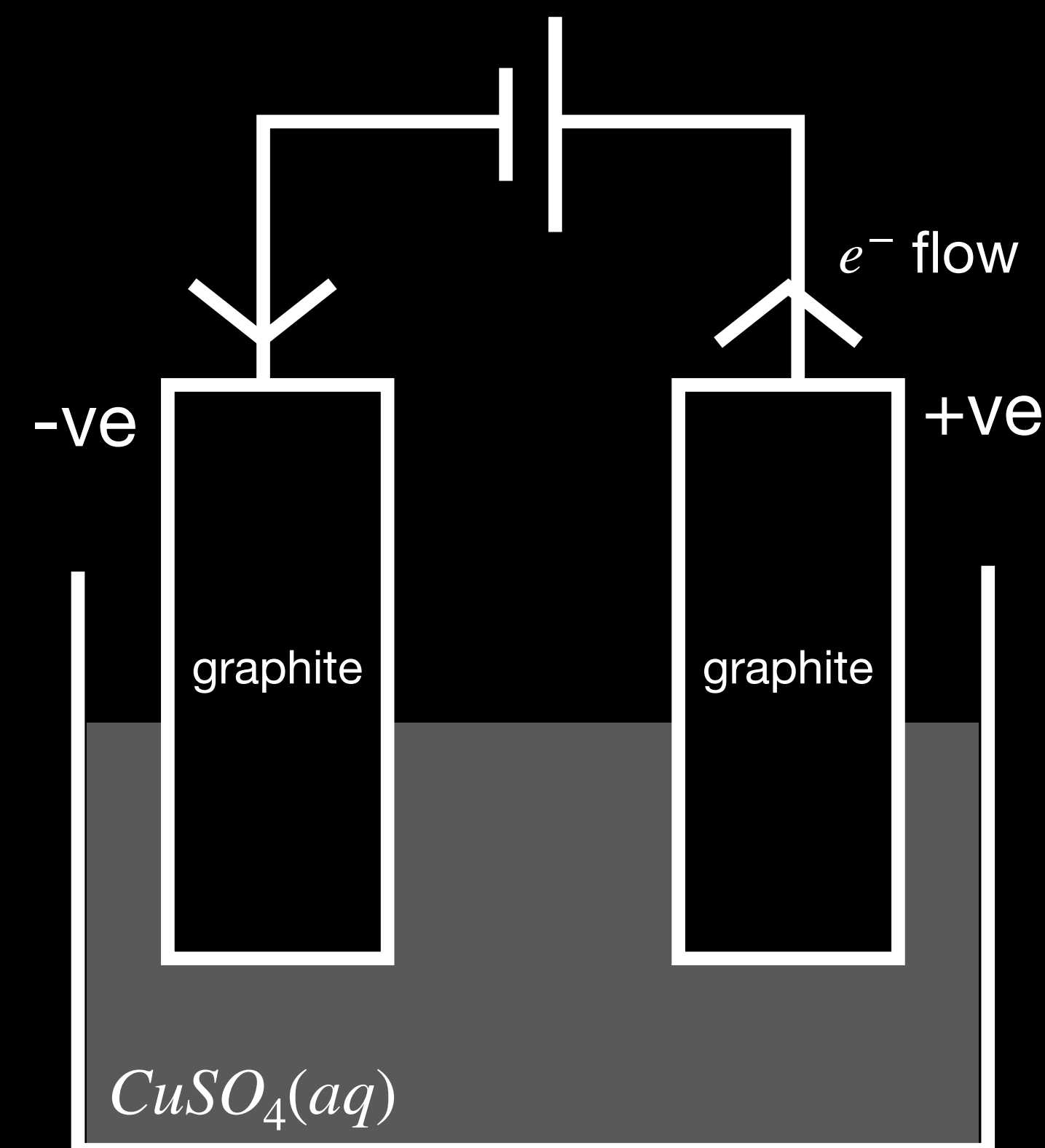
At the negative electrode/cathode:

Both Cu^{2+} and H^+ are attracted, however only Cu^{2+} is discharged



PRODUCTS FORMED: Copper metal and O_2 gas

Blue colour of electrolyte will fade & becomes more acidic



Reactive Electrodes

rea

Setup: COPPER electrodes with dilute aqueous $\text{CuSO}_4(aq)$ as electrolyte

Ions present: Cu^{2+} , SO_4^{2-} , H^+ , OH^-

At the positive electrode/anode:

The copper from the electrode is oxidised to form Cu^{2+} ions

$\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2e^-$ (loses electrons, oxidised)

Thus, anode shrinks in size/mass decreases

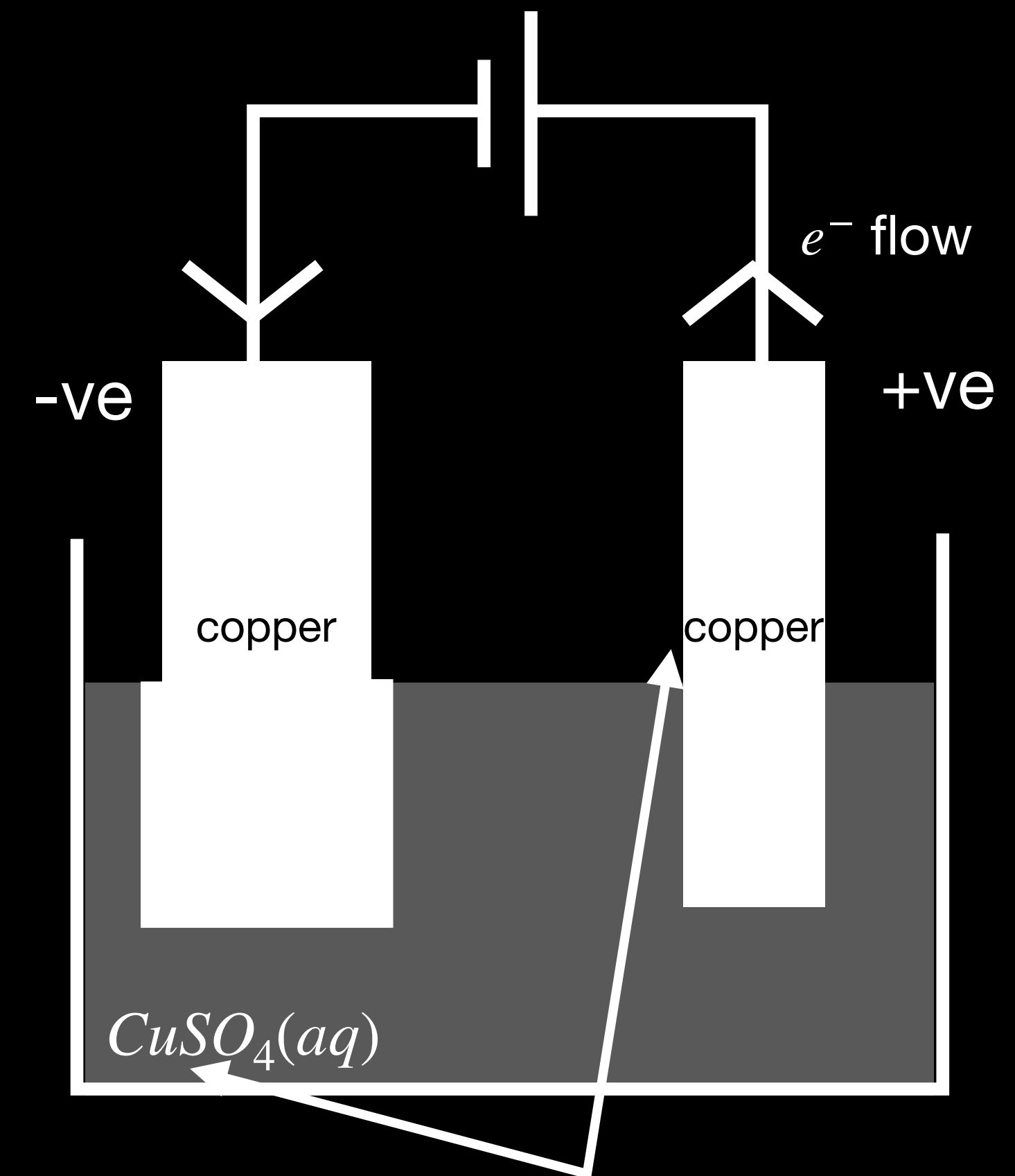
At the negative electrode/cathode:

The Cu^{2+} is attracted to the cathode and discharged, causing copper to deposit on the electrode

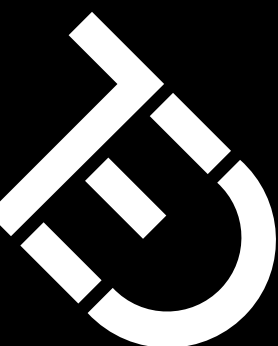
$\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$ (gains electrons, reduced)

Thus, cathode grows in size/mass increases

Blue colour of electrolyte does not fade and red-brown solid deposited onto electrode

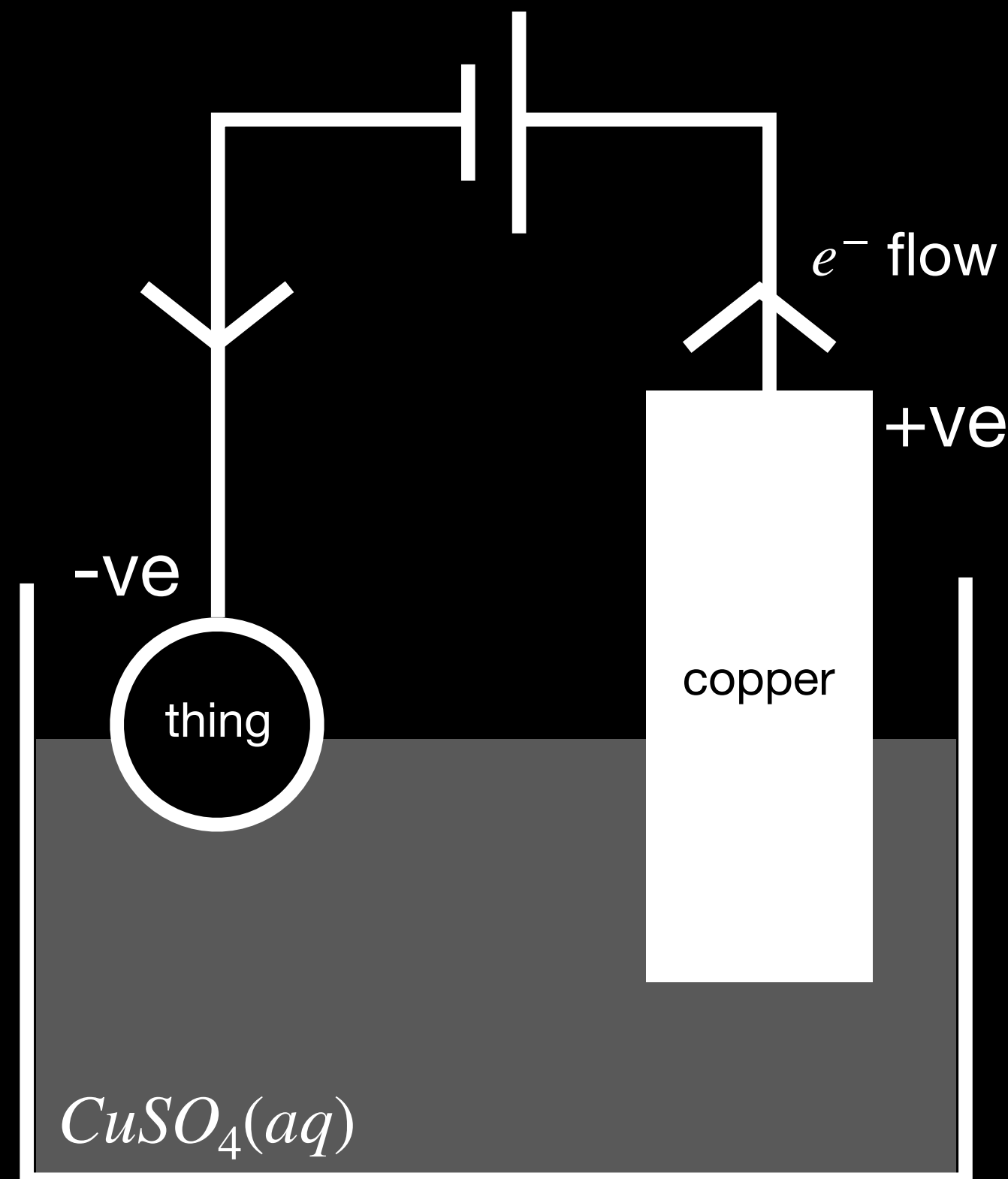


Electrolyte must contain metal cation of metal at anode
e.g. Cu^{2+} , Ag^+



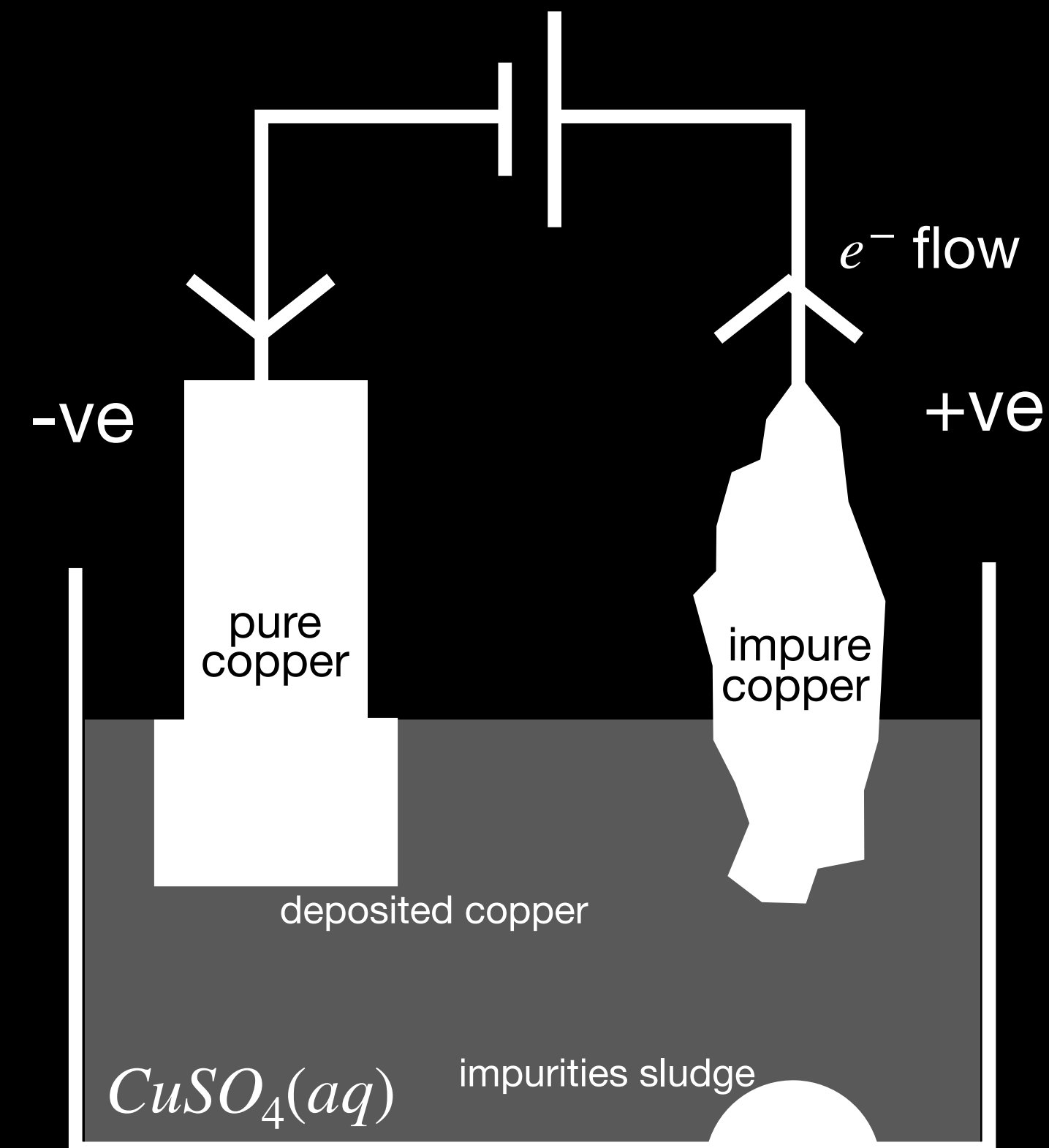
Applications of Electrolysis

a



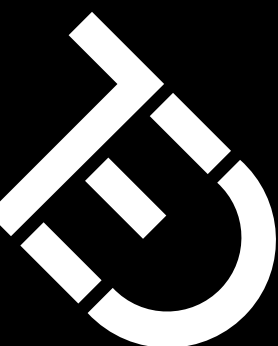
Electroplating

metallic object at cathode is
electroplated with metal at anode
used for protective coating/aesthetics



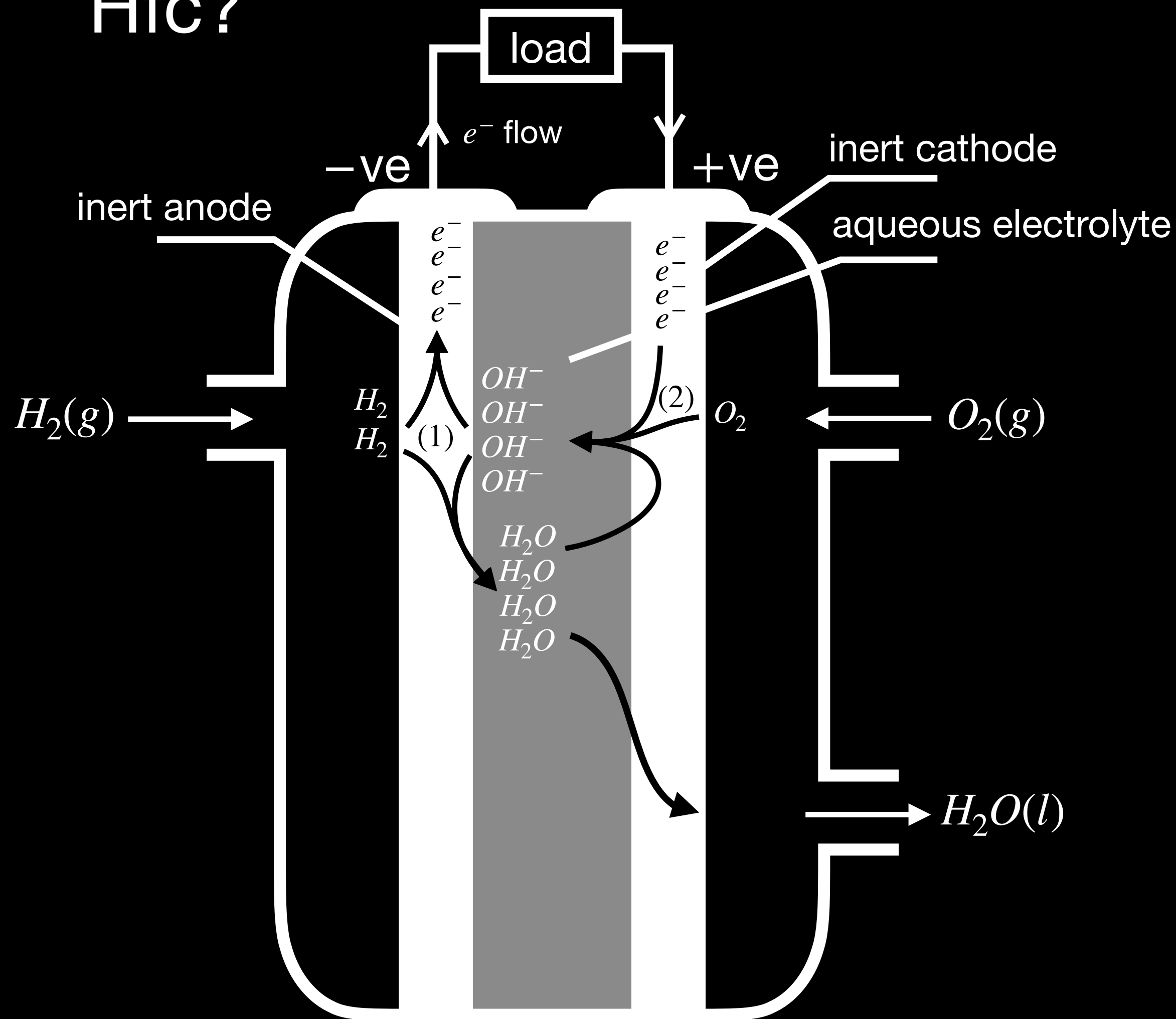
Metal Purification

impure metal at anode is purified



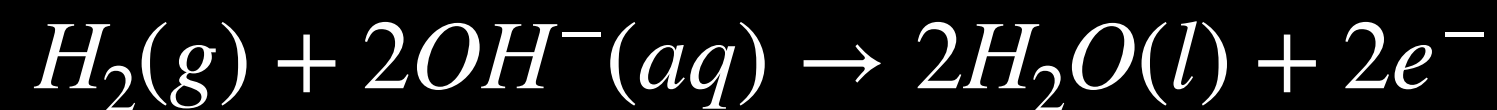
Hydrogen Fuel Cells

Hfc?



Hydrogen fuel cells (HFCs) are simple cells that use hydrogen as a fuel (with oxygen) to generate electricity. The fuel cell has inert graphite electrodes which allows substances to pass through the electrodes as it is porous

(1) At the negative electrode/anode: hydrogen reacts with the hydroxide ions in the electrolyte to form water and losing electrons, while also generating heat:



The lost electrons flow along the anode and through the electrical load (e.g. motor)

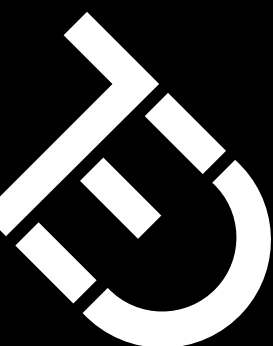
(2) At the positive electrode/cathode: oxygen reacts with the water produced to gain electrons to produce hydroxide ions, which allow the hydrogen to react with them again:



Excess water and heat then leaves from the exhaust.

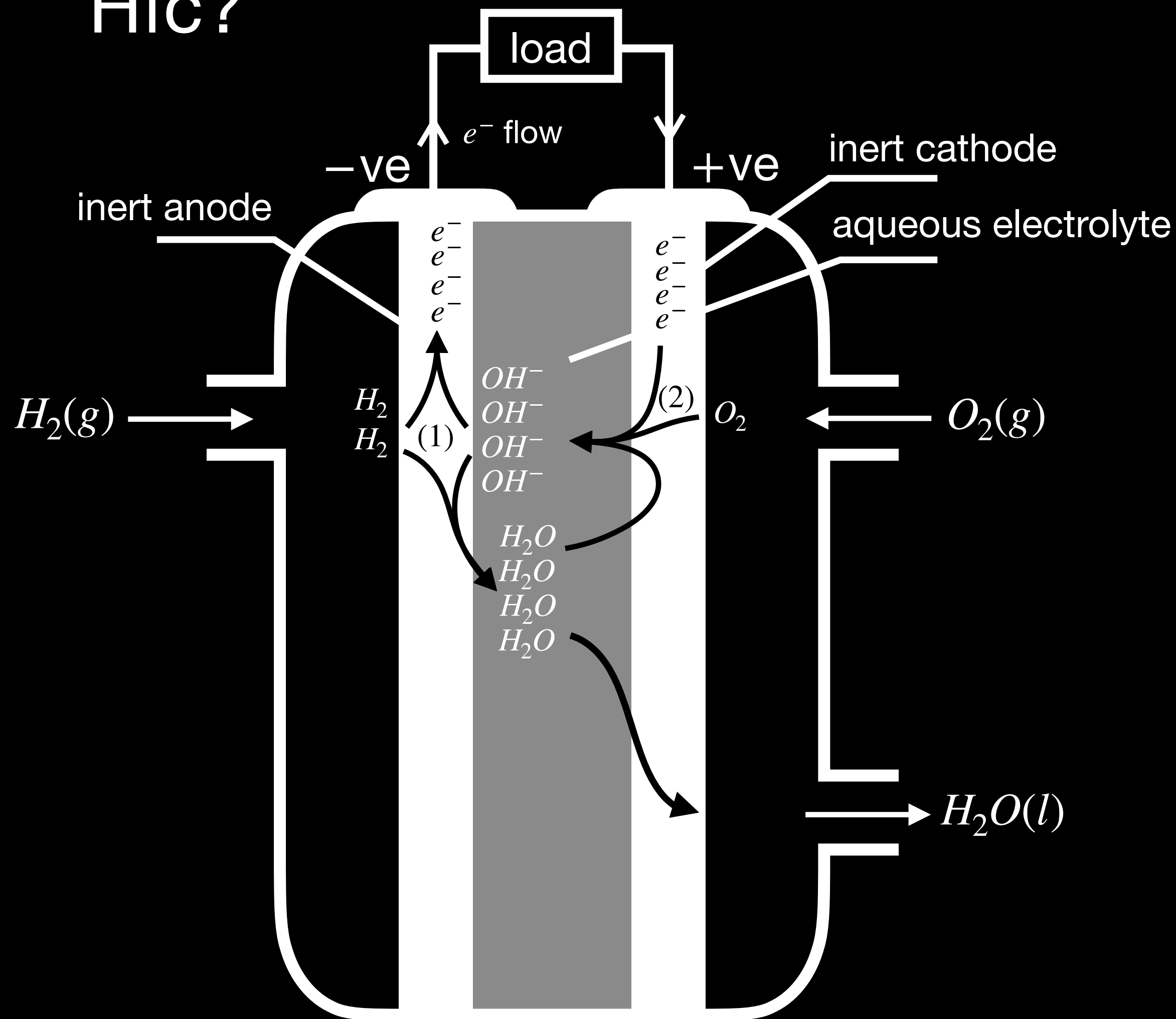
Thus, the flow of electrons from the anode to the cathode allows electricity to be produced.

The overall equation of the HFC is $2H_2 + O_2 \rightarrow 2H_2O$, showing that water is the byproduct of this reaction.



Hydrogen Fuel Cells

Hfc?



Advantages of hydrogen as a fuel/HFCs:

Renewable; obtained by electrolysis of water/cracking of hydrocarbons

Produces non-harmful byproducts (water)

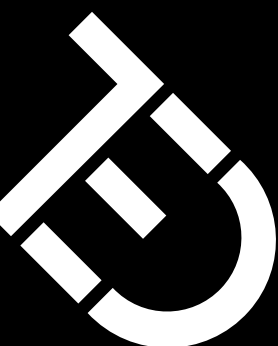
More efficient than fuel-burning electricity

Disadvantages of hydrogen as a fuel/HFCs:

Difficult to store & transport (highly flammable, usually handled in high pressure cylinders which can be dangerous)

Large amount of energy needed for electrolysis of water to produce hydrogen.

Memorise at least 1 advantage and disadvantage



Organic Chemistry



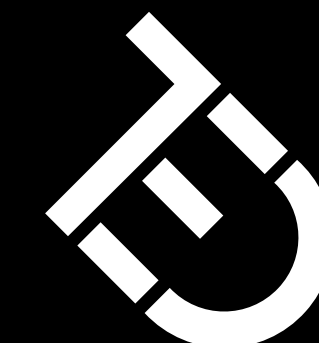
Organic Compounds & Homologous Series

get real

Organic compounds tend to contain carbon and mostly hydrogen, with some exceptions (CO_2)
Organic compounds can be grouped together in the same homologous series.

A homologous series is a family of compounds with the same general formula and similar chemical properties

Each homologous series has its own functional group; an atom/ a group of atoms that are bonded together that gives the compound's unique chemical properties in the homologous series.

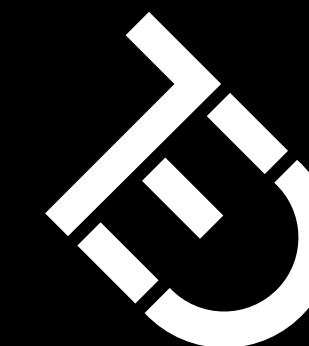


Homologous Series

hg

Alkanes			Notes
Name	Full structural formula	Chemical formula	Alkanes have a suffix of -ane and its general formula is C_nH_{2n+2} Alkanes do not have a specific functional group as even though alkanes have carbon bonds that are only C-C, all other organic molecules have C-C bonds. A Molecule with carbon bonds that are only C-C bonds are referred to as saturated (Thus Alkanes are saturated)
Methane	$\begin{array}{c} H \\ \\ H-C-H \\ \\ H \end{array}$	CH_4	
Ethane	$\begin{array}{c} H & H \\ & \\ H-C & -C-H \\ & \\ H & H \end{array}$	C_2H_6	
Propane	$\begin{array}{c} H & H & H \\ & & \\ H-C & -C & -C-H \\ & & \\ H & H & H \end{array}$	C_3H_8	

Alkenes			Notes
Name	Full structural formula	Chemical formula	Alkenes have a suffix of -ene and its general formula is C_nH_{2n} Alkenes have a -C=C- functional group. C=C double bonds are also referred to as unsaturated (Alkenes are unsaturated) Only ONE double bond is present in the molecule
Ethene	$\begin{array}{c} H & H \\ & \\ C & =C \\ & \\ H & H \end{array}$	C_2H_4	
Propene	$\begin{array}{c} H & H & H \\ & & \\ C & =C & -C-H \\ & & \\ H & & H \end{array}$	C_3H_6	
Butene	$\begin{array}{c} H & H & H & H \\ & & & \\ C & =C & -C & -C-H \\ & & & \\ H & & H & H \end{array}$	C_4H_8	

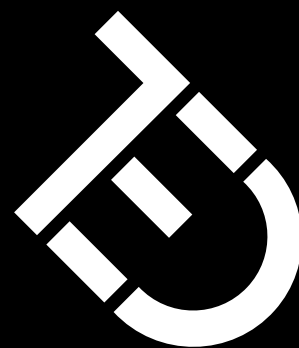


Homologous Series

hg

Alcohols			Notes
Name	Full structural formula	Chemical formula	Alcohols have a suffix of -ol and its general formula is $C_nH_{2n+2}O$ Alkanes have a -O-H functional group
Methanol	$\begin{array}{c} H \\ \\ H-C-O-H \\ \\ H \end{array}$	CH_4O or CH_3OH	
Ethanol	$\begin{array}{c} H & H \\ & \\ H-C-C-O-H \\ & \\ H & H \end{array}$	C_2H_6O or C_2H_5OH	
Propanol	$\begin{array}{c} H & H & H \\ & & \\ H-C-C-C-O-H \\ & & \\ H & H & H \end{array}$	C_3H_8O or C_3H_7OH	

Carboxylic Acids			Notes
Name	Full structural formula	Chemical formula	Carboxylic Acids have a suffix of -oic acid and its general formula is $C_nH_{2n+1}COOH$ (n starts from 0) Carboxylic acids have the -COOH functional group
Methanoic Acid	$\begin{array}{c} O \\ \\ H-C-O-H \end{array}$	$HCOOH$	
Ethanoic Acid	$\begin{array}{c} H & O \\ & \\ H-C-C-O-H \\ \\ H \end{array}$	CH_3COOH	
Propanoic Acid	$\begin{array}{c} H & H & O \\ & & \\ H-C-C-C-O-H \\ & \\ H & H \end{array}$	C_2H_5COOH	

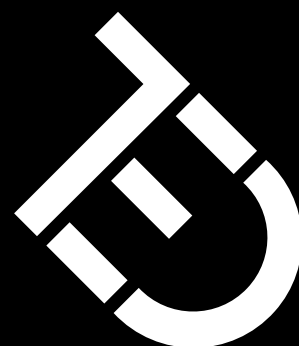


Homologous Series Trends

g

Alkanes (Naming)			
Name	Chemical Formula	No. Of C	Prefix
Methane	CH ₄	1	Meth-
Ethane	C ₂ H ₆	2	Eth-
Propane	C ₃ H ₈	3	Prop-
Butane	C ₄ H ₁₀	4	But-
Pentane	C ₅ H ₁₂	5	Pent-
Hexane	C ₆ H ₁₄	6	Hex-

As relative molecular mass / number of C increases	Explanation
m.p and b.p increases	Molecules become heavier and larger → stronger intermolecular forces of attraction → more energy needed to overcome → higher m.p./b.p.
Flammability decreases/ Flashpoint (lowest temp needed to ignite with air) increases	Larger molecules → harder to evaporate → flammability decreases/flashpoint increases Higher percentage of C → more O ₂ needed to burn → flammability decreases/flashpoint increases
Viscosity increases	Larger molecules → stronger intermolecular forces of attraction → harder to flow



Homologous Series - Combustion

g

Alkanes, Alkenes and Alcohols can all undergo combustion

There are two types of combustion, complete and incomplete

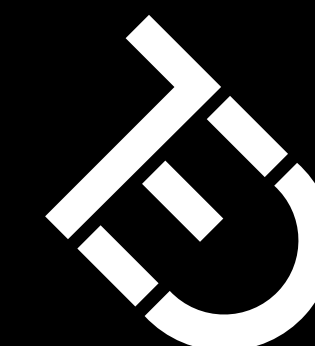
Complete combustion has CO_2 and H_2O as products while Incomplete combustion has CO and H_2O as products and will have a sooty flame (carbon).

Incomplete combustion tends to occur under limited oxygen.

The more carbon there is, the more likely it will undergo incomplete combustion than complete combustion (requires more oxygen to undergo complete combustion)

Complete	Incomplete
$2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$	$2\text{C}_2\text{H}_6 + 5\text{O}_2 \rightarrow 4\text{CO} + 2\text{H}_2\text{O}$
$\text{C}_2\text{H}_4 + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$	$\text{C}_2\text{H}_4 + 2\text{O}_2 \rightarrow 2\text{CO} + 2\text{H}_2\text{O}$
$\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$	$\text{C}_2\text{H}_5\text{OH} + 2\text{O}_2 \rightarrow 2\text{CO} + 3\text{H}_2\text{O}$

notice how there is less O_2 needed for incomplete than complete

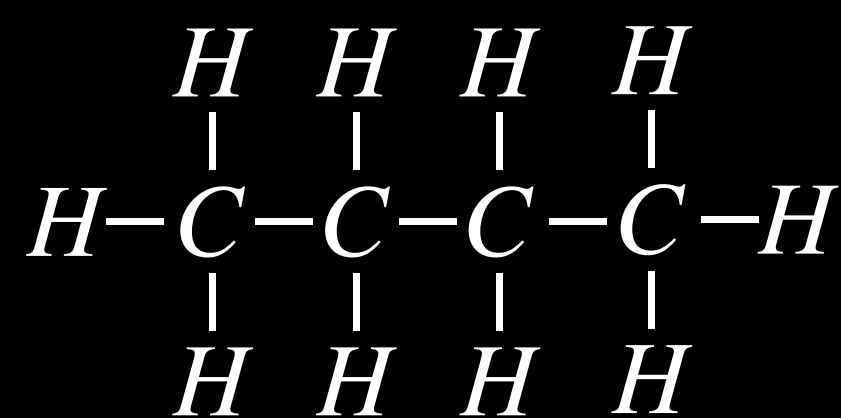


Isomerism

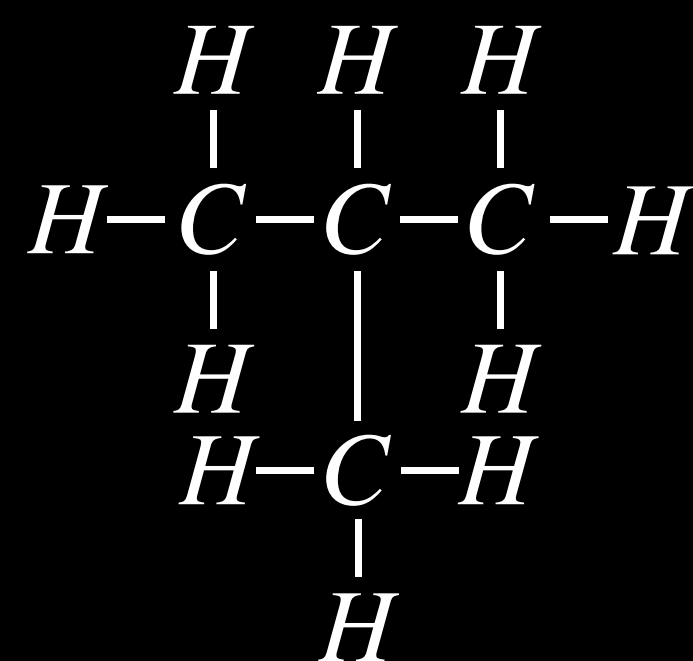
h

Isomerism is when molecules have the same chemical formula but have different structural formulas

C_4H_{10} isomers

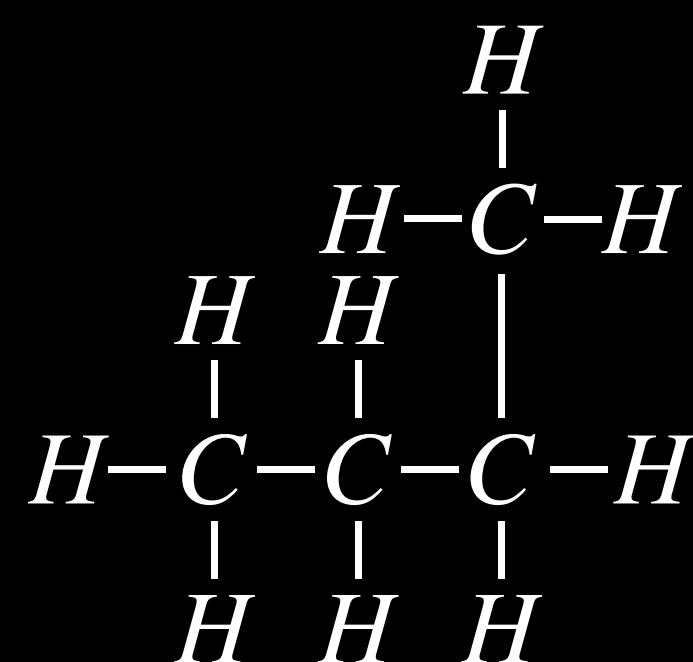


C_4H_{10} isomer



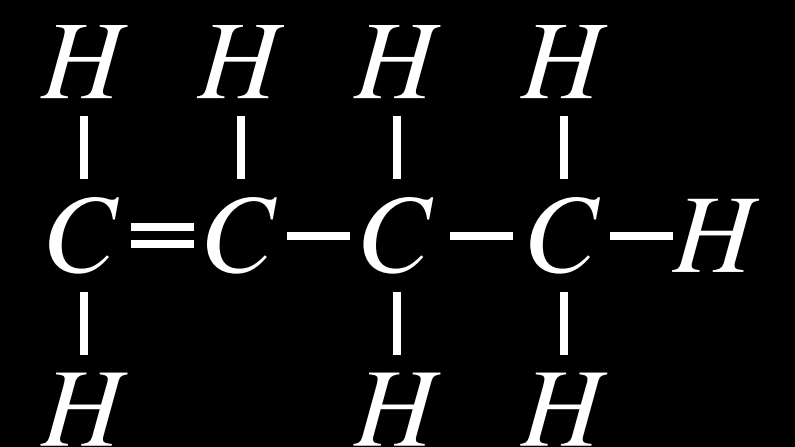
is a C_4H_{10} isomer

=

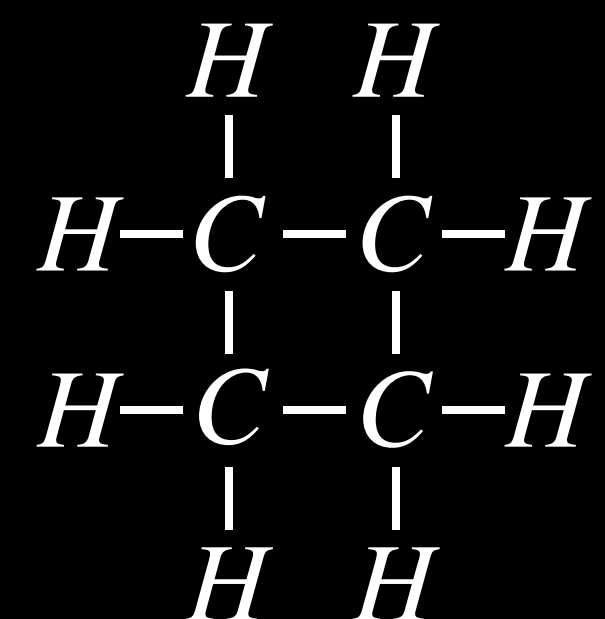


is not a C_4H_{10} isomer
(already existing isomer)

isomers of C_4H_8

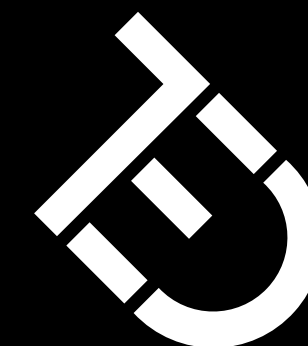


alkene (C=C bonds)



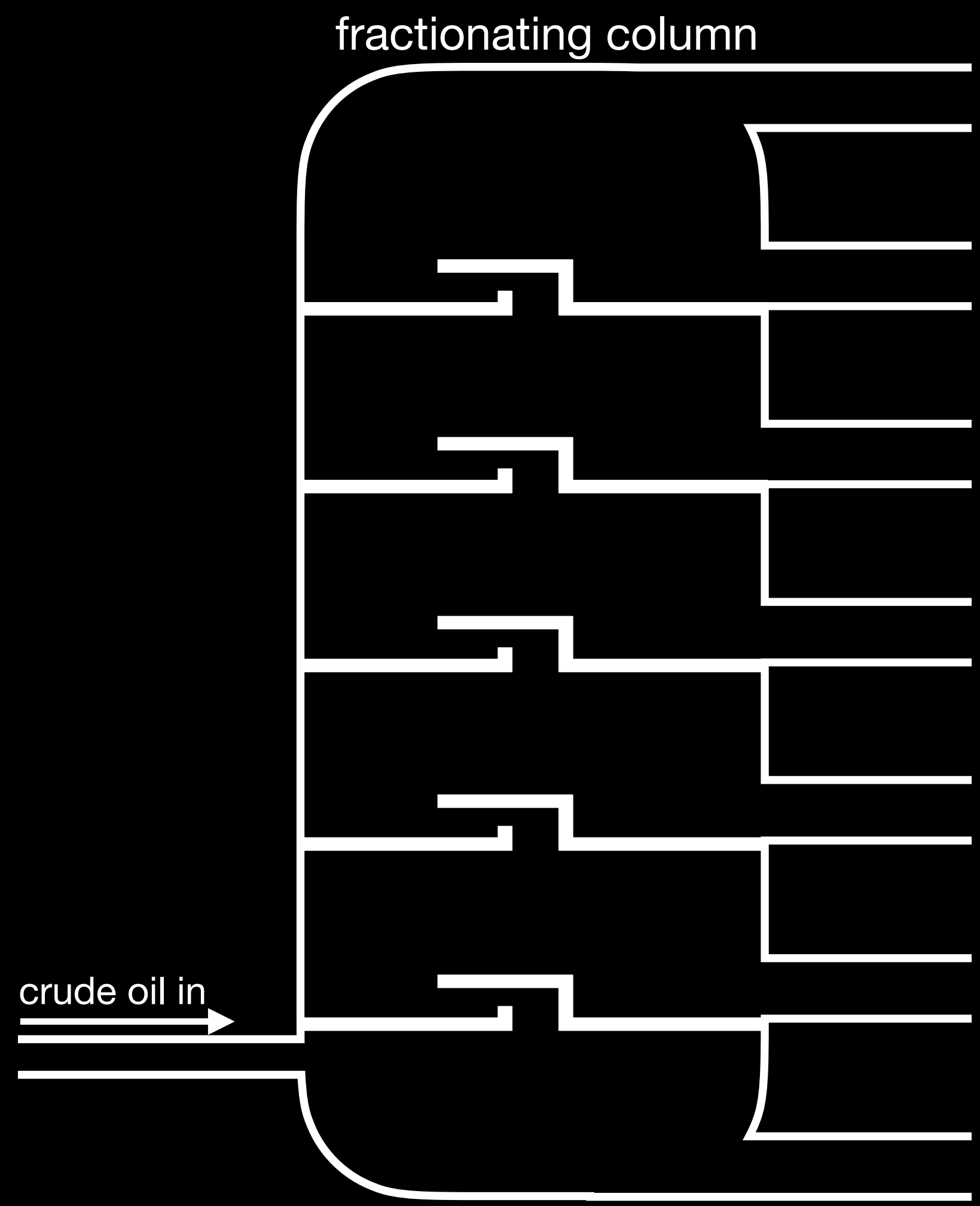
alkane (C-C bonds)

isomers need not be in the same homologous series



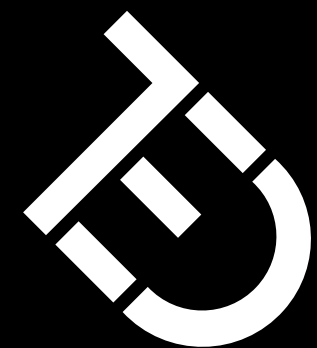
Fuels & Crude Oil

j



Fraction Name	No. Of C	Uses
Petroleum gas	1-4	Cooking fuel
Petrol	5-10	Vehicle fuel
Naphtha	7-14	Plastic production
Paraffin/ Kerosene	9-16	Jet fuel
Diesel Oil	15-25	Larger vehicle fuel
Lubricating Oil	20-35	Lubrication of parts
Bitumen	>70	Road surfacing

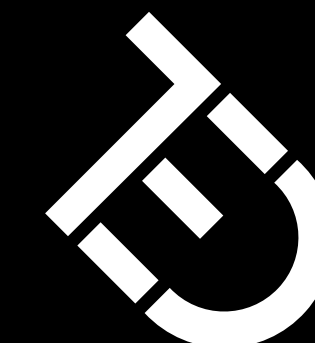
Different fractions have different boiling points which allow them to condense and collect at different regions along the column, separating the fractions.



Reaction Types

r

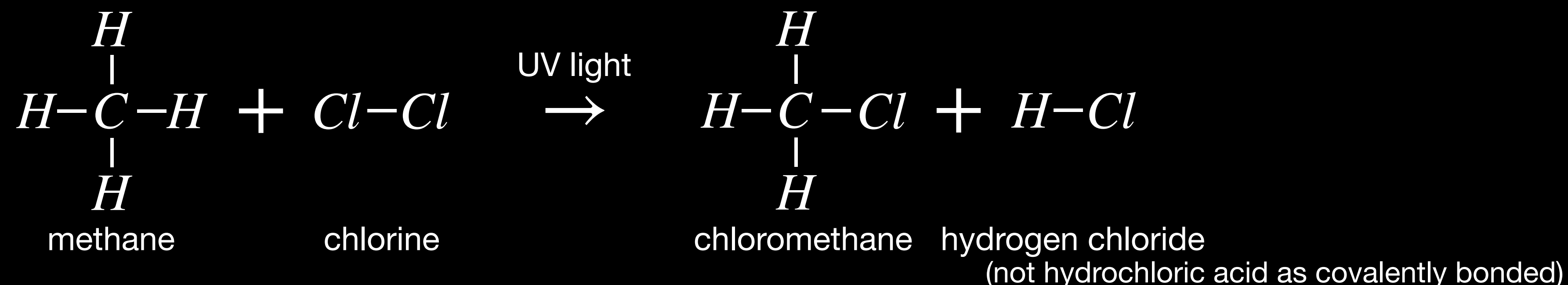
Classification	Definition/How to recognise
Substitution	One atom/a group of atoms being replaced by another atom/a group of atoms
Addition	Two or more substances combined across a carbon-carbon multiple bond to have only ONE product
Condensation	Two substances combine to have water or a small molecule (e.g. HCl) removed(aka as a product)
Hydrolysis	Water reacting with a molecule which causes the molecule to break down



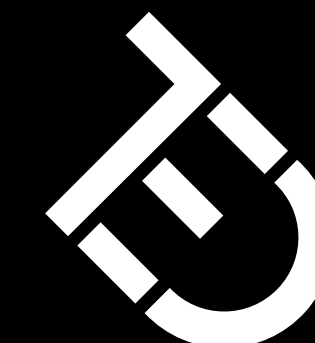
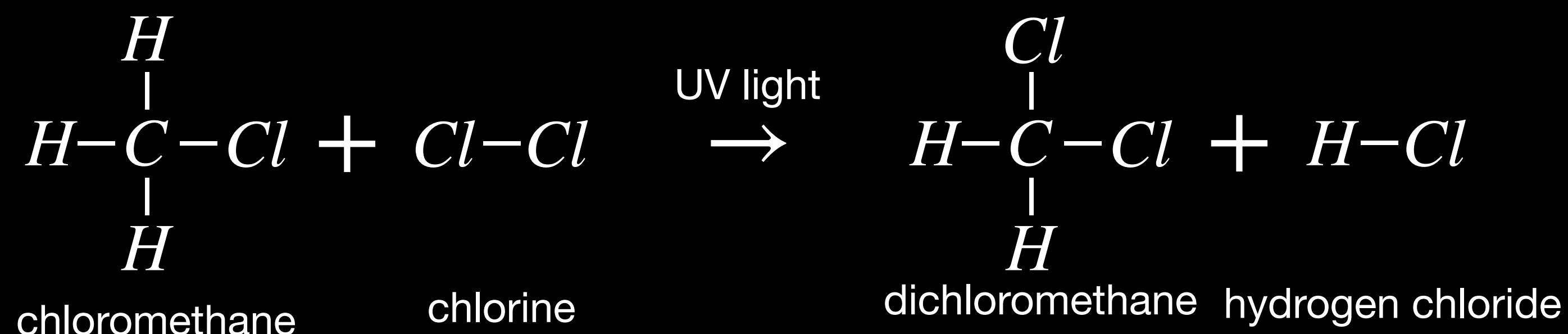
Alkanes - Substitution

S

Alkanes can react with a halogen to form the same alkane but with a hydrogen atom replaced with an atom of the halogen. The reaction needs ultraviolet light to undergo.



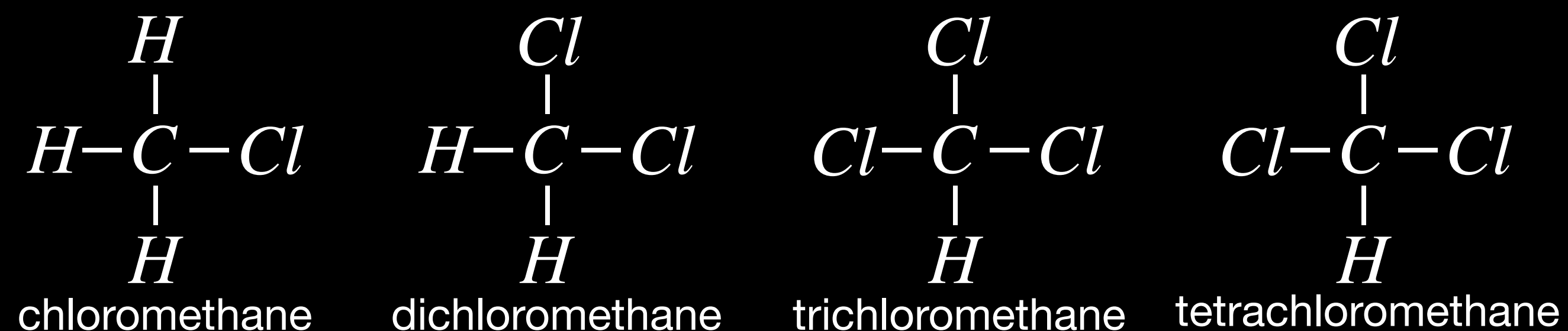
The resulting chloromethane belongs in the homologous series of halogenoalkanes. The substitutions can occur multiple times as shown:



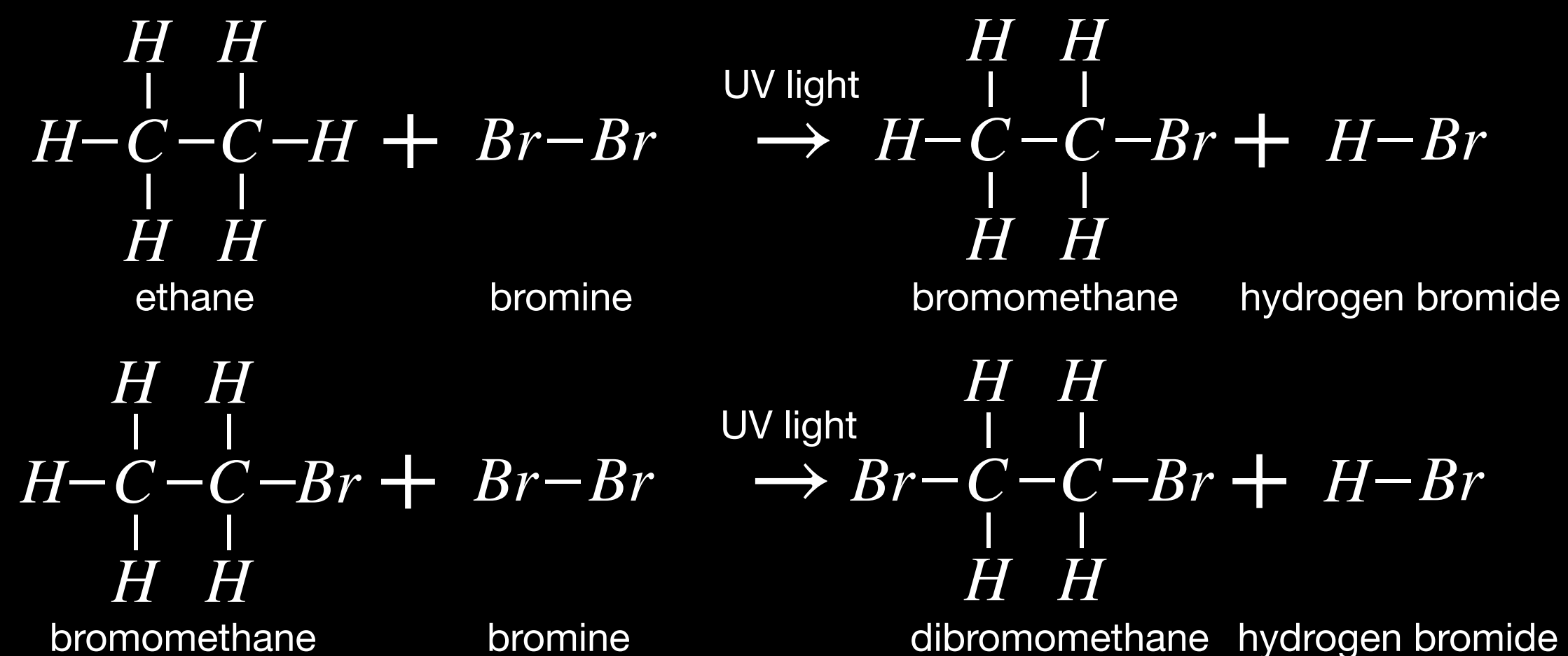
Alkanes - Substitution

S

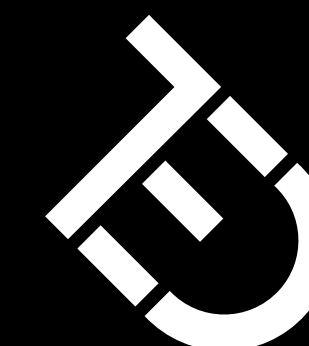
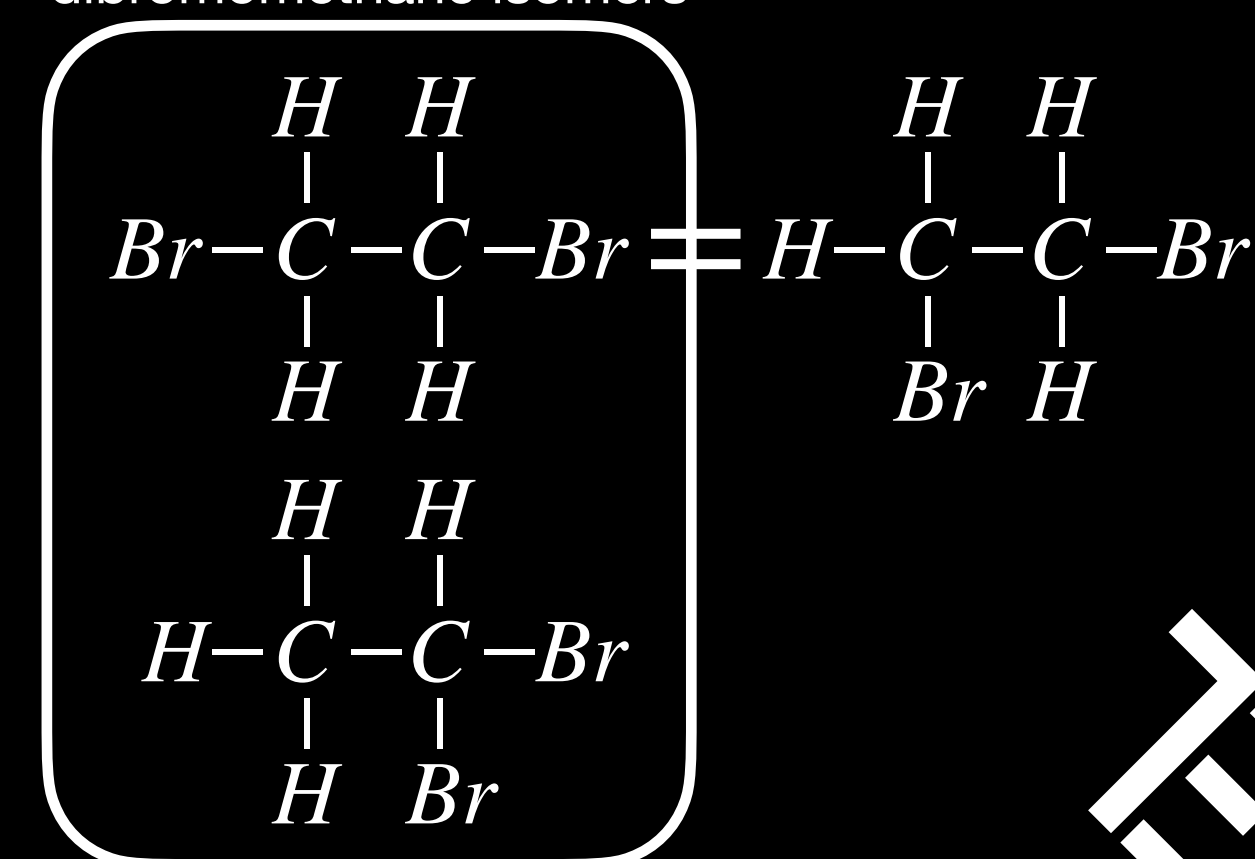
Under excess chlorine, substitutions will keep occurring until all of the hydrogen in one molecule is replaced with chlorine



Likewise, Alkanes can also react with different halogens:



dibromomethane isomers



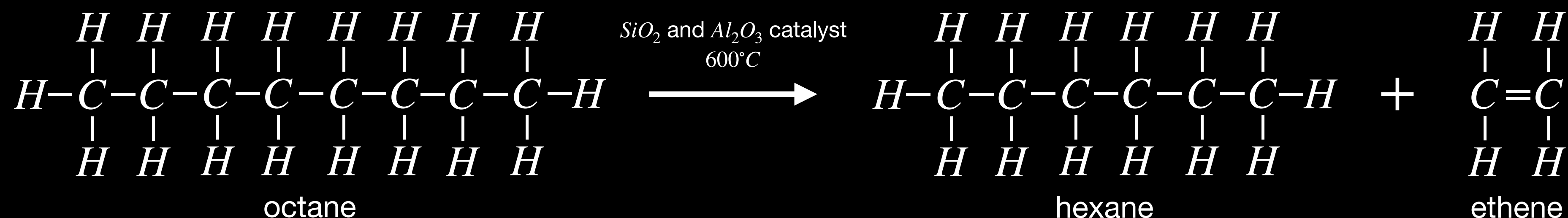
Alkanes - Cracking

C

Cracking is a process where long chain alkanes produce short chain alkanes and alkenes as products, with some reactions having hydrogen as a product.

Conditions for cracking: **600°C, with aluminium oxide (Al_2O_3) and silicon dioxide (SiO_2) catalysts** (NEED TO MEMORISE)

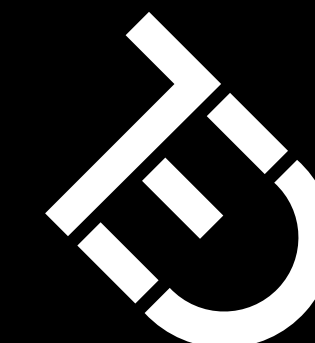
Shown is the catalytic cracking of octane:



Uses:

Produce smaller molecules from heavier fractions (smaller molecules of alkanes/alkenes are more useful as fuel, plastic production, petrol, etc.)

Produces hydrogen for Haber Process (cracking of hydrocarbons).



Alkanes - Cracking

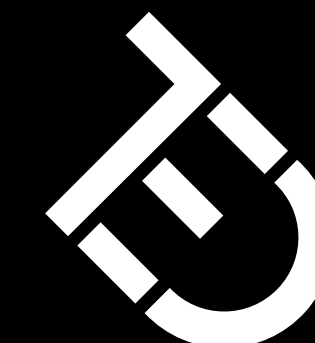
C

Write a balanced chemical equation to show the cracking of dodecane, $C_{12}H_{26}$, to form butene, hydrogen gas, and another alkene.

Ans:



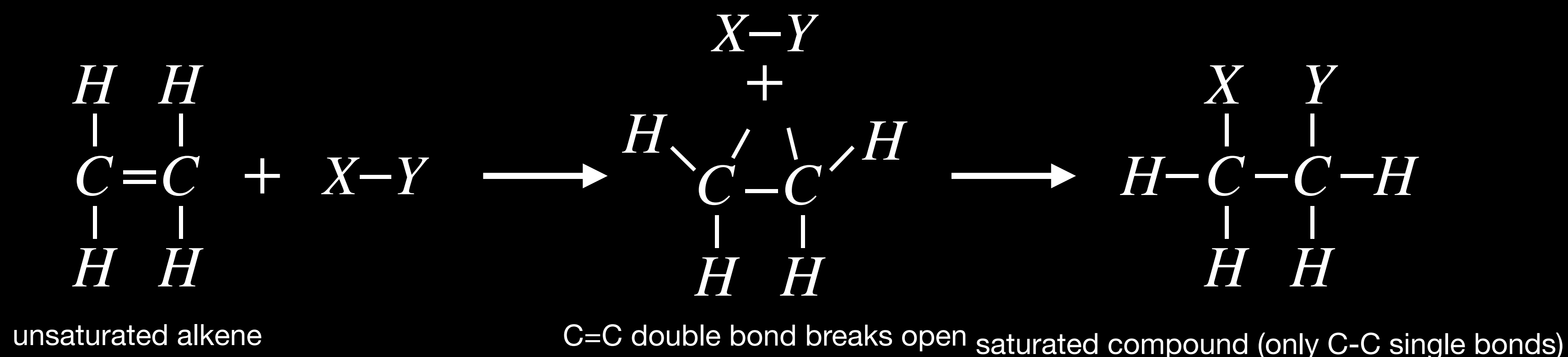
Make sure than the number of C matches (Dodecane: 12 C = butene: 4 C + octene 8 C) and hydrogen matches (26H in Dodecane = 8H in butene + 2H in hydrogen gas + 16H in octene)



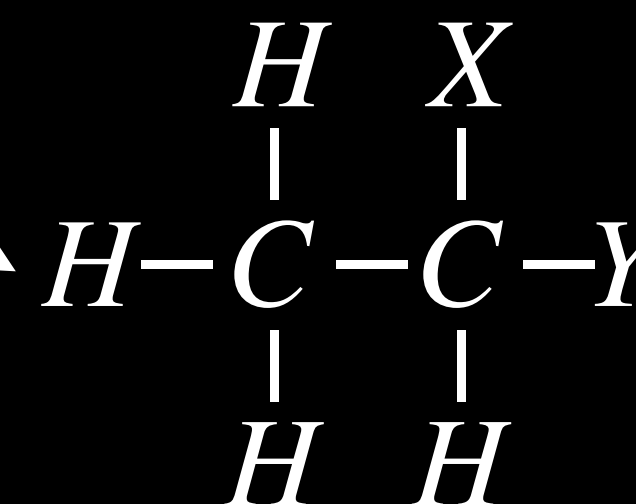
Alkenes - Addition

A

Alkenes tend to undergo addition reactions due to the highly reactive C=C double bonds present.
In general:

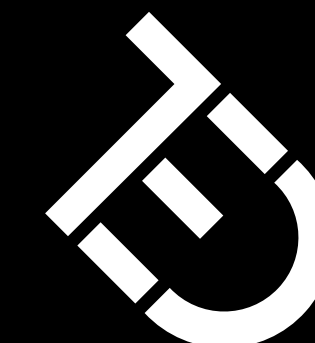


product is NOT



double bond exists is between carbon
X and Y cannot be bonded to the same carbon atom

****+ take note of structure



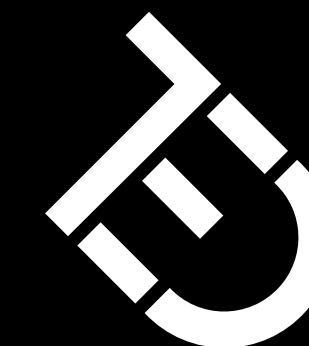
Alkenes - Addition

A

There are 4 addition reactions with alkenes:

With [reagent]	Chemical Reaction
With hydrogen (Hydrogenation)	$ \begin{array}{c} H & H \\ & \\ C & = & C \\ & \\ H & H \end{array} + H-H \xrightarrow[150^{\circ}C]{\text{nickel catalyst}} \begin{array}{c} H & H \\ & \\ H-C & -C-H \\ & \\ H & H \end{array} $ ethene hydrogen ethane
With halogens (Halogenation)	$ \begin{array}{c} H & H & H \\ & & \\ C & = & C-C-H \\ & & \\ H & & H \end{array} + Cl-Cl \xrightarrow{\text{r.t.p}} \begin{array}{c} H & H & H \\ & & \\ H-C & -C-C-H \\ & & \\ Cl & Cl & H \end{array} $ propene chlorine dichloropropane belongs in halogenoalkanes
With STEAM (DO NOT WRITE WATER) (Hydration)	$ \begin{array}{c} H & H & H \\ & & \\ C & = & C-C-H \\ & & \\ H & & H \end{array} + H-O-H \xrightarrow[300^{\circ}C, 60 \text{ atm}]{H_3PO_4 \text{ acid catalyst}} \begin{array}{c} H & H & H \\ & & \\ H-C & -C-C-H \\ & & \\ H & O & H \\ & & \\ & H & \end{array} \quad \begin{array}{c} H & H & H \\ & & \\ H-O-C & -C-C-H \\ & & \\ H & H & H \end{array} $ propene steam(high temp) isomers of propanol
With hydrogen halides	$ \begin{array}{c} H & H & H \\ & & \\ C & = & C-C-H \\ & & \\ H & & H \end{array} + H-Cl \longrightarrow \begin{array}{c} H & H & H \\ & & \\ H-C & -C-C-H \\ & & \\ H & Cl & H \end{array} \quad \begin{array}{c} H & H & H \\ & & \\ H-C & -C-C-H \\ & & \\ Cl & H & H \end{array} $ propene hydrogen chloride (IONIC) (not hydrochloric acid as covalently bonded) isomers of chloropropane belongs in halogenoalkanes

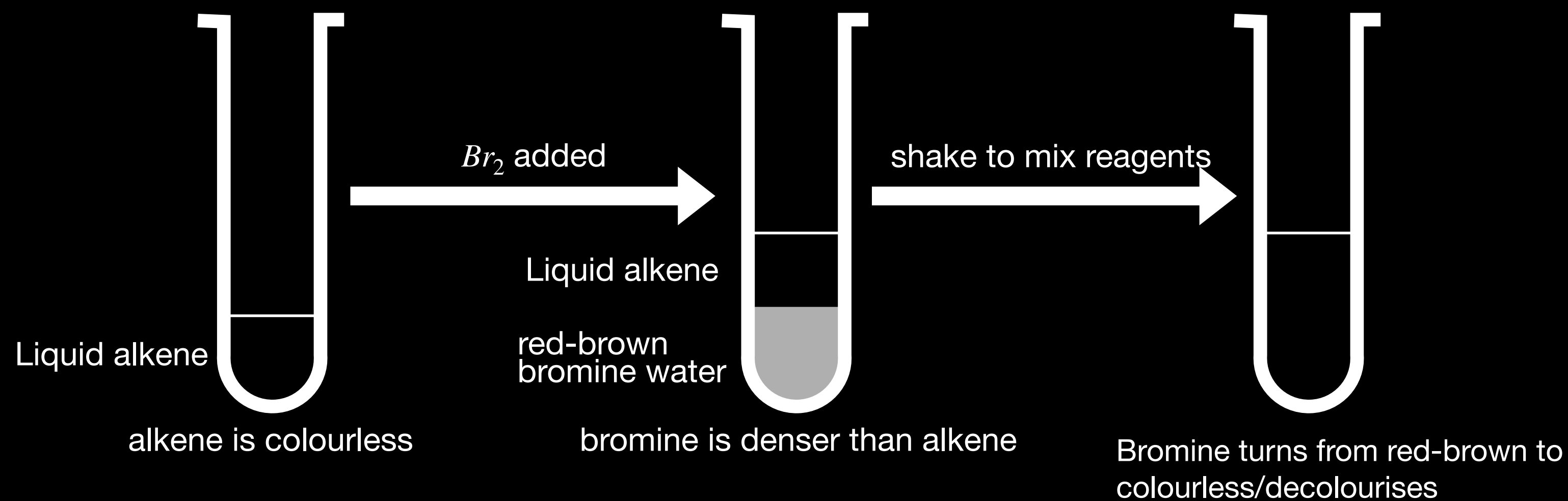
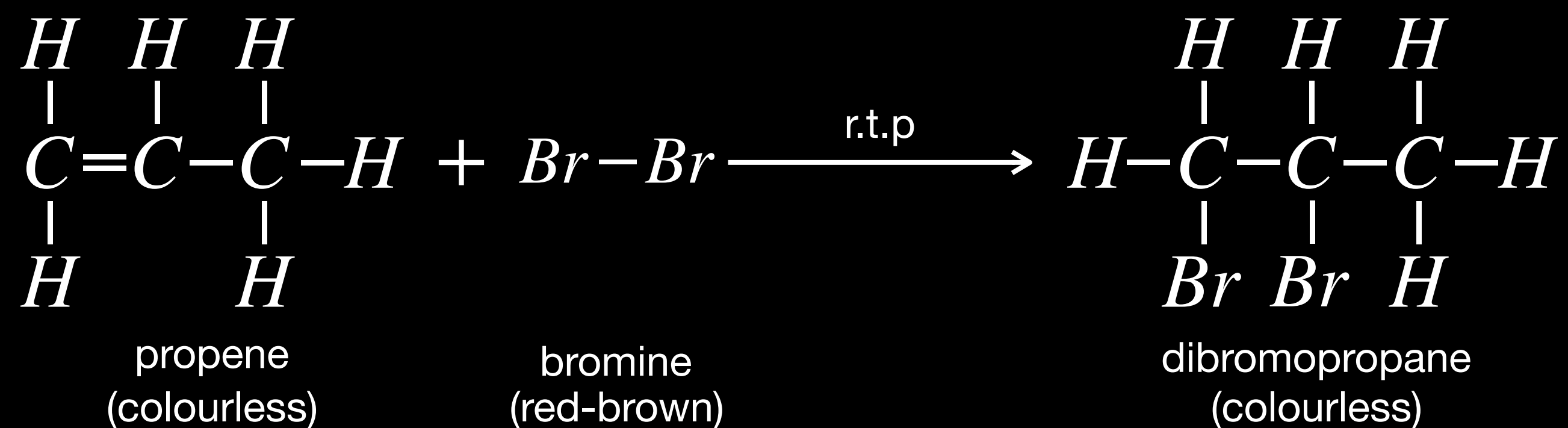
Conditions HAVE TO BE MEMORISED !!!!!!!



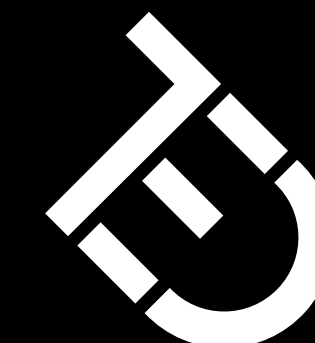
Alkenes - Addition

A

Alkenes can be tested as so:



When bromine turns from red-brown to colourless, alkenes are present

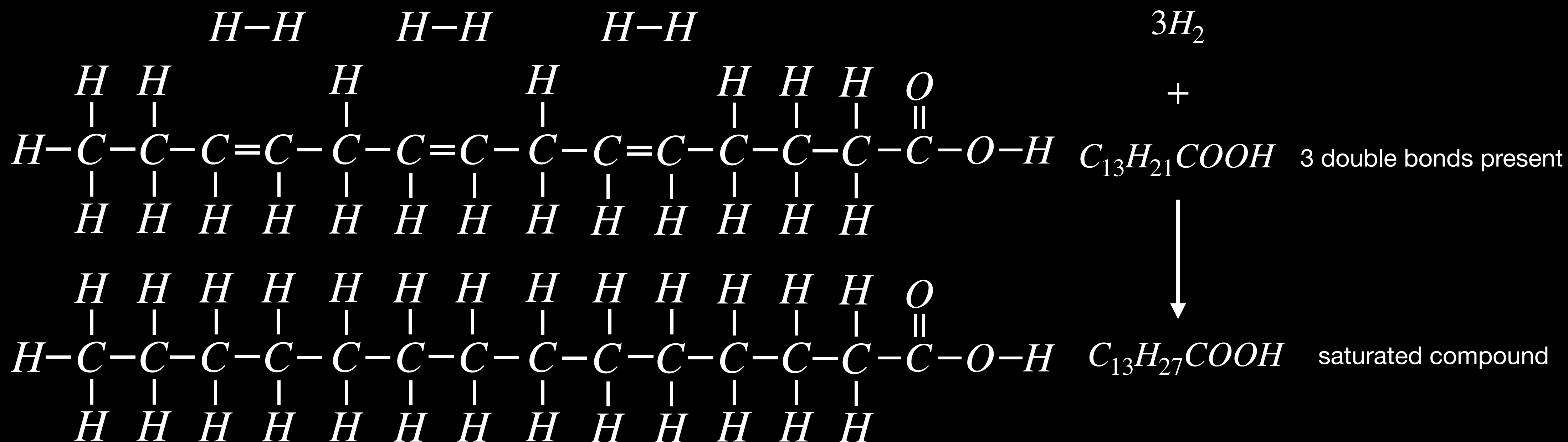


Alkenes - Polyunsaturated Compounds

A

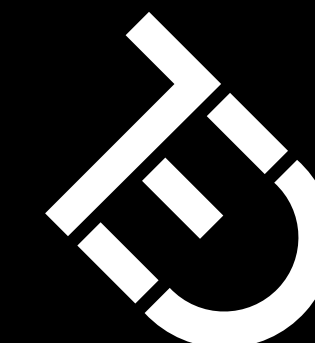
Polyunsaturated compounds refer to compounds with multiple(poly) C=C double bonds (unsaturated)

Polyunsaturated compounds can undergo hydrogenation (addition) reactions to be saturated, like from vegetable oil to produce saturated fats like margarine



From the number of moles of hydrogen used for 1 mol of polyunsaturated compound, we can deduce the number of C=C double bonds present in the alkene

e.g. as 3 moles of hydrogen were used to react with 1 mole of the alkene, the alkene has 3 C=C double bonds. (However number of other double bonds cannot be found through this!!)



Alcohols - Production

alc

Alcohols are produced by the fermentation of glucose with yeast or by the hydration of alkenes:



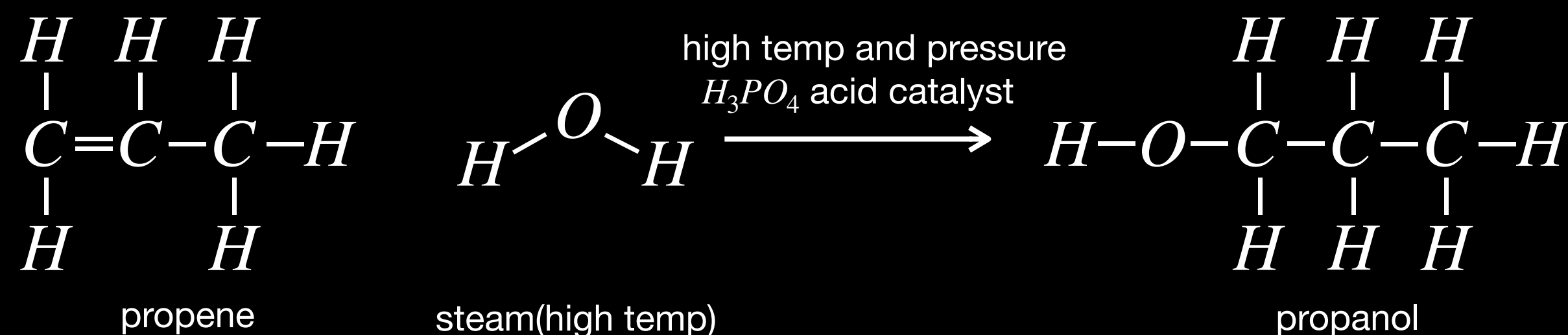
Glucose is obtained from crops(which contain sugar) and sawdust/woodchips (which contain cellulose, which has glucose)

Yeast requires these conditions to break down glucose:

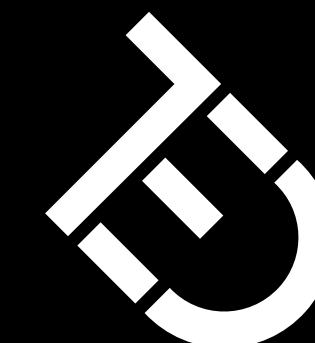
Temp kept at 37°C (too high and yeast denatures(do not write die off), too low and yeast does not break down glucose fast enough/does not break down)

Absence of oxygen (which can be done by using boiled water, which removes oxygen, or using an air lock)

Alkenes can be hydrated to form alcohols as so:



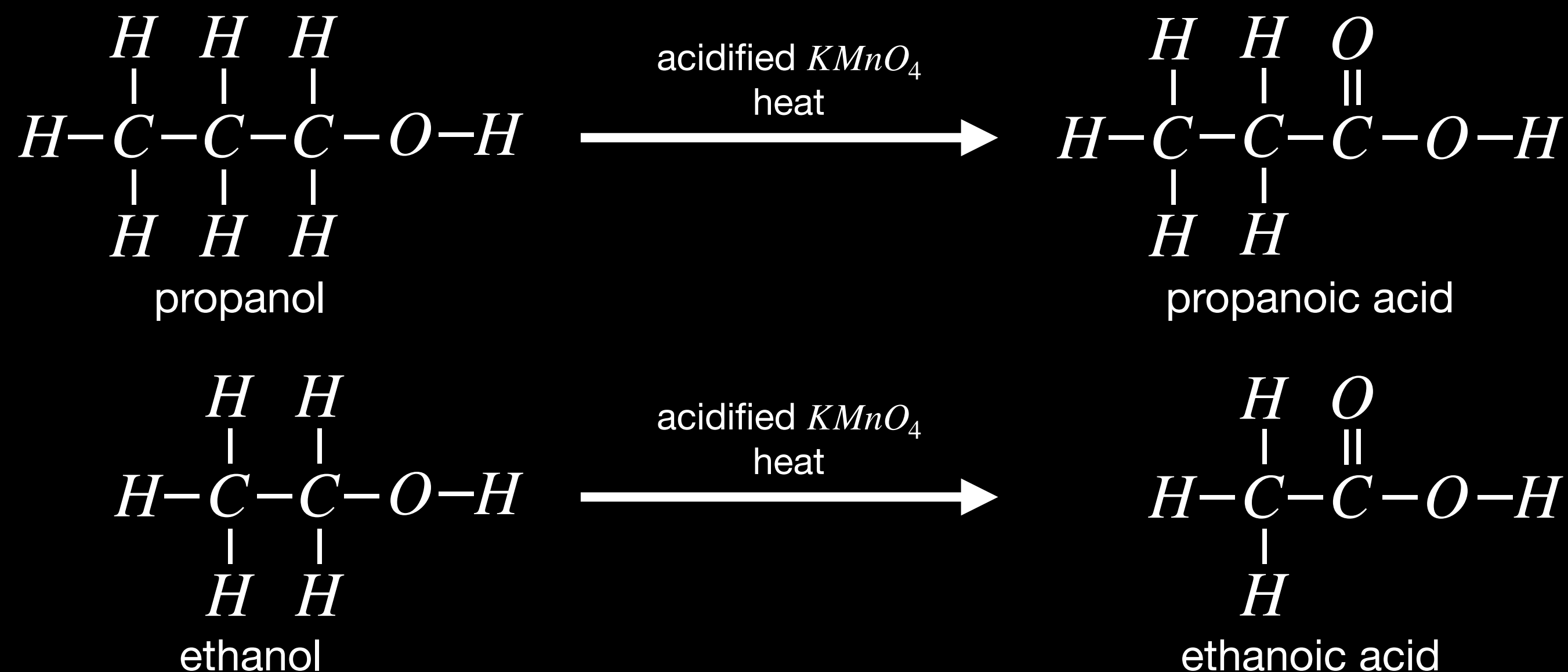
Alcohols are also soluble in water, thus ethanol has to be fractionally distilled from water to be purified



Alcohols - Oxidisation

O

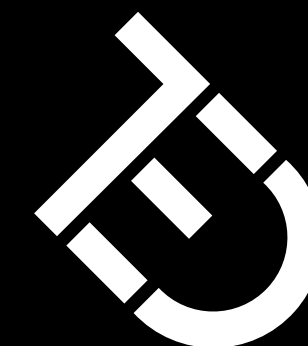
Alcohols can undergo oxidation to form carboxylic acids (with the help of acidified $KMnO_4$ and heat)



Equation shows oxidation as oxygen is gained and hydrogen is lost

Reaction will have colour change of purple to colourless ($KMnO_4$ is reduced)

Typically, alcohols are more reactive than alkanes as the C-O bond in alcohols is more reactive than the C-C bonds in alkanes



Alcohols - Uses

U

Alcohols have many uses, with ethanol (C_2H_5OH) being the most commonly used

Drinks:

Ethanol is used in all alcoholic drinks

Solvents:

Ethanol is used to dissolve substances that are insoluble in water

Used for paints, varnishes, deodorants, perfumes, etc.

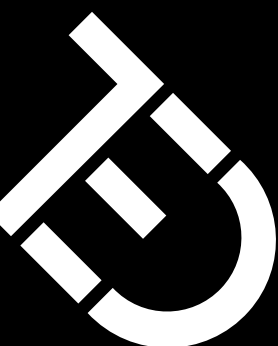
Fuel:

Used in methylated spirit (85% ethanol, 15% water) for cooking burgers or spirit lamps

Can also be added to petrol to be used for vehicle fuel in some countries (US, China, Canada, etc.)

Ethanol as a fuel is nontoxic (combustion produces CO_2 and H_2O), biodegradable and is renewable

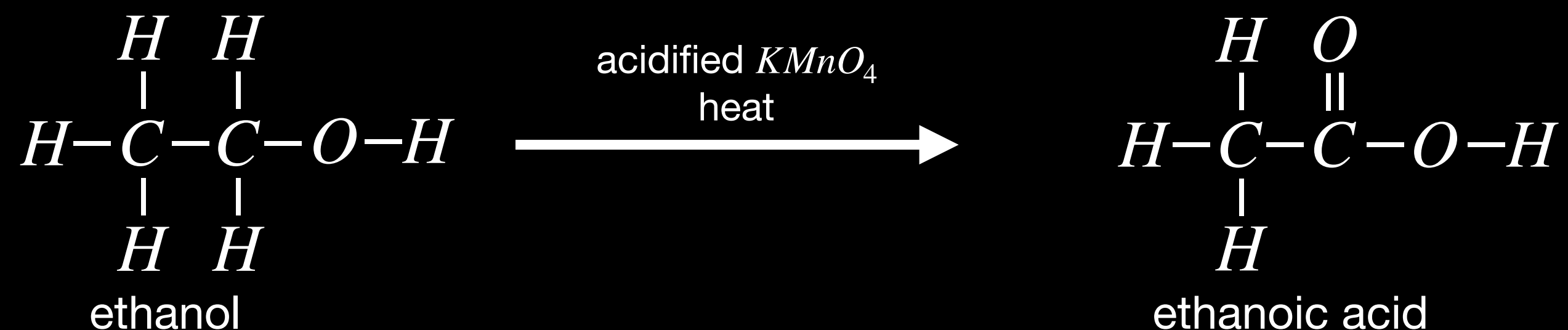
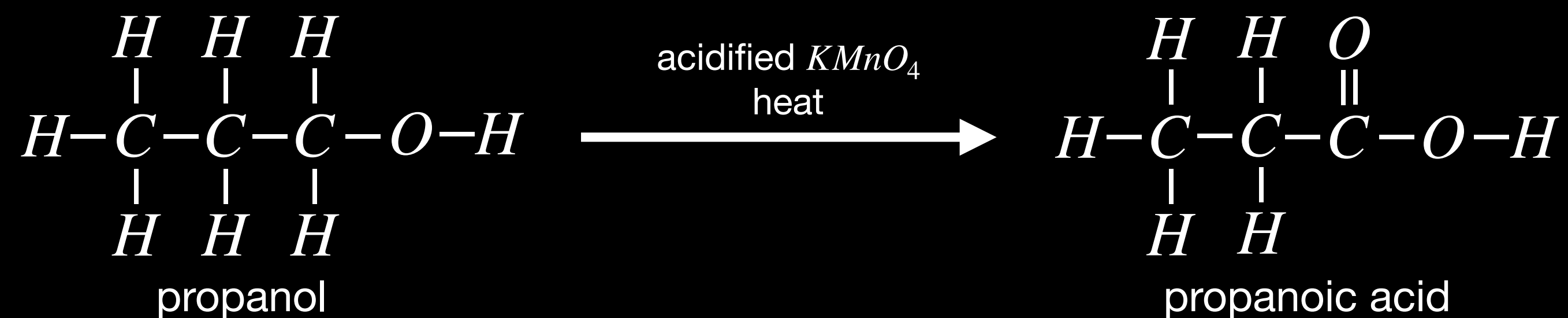
However production of ethanol could prevent crops from being directed to those who need food



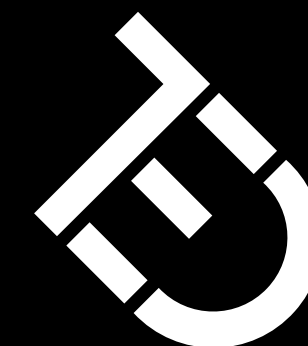
Carboxylic Acids - Production

ca

Carboxylic acids are produced by the oxidation of alcohols (with the help of acidified $KMnO_4$ and heat)



Carboxylic acids are also soluble in water like alcohols



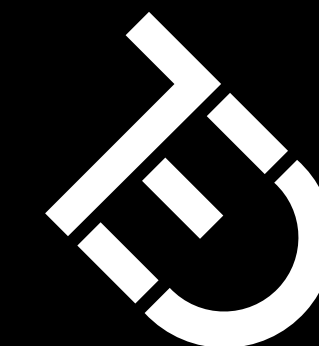
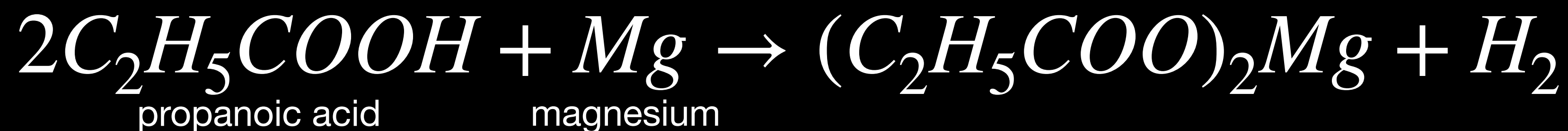
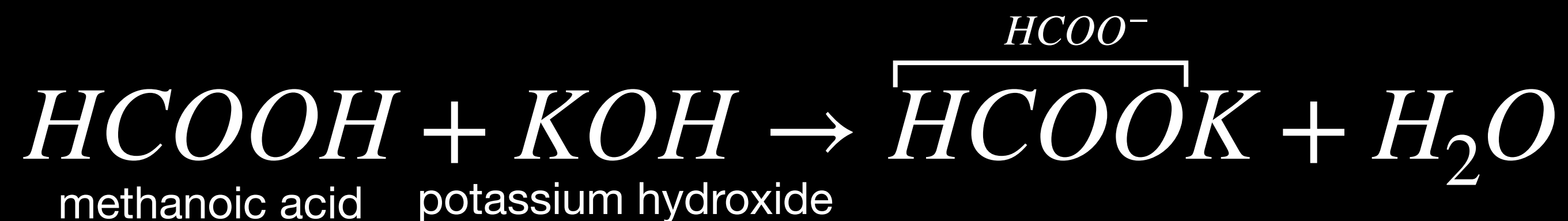
Carboxylic Acids - Acid Reactions

ac

Carboxylic acids are weak acids as they dissociate partially:



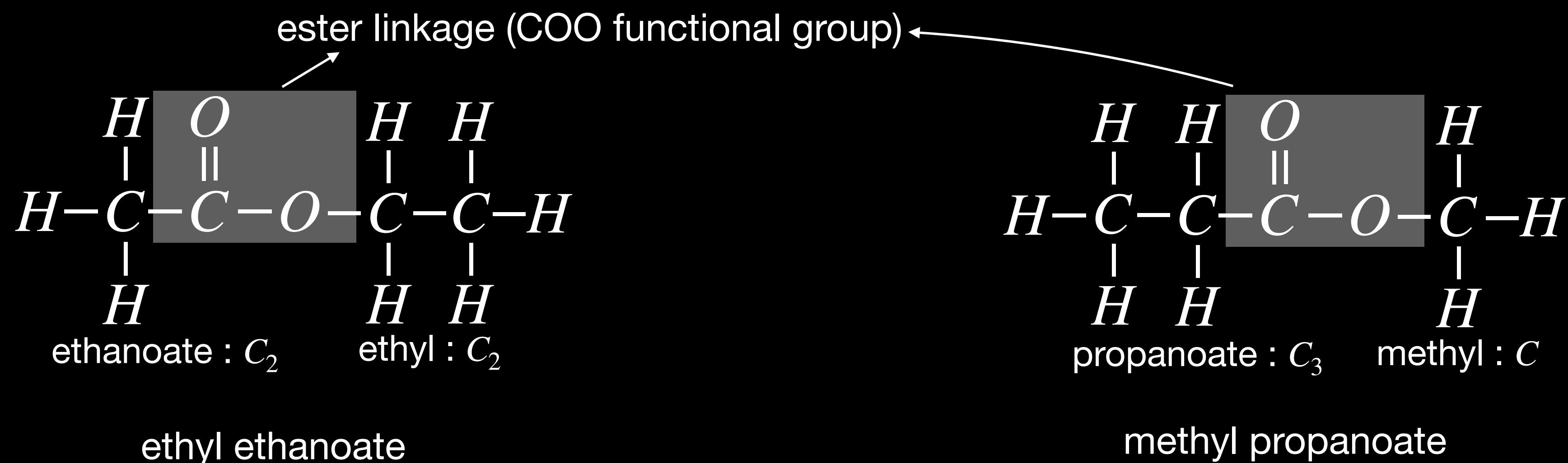
As they are acids, carboxylic acids also react with bases, carbonates and (reactive) metals



Carboxylic Acids - Esterification

es

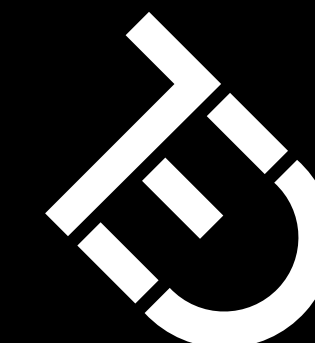
Carboxylic acids react with alcohols to form esters, a homologous series with the COO functional group.



General formula: $C_nH_{2n+1}COOC_mH_{2m+1}$ or $RCOOR'$, where R and R' are two different group of atoms

Naming: -yl -anoate

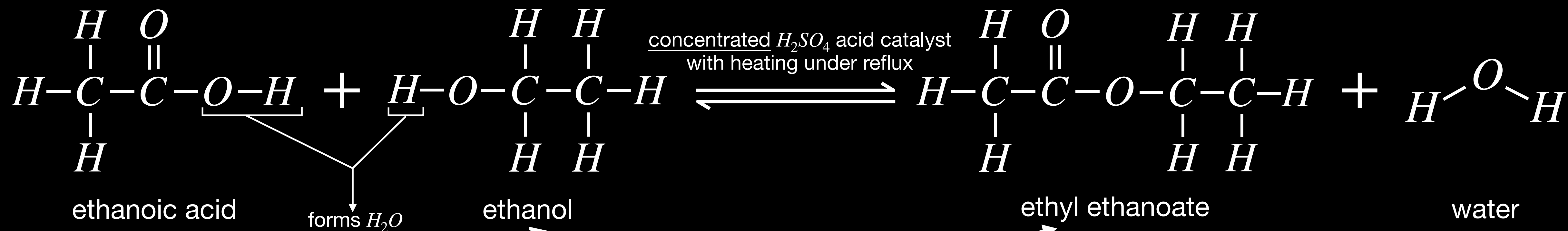
Esters are used as solvent for cosmetics and glues, and have a sweet fruity smell, thus making it useful for perfumes and artificial food flavourings



Carboxylic Acids - Esterification

es

Carboxylic acids react with alcohols to form esters and water, thus the forward reaction is a condensation reaction.

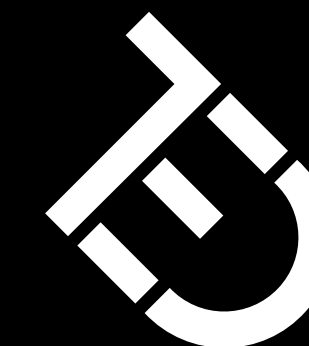
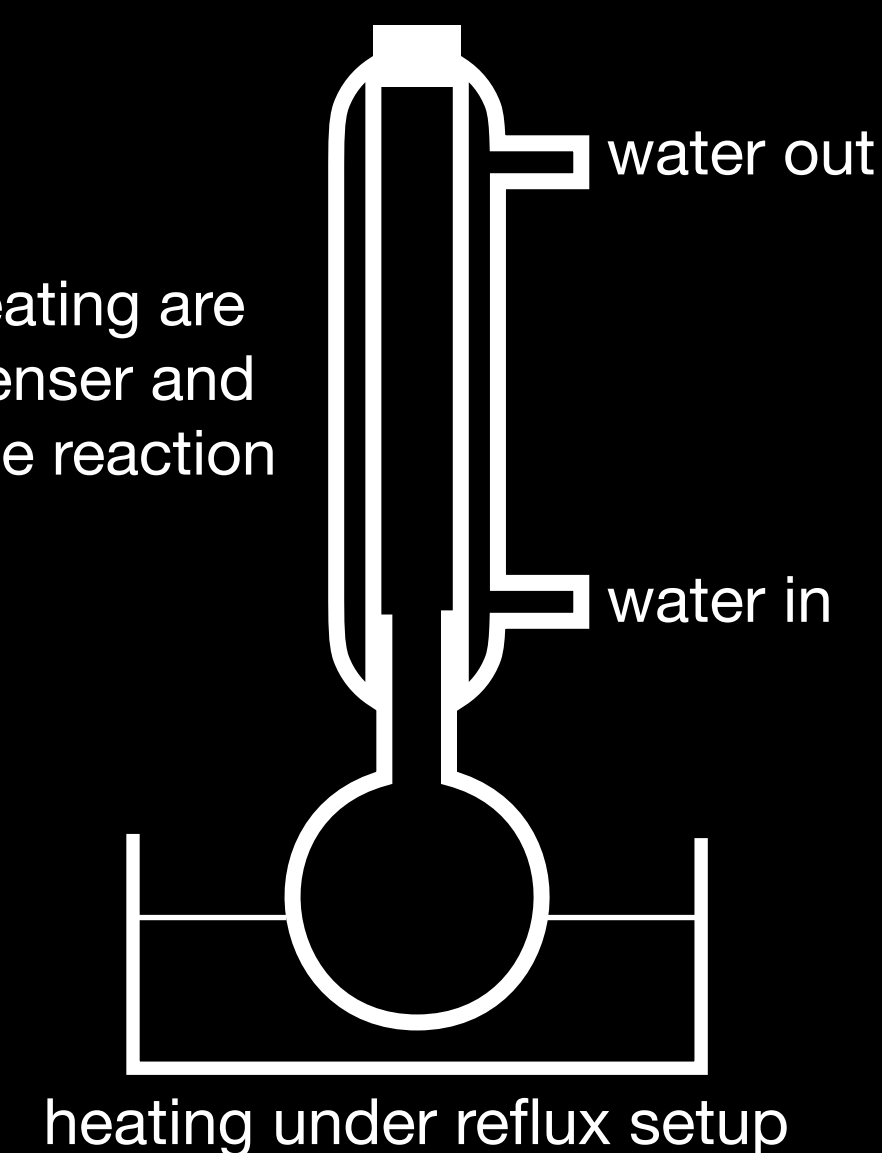


The reaction needs heat with
CONCENTRATED sulfuric acid to
react, where it is heated under reflux

(heated with a condenser to condense vapours so they return back to the condenser)

This prevents the volatile reagents from escaping the reaction vessel

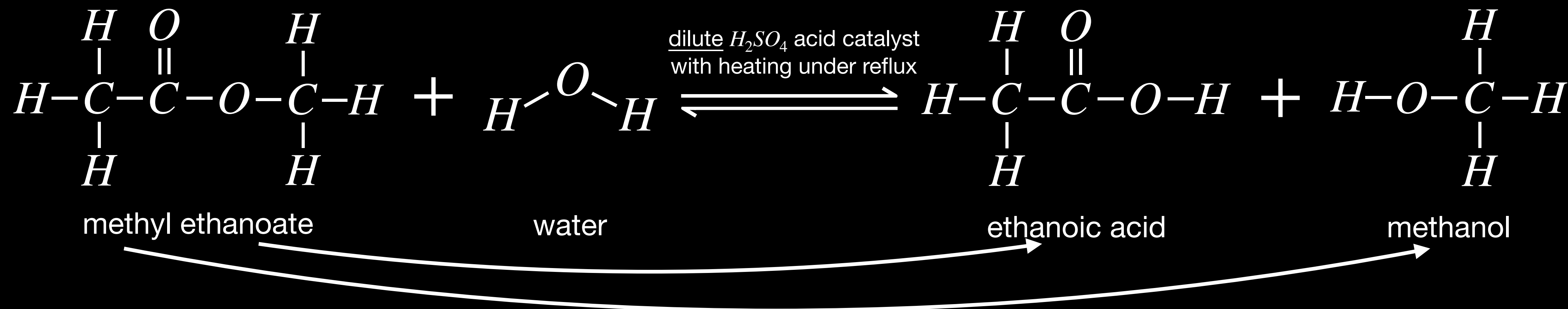
Vapours from heating are cooled by condenser and drip back into the reaction vessel



Carboxylic Acids - Esterification

es

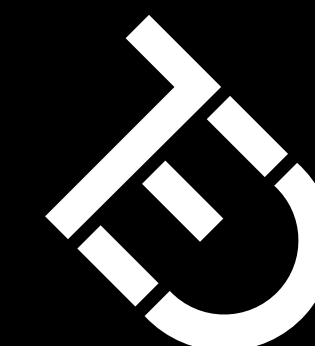
As esterification is a reversible reaction, esters can be broken down (hydrolysed when warmed with dilute sulfuric acid under reflux)



Dilute sulfuric acid is used because it has more water, thus there is more water on the left of the reaction, thus the reaction will favour the forward reaction to remove water.

Likewise, concentrated sulfuric acid is used as it has less water to favour the condensation reaction to produce more water

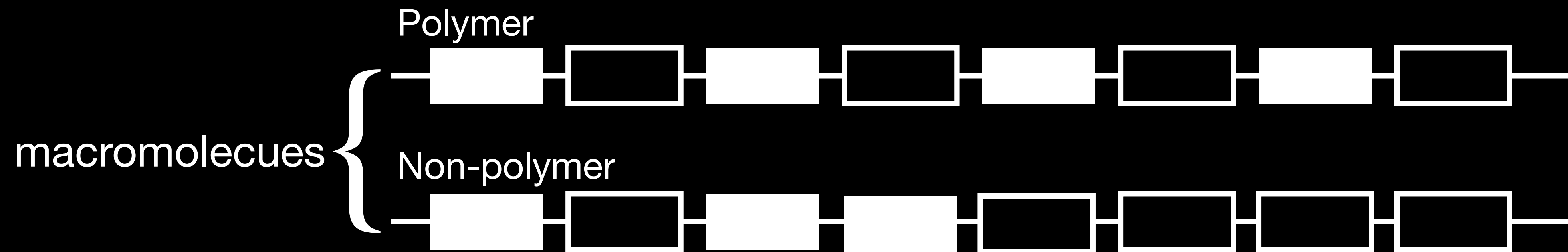
[Le Chatlier's Principle]



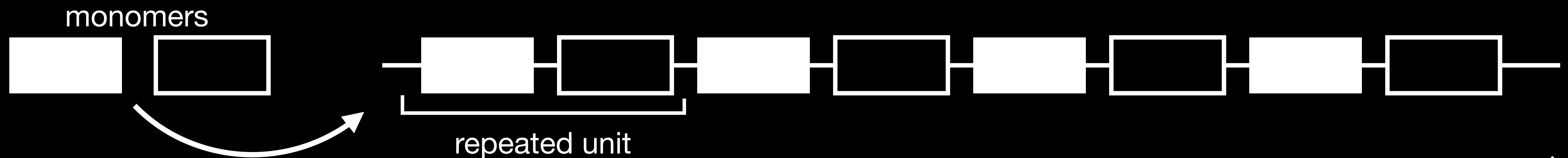
Macromolecules & Polymers

ma

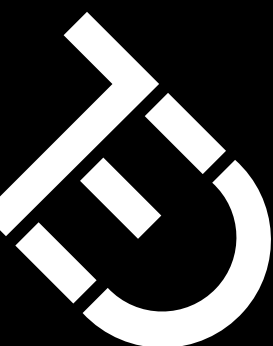
Macromolecules are larger molecules that are made from smaller units whereas polymers are a subset of macromolecules whose subunits are the same type



Polymers are made out of smaller molecules called monomers:



Monomers are not the same as repeated units, monomers are its own molecules while the repeated unit is part of a molecule

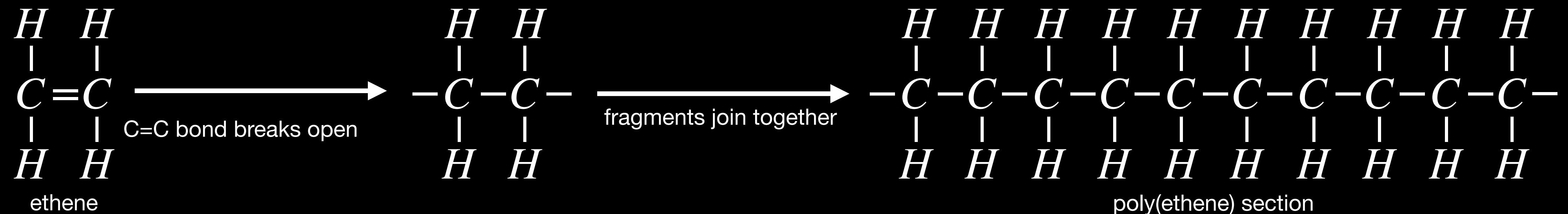


Addition Polymerisation

ap

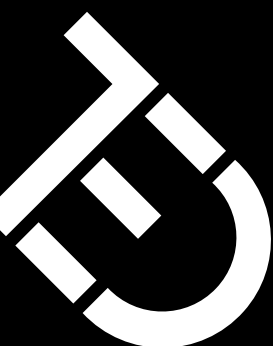
For addition polymerisation, the monomers need to be unsaturated, and have to occur at **high temp and pressure**. As the reaction is an addition reaction, there will be no other products other than the polymer formed.

For example, shown is the addition polymerisation of ethene



This causes the polymer to be saturated, as there are no more C=C double bonds present.

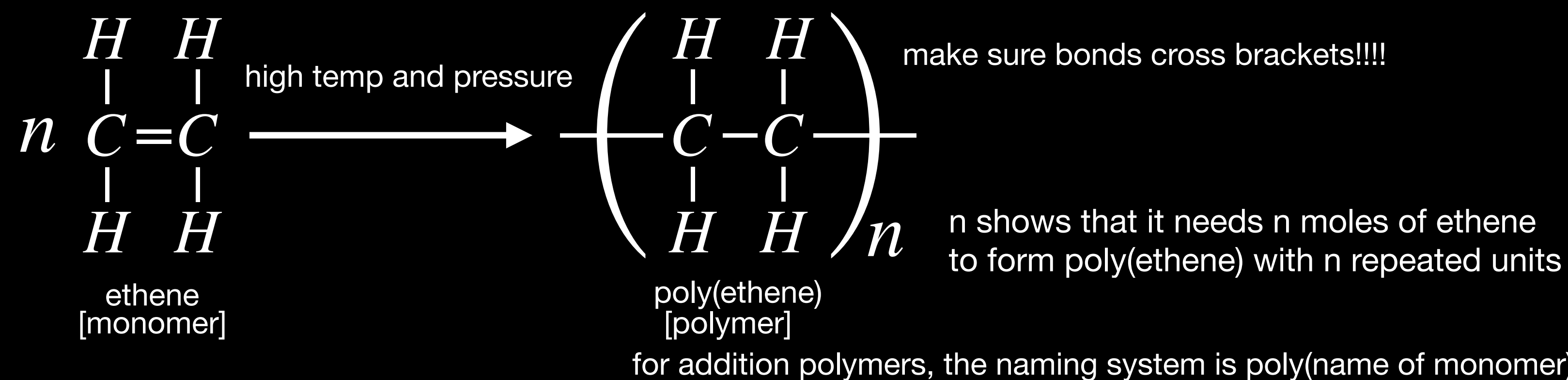
poly(ethene) is commonly used for plastic bags and cling film !!!! expldoing



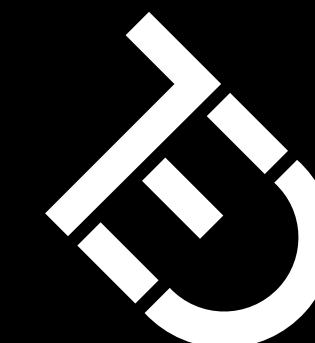
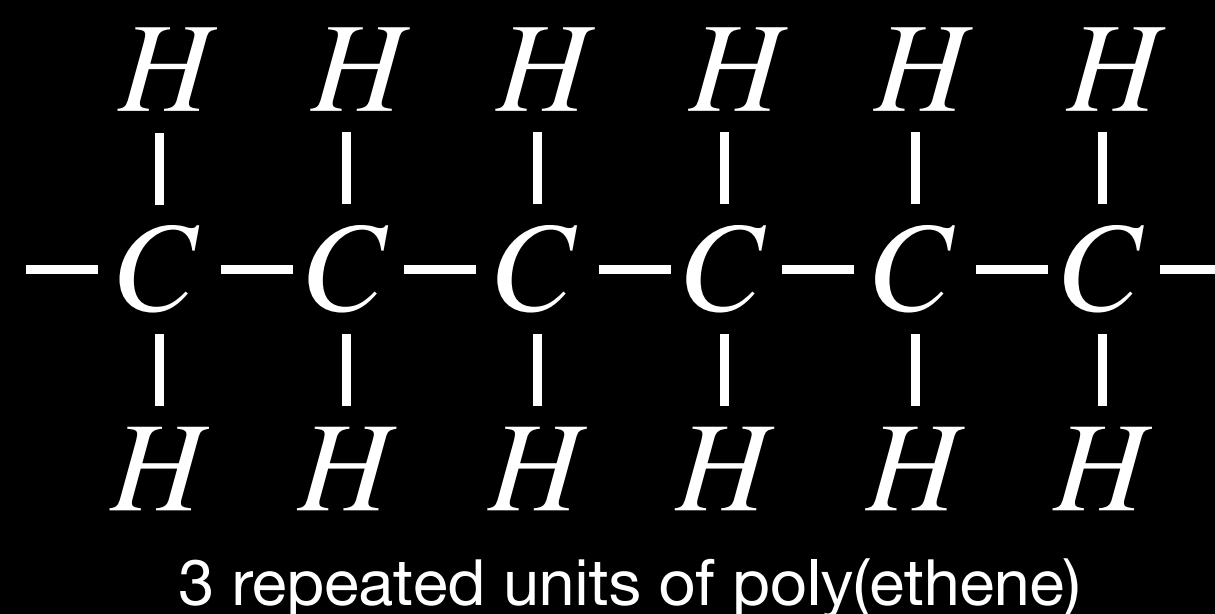
Addition Polymerisation

ap

The chemical equation of the addition polymerisation of ethene is



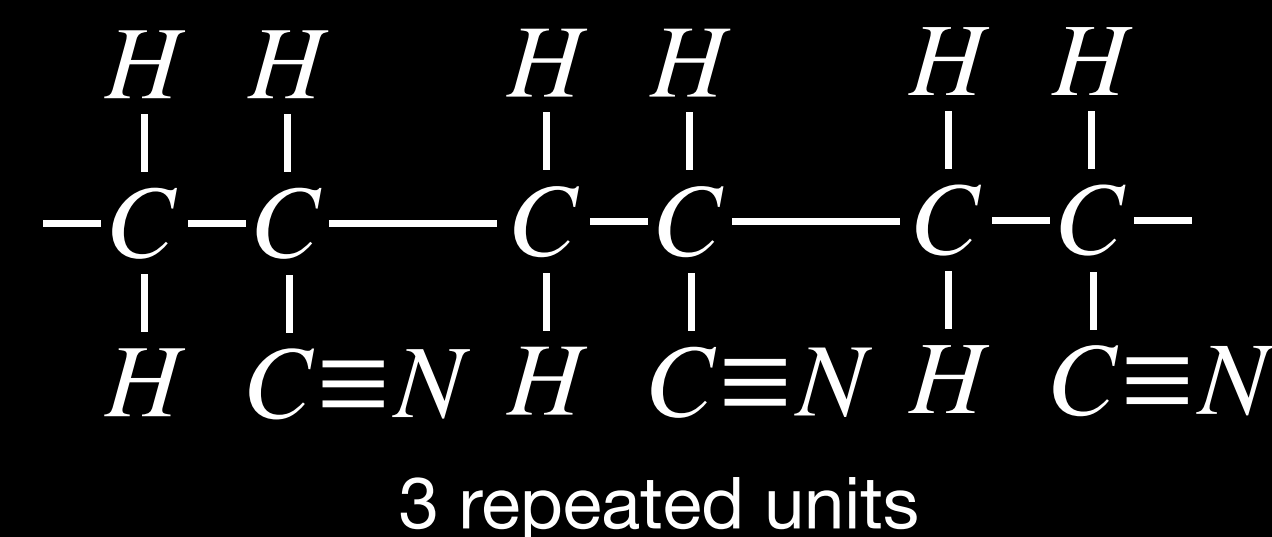
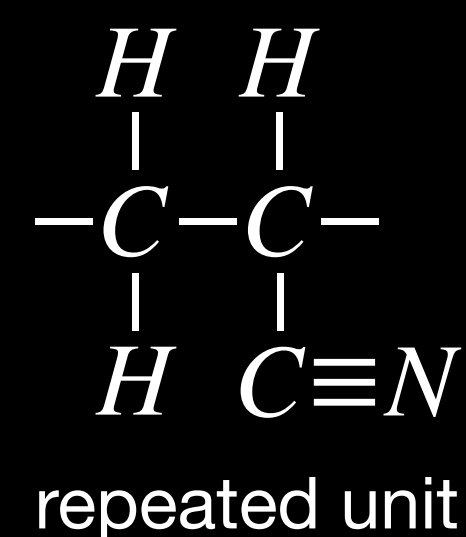
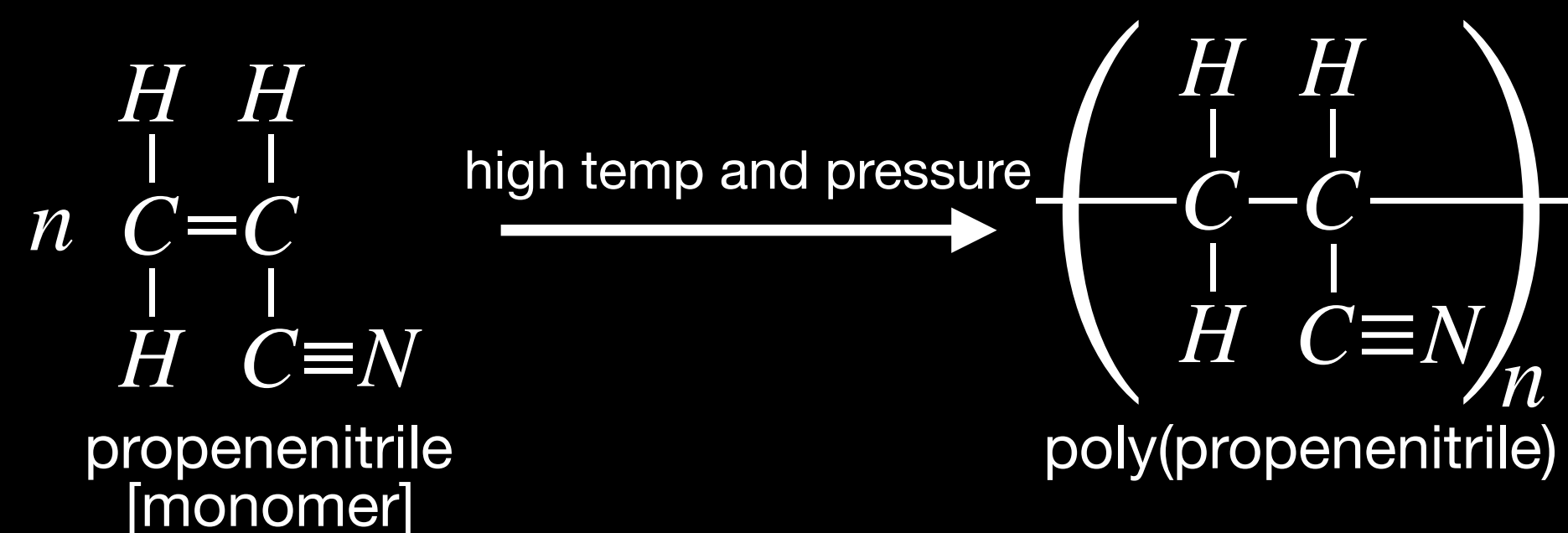
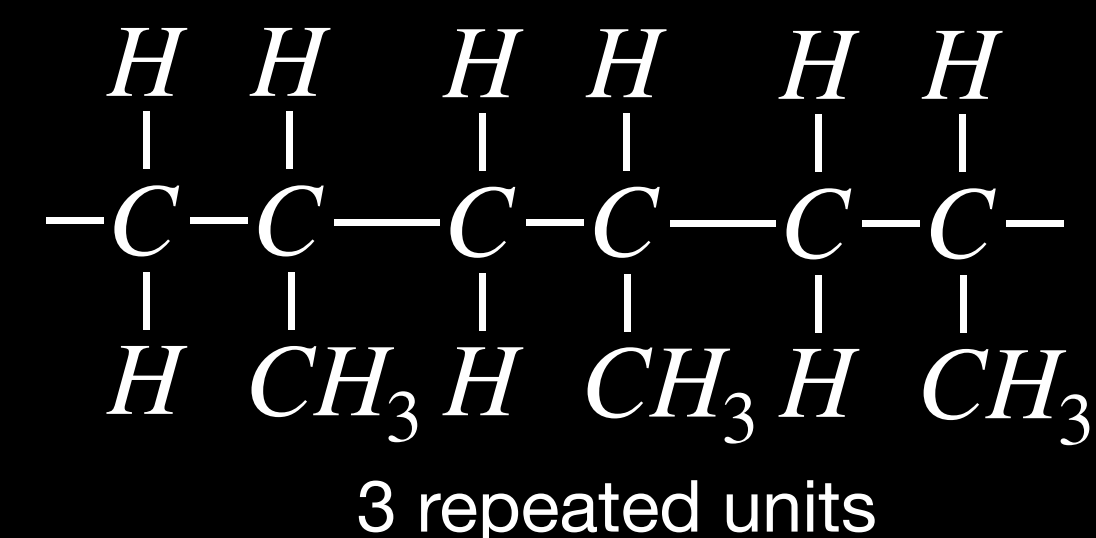
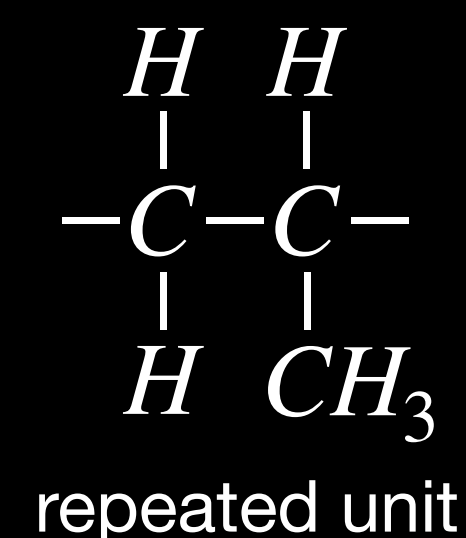
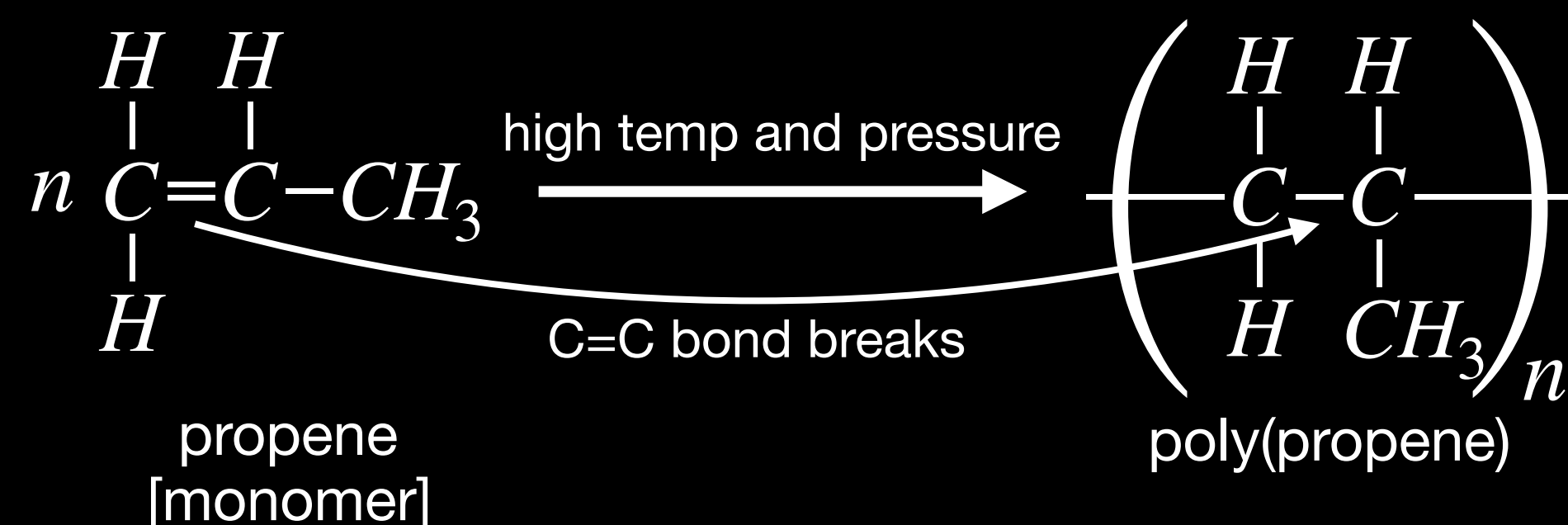
Polymers are drawn with a bracket around the repeated unit with n, to show that there are multiple units. Questions may ask you to draw the monomer, the repeated unit, or the polymer: e.g. Draw 3 repeated units of poly(ethene)



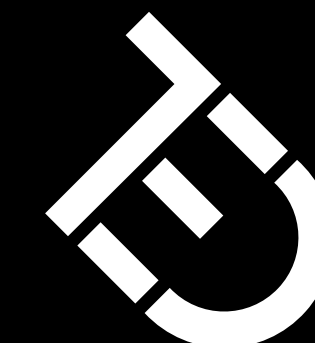
Addition Polymerisation

ap

Only the C=C double bond breaks and anything else stays bonded to its respective C atom.



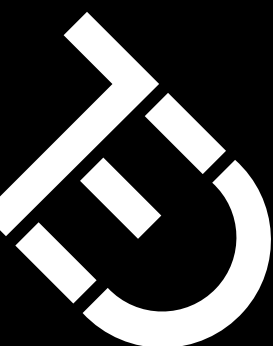
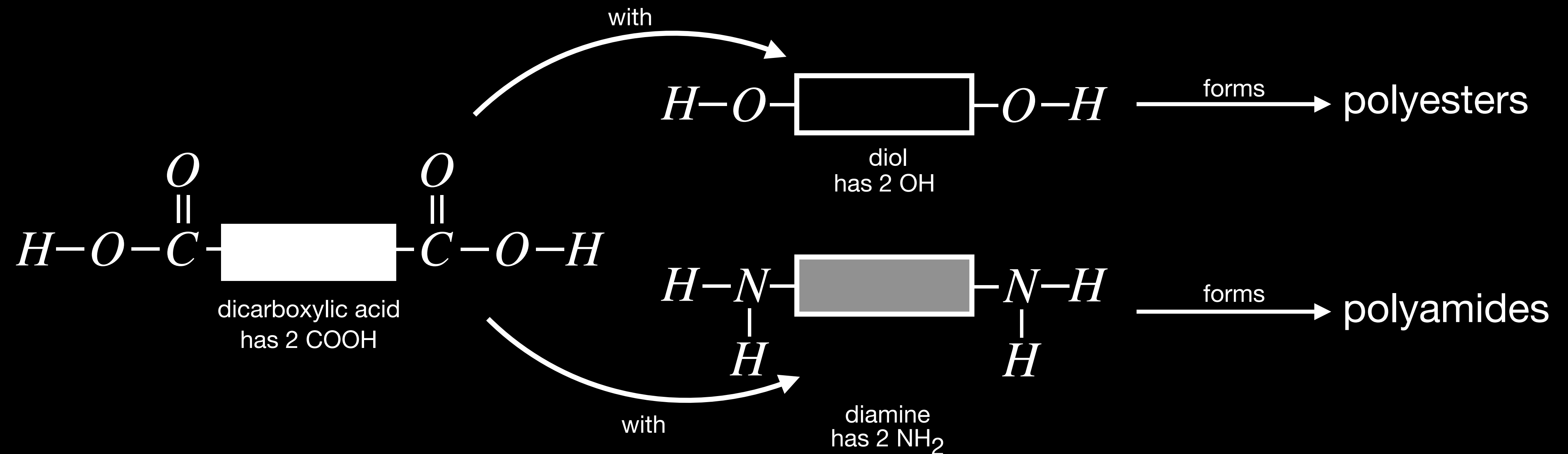
Note that other double bonds are not disturbed in the reaction



Condensation Polymerisation

CO

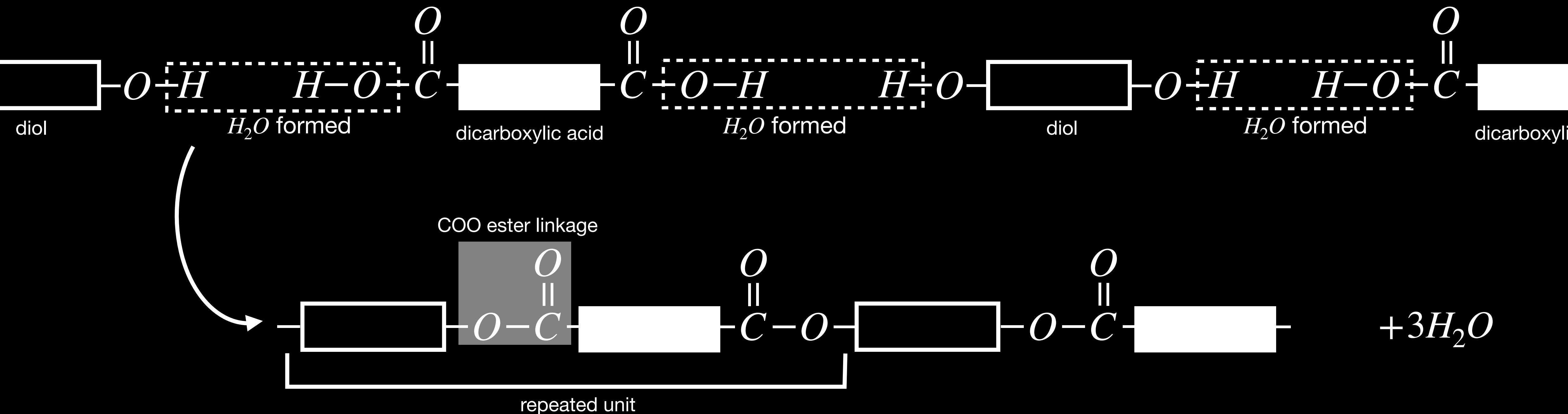
Condensation polymerisation has monomers of different functional groups and join together with the loss of a small molecule (H_2O , HCl , etc.)



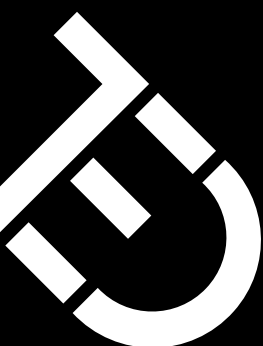
Condensation Polymerisation - Polyesters

CO

Polyesters are polymers that are linked with the ester linkage (COO). Polyesters are made with monomers containing the -COOH(carboxylic acid) and -OH(alcohol) functional groups.(e.g. dicarboxylic acid and diol)



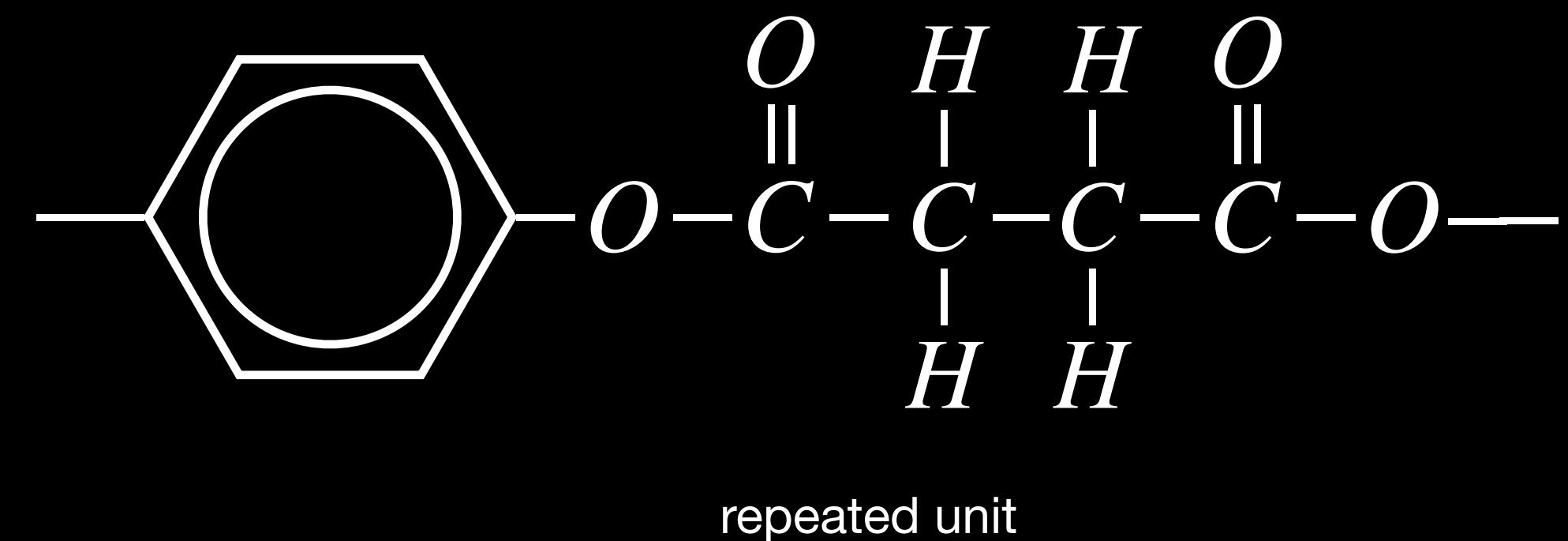
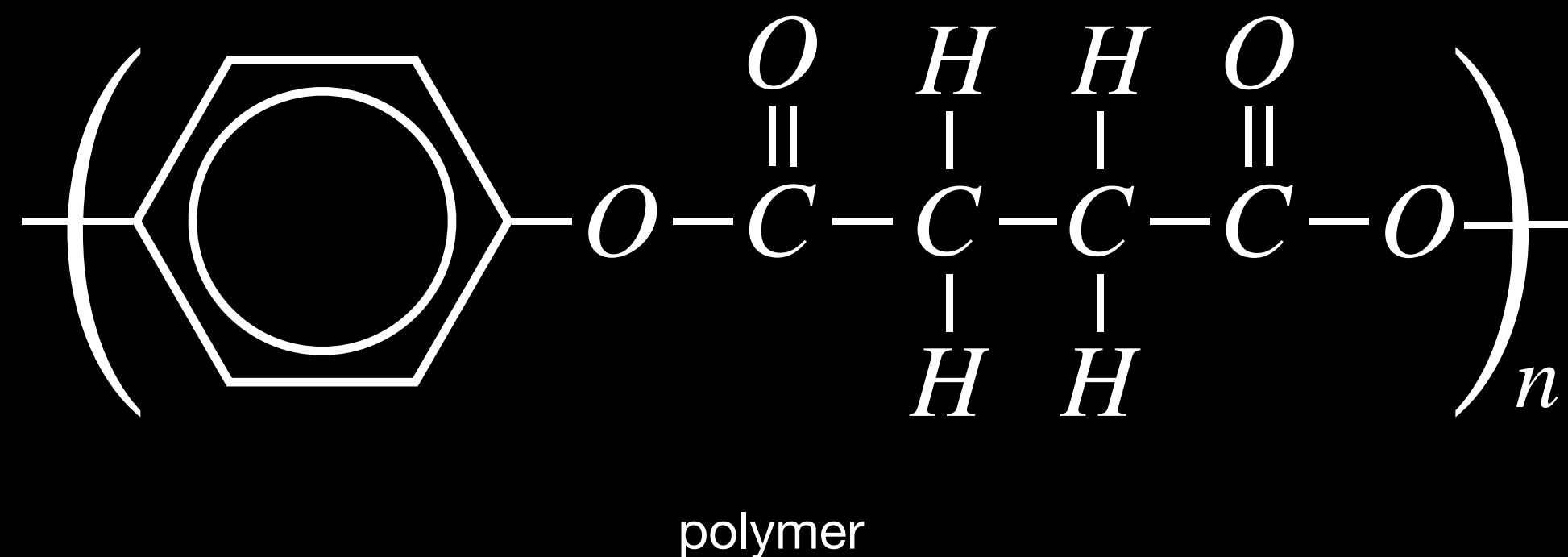
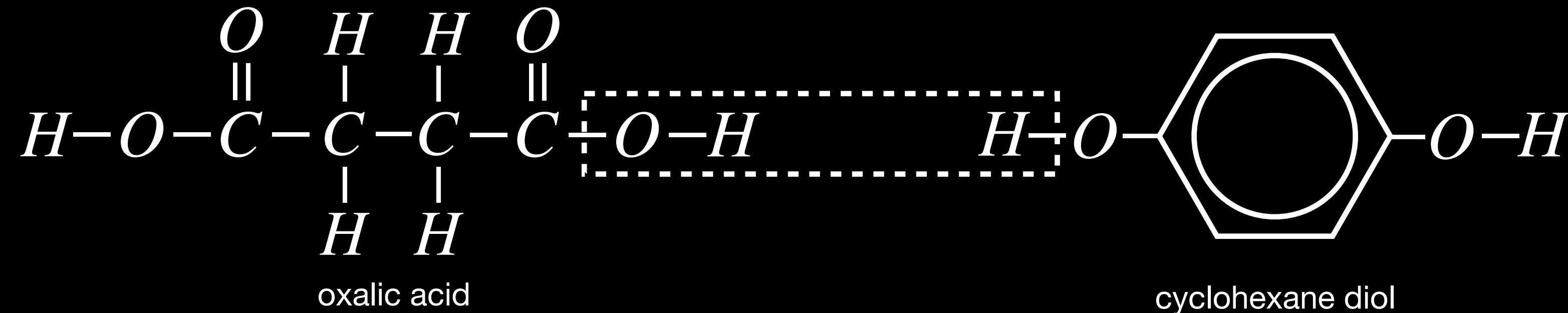
A common polyester used is Terylene, which is used for clothing, curtains, parachutes, sails, as it is strong, elastic and light.



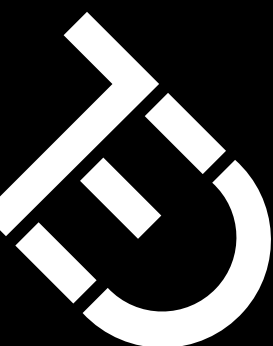
Condensation Polymerisation - Polyesters

CO

The group of atoms in the diol or dicarboxylic acid does not matter as they do not participate in the reaction.
For example:



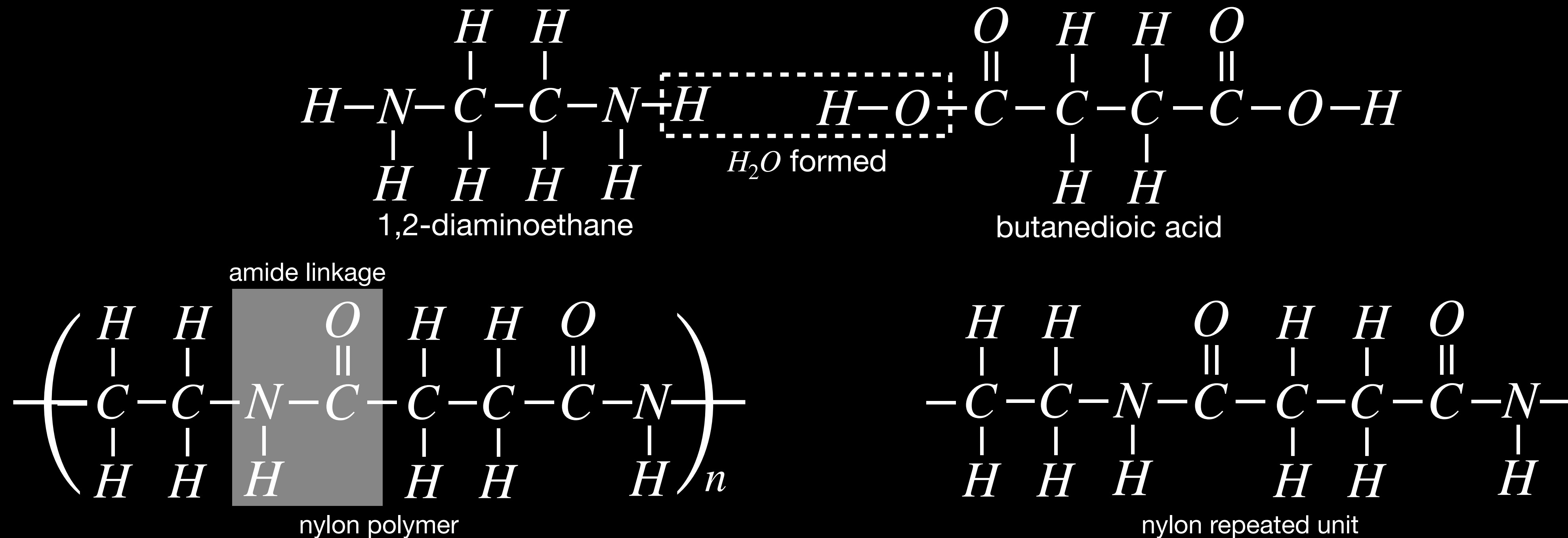
Only the COOH and OH function groups are changed to form the ester linkage, and the atoms in the middle does not matter.
The repeated unit should also contain 2 ester linkages that are flipped, and not 1



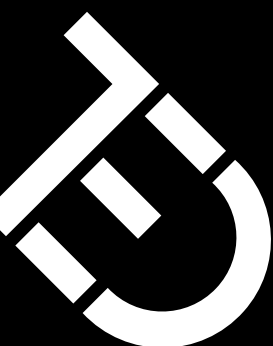
Condensation Polymerisation - Polyamides

CO

Polyamides are also formed via condensation but are linked with the amide linkage (CONH), which is formed with the -COOH functional group and the -NH₂ functional group. One example is nylon, made with 1,2-diaminoethane and butanedioic acid. (No need to memo names of monomers)



Nylon is strong, light and stretchable, making it useful for clothing and fishing line

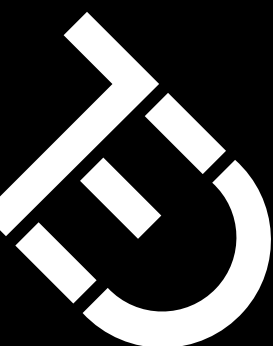
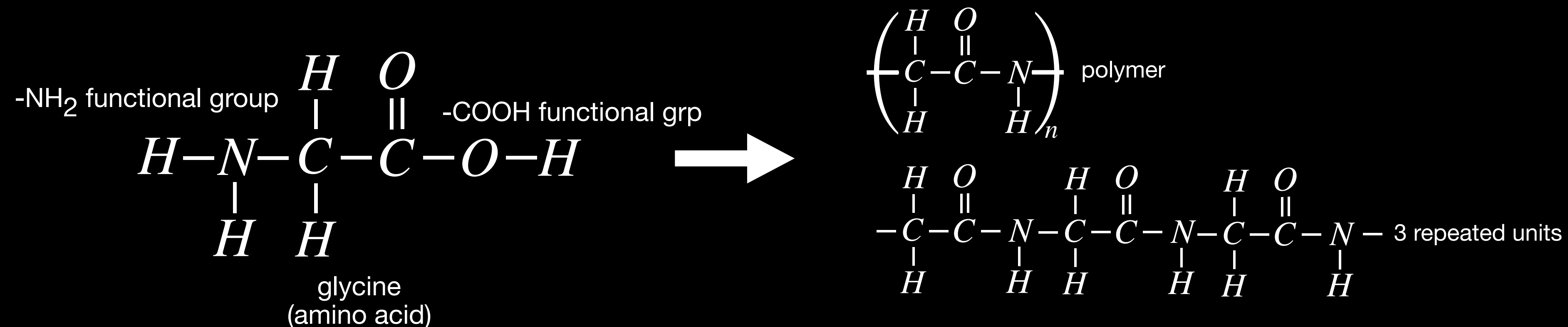
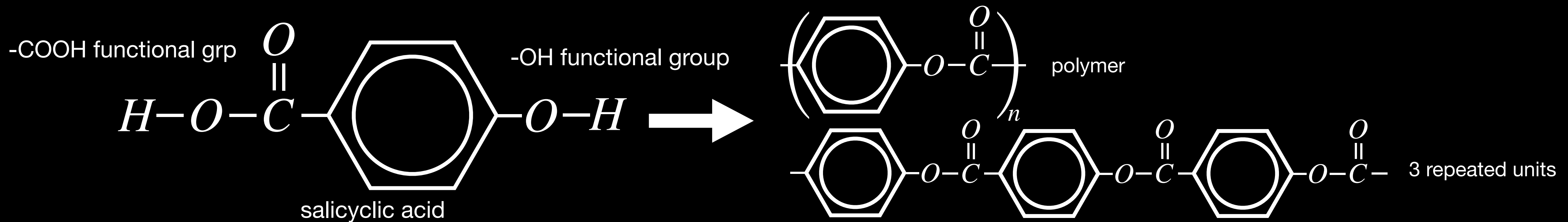


Condensation Polymerisation

CO

Note that condensation polymers do not need two monomers to form the polymer, and can have only one monomer with both functional groups needed to form the polymer.

For example, Salicylic acid has both -OH and -COOH functional groups, thus it can undergo condensation to form ester linkages. Amino acids also have -COOH and -NH₂ functional groups to form amide linkages under condensation.

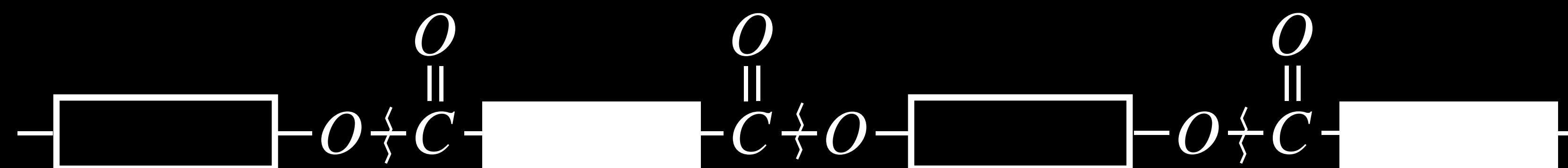


Hydrolysis of Condensation Polymers

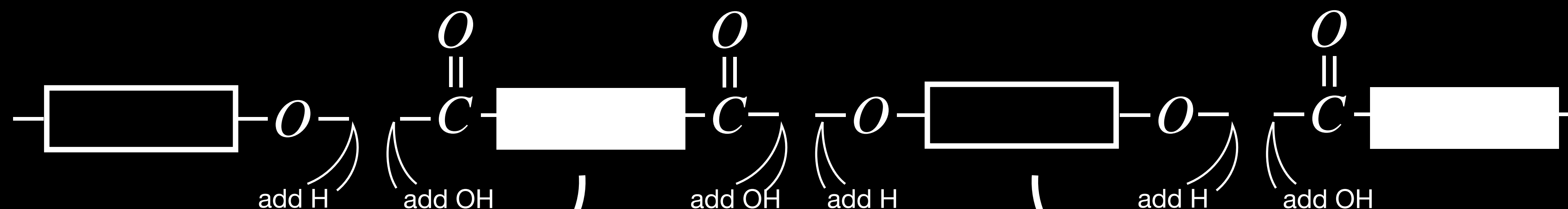
CO

Condensation polymers can also be broken down by water (hydrolysed) to form the original monomers, **by warming with a dilute acid or a base as a catalyst.** For example, the hydrolysis of esters is shown:

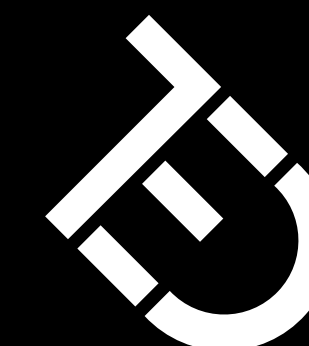
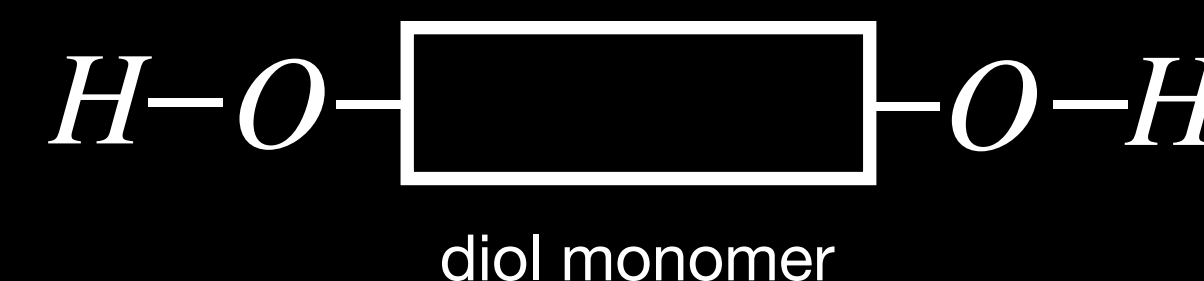
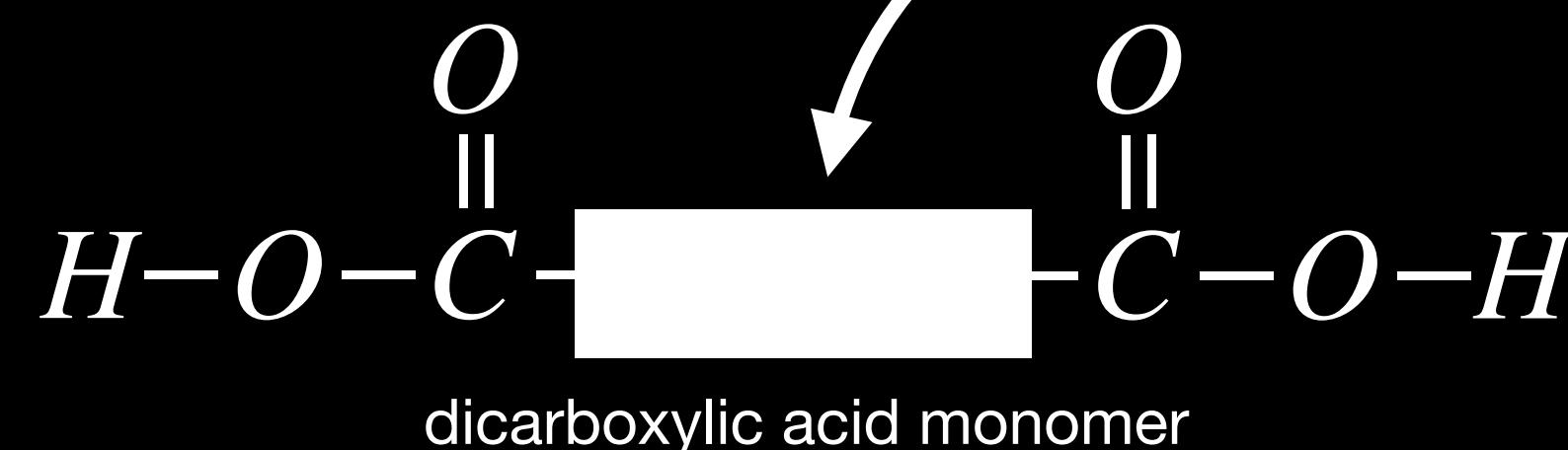
1. The C-O in the ester linkage is broken



2. Fragments are formed which then have H or OH added to them



3. The added O and H can form COOH and OH functional groups

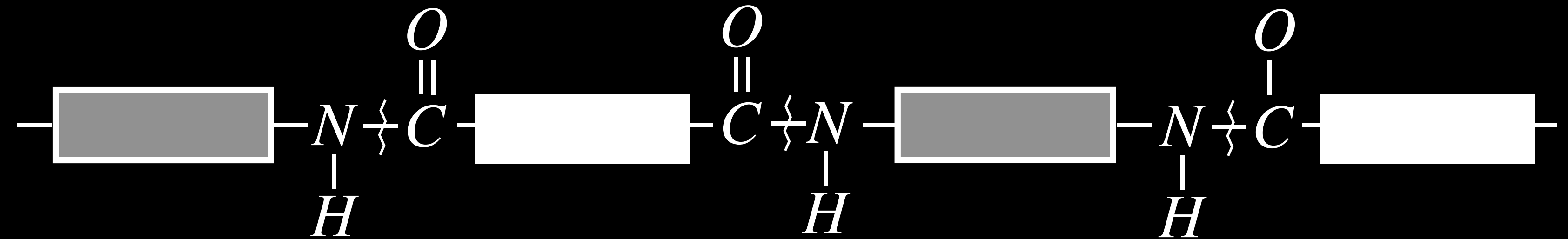


Hydrolysis of Condensation Polymers

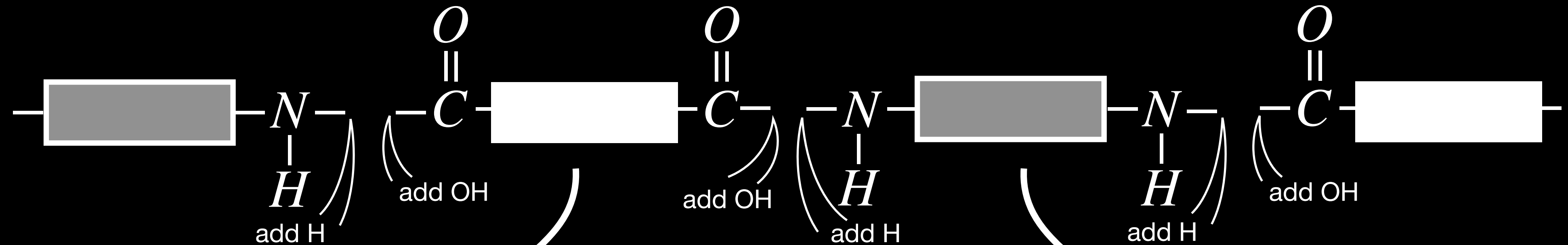
CO

Likewise, polyamides can also be hydrolysed similarly under the same conditions to form the original monomers:

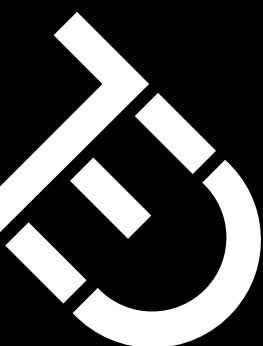
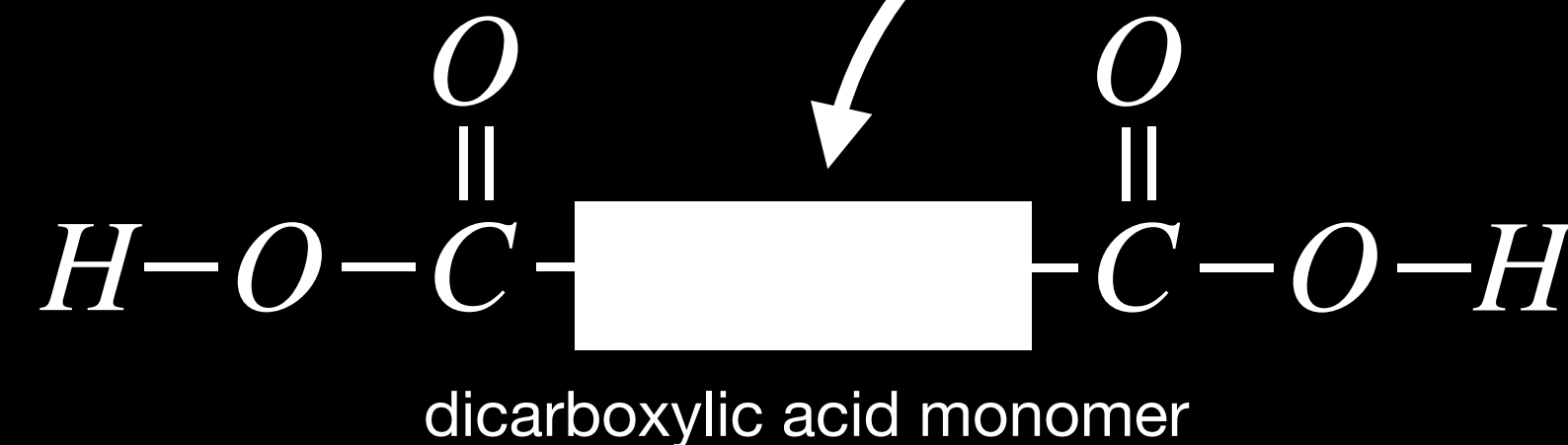
1. The C-N in the amide linkage is broken



2. Fragments are formed which then have H or OH added to them



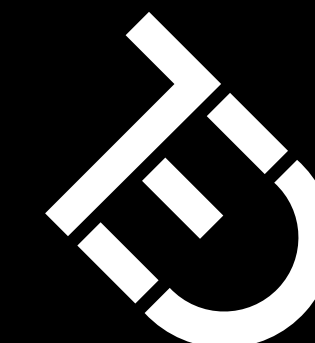
3. The added O and H can form COOH and NH₂ functional groups



Reaction Conditions Summary

rCS

Reaction	Conditions
<u>Alkane halogen substitution</u>	UV Light
<u>Alkane cracking</u>	SiO ₂ and Al ₂ O ₃ catalyst with 600°C
<u>Alkene addition with hydrogen</u>	Nickel catalyst at 150°C
<u>Alkene addition with halogen</u>	r.t.p
<u>Alkene addition with STEAM</u>	H ₃ PO ₄ catalyst, 300°C, 60 atm
<u>Alcohol production by fermentation</u>	37°C in the absence of oxygen
<u>Alcohol oxidation</u>	Heat with acidified KMnO ₄
<u>Carboxylic acid esterification</u>	Concentrated H ₂ SO ₄ with heating under reflux
<u>Breaking down esters</u>	Dilute H ₂ SO ₄ with heat
<u>Addition Polymerisation</u>	High temp and pressure
<u>Hydrolysis of condensation polymers</u>	Warmed with dilute acid or base



Plastic Usages & Impacts

pla

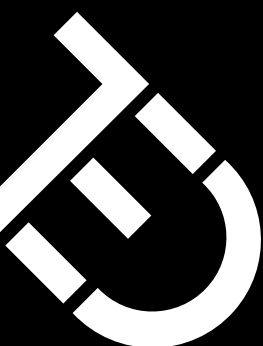
Most plastics are non-biodegradable, which means that they cannot be broken down by living organisms. As such, plastics are either dumped in landfills or burnt

Plastics in landfills fill up quickly due to them being non-biodegradable and take up a lot of space

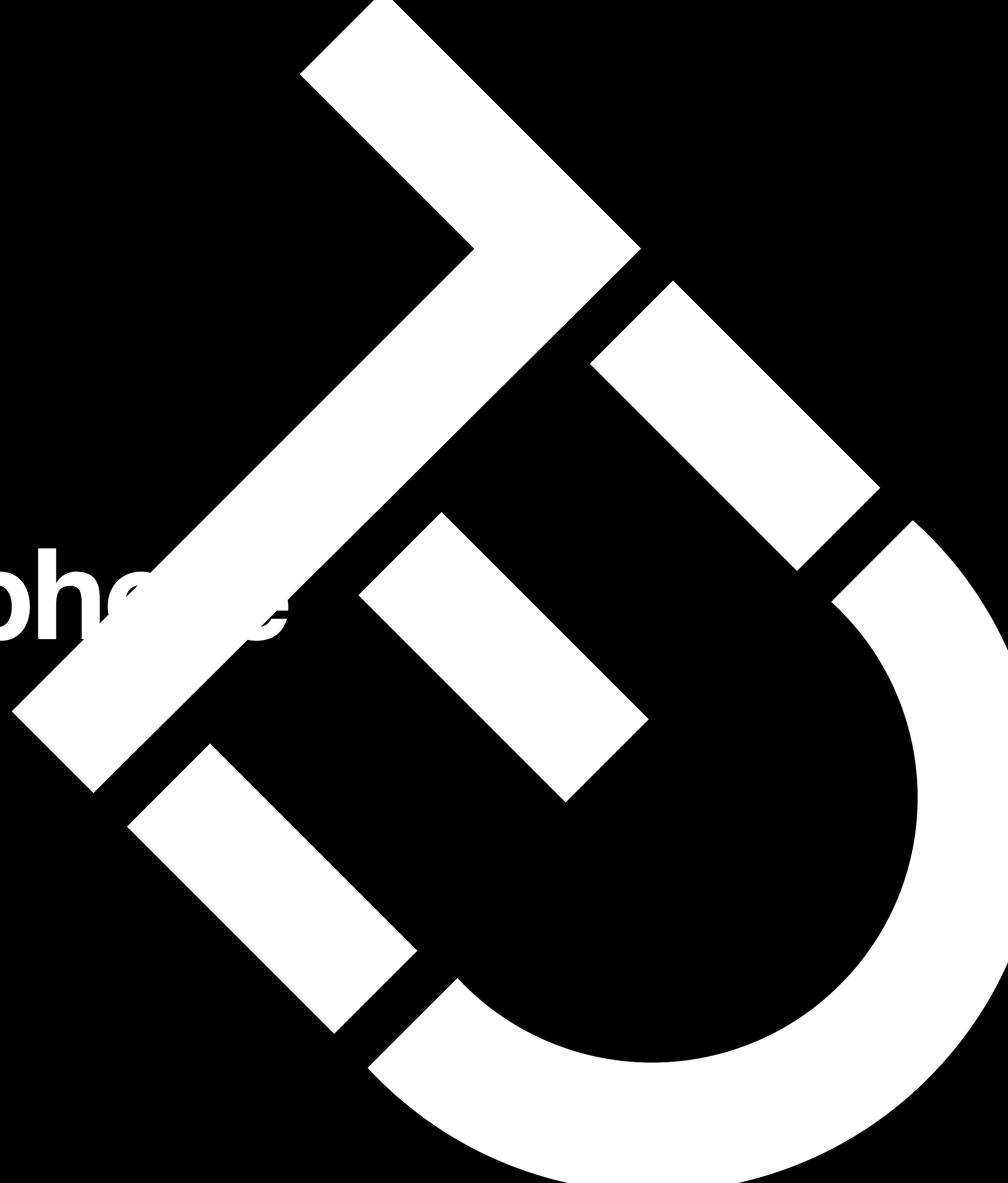
Plastics in oceans cause marine wildlife to accidentally choke on them

Incinerated plastics also release gases like carbon dioxide when completely combusted, which traps more heat as it is a greenhouse gas, causing global warming. Incompletely combusted plastics produce carbon monoxide, which is a toxic gas that is lethal in small concentrations [Air and Atmosphere]

(Questions usually need you to state one or two reasons the negative impacts of plastic pollution, does not really need you to be very specific)



Air & The Atmosphere



Composition of Air

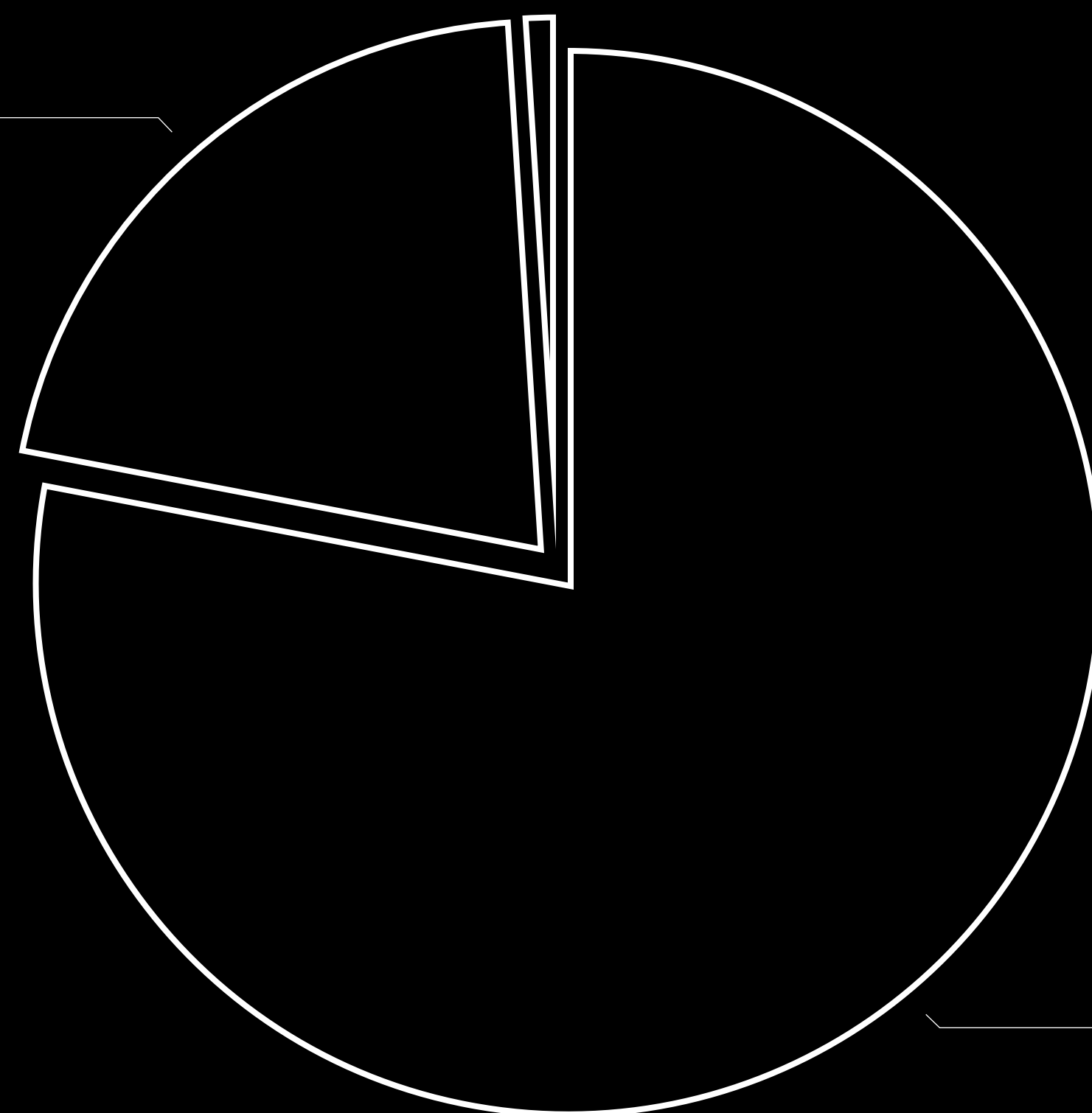
cool

Other gases, mostly Argon [0.93%], then carbon dioxide [0.04%] and water [varies with humidity]

1%

21%
Oxygen [O₂]

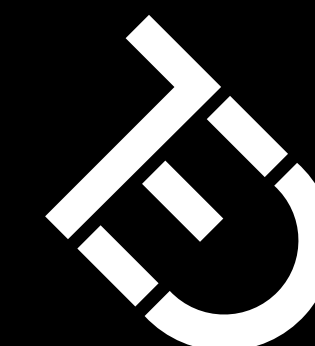
You must know the approximate percentages of nitrogen and oxygen, and that the remaining 1% is mostly argon



78%
Nitrogen [N₂]

As air is a mixture of gases as shown, when air is passed through a substance to cause a reaction, only specific gases in air actually take part in the reaction (usually oxygen)

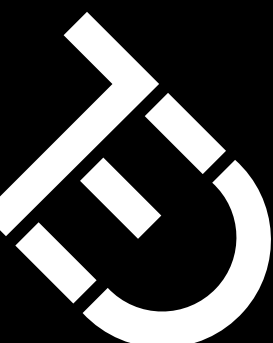
E.g. if air is passed through heated copper, only the oxygen reacts with it to form copper(II) oxide, thus 21% of the air passed through reacts with the copper



Air Pollutants & Greenhouse Gases

aegg

	Substance	Sources	Effects	Reducing Pollutants
Pollutants	Carbon Monoxide [CO]	<u>Incomplete combustion</u> of carbon containing fuels	Forms irreversible bond with haemoglobin which causes breathing difficulty	Surveillance of CO levels
	Nitrous Oxides [NO,NO2]	Naturally from <u>lighting</u> , however also produced from <u>combustion engines</u>	Irritates eyes, lungs, and causes breathing difficulties Forms acid rain (nitric acid, sulfuric acid, and sulfurous acid[H2SO3] are formed)	Use less Nitrogen Fertilisers. Catalytic converter
	Sulfur Dioxide [SO2]	Naturally from <u>volcanic eruptions</u> , however also from <u>combustion of fossil fuels</u>		Scrubbers to remove it from emissions
Greenhouse Gases	Methane [CH4]	<u>Living matter decay</u> , <u>cow flatulence</u>	Greenhouse effect which leads to global warming	Combustion of Methane to form Carbon Dioxide
	Carbon Dioxide [CO2]	<u>Complete combustion</u> of carbon contains fuels		Use renewable energy



Catalytic Converters

C

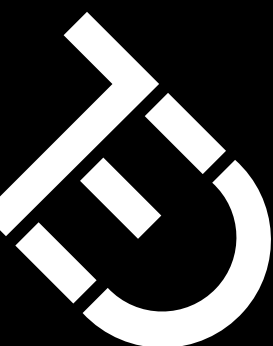
Catalytic converters use catalysts such as platinum, palladium, and rhodium to remove harmful gases emitted from car exhausts to form less harmful gases

the reactions remove oxides of nitrogen, carbon monoxide, and unburnt hydrocarbons

reduction of oxides of nitrogen: $NO_x \rightarrow N_x + O_x$

oxidation of carbon monoxide: $2CO + O_2 \rightarrow 2CO_2$

oxidation of unburnt hydrocarbons: $C_xH_x + O_2 \rightarrow CO_2 + H_2O$

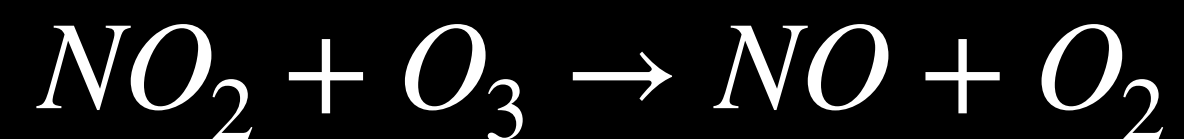
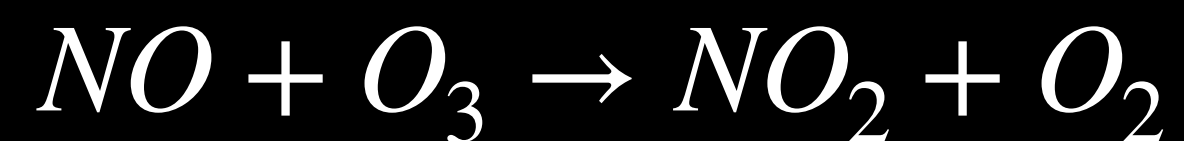


The Ozone Layer

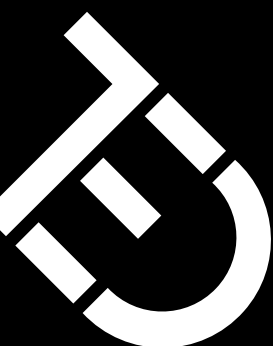
what?

The ozone layer is made up of ozone or O_3 . The ozone layer protects earth from UV light. It is essential for life to exist on earth.

Oxides of Nitrogens and Chloroflourocarbons are able to destroy the ozone layer by reacting with the O_3 to form O_2 .



As shown by these 2 equations, even 1 molecule of NO can destroy thousands of atoms of O_3 due to its ability



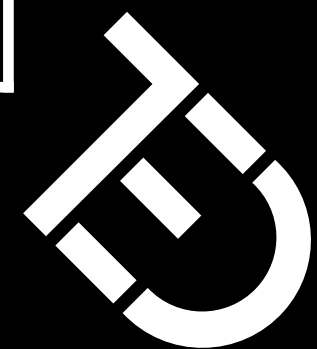
Carbon Cycle

C

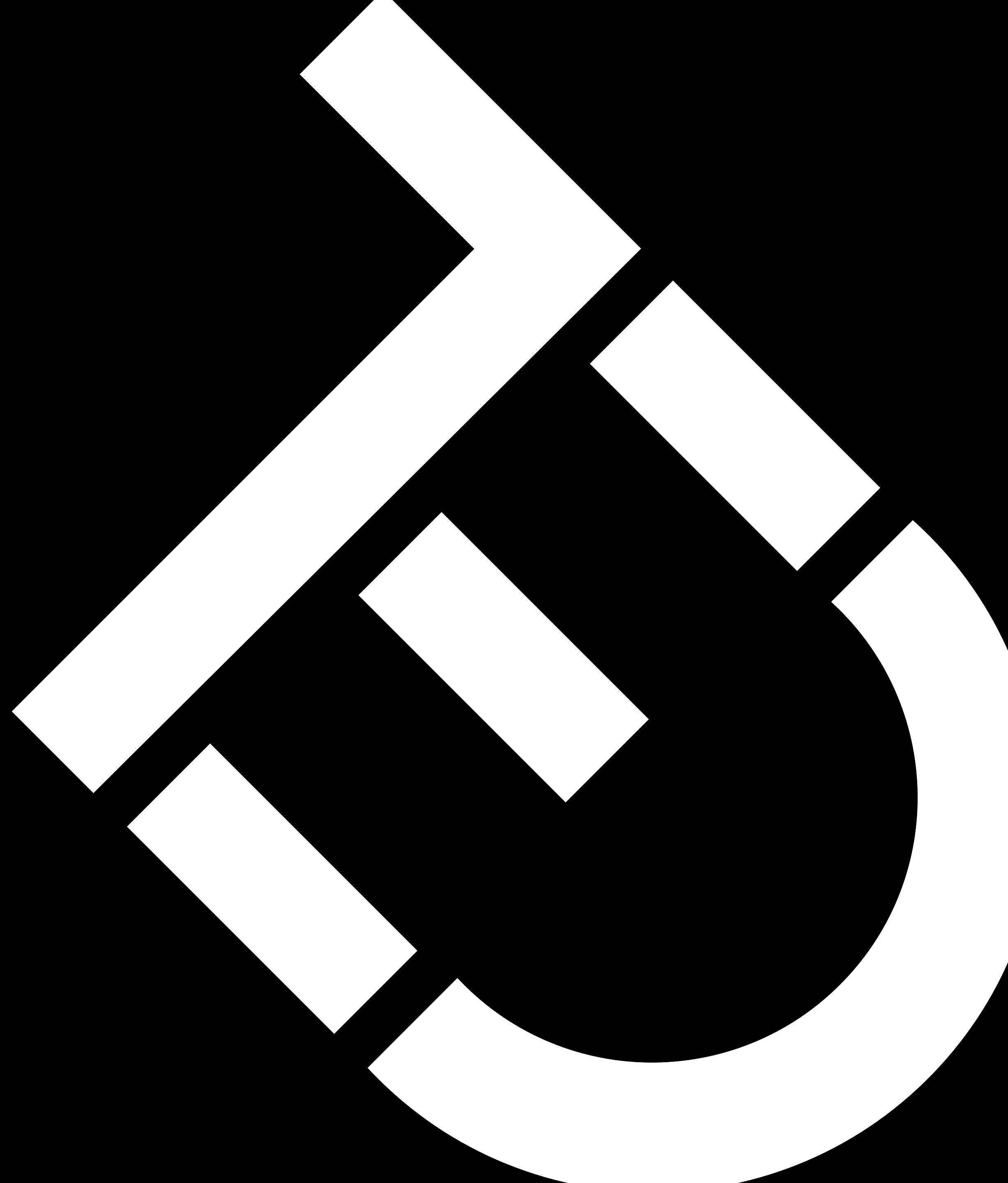
the carbon cycle is made up of various different processes that take in carbon from the atomsphere and emit carbon to the atmosphere

The processes in the carbon cycle work together to maintain a stable amount of carbon in the atmosphere

take in carbon	emit carbon
photosynthesis	respiration of animals
ocean uptake of carbon dioxide	factory emissions
	diffusion of carbon dioxide from ocean
	decaying organisms



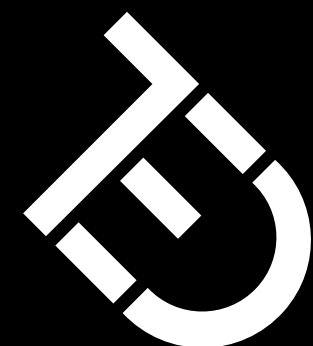
Practical Skills



Instrument Accuracies

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Apparatus	Unit	Smallest division	Precision	Example
Burette	cm ³	0.1cm ³	0.05cm ³	21.00cm ³ , 16.35cm ³
Electronic mass balance	g		0.01g	7.92g, 2.01g
Measuring cylinder	cm ³	1cm ³	1cm ³	1cm ³
Pipette	cm ³			25.0cm ³
Stopwatch	s		1s	5s, 24s
Thermometer	°C	1°C	0.5°C	23.5°C, 31.0°C



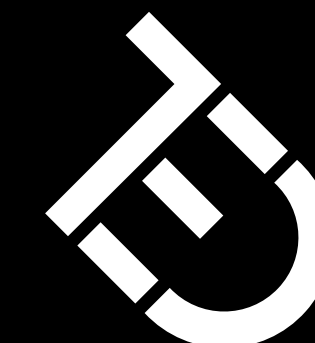
QA Tips

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For most substances, pour about 1cm depth. For limewater, pour 2cm depth

To check if precipitate is soluble in excess, firmly knock test tube against wrist/hand to mix properly. If precipitate still accumulates, insoluble in excess

Make sure to conduct any gas tests if there is significant effervescence, by using your thumb to cover the mouth of the test tube to prevent the gas from escaping



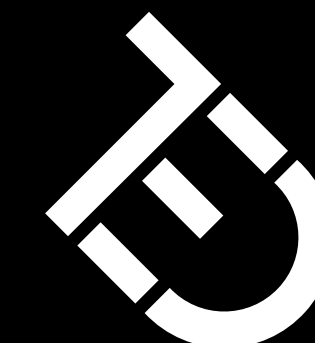
QA Tips - Observation Writing

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	Solids & precipitates		Solutions & Liquids		Gases & Effervescence	
Test result	+ve	-ve	+ve	-ve	+ve	-ve
Template	[colour of ppt.] ppt observed [soluble/insoluble] in excess to give a [colour of solution] solution	No ppt. formed, solution remained [colour]	Solution/Mixture changed from [initial colour] to [final colour] **note that for decolorises/fades, must state original colour, e.g. blue solution fades	Solution remained [colour]	Effervescence produced, forming a [colour & odour] gas. When [gas test], [positive test result] [identity of gas] produced	No gas formed
Example	[Cu ²⁺ under NH ₃ (aq)] Light blue ppt observed Soluble in excess to give a dark blue solution	No ppt. formed, solution remained colourless	Mixture changed from purple to colourless	Solution remained colourless	[NH ₃ gas] Effervescence produced, forming a colourless & pungent gas When damp red litmus paper is placed at mouth of test tube, it turns blue ammonia/NH ₃ produced	No gas formed

Observations like temperature changes should be written e.g. test tube feels warmer, etc.

****Make sure you write ALL observations after every step, even when no change is observed. e.g. for anion tests: “no effervescence observed when HNO₃ added, solution remained colourless”**



QA Tips - Observation Writing

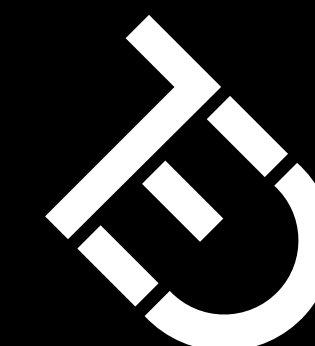
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When qns ask about the identity of the unknown substance that you have done tests on, write as so:

When test [test number] is conducted, as [positive test result] when [test], [cation/anion present].

e.g. When test 3 is conducted, as damp red litmus paper turns blue when X is gently heated with NaOH and aluminium foil, NO_3^- is present.

When test 5 is conducted, as an insoluble in excess blue ppt. of $Cu(OH)_2$ was formed dropwise, Cu^{2+} is present



Titration Tips

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Make sure you remember:

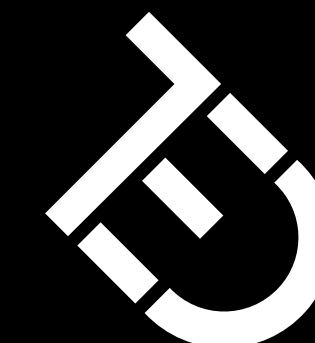
the titration table & headings

how to use a pipette, pipette filler and burette

Placing the white tile under the flask !!!!!!! Please

Removing the funnel from the burette after filling it !!!!!

Your first titration should be a rough estimation/overestimation to have a rough gauge of how much of the substance in the burette should be used
The next two titrations should be as accurate as possible, with the titrated substance being added drop-wise



Graphing Tips

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Graphing has two parts; the table and the graph

Make sure your table has:

Appropriate **headings** in the form of quantity/unit, e.g. time taken/s (qn usually states what headings and how many you need to write)

Correct **d.p** for the values (depends on instrument used)

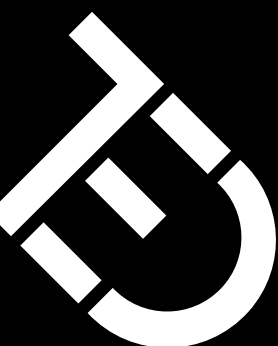
Make sure that your graph has:

Appropriate scale; allows graph to cover minimum $\frac{1}{2}$ of the graph paper

Appropriate axes; with equal intervals of correct d.p. with headings at the end of the axes

Smooth line; no double lines, no gaps, no sharp corners

Clear points plotted; clear X marking each point.



Planning Tips

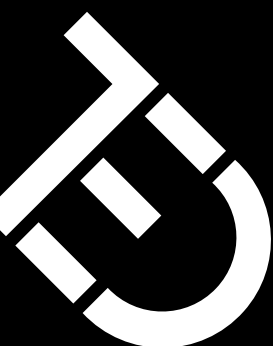
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For planning questions:

Make sure you know what they are asking for, is it obtaining a specific substance, determining a value from an experiment, or comparing different substances.

Make sure each step has CLEAR instructions with apparatus referenced and DEFINITE amounts written. (Including volumes, mass, concentration, time, etc.)
(e.g. Measure out 2.50g of $CuSO_4$ with a electronic mass balance. Transfer to a conical flask)

Check if the question needs anything specific; does it require a specific method you need to use/ is there another part that you have to answer



Planning Tips

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Example 1: Electrolysis is a method used to purify certain types of metals, like magnesium. By using another reaction other than electrolysis, outline a method to determine the percentage purity of a given piece of impure magnesium metal.

Example 2: Magnesium sulfate is a salt commonly used in the medical, agriculture and food industry. Outline a method to obtain magnesium sulfate using a suitable chemical reaction and method.

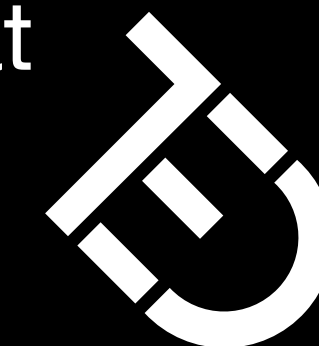
Example 3: Two solid catalysts, X and Y, are said to speed up the decomposition of hydrogen peroxide $[2H_2O_2 \rightarrow 2H_2O + O_2]$. Outline a method to find which catalyst speeds up the reaction more.

These examples are possible planning questions.

Example 1 talks about finding percentage purity, so your final step should show how you use your recordings to calculate percentage purity. It also says you cannot use electrolysis, so any mention of electrolysis will be marked down.

Example 2 talks about obtaining a magnesium sulfate salt, so the final step should be how you obtain the salt from the reaction once the reaction is done. Note that reactions have to be practical; i.e. safe enough to be conducted in a school laboratory.

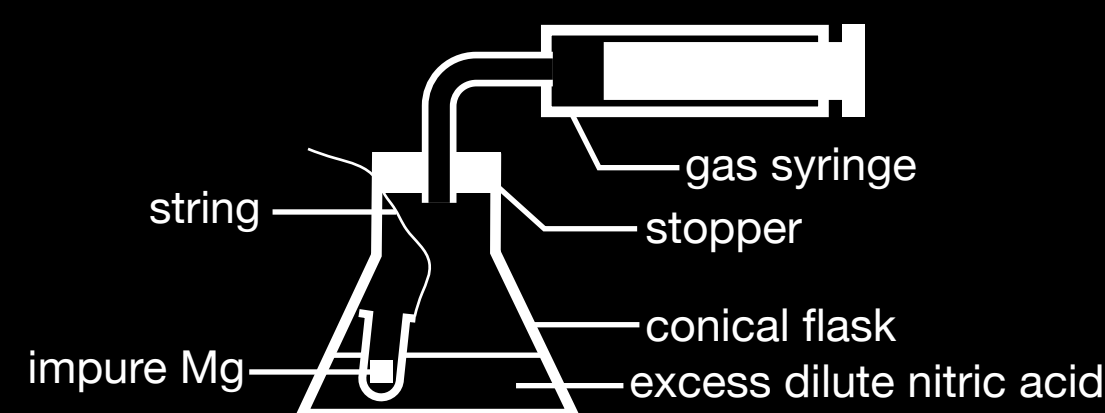
Example 3 talks about comparing two catalysts, so the final step should be a conclusion about what the recorded data says about X and Y.



Planning Tips

House makes me feel at home

Example 1: Electrolysis is a method used to purify certain types of metals, like magnesium. By using another reaction other than electrolysis, outline a method to determine the percentage purity of a given piece of impure magnesium metal.



1. Use an electronic mass balance to measure the mass of the impure magnesium. Record the mass.
2. Set up the experiment as shown. Pull the string and wait for the reaction to be completed when no more effervescence is observed. Record the volume of gas produced.
3. Thus, to find the percentage purity,

$$\text{Mg} + 2\text{HNO}_3 \rightarrow \text{Mg}(\text{NO}_3)_2 + \text{H}_2$$

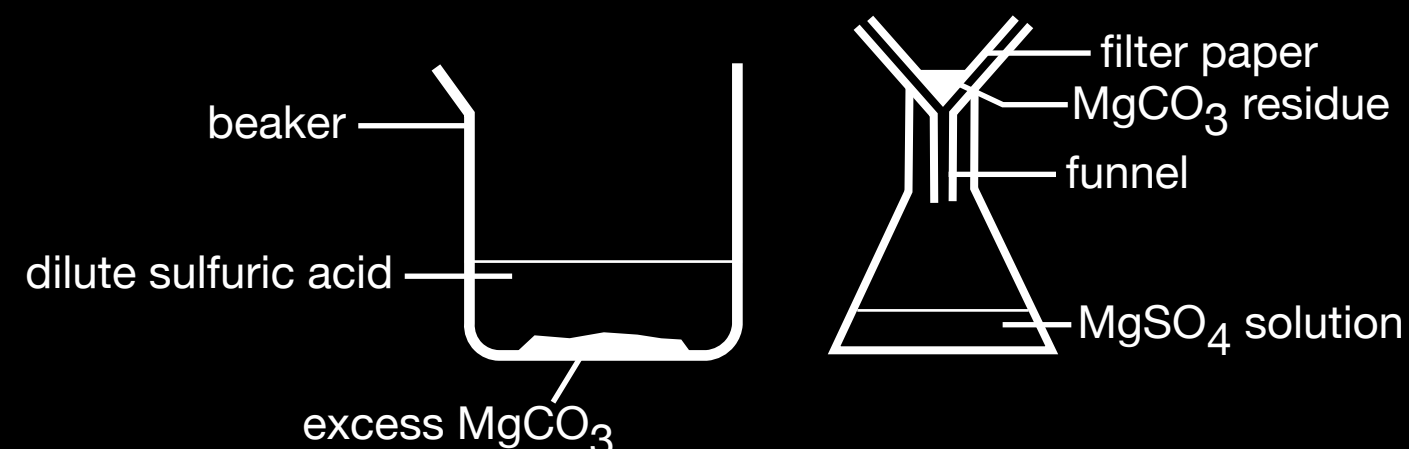
$$\text{Moles of H}_2 = \text{Volume of gas} / 24$$

$$\text{Moles of Mg} = \text{Moles of H}_2$$

$$\text{Mass of pure Mg} = \text{Moles of Mg} \times \text{Mr of Mg}$$

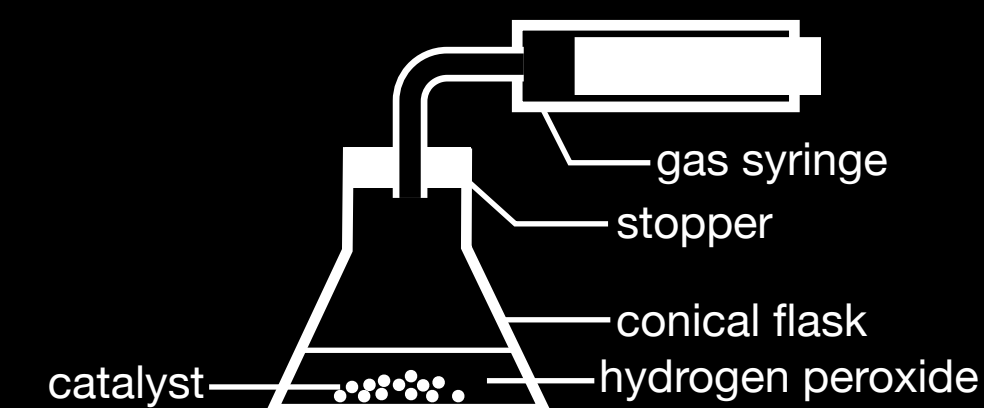
$$\text{Percentage purity} = (\text{mass of Mg}) / (\text{mass of impure Mg}) \times 100\%$$

Example 2: Magnesium sulfate is a salt commonly used in the medical, agriculture and food industry. Outline a method to obtain magnesium sulfate using a suitable chemical reaction and method.

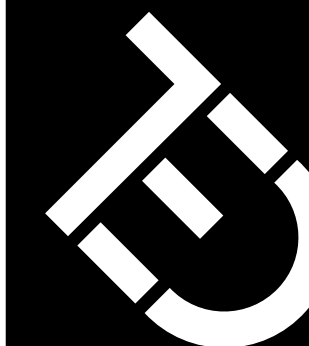


1. Measure out 15.0cm³ of dilute sulfuric acid with a measuring cylinder. Transfer to a beaker
2. Add magnesium carbonate in excess to the sulfuric acid until an excess magnesium carbonate forms at the bottom of the beaker
3. Filter out the excess using filter paper and a filter funnel and obtain the filtrate.
4. Crystallise the filtrate to obtain the magnesium sulfate salt. Wash with a little deionised water and dry the salt between sheets of filter paper.

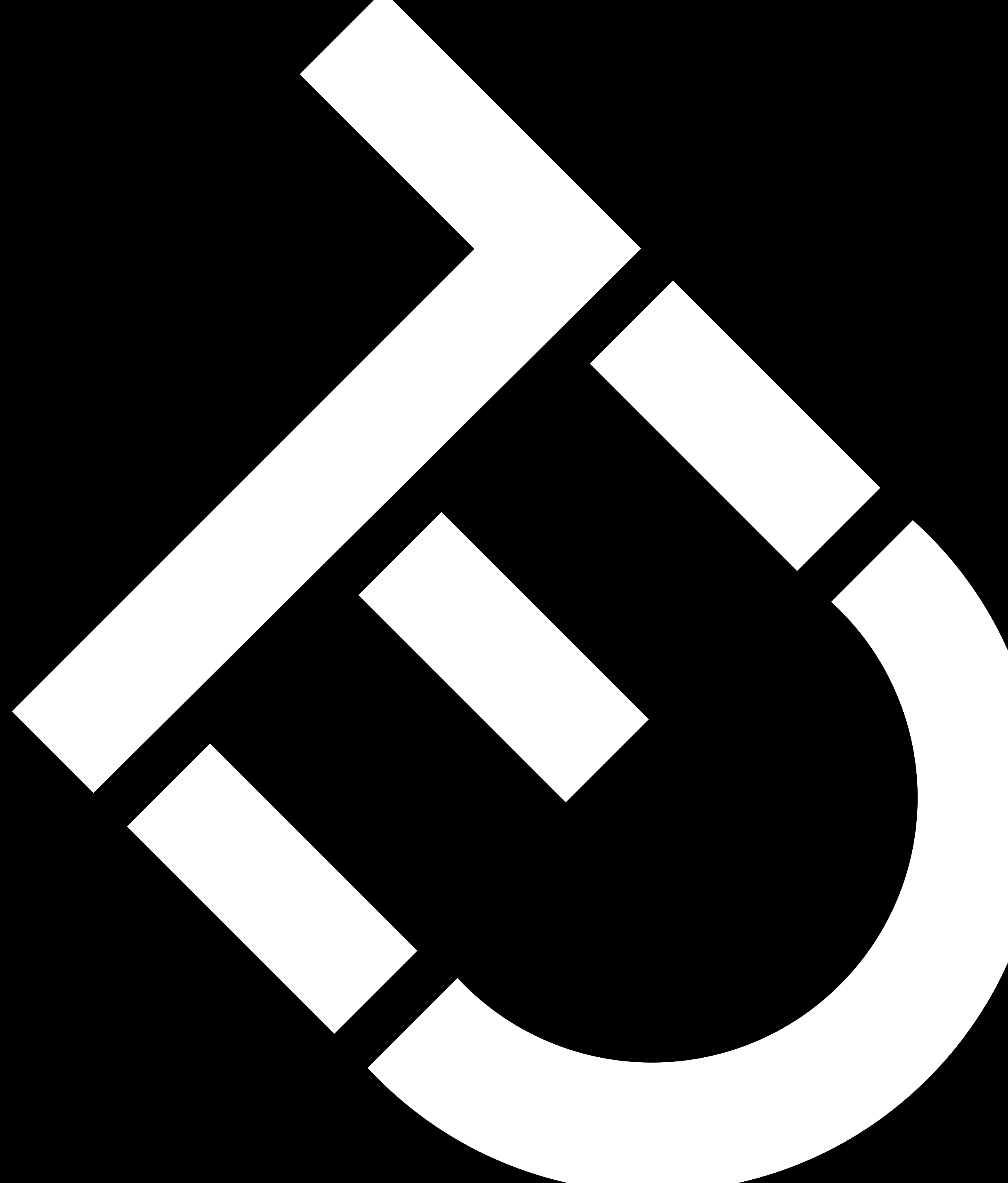
Example 3: Two solid catalysts, X and Y, are said to speed up the decomposition of hydrogen peroxide. Outline a method to find which catalyst speeds up the reaction more.



1. Measure out 15.0cm³ of hydrogen peroxide with a measuring cylinder. Transfer to the setup shown.
2. Measure out 3.0g of catalyst X with an electronic mass balance and add to the hydrogen peroxide.
3. Use a stopwatch to record the time taken for 5cm³ of gas to be formed. Record the elapsed time
4. Repeat steps 1 and 2, except with catalyst Y added.
5. Conclusion: If catalyst X takes a shorter time to form 5cm³ of gas than Y, catalyst X speeds up the reaction more. Else, catalyst Y speeds up the reaction more.



Exam Skills

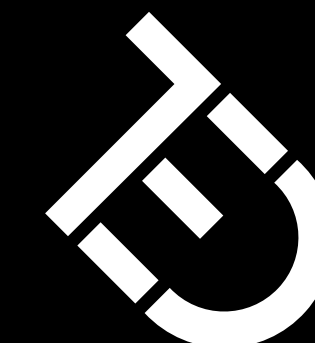


Calculations

ca

Intermediate steps should be written clearly (preferably showing what you're calculating, e.g. Moles of X = $0.1\text{dm}^3 \times 0.25\text{mol/dm}^3 = \dots$), with non exact values to 5.s.f

For the final answer, if non exact, leave as 3.s.f. Otherwise, leave as exact decimal (or to specified accuracy)



Data Based Questions

ba

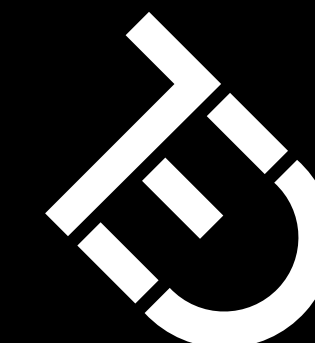
Information questions tend to make use of unfamiliar information to overwhelm you

For these, first identify what the topic(s) is/what topic(s) is most related to it (usually the s4 topics, organic chem, electrolysis, as they can extent the o lvl syllabus to outside topics)

Afterwards, you can read the first few lines (to understand what the general context of the info is) and then jump to the questions and find its corresponding info needed there (this works as each part helps to lead you into the next part of the question)

This helps to reduce time wasted on reading parts not relevant to the question part you are working on

If you find yourself stuck, try reading through other parts that you may not have read yet as info may be there that can help you to answer the qn.



Data Based Questions

da

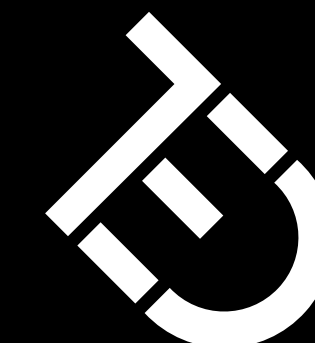
Data based questions can be identified by “Using the data shown”/“Using the table given”, “Using Fig X”.

Thus, MAKE SURE TO CITE DATA otherwise marks will not be awarded.
e.g. As the boiling point of X is higher than Y ($100^{\circ}\text{C} > 87^{\circ}\text{C}$), Y will be first collected in the distillate vessel.

OR

As the boiling point of X is 100°C , which is higher than the boiling point of Y of 87°C , Y will be first collected in the distillate vessel

Qns like these are usually 2 marks, 1 for data citation + 1 for concept



done

